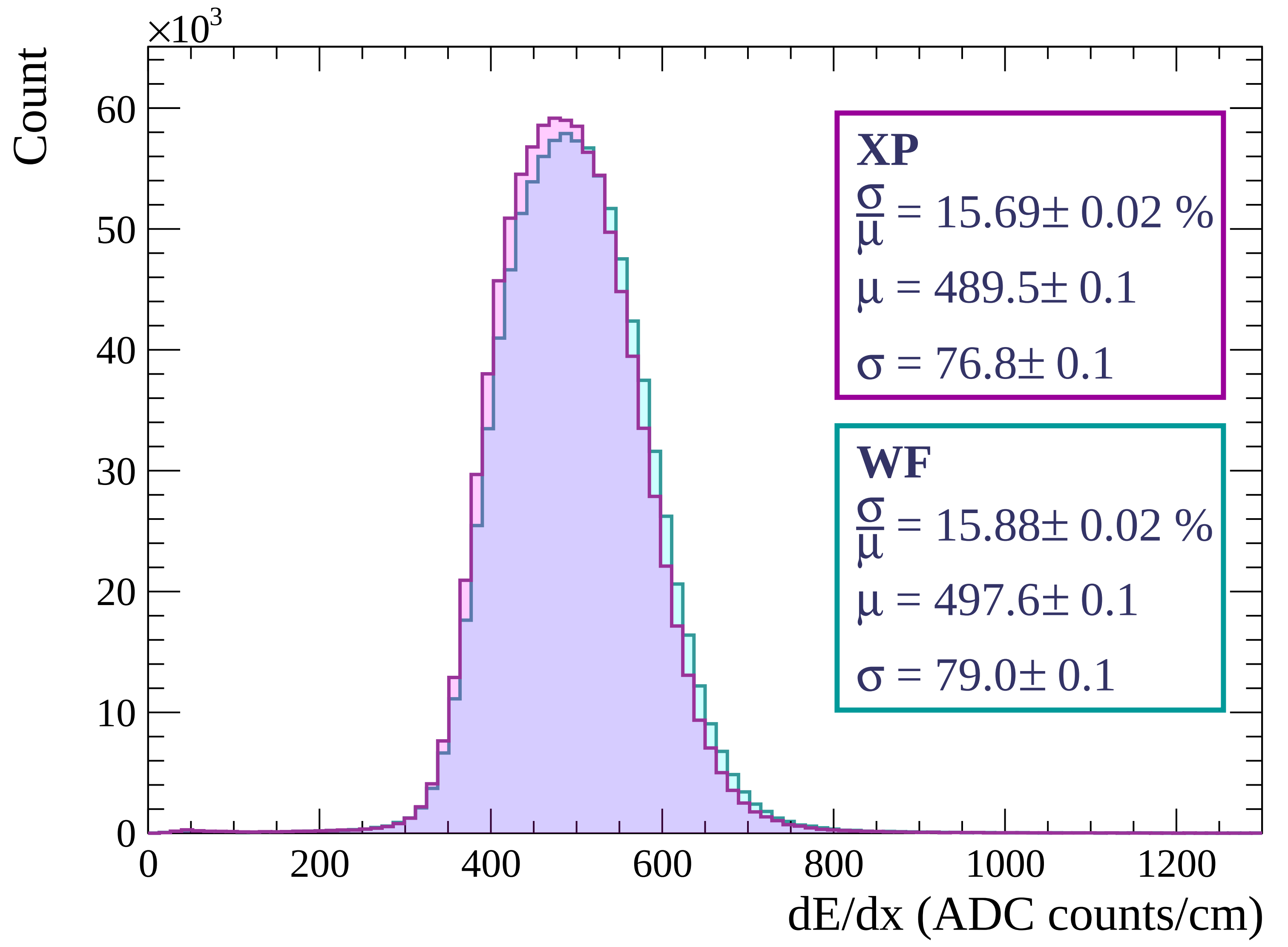
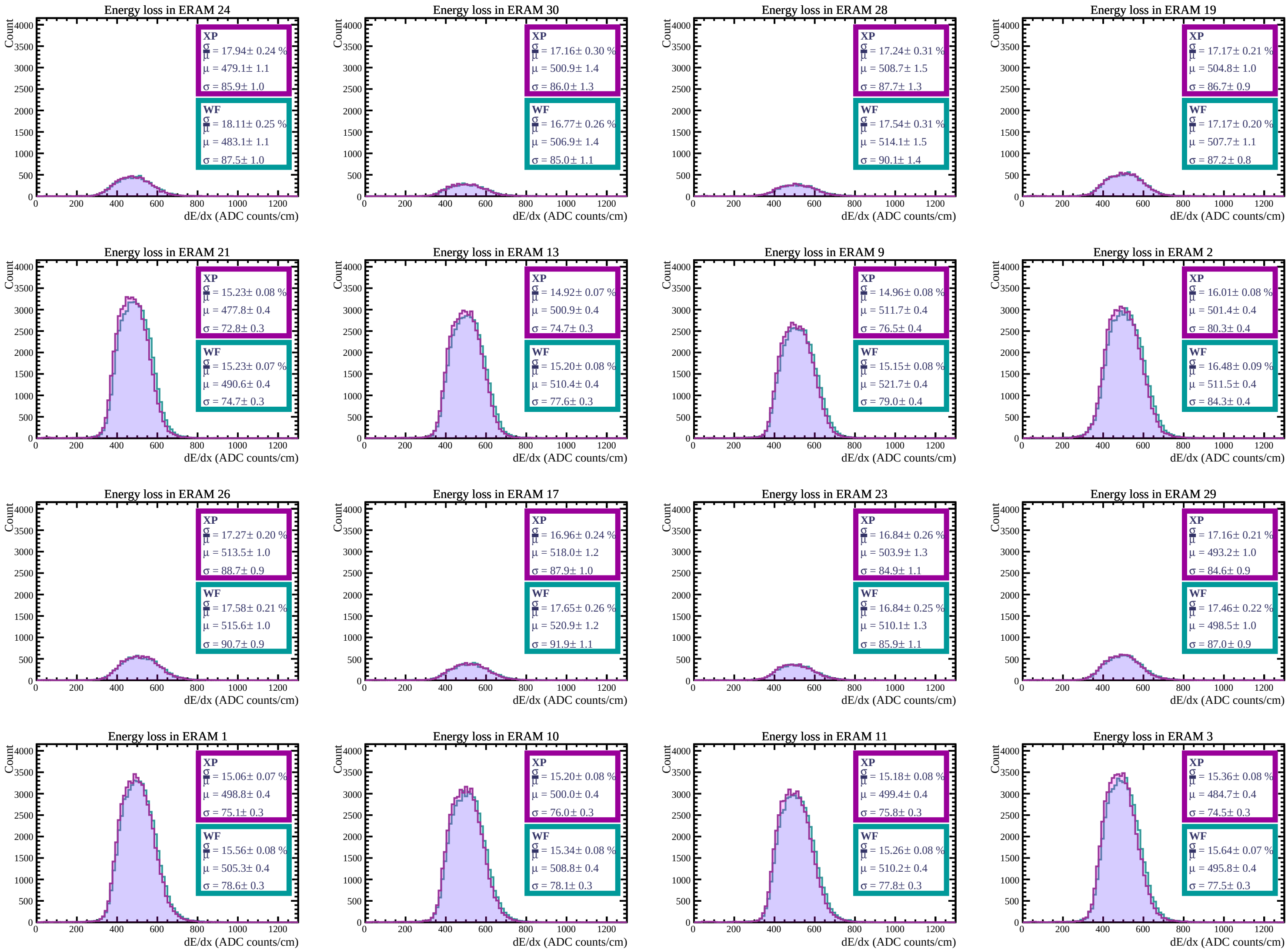
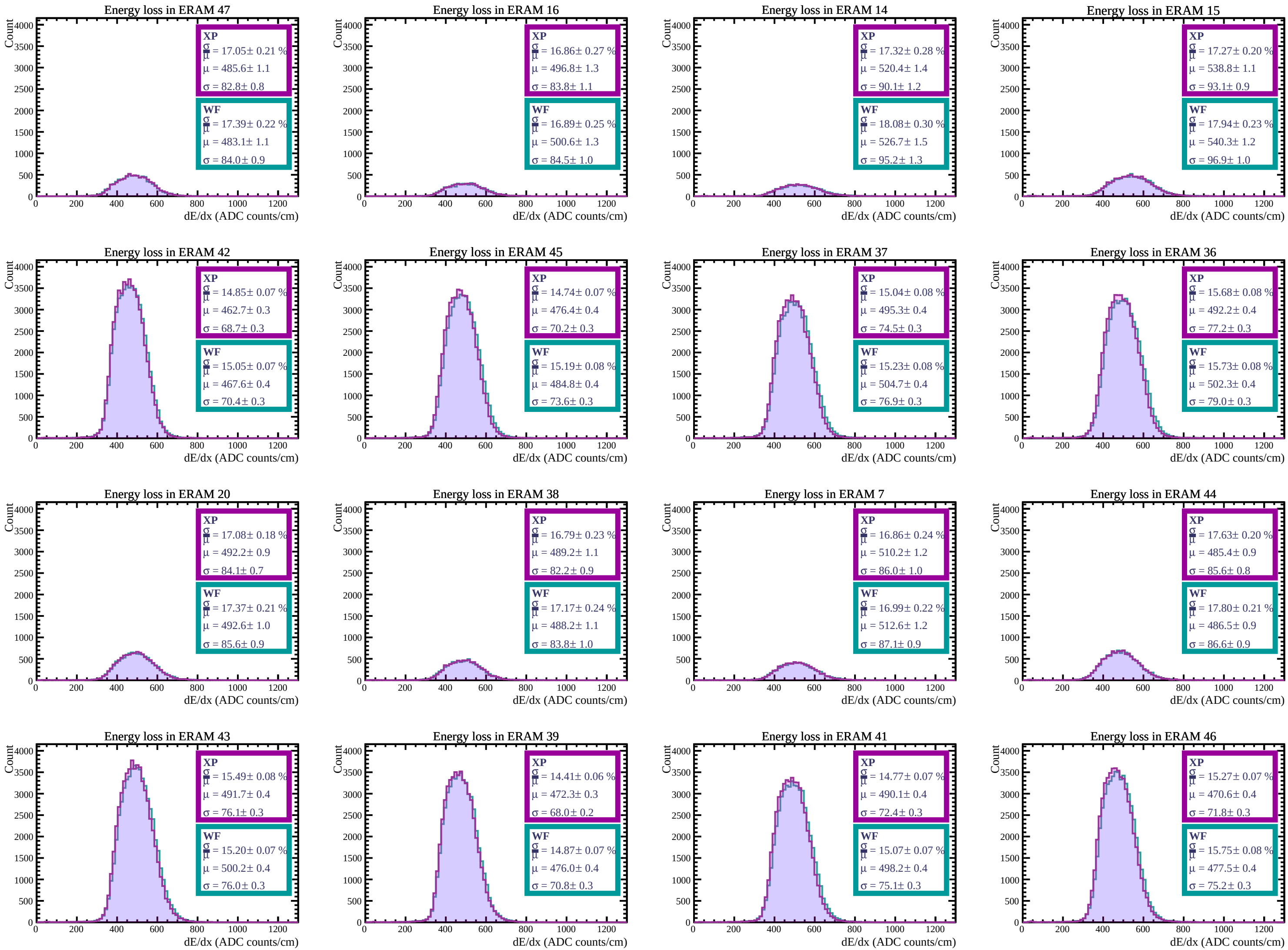
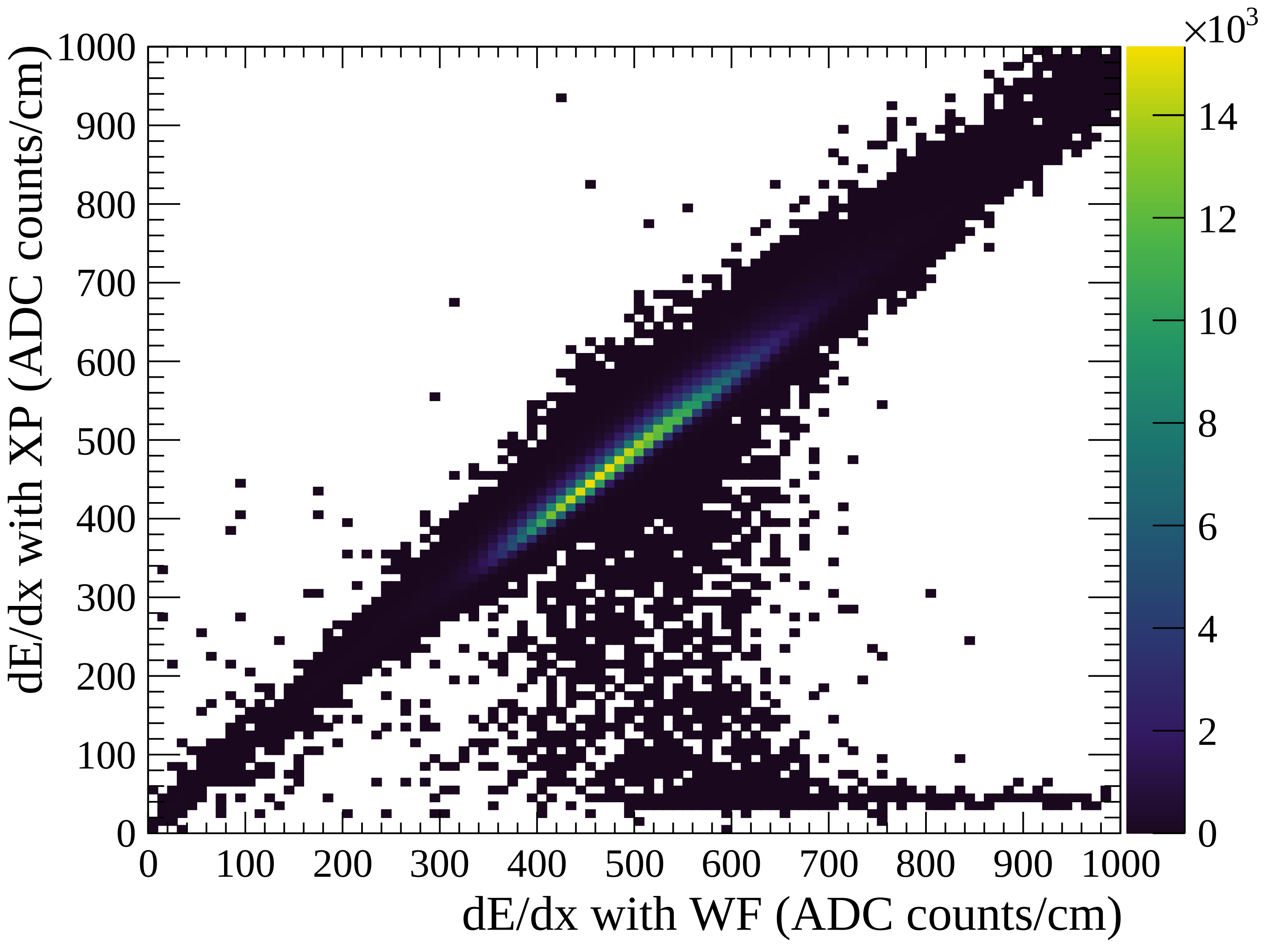
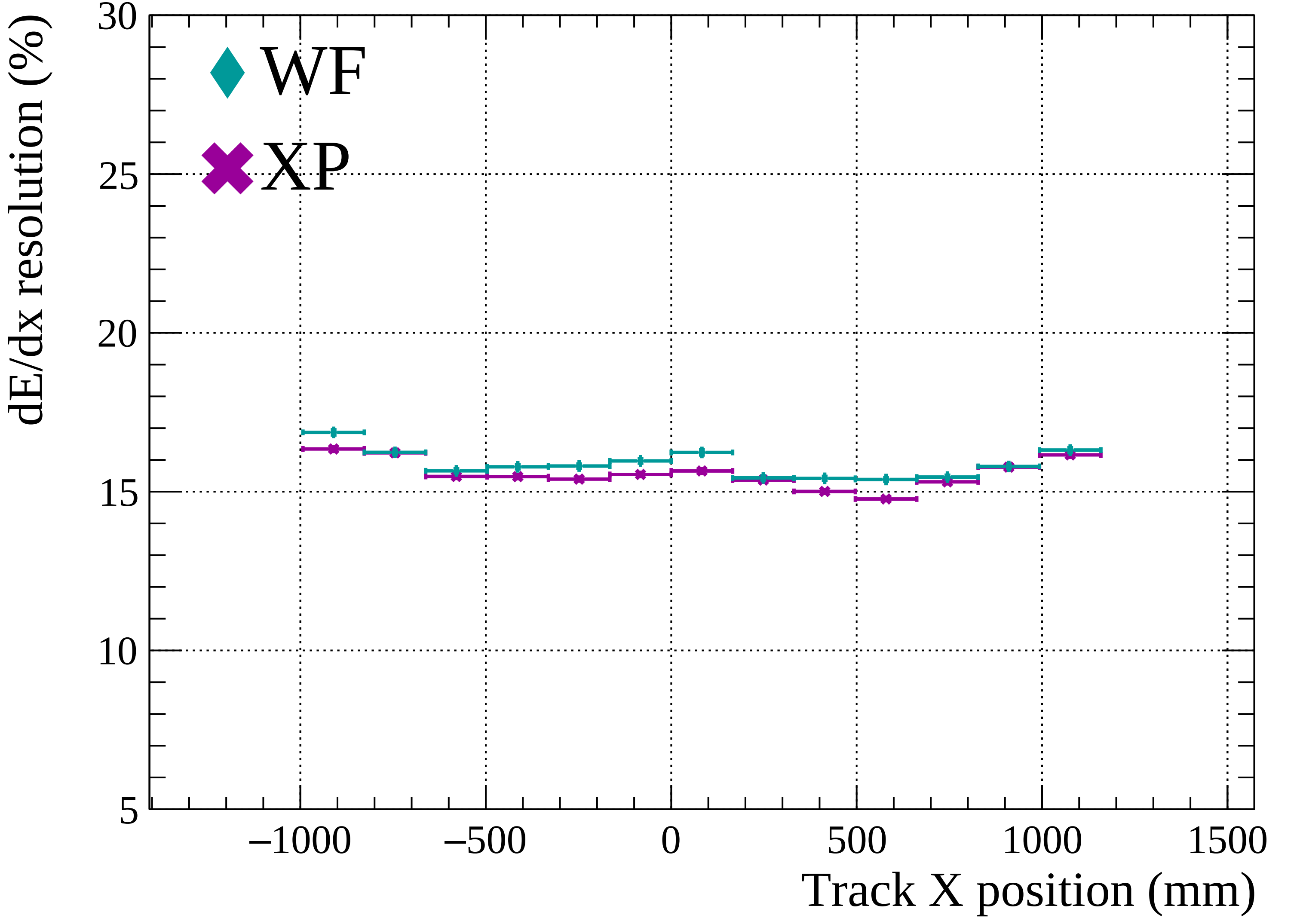

Count
×10^3
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.69± 0.02 %
μ = 489.5± 0.1
σ = 76.8± 0.1
WF
σ/μ = 15.88± 0.02 %
μ = 497.6± 0.1
σ = 79.0± 0.1



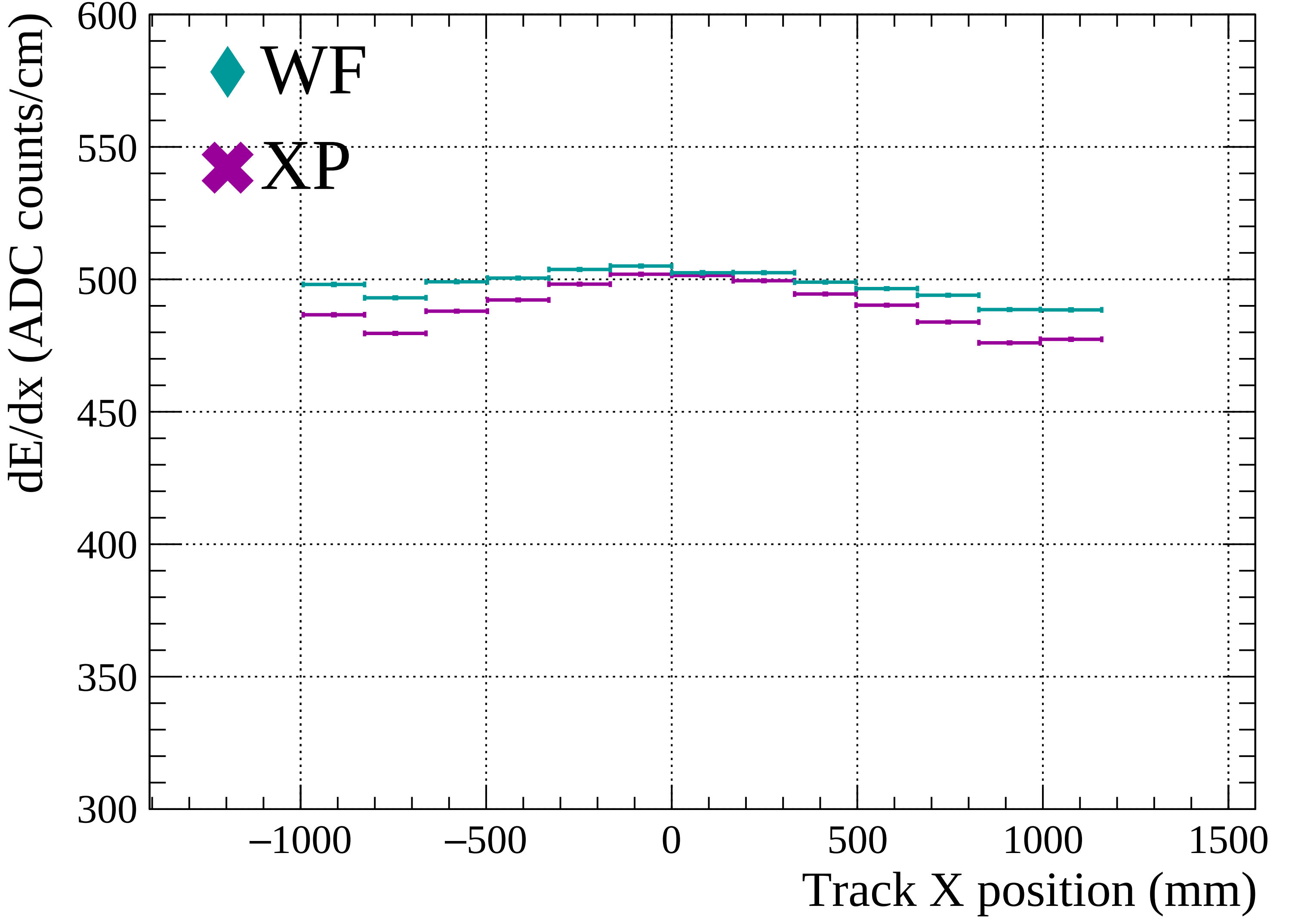
Energy loss in ERAM 24
XP σ/μ = 17.94± 0.24 % μ = 479.1± 1.1 σ = 85.9± 1.0
WF σ/μ = 18.11± 0.25 % μ = 483.1± 1.1 σ = 87.5± 1.0
Energy loss in ERAM 30
XP σ/μ = 17.16± 0.30 % μ = 500.9± 1.4 σ = 86.0± 1.3
WF σ/μ = 16.77± 0.26 % μ = 506.9± 1.4 σ = 85.0± 1.1
Energy loss in ERAM 28
XP σ/μ = 17.24± 0.31 % μ = 508.7± 1.5 σ = 87.7± 1.3
WF σ/μ = 17.54± 0.31 % μ = 514.1± 1.5 σ = 90.1± 1.4
Energy loss in ERAM 19
XP σ/μ = 17.17± 0.21 % μ = 504.8± 1.0 σ = 86.7± 0.9
WF σ/μ = 17.17± 0.20 % μ = 507.7± 1.1 σ = 87.2± 0.8
Energy loss in ERAM 21
XP σ/μ = 15.23± 0.08 % μ = 477.8± 0.4 σ = 72.8± 0.3
WF σ/μ = 15.23± 0.07 % μ = 490.6± 0.4 σ = 74.7± 0.3
Energy loss in ERAM 13
XP σ/μ = 14.92± 0.07 % μ = 500.9± 0.4 σ = 74.7± 0.3
WF σ/μ = 15.20± 0.08 % μ = 510.4± 0.4 σ = 77.6± 0.3
Energy loss in ERAM 9
XP σ/μ = 14.96± 0.08 % μ = 511.7± 0.4 σ = 76.5± 0.4
WF σ/μ = 15.15± 0.08 % μ = 521.7± 0.4 σ = 79.0± 0.4
Energy loss in ERAM 2
XP σ/μ = 16.01± 0.08 % μ = 501.4± 0.4 σ = 80.3± 0.4
WF σ/μ = 16.48± 0.09 % μ = 511.5± 0.4 σ = 84.3± 0.4
Energy loss in ERAM 26
XP σ/μ = 17.27± 0.20 % μ = 513.5± 1.0 σ = 88.7± 0.9
WF σ/μ = 17.58± 0.21 % μ = 515.6± 1.0 σ = 90.7± 0.9
Energy loss in ERAM 17
XP σ/μ = 16.96± 0.24 % μ = 518.0± 1.2 σ = 87.9± 1.0
WF σ/μ = 17.65± 0.26 % μ = 520.9± 1.2 σ = 91.9± 1.1
Energy loss in ERAM 23
XP σ/μ = 16.84± 0.26 % μ = 503.9± 1.3 σ = 84.9± 1.1
WF σ/μ = 16.84± 0.25 % μ = 510.1± 1.3 σ = 85.9± 1.1
Energy loss in ERAM 29
XP σ/μ = 17.16± 0.21 % μ = 493.2± 1.0 σ = 84.6± 0.9
WF σ/μ = 17.46± 0.22 % μ = 498.5± 1.0 σ = 87.0± 0.9
Energy loss in ERAM 1
XP σ/μ = 15.06± 0.07 % μ = 498.8± 0.4 σ = 75.1± 0.3
WF σ/μ = 15.56± 0.08 % μ = 505.3± 0.4 σ = 78.6± 0.3
Energy loss in ERAM 10
XP σ/μ = 15.20± 0.08 % μ = 500.0± 0.4 σ = 76.0± 0.3
WF σ/μ = 15.34± 0.08 % μ = 508.8± 0.4 σ = 78.1± 0.3
Energy loss in ERAM 11
XP σ/μ = 15.18± 0.08 % μ = 499.4± 0.4 σ = 75.8± 0.3
WF σ/μ = 15.26± 0.08 % μ = 510.2± 0.4 σ = 77.8± 0.3
Energy loss in ERAM 3
XP σ/μ = 15.36± 0.08 % μ = 484.7± 0.4 σ = 74.5± 0.3
WF σ/μ = 15.64± 0.07 % μ = 495.8± 0.4 σ = 77.5± 0.3
Count
dE/dx (ADC counts/cm)

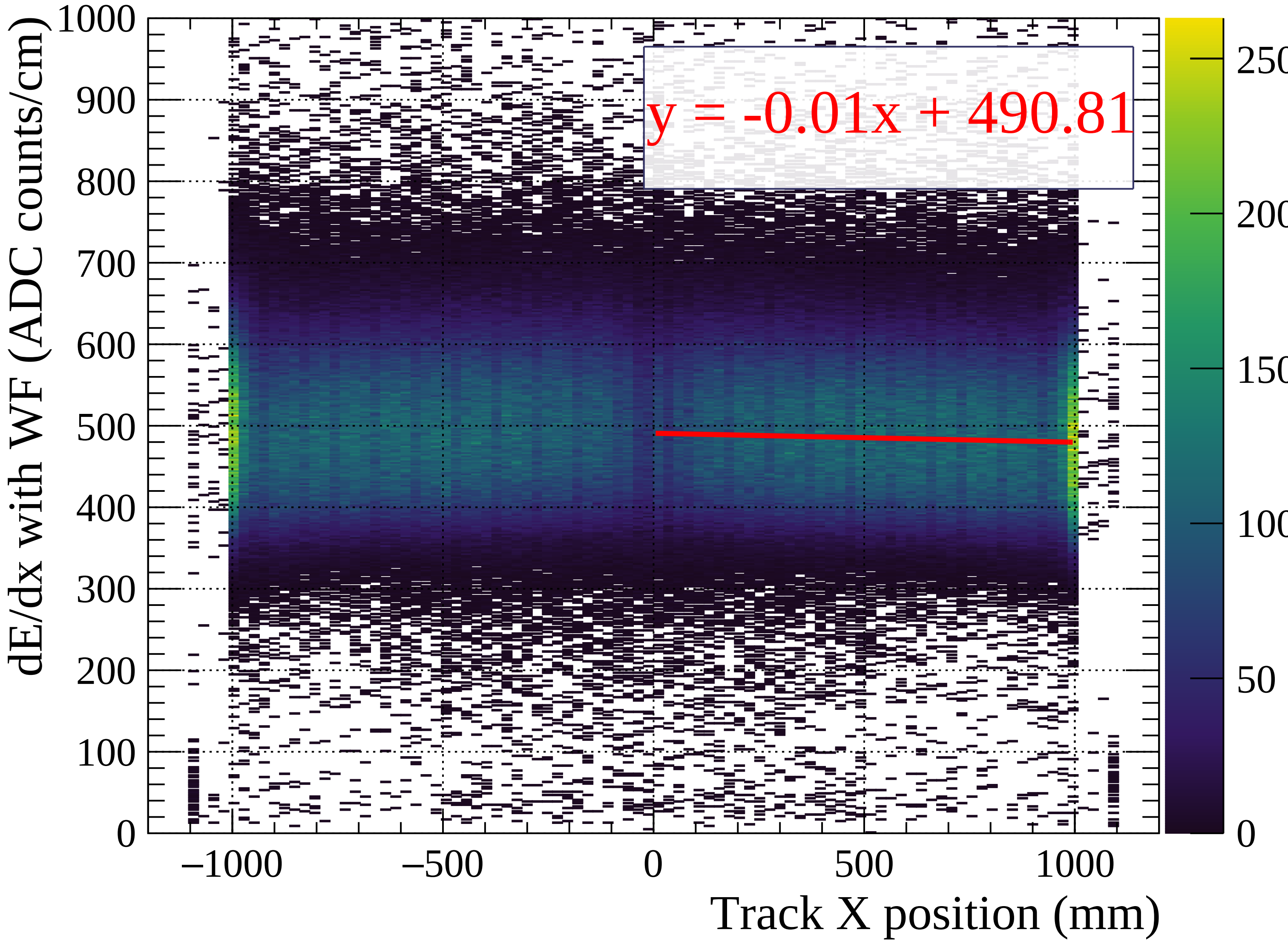

Energy loss in ERAM 47
XP
σ/μ = 17.05± 0.21 %
μ = 485.6± 1.1
σ = 82.8± 0.8
WF
σ/μ = 17.39± 0.22 %
μ = 483.1± 1.1
σ = 84.0± 0.9
Energy loss in ERAM 16
XP
σ/μ = 16.86± 0.27 %
μ = 496.8± 1.3
σ = 83.8± 1.1
WF
σ/μ = 16.89± 0.25 %
μ = 500.6± 1.3
σ = 84.5± 1.0
Energy loss in ERAM 14
XP
σ/μ = 17.32± 0.28 %
μ = 520.4± 1.4
σ = 90.1± 1.2
WF
σ/μ = 18.08± 0.30 %
μ = 526.7± 1.5
σ = 95.2± 1.3
Energy loss in ERAM 15
XP
σ/μ = 17.27± 0.20 %
μ = 538.8± 1.1
σ = 93.1± 0.9
WF
σ/μ = 17.94± 0.23 %
μ = 540.3± 1.2
σ = 96.9± 1.0
Energy loss in ERAM 42
XP
σ/μ = 14.85± 0.07 %
μ = 462.7± 0.3
σ = 68.7± 0.3
WF
σ/μ = 15.05± 0.07 %
μ = 467.6± 0.4
σ = 70.4± 0.3
Energy loss in ERAM 45
XP
σ/μ = 14.74± 0.07 %
μ = 476.4± 0.4
σ = 70.2± 0.3
WF
σ/μ = 15.19± 0.08 %
μ = 484.8± 0.4
σ = 73.6± 0.3
Energy loss in ERAM 37
XP
σ/μ = 15.04± 0.08 %
μ = 495.3± 0.4
σ = 74.5± 0.3
WF
σ/μ = 15.23± 0.08 %
μ = 504.7± 0.4
σ = 76.9± 0.3
Energy loss in ERAM 36
XP
σ/μ = 15.68± 0.08 %
μ = 492.2± 0.4
σ = 77.2± 0.3
WF
σ/μ = 15.73± 0.08 %
μ = 502.3± 0.4
σ = 79.0± 0.3
Energy loss in ERAM 20
XP
σ/μ = 17.08± 0.18 %
μ = 492.2± 0.9
σ = 84.1± 0.7
WF
σ/μ = 17.37± 0.21 %
μ = 492.6± 1.0
σ = 85.6± 0.9
Energy loss in ERAM 38
XP
σ/μ = 16.79± 0.23 %
μ = 489.2± 1.1
σ = 82.2± 0.9
WF
σ/μ = 17.17± 0.24 %
μ = 488.2± 1.1
σ = 83.8± 1.0
Energy loss in ERAM 7
XP
σ/μ = 16.86± 0.24 %
μ = 510.2± 1.2
σ = 86.0± 1.0
WF
σ/μ = 16.99± 0.22 %
μ = 512.6± 1.2
σ = 87.1± 0.9
Energy loss in ERAM 44
XP
σ/μ = 17.63± 0.20 %
μ = 485.4± 0.9
σ = 85.6± 0.8
WF
σ/μ = 17.80± 0.21 %
μ = 486.5± 0.9
σ = 86.6± 0.9
Energy loss in ERAM 43
XP
σ/μ = 15.49± 0.08 %
μ = 491.7± 0.4
σ = 76.1± 0.3
WF
σ/μ = 15.20± 0.07 %
μ = 500.2± 0.4
σ = 76.0± 0.3
Energy loss in ERAM 39
XP
σ/μ = 14.41± 0.06 %
μ = 472.3± 0.3
σ = 68.0± 0.2
WF
σ/μ = 14.87± 0.07 %
μ = 476.0± 0.4
σ = 70.8± 0.3
Energy loss in ERAM 41
XP
σ/μ = 14.77± 0.07 %
μ = 490.1± 0.4
σ = 72.4± 0.3
WF
σ/μ = 15.07± 0.07 %
μ = 498.2± 0.4
σ = 75.1± 0.3
Energy loss in ERAM 46
XP
σ/μ = 15.27± 0.07 %
μ = 470.6± 0.4
σ = 71.8± 0.3
WF
σ/μ = 15.75± 0.08 %
μ = 477.5± 0.4
σ = 75.2± 0.3
Count
dE/dx (ADC counts/cm)


dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3
0
100
200
300
400
500
600
700
800
900
1000
2
4
6
8
10
12
14

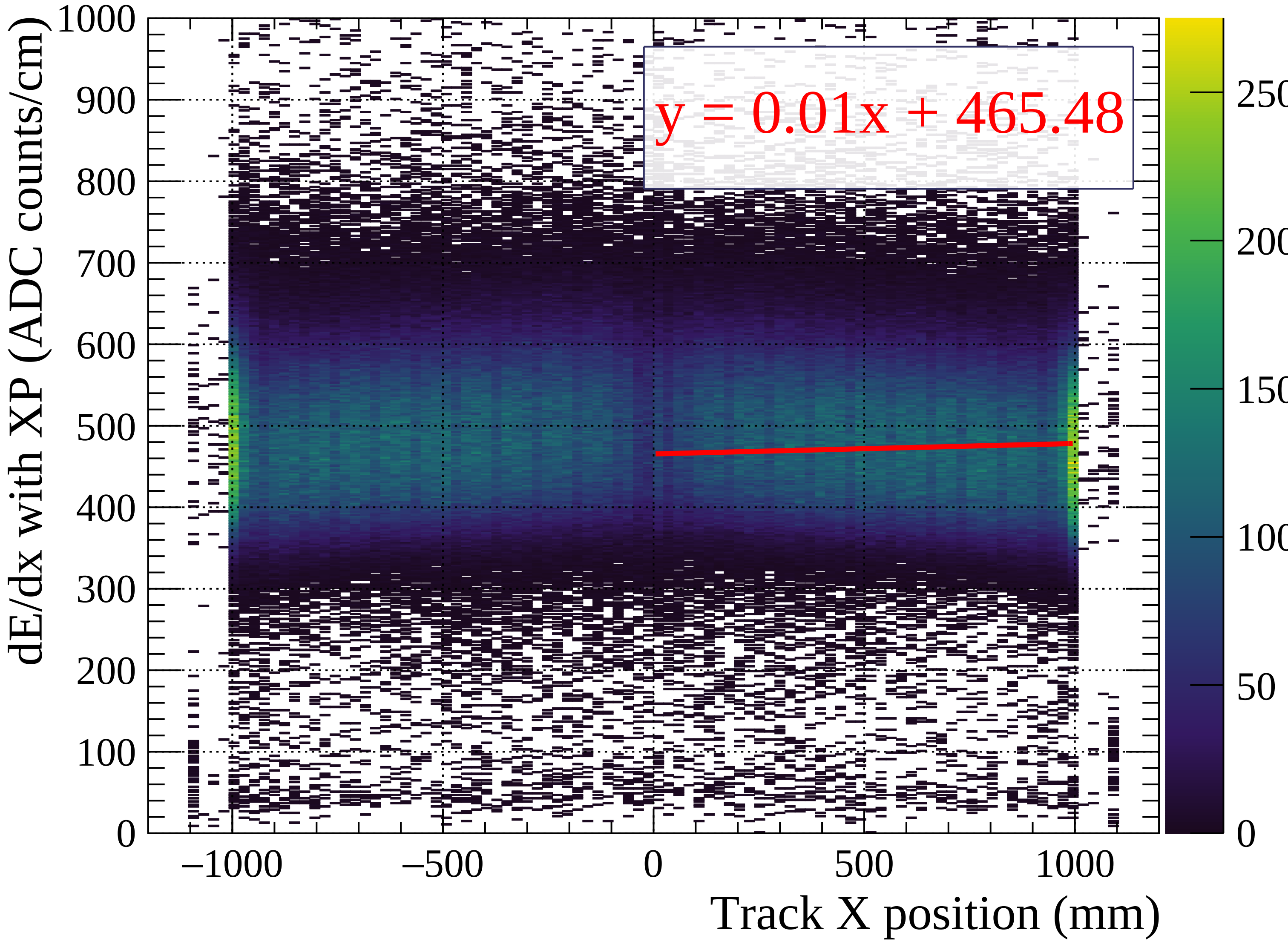


dE/dx resolution (%)
Track X position (mm)
WF
XP
30
25
20
15
10
5
−1000
−500
0
500
1000
1500

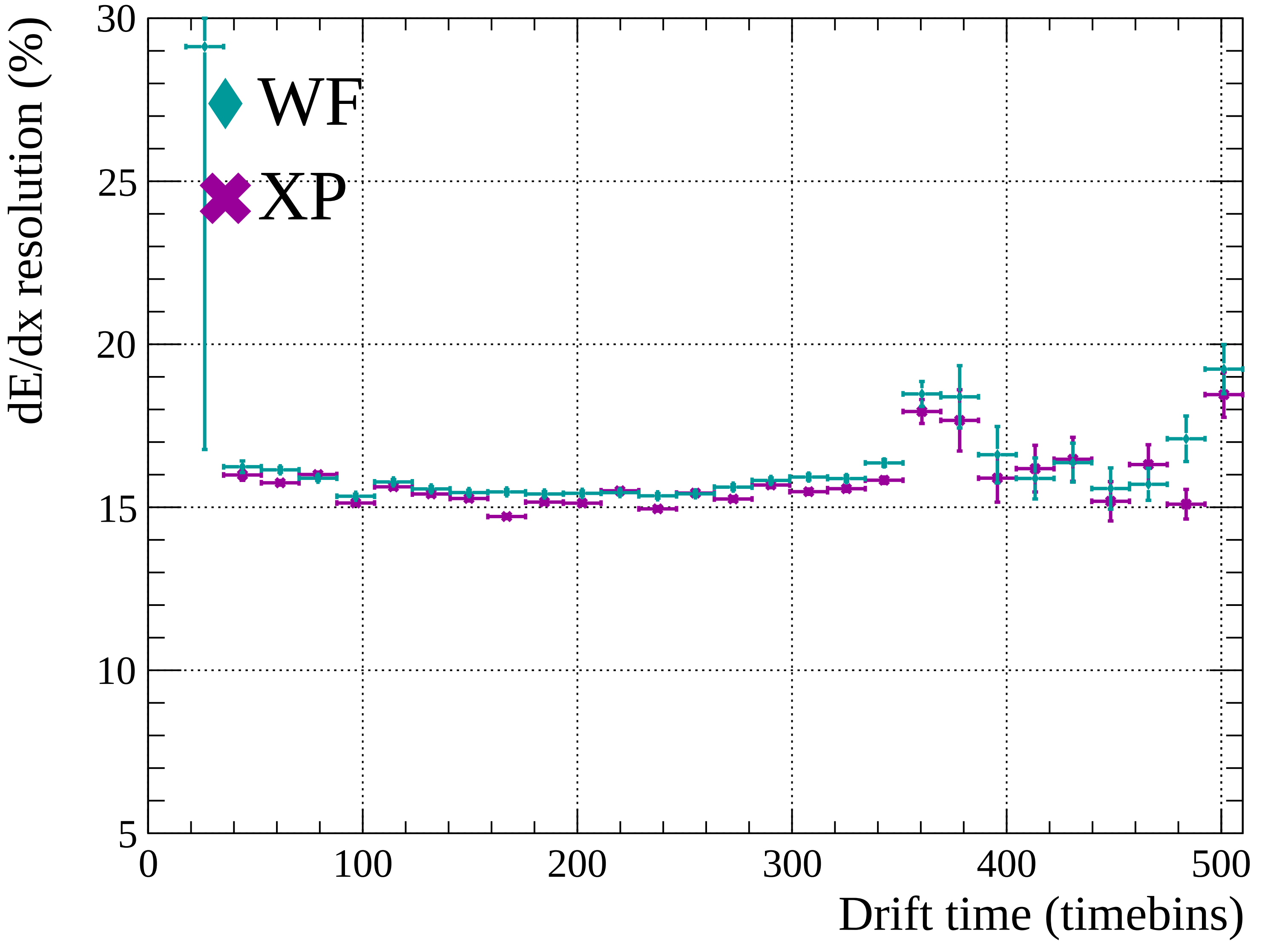

WF
XP
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
1500
Track X position (mm)



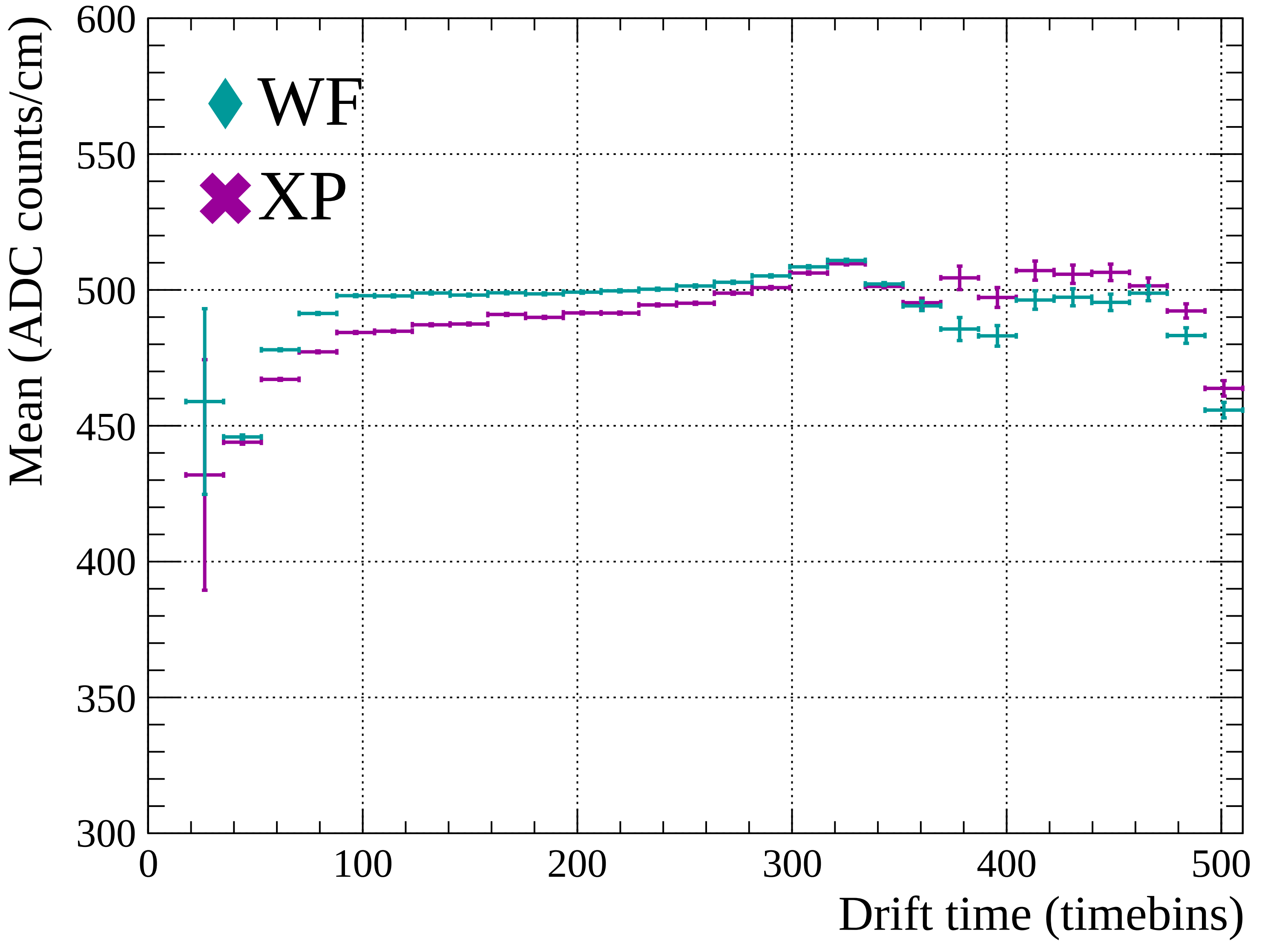
y = -0.01x + 490.81
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
50
100
150
200
250

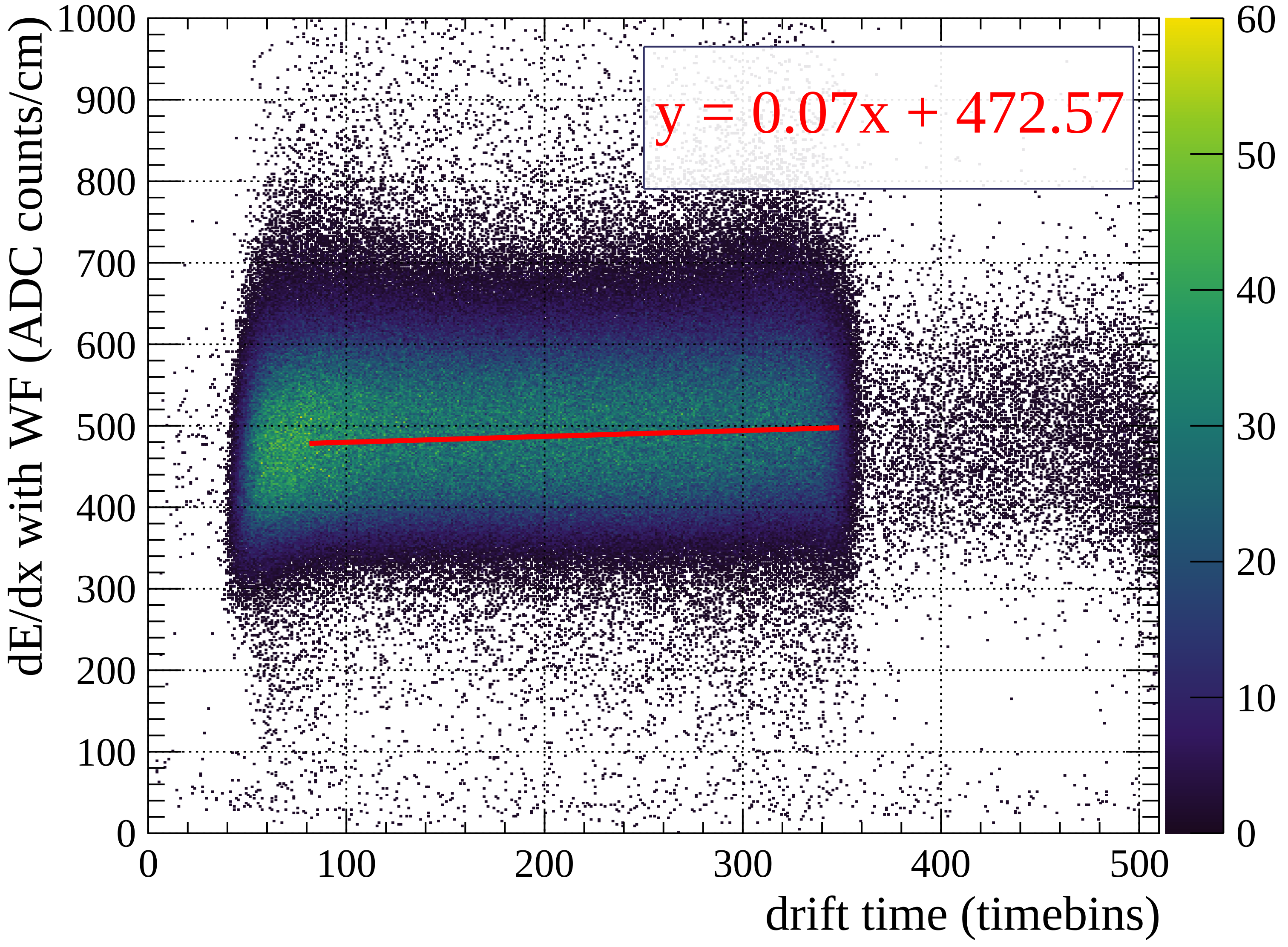


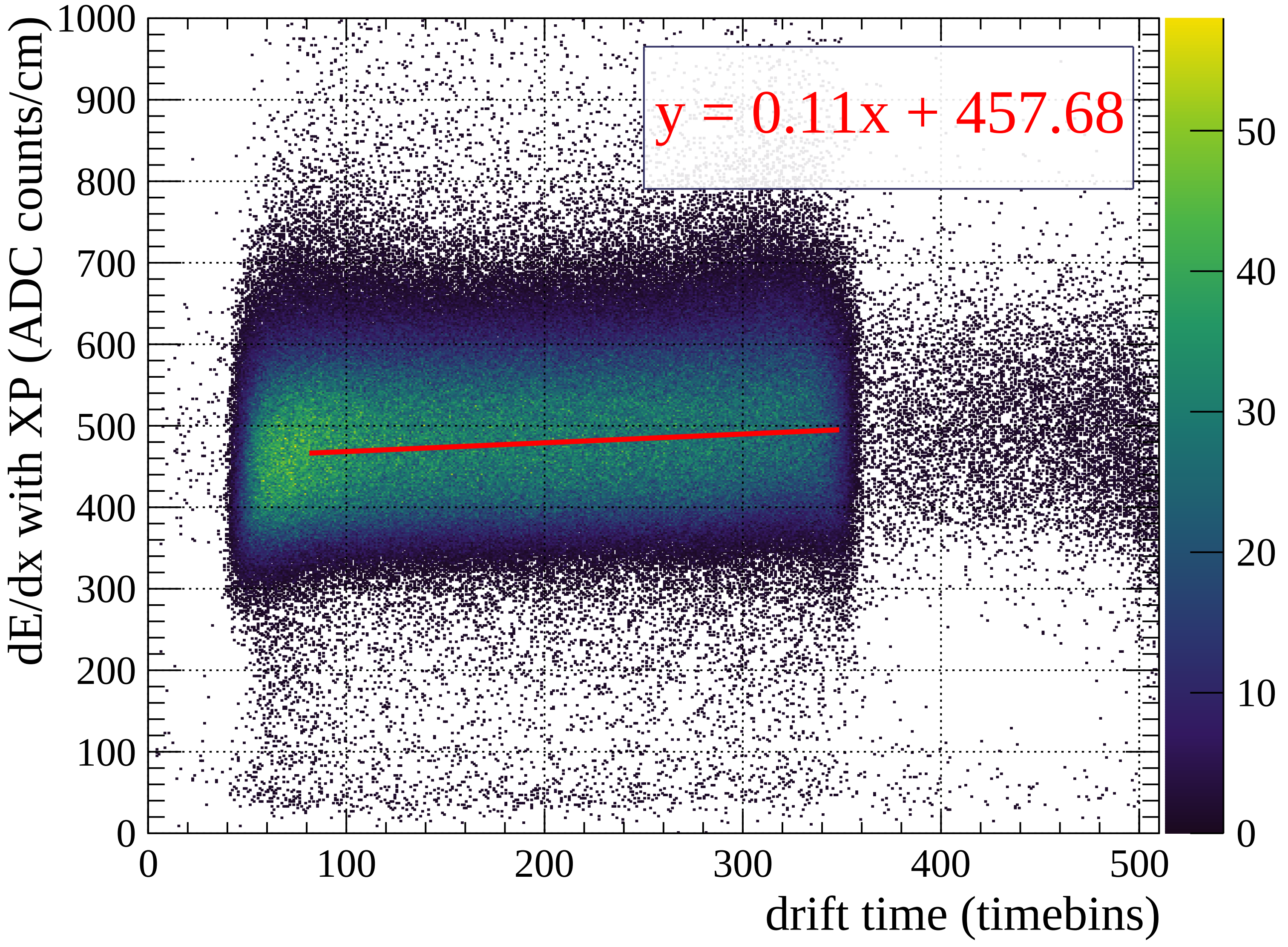
y = 0.01x + 465.48
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
50



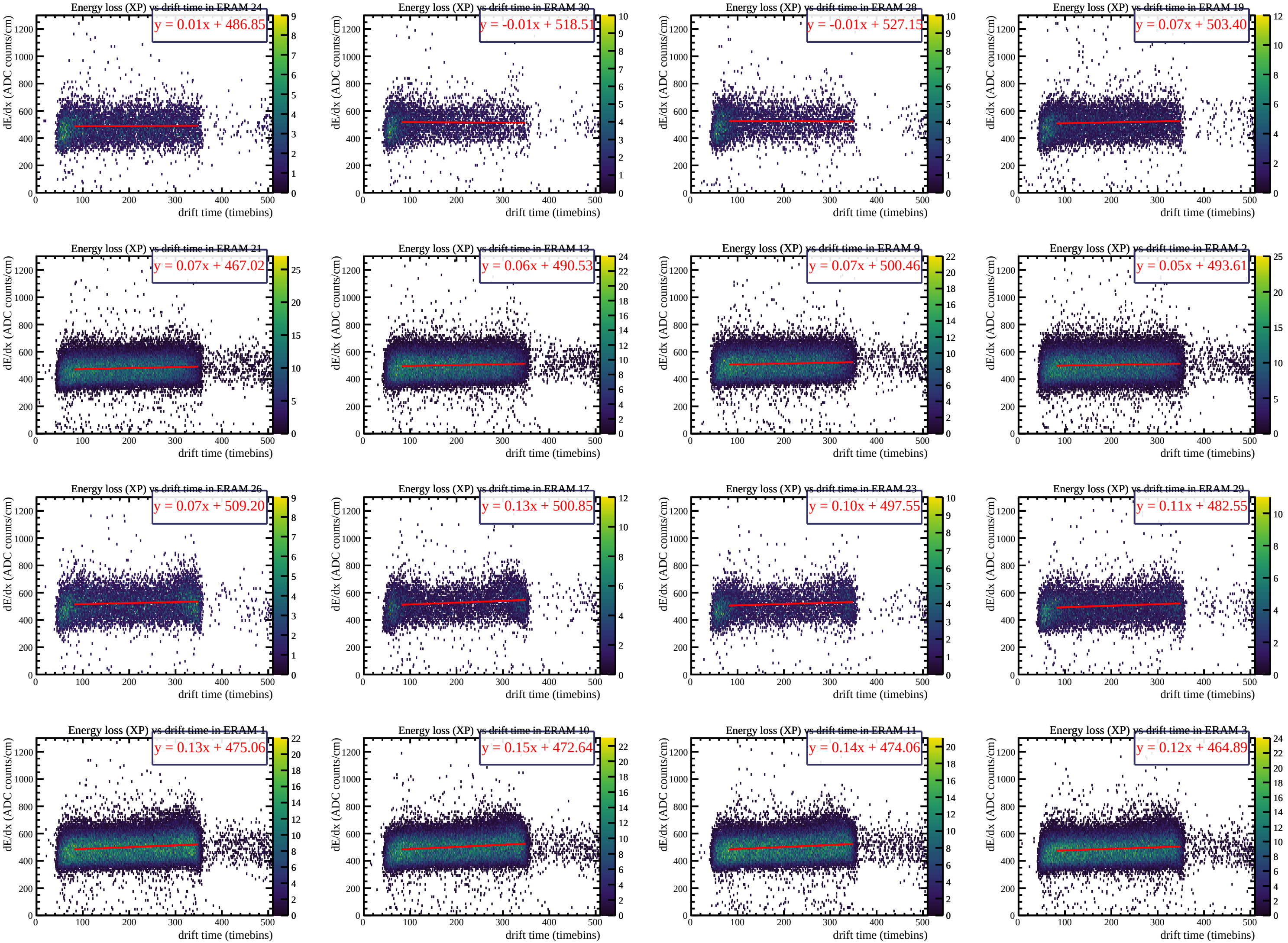
WF
XP
dE/dx resolution (%)
Drift time (timebins)
30
25
20
15
10
5
0
100
200
300
400
500

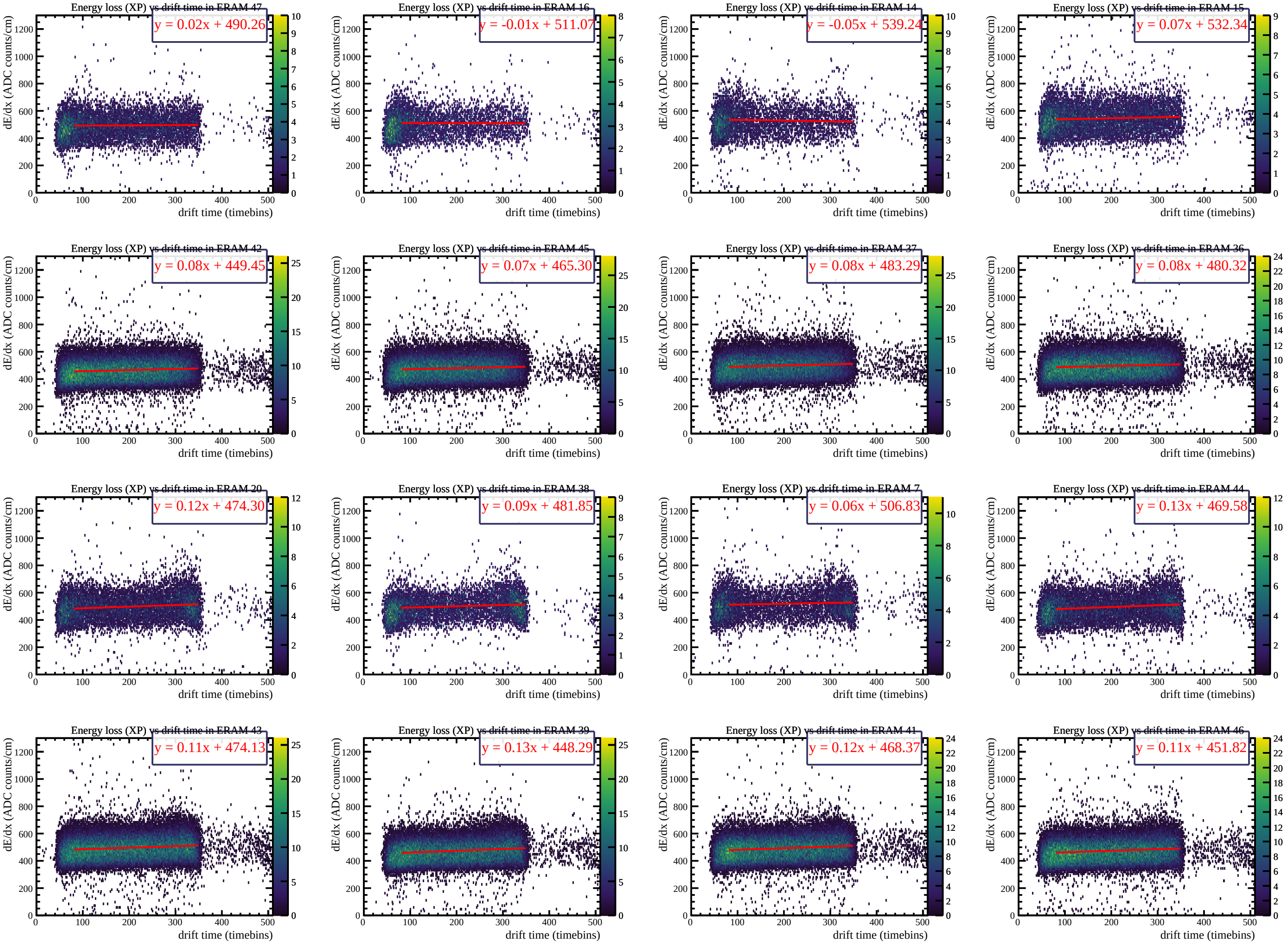

WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500

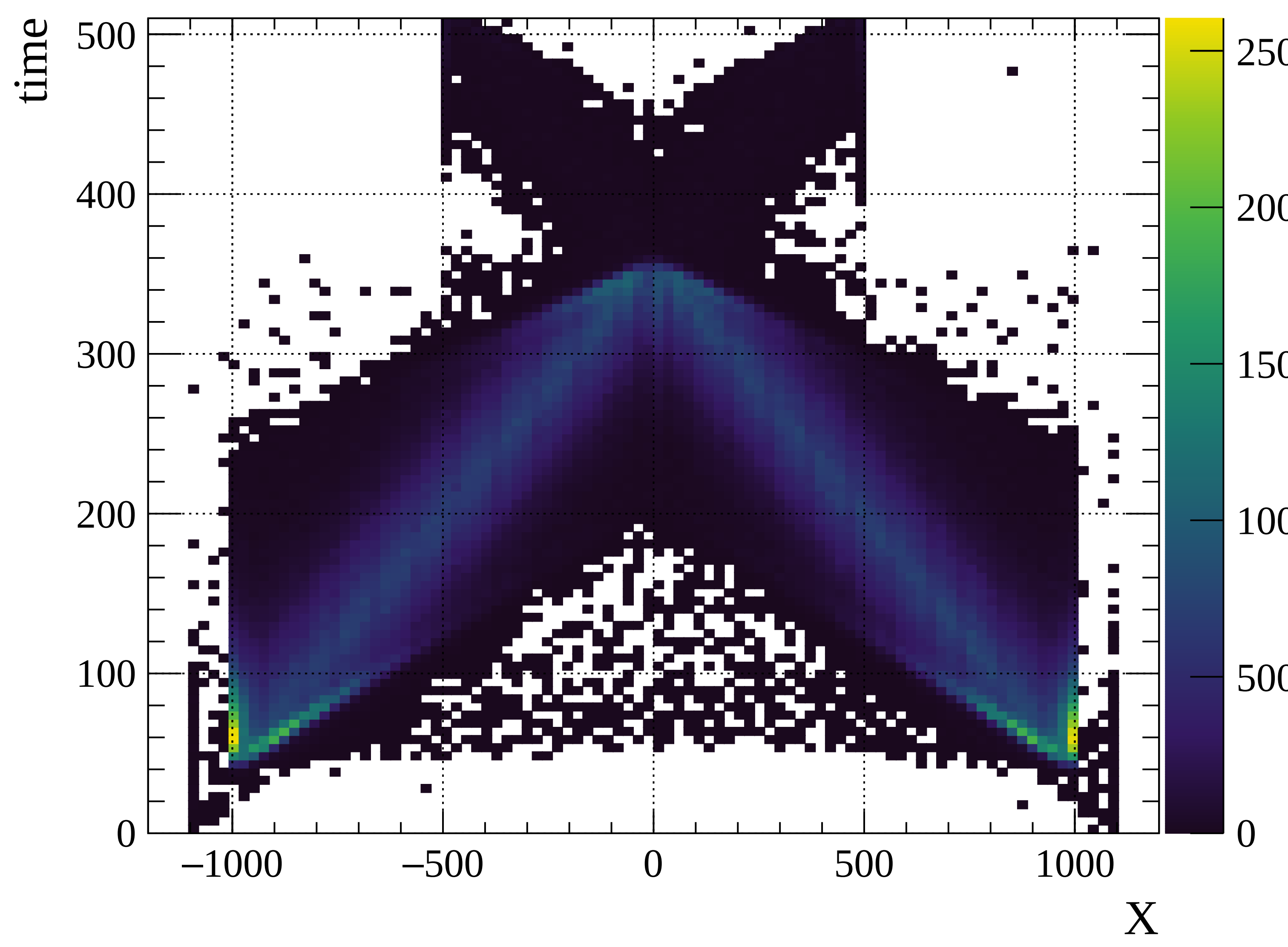

y = 0.07x + 472.57
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
10
20
30
40
50
60



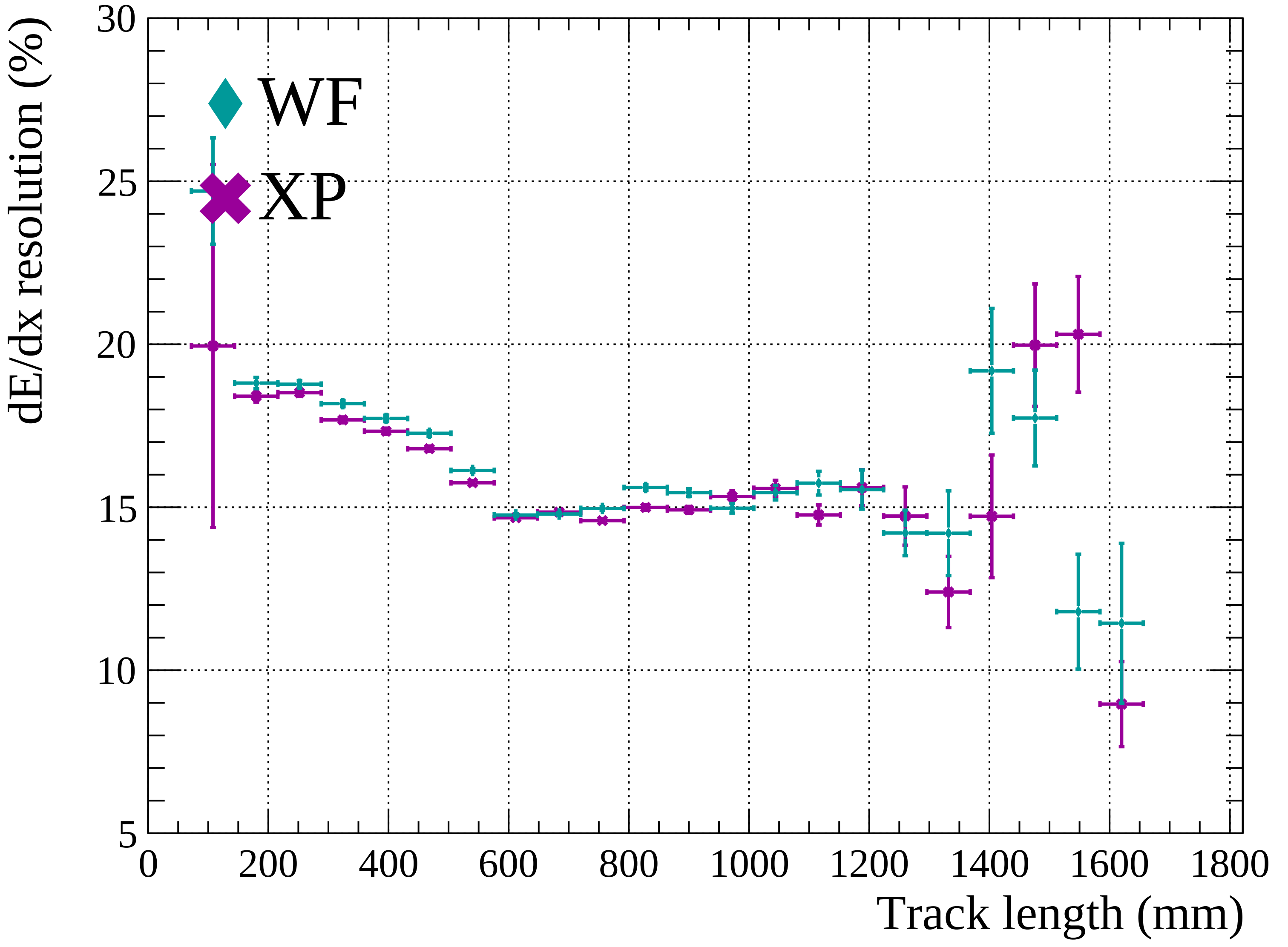
y = 0.11x + 457.68
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
10
20
30
40
50


Energy loss (XP) vs drift time in ERAM 24
y = 0.01x + 486.85
Energy loss (XP) vs drift time in ERAM 30
y = -0.01x + 518.51
Energy loss (XP) vs drift time in ERAM 28
y = -0.01x + 527.15
Energy loss (XP) vs drift time in ERAM 19
y = 0.07x + 503.40
Energy loss (XP) vs drift time in ERAM 21
y = 0.07x + 467.02
Energy loss (XP) vs drift time in ERAM 13
y = 0.06x + 490.53
Energy loss (XP) vs drift time in ERAM 9
y = 0.07x + 500.46
Energy loss (XP) vs drift time in ERAM 2
y = 0.05x + 493.61
Energy loss (XP) vs drift time in ERAM 26
y = 0.07x + 509.20
Energy loss (XP) vs drift time in ERAM 17
y = 0.13x + 500.85
Energy loss (XP) vs drift time in ERAM 23
y = 0.10x + 497.55
Energy loss (XP) vs drift time in ERAM 29
y = 0.11x + 482.55
Energy loss (XP) vs drift time in ERAM 1
y = 0.13x + 475.06
Energy loss (XP) vs drift time in ERAM 10
y = 0.15x + 472.64
Energy loss (XP) vs drift time in ERAM 11
y = 0.14x + 474.06
Energy loss (XP) vs drift time in ERAM 3
y = 0.12x + 464.89
dE/dx (ADC counts/cm)
drift time (timebins)

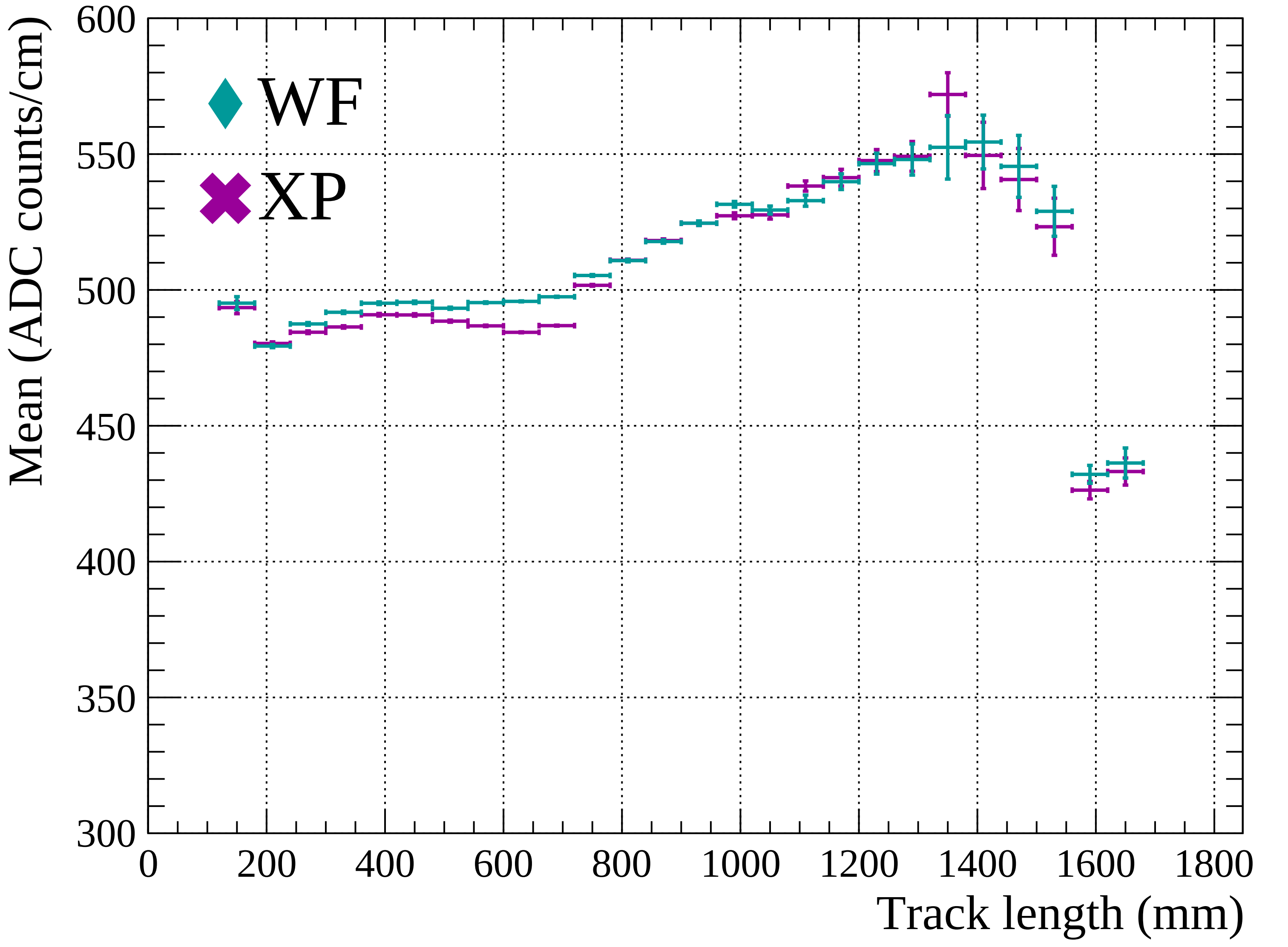

Energy loss (XP) vs drift time in ERAM 47
y = 0.02x + 490.26
Energy loss (XP) vs drift time in ERAM 16
y = -0.01x + 511.07
Energy loss (XP) vs drift time in ERAM 14
y = -0.05x + 539.24
Energy loss (XP) vs drift time in ERAM 15
y = 0.07x + 532.34
Energy loss (XP) vs drift time in ERAM 42
y = 0.08x + 449.45
Energy loss (XP) vs drift time in ERAM 45
y = 0.07x + 465.30
Energy loss (XP) vs drift time in ERAM 37
y = 0.08x + 483.29
Energy loss (XP) vs drift time in ERAM 36
y = 0.08x + 480.32
Energy loss (XP) vs drift time in ERAM 20
y = 0.12x + 474.30
Energy loss (XP) vs drift time in ERAM 38
y = 0.09x + 481.85
Energy loss (XP) vs drift time in ERAM 7
y = 0.06x + 506.83
Energy loss (XP) vs drift time in ERAM 44
y = 0.13x + 469.58
Energy loss (XP) vs drift time in ERAM 43
y = 0.11x + 474.13
Energy loss (XP) vs drift time in ERAM 39
y = 0.13x + 448.29
Energy loss (XP) vs drift time in ERAM 41
y = 0.12x + 468.37
Energy loss (XP) vs drift time in ERAM 46
y = 0.11x + 451.82
dE/dx (ADC counts/cm)
drift time (timebins)



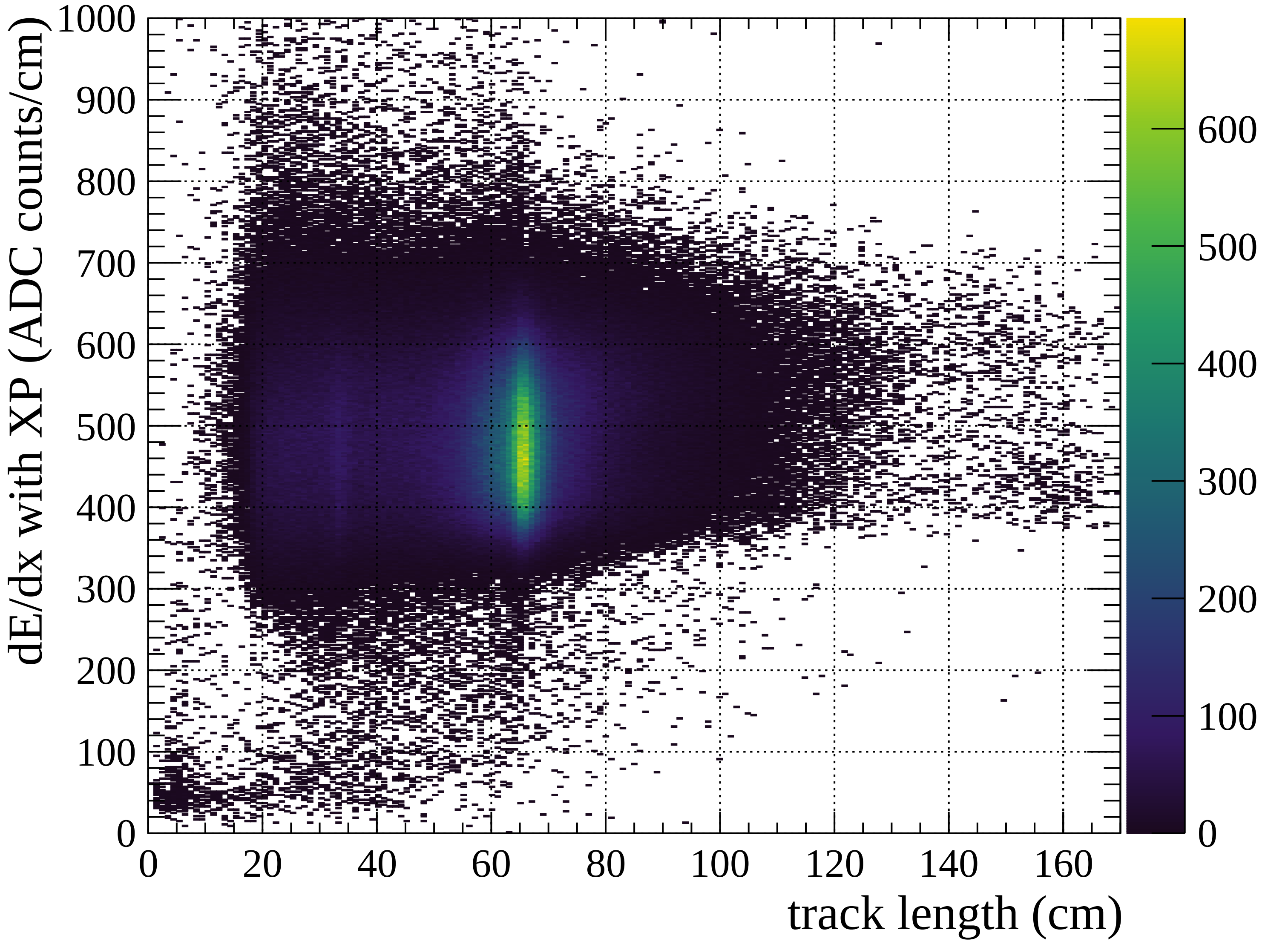
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
250
200
150
100
500
0

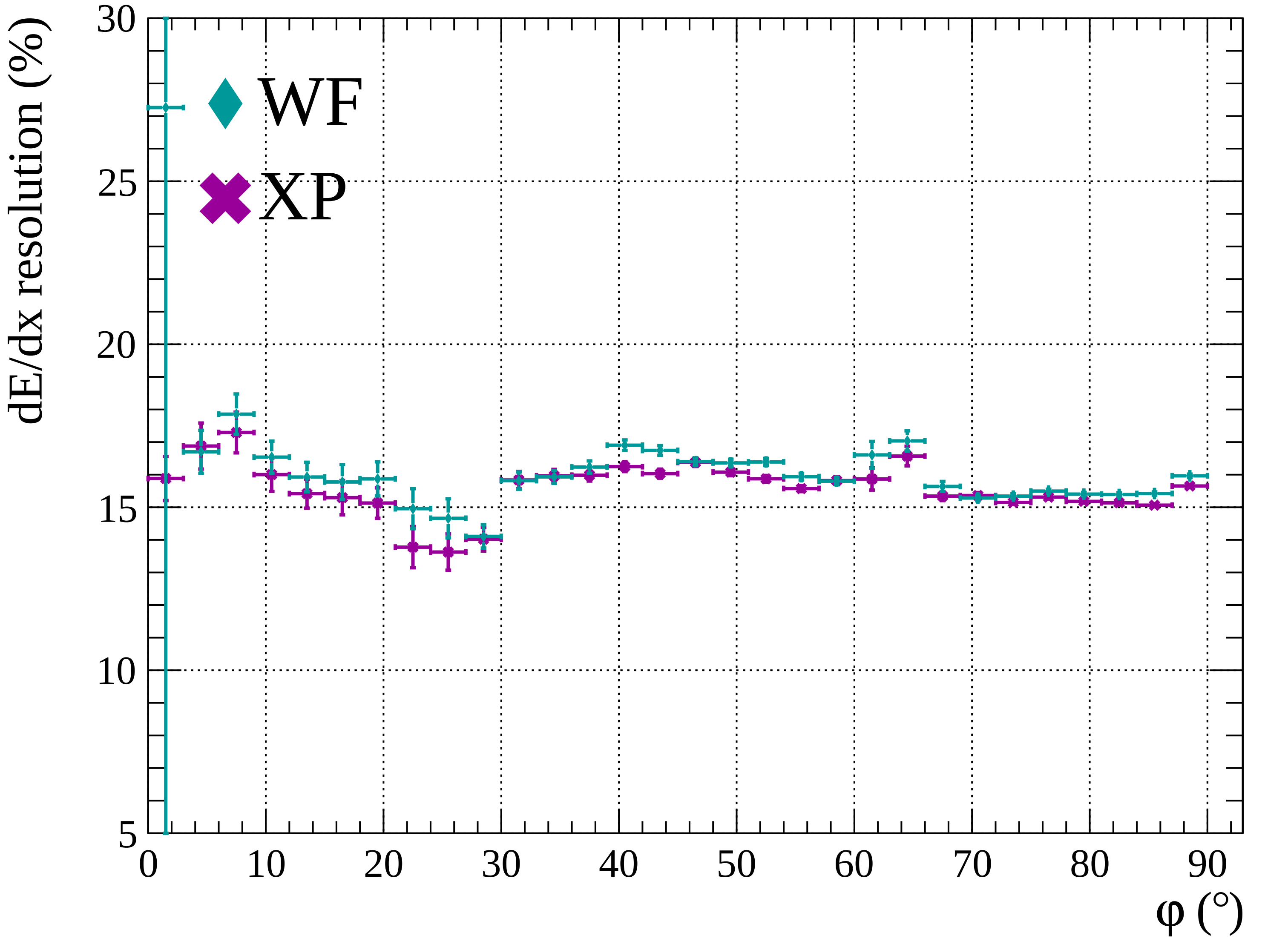


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
Track length (mm)

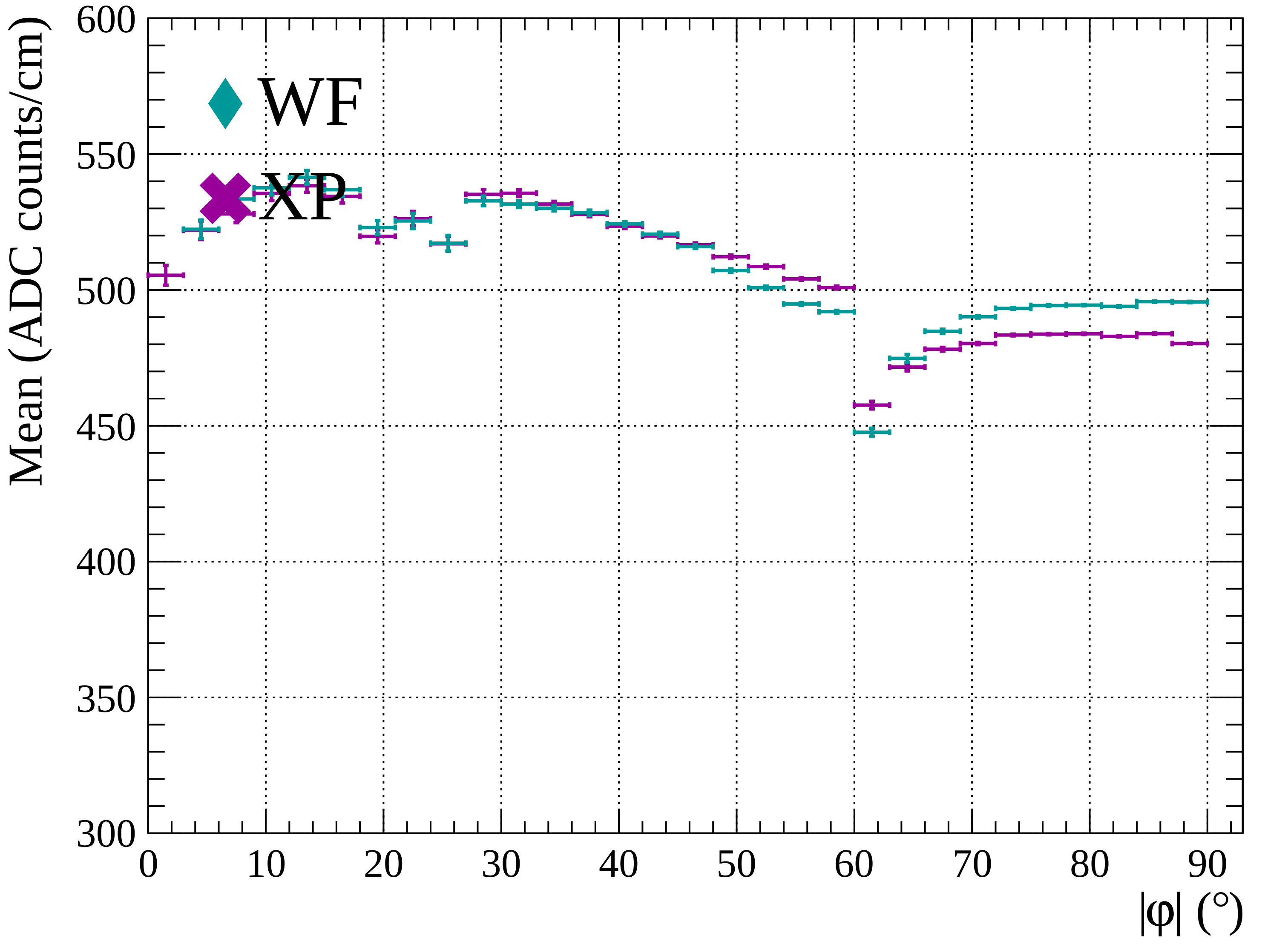


Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
Track length (mm)
WF
XP

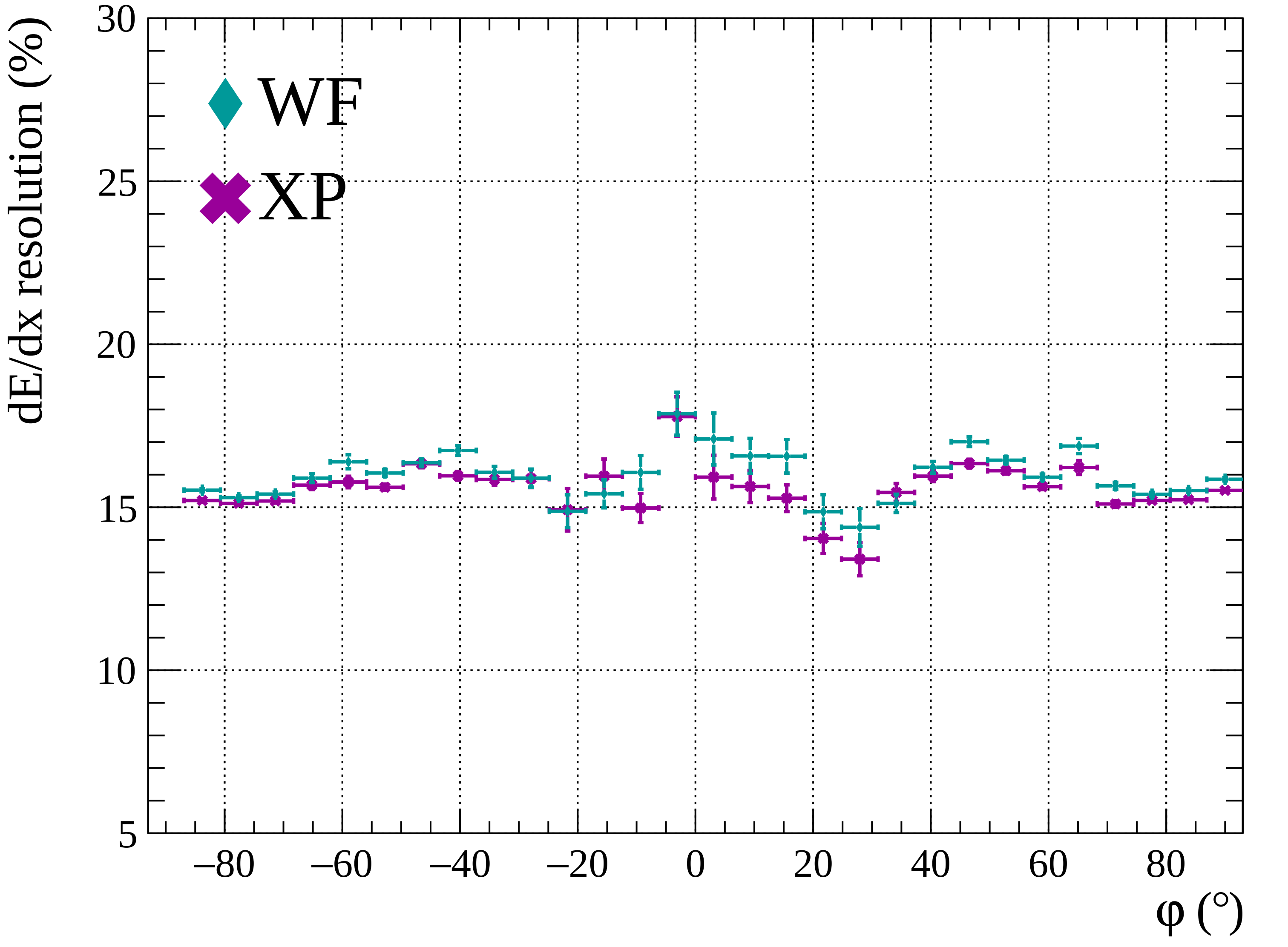

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600



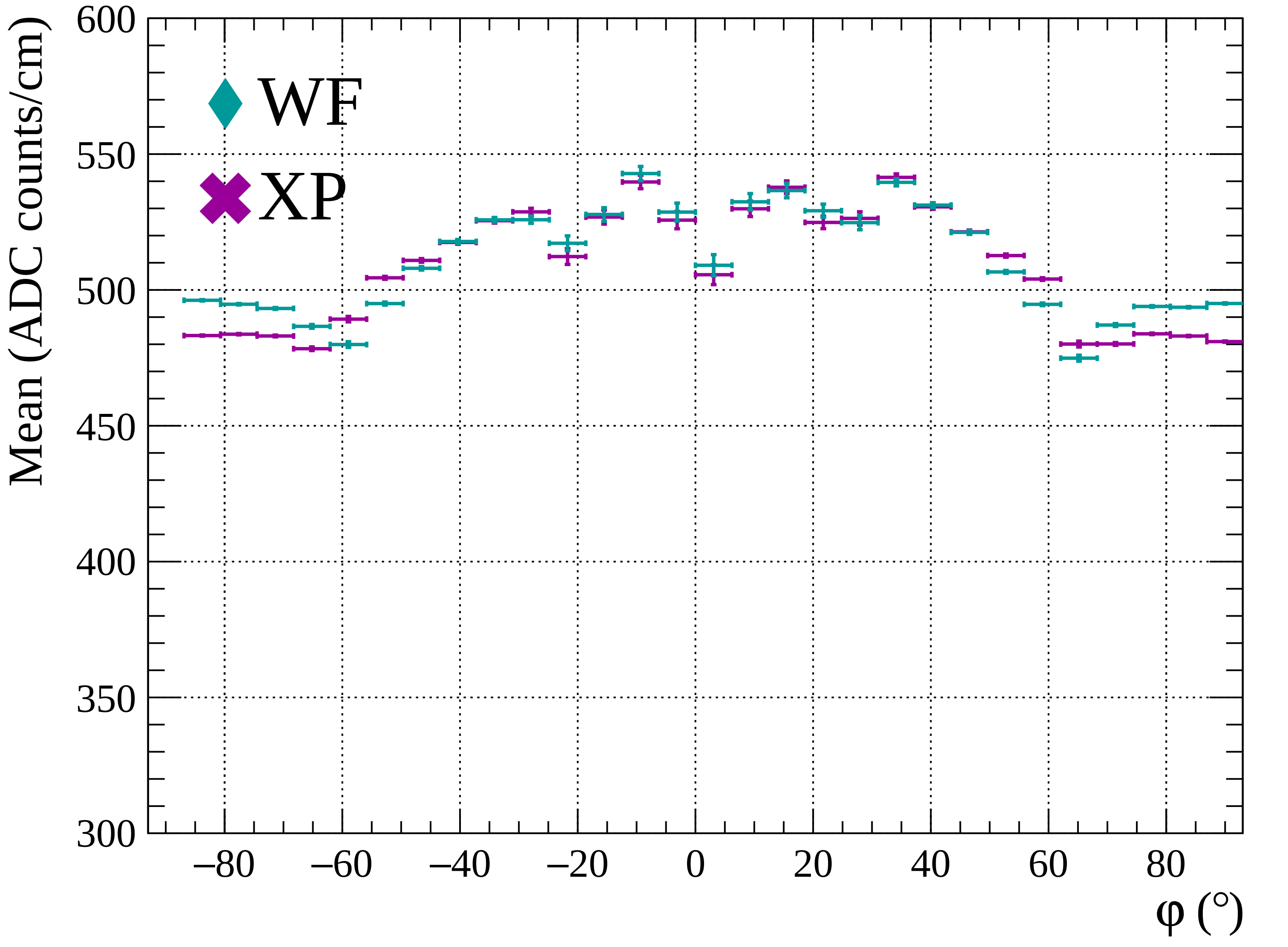
dE/dx resolution (%)
φ (°)
WF
XP



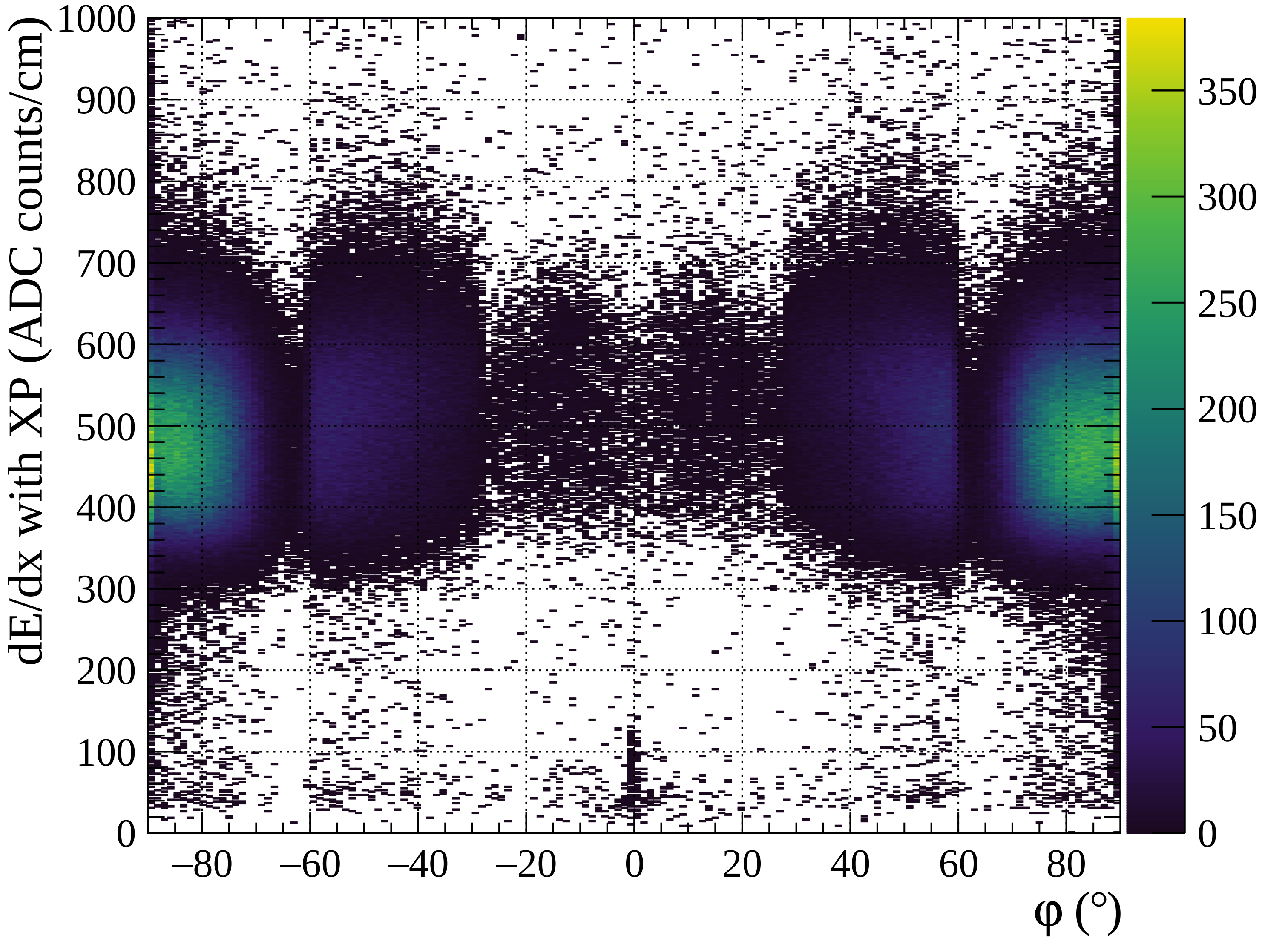
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
|φ| (°)

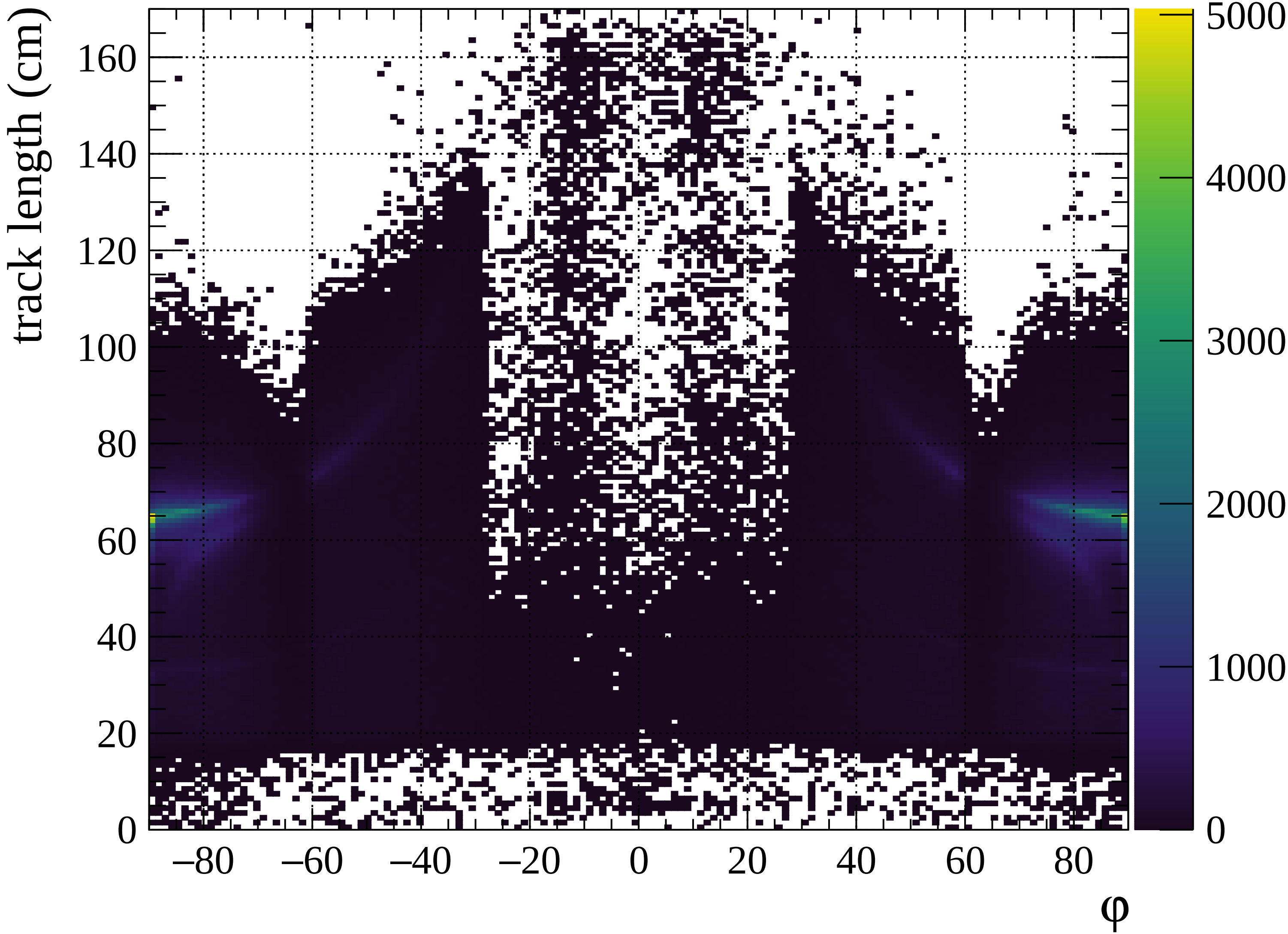


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)

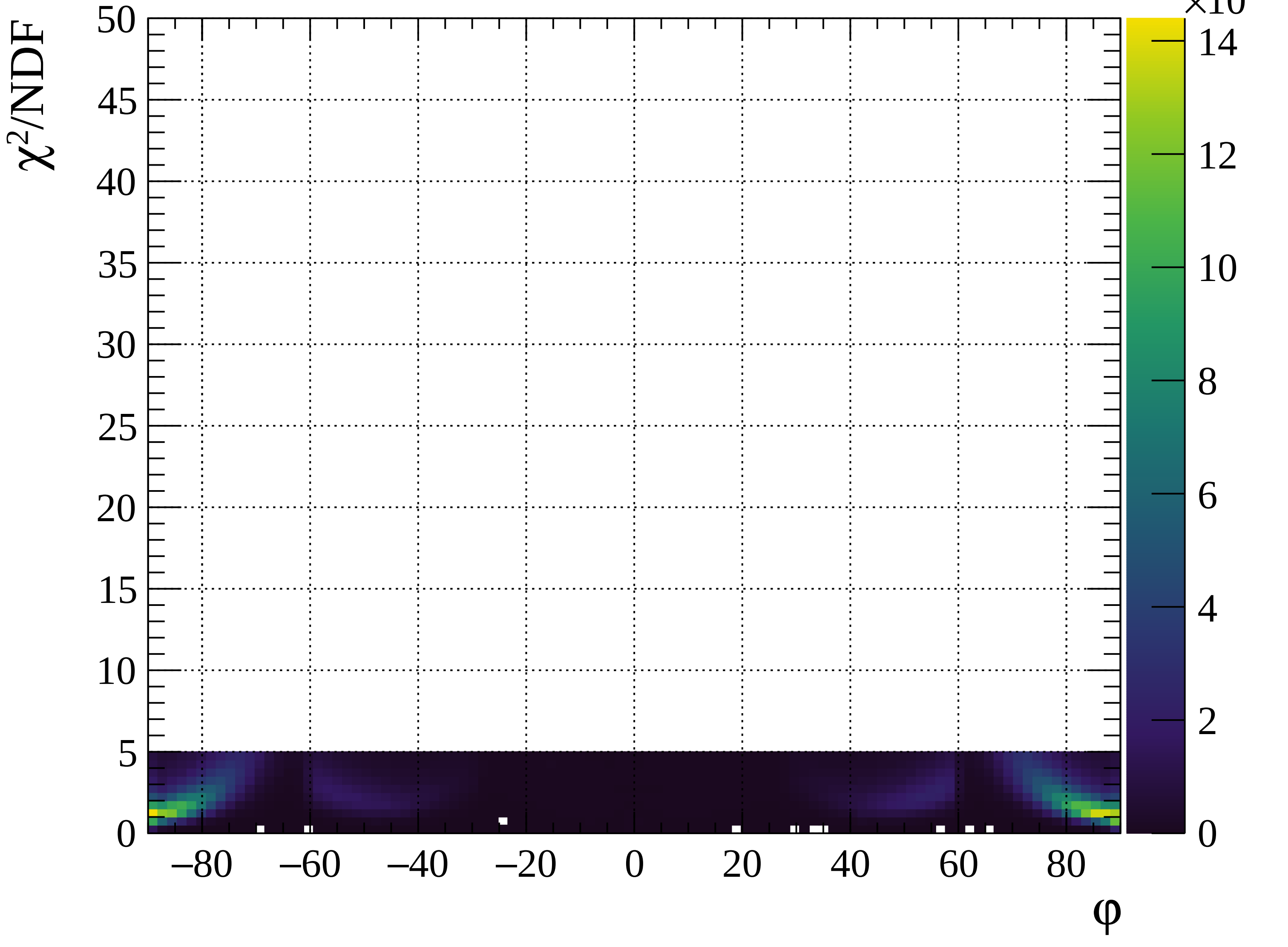


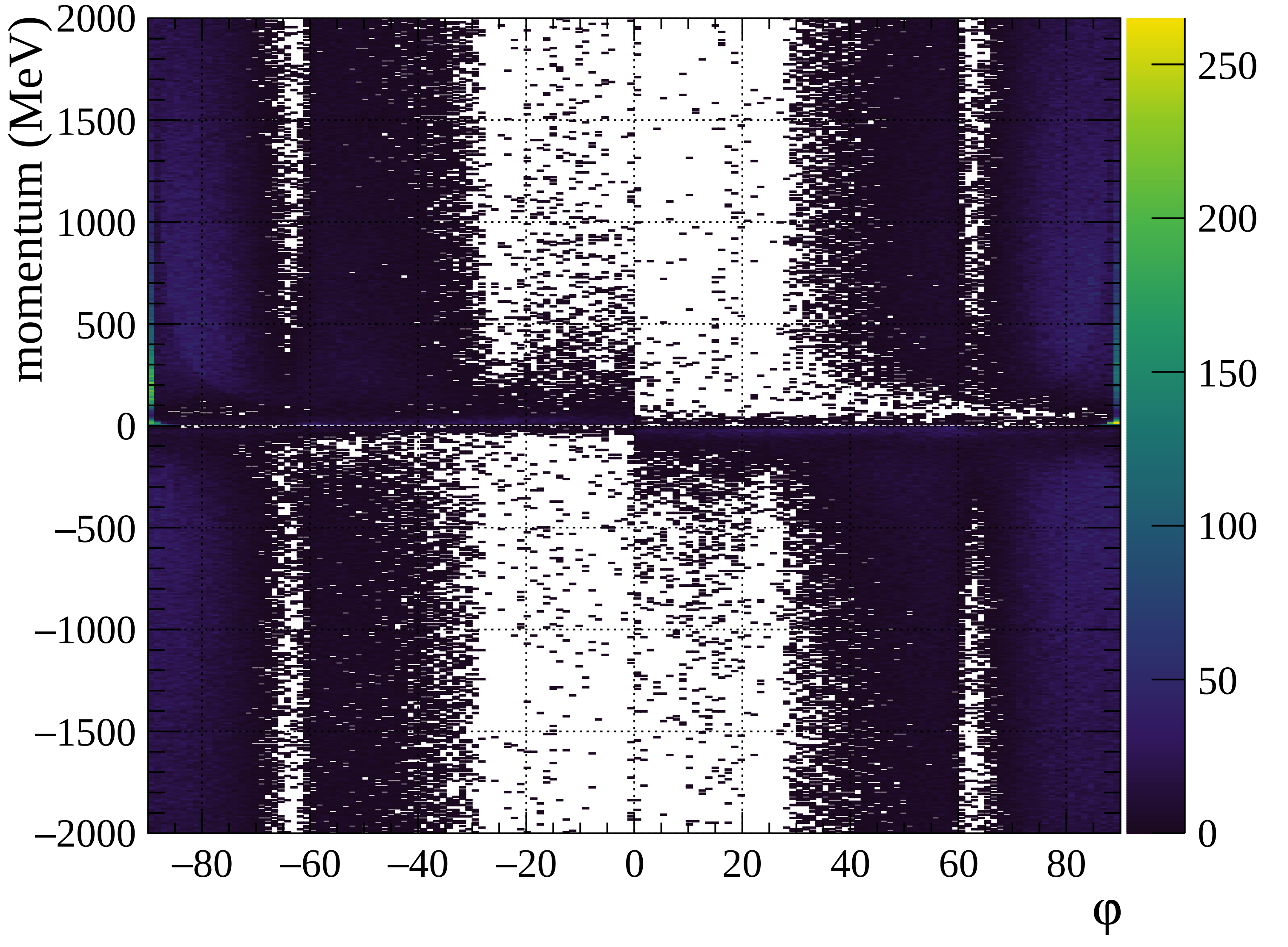
WF
XP
Mean (ADC counts/cm)
φ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80

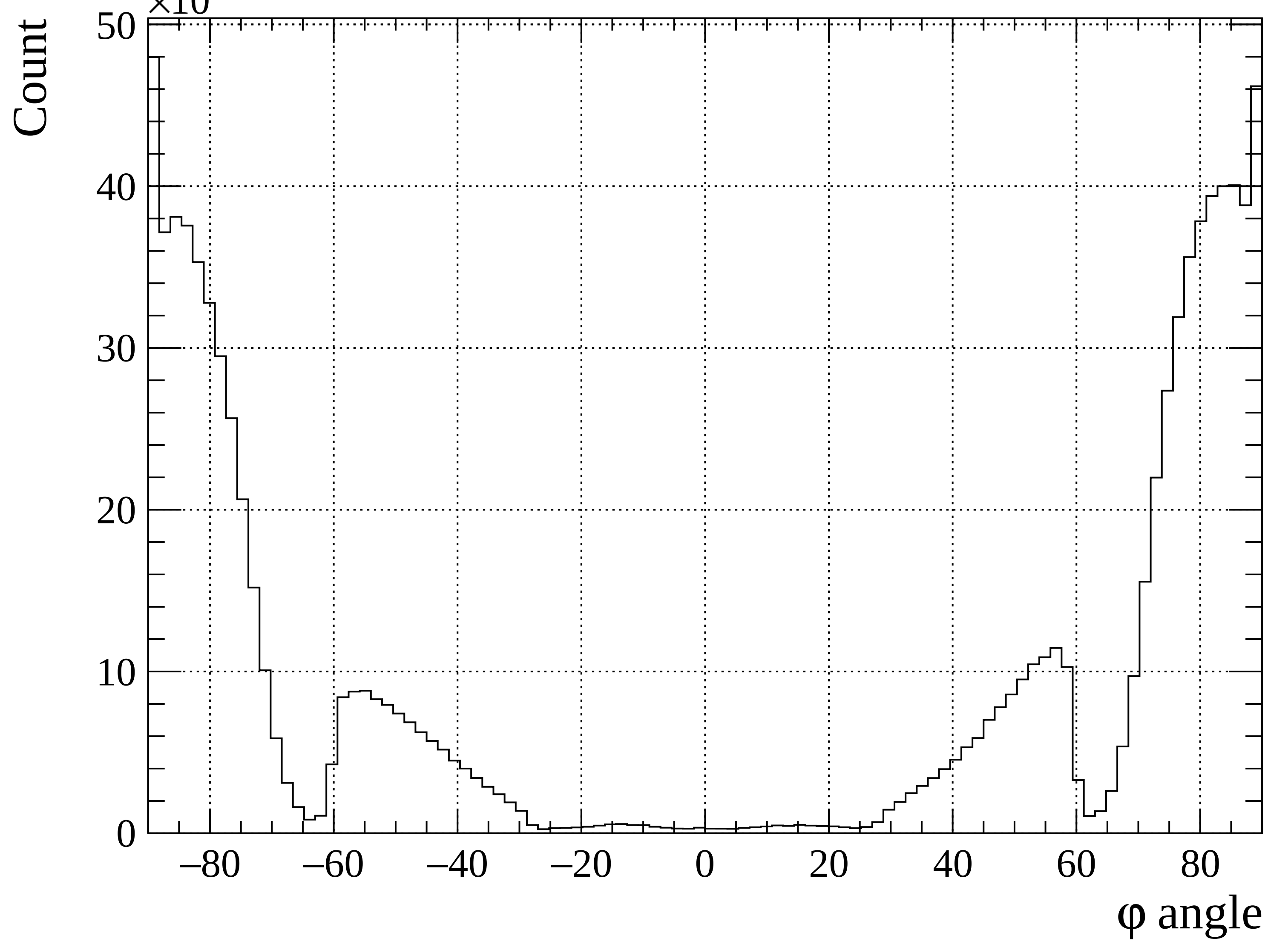

dE/dx with XP (ADC counts/cm)
φ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
50
100
150
200
250
300
350



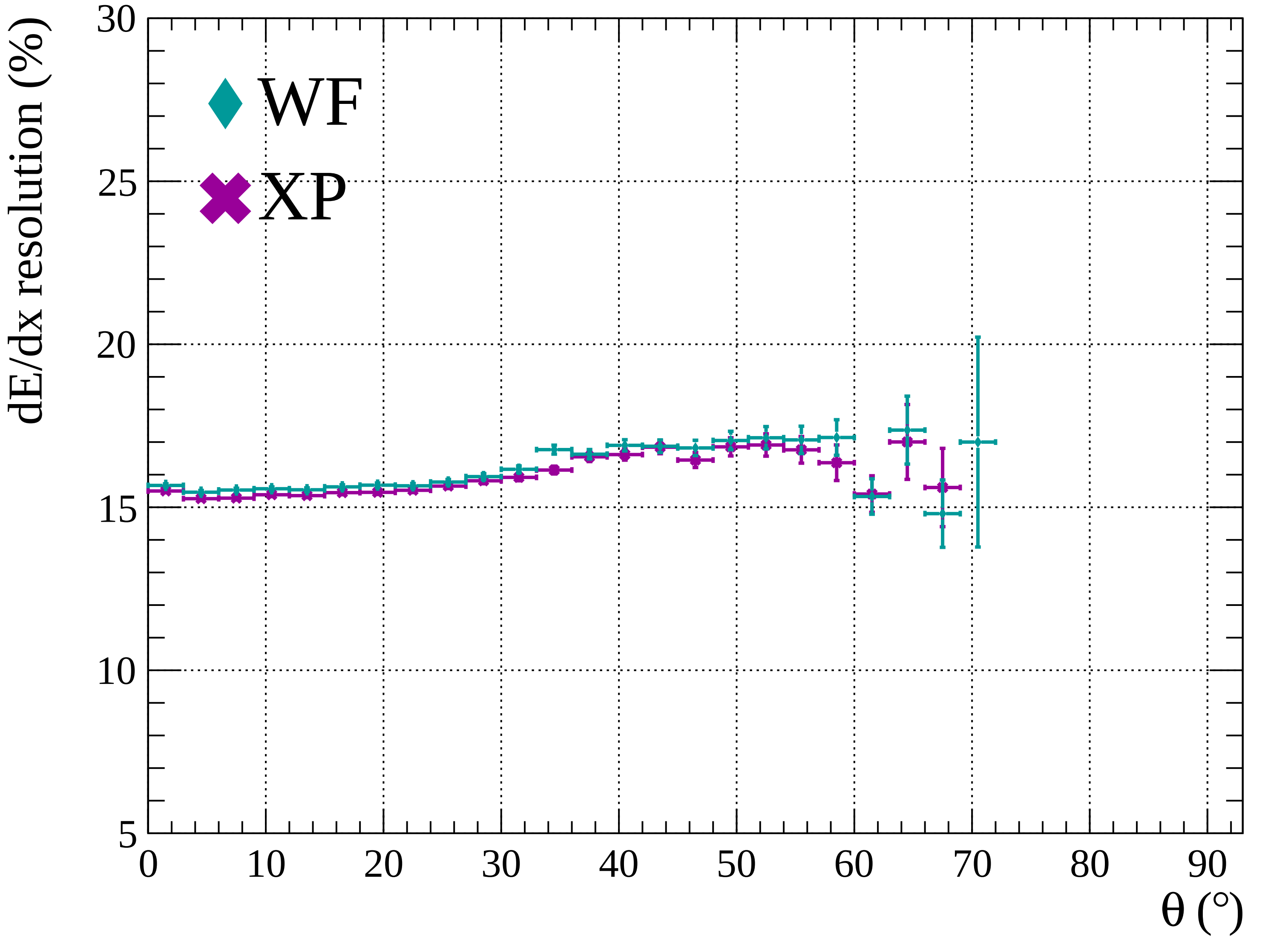
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
5000
4000
3000
2000
1000
0

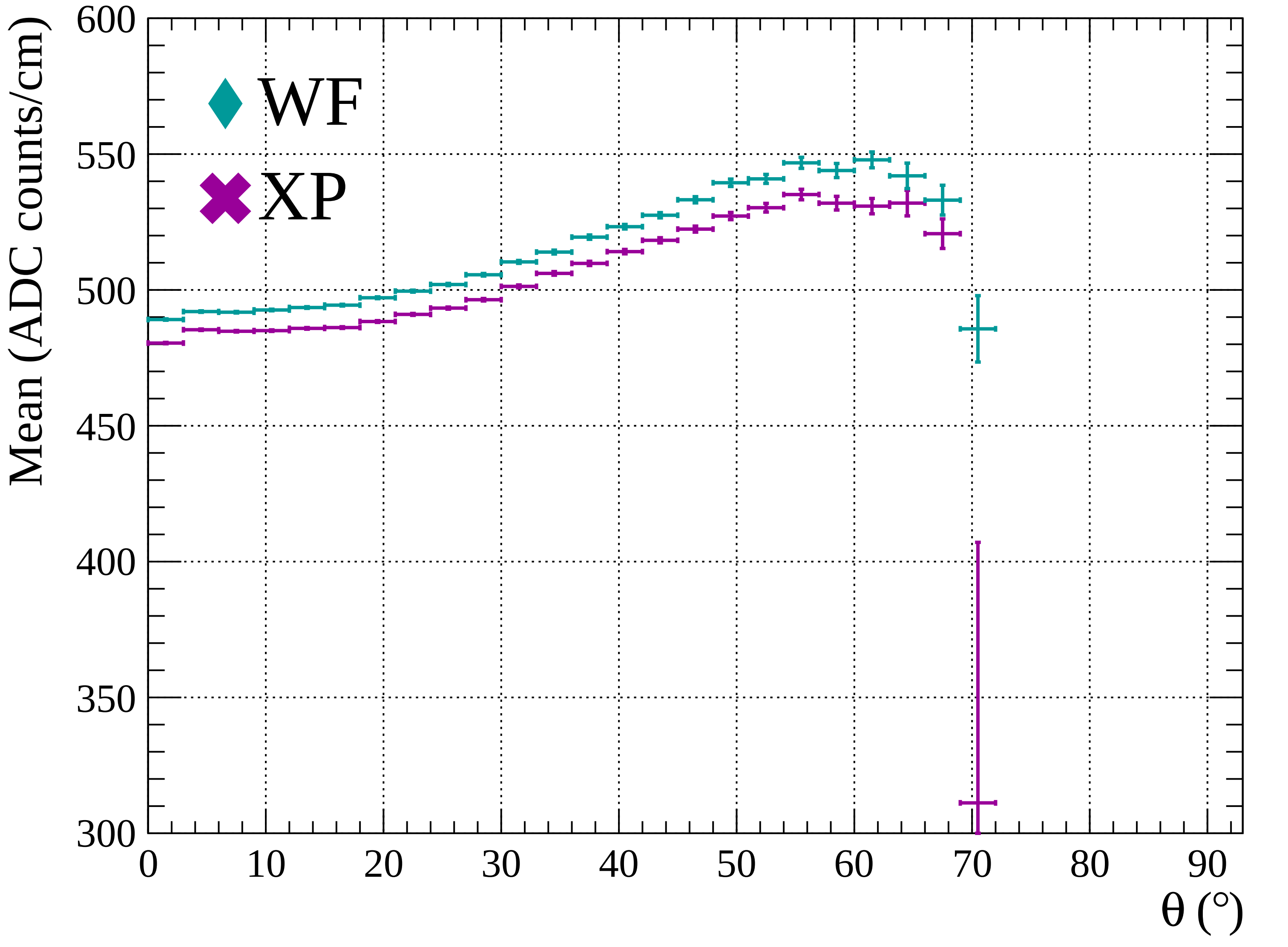

χ^2/NDF
φ
×10
0
5
10
15
20
25
30
35
40
45
50
−80
−60
−40
−20
0
20
40
60
80
2
4
6
8
10
12
14


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
250
200
150
100
50
0

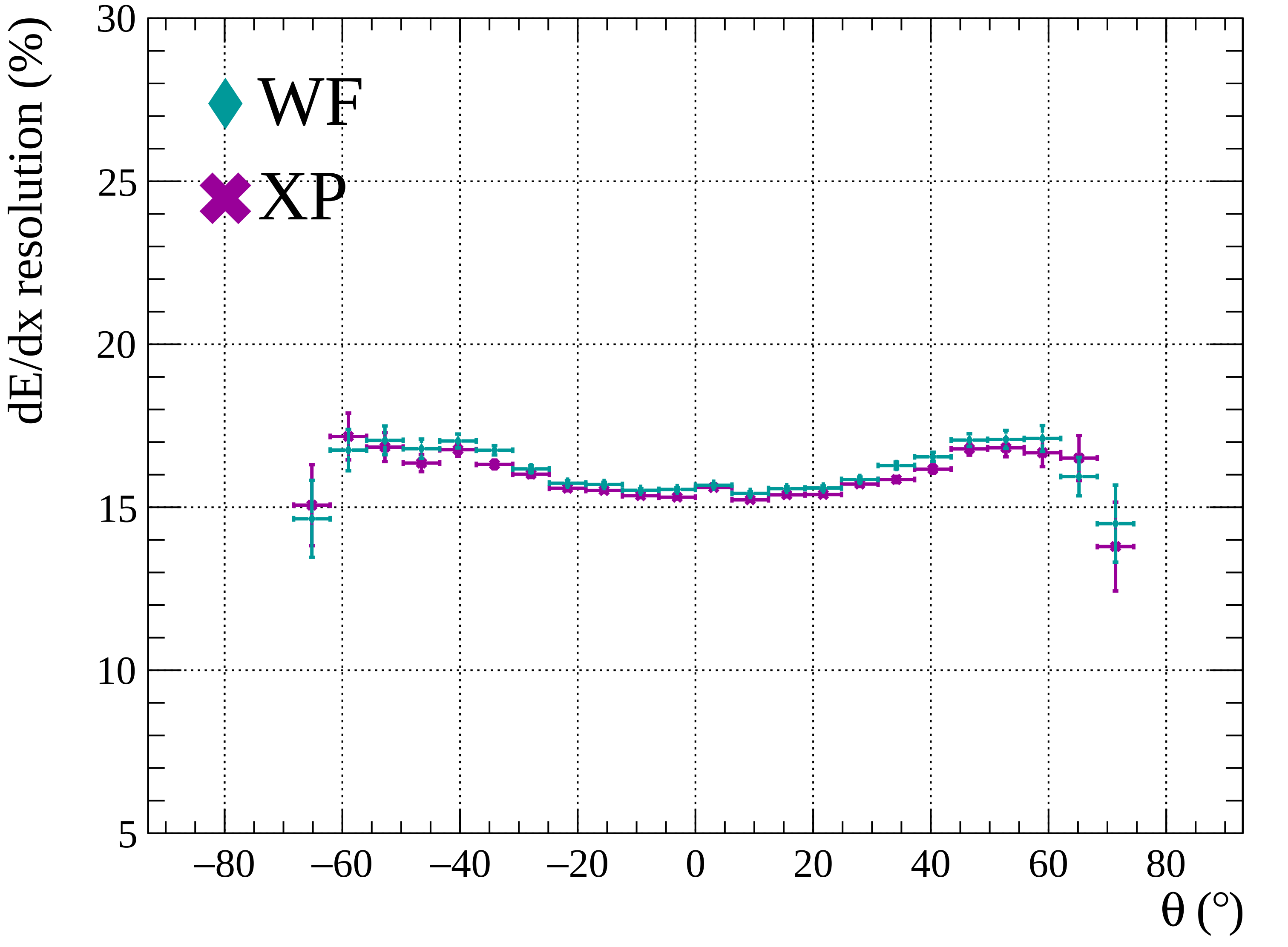


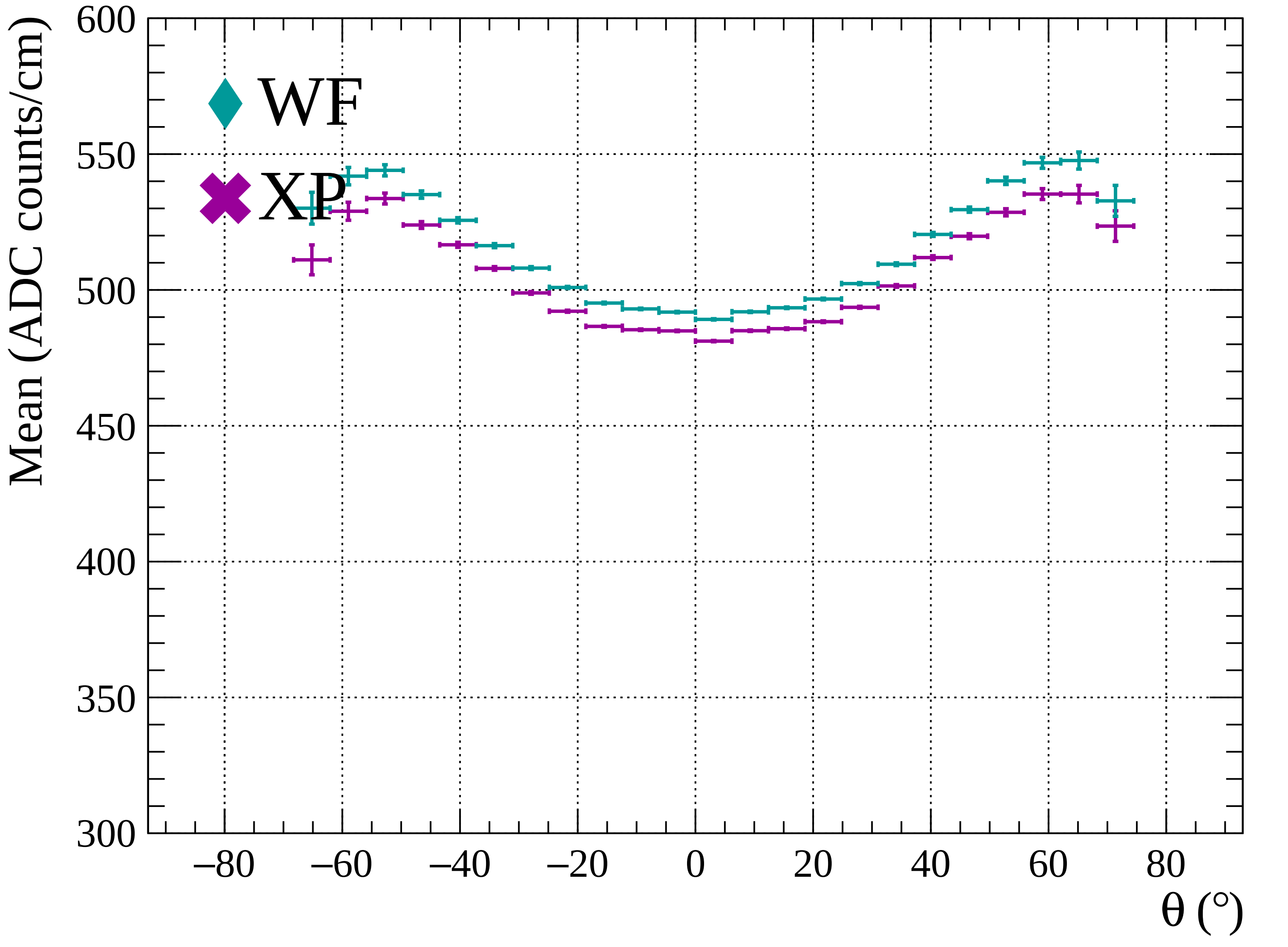
×10
Count
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

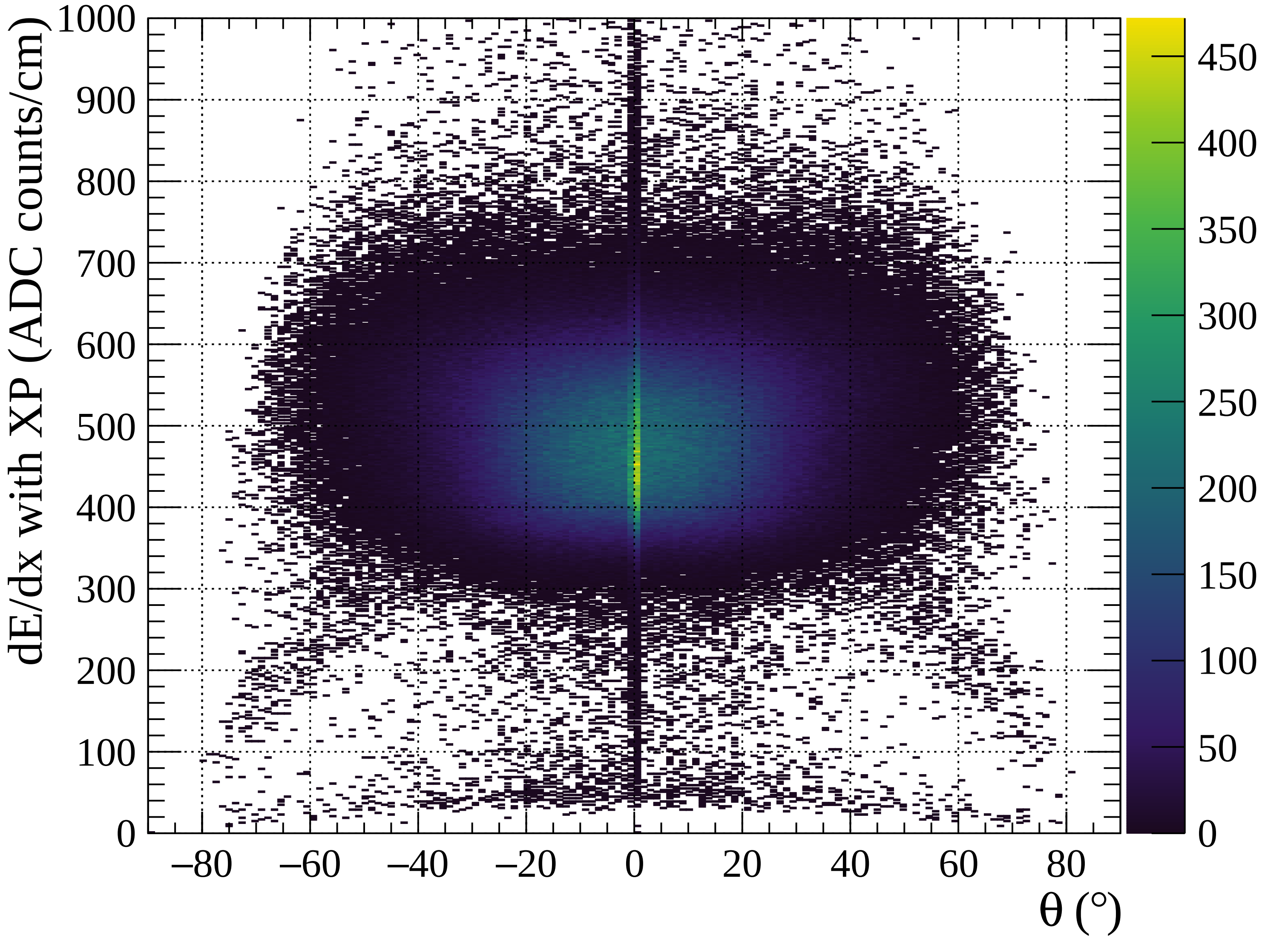

WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90



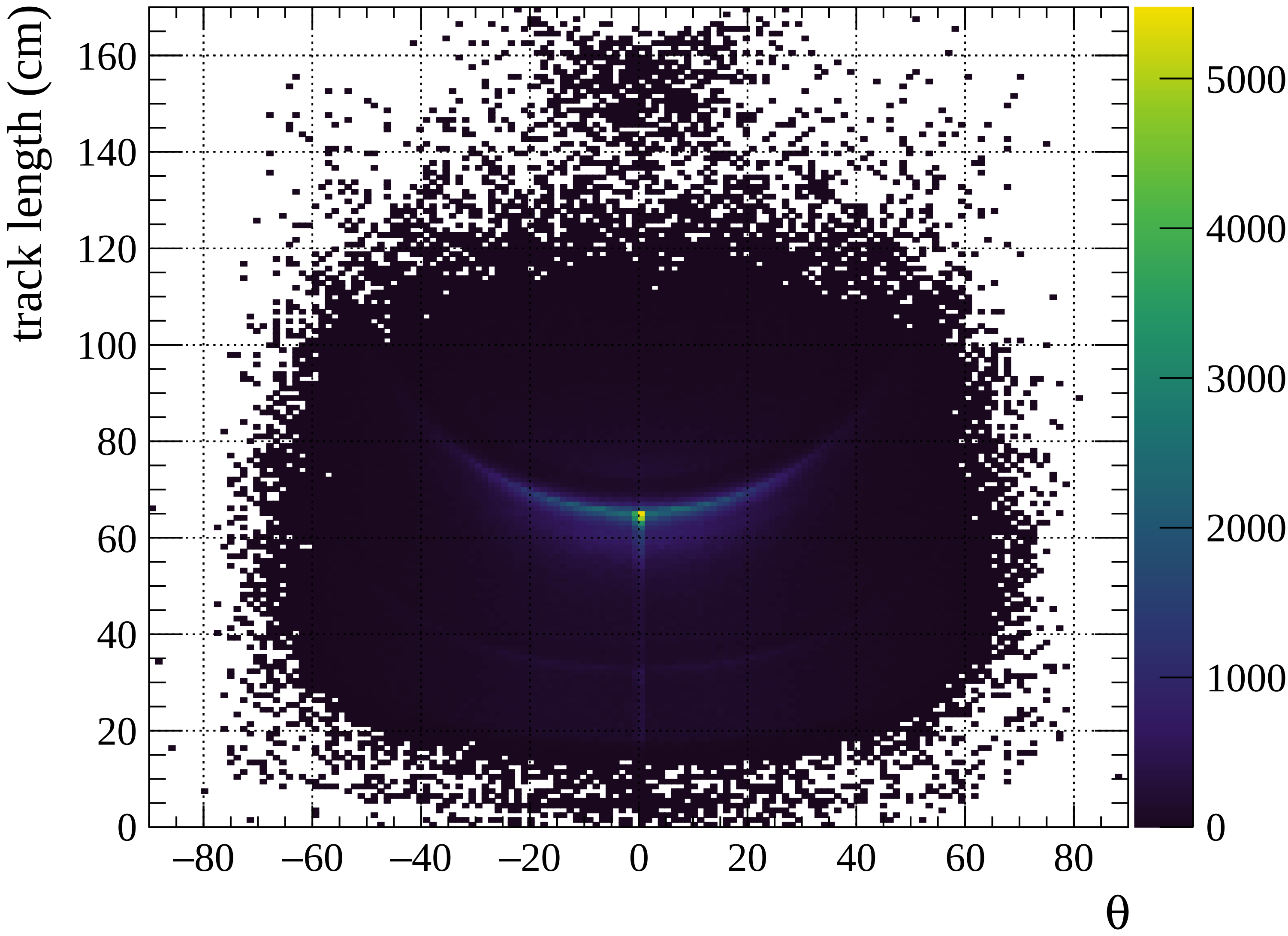
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
10
20
30
40
50
60
70
80
90
θ (°)


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)

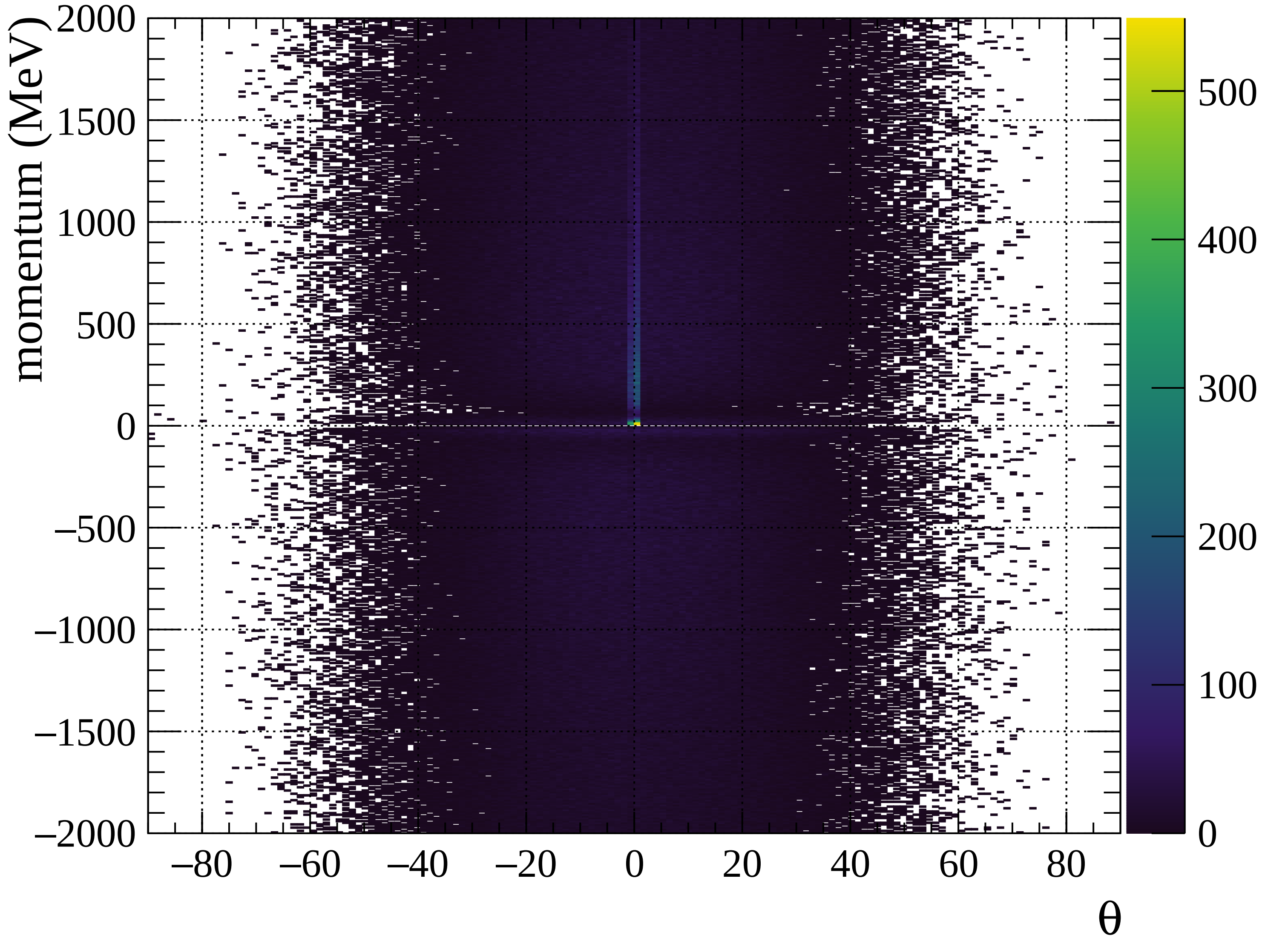

Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)



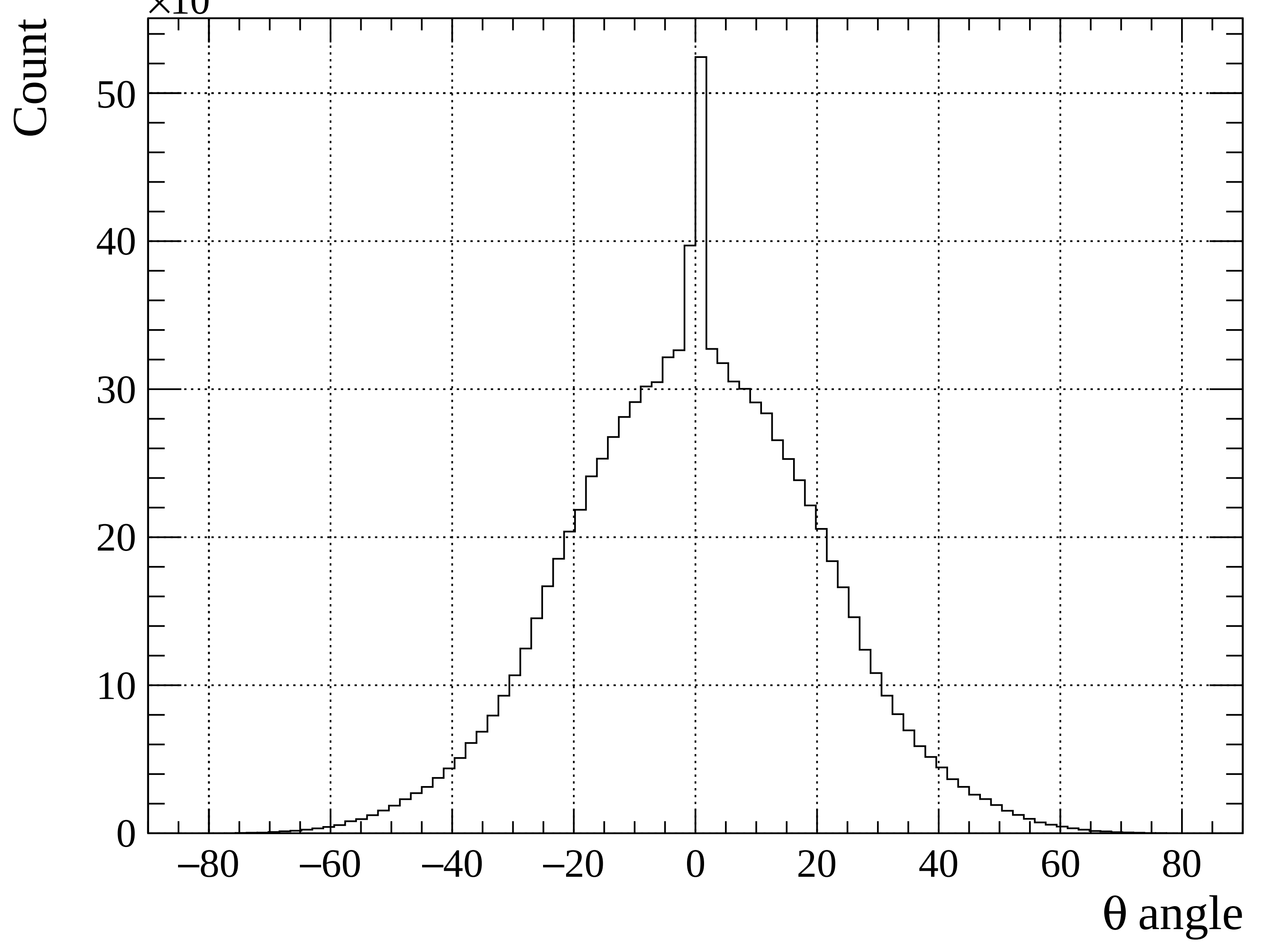
dE/dx with XP (ADC counts/cm)
θ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
50
100
150
200
250
300
350
400
450



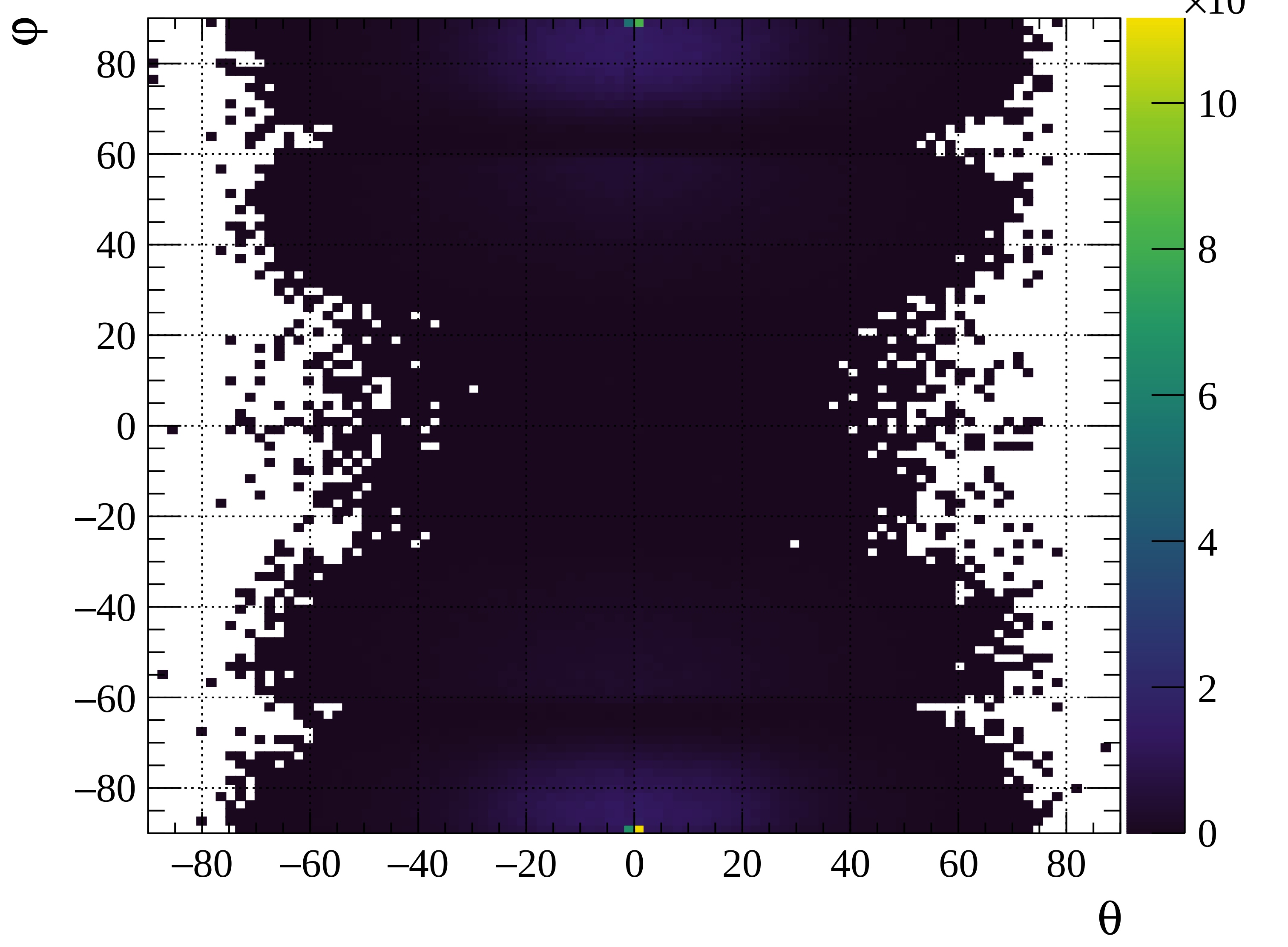
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
5000
4000
3000
2000
1000
0



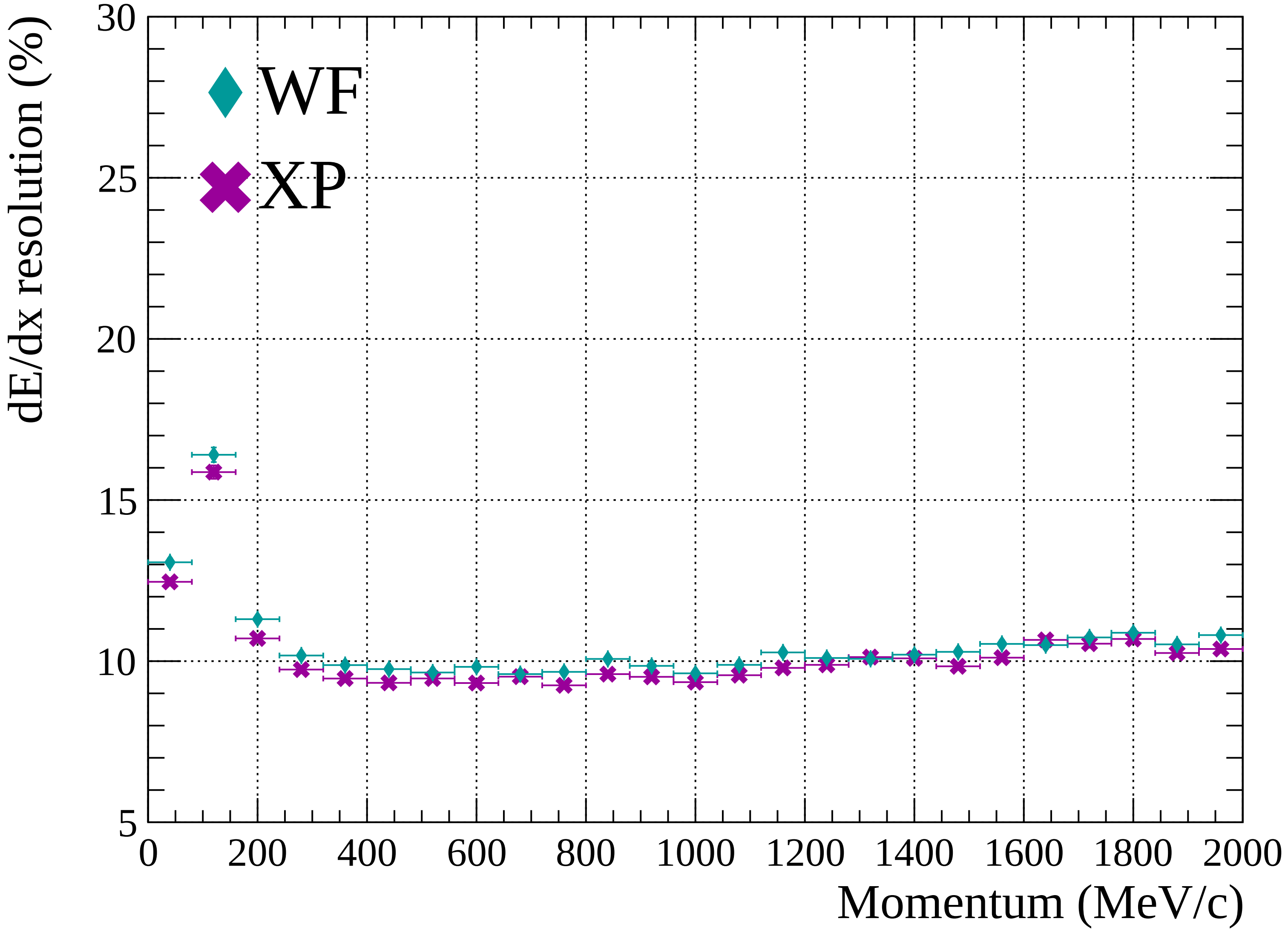
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
500
400
300
200
100
0



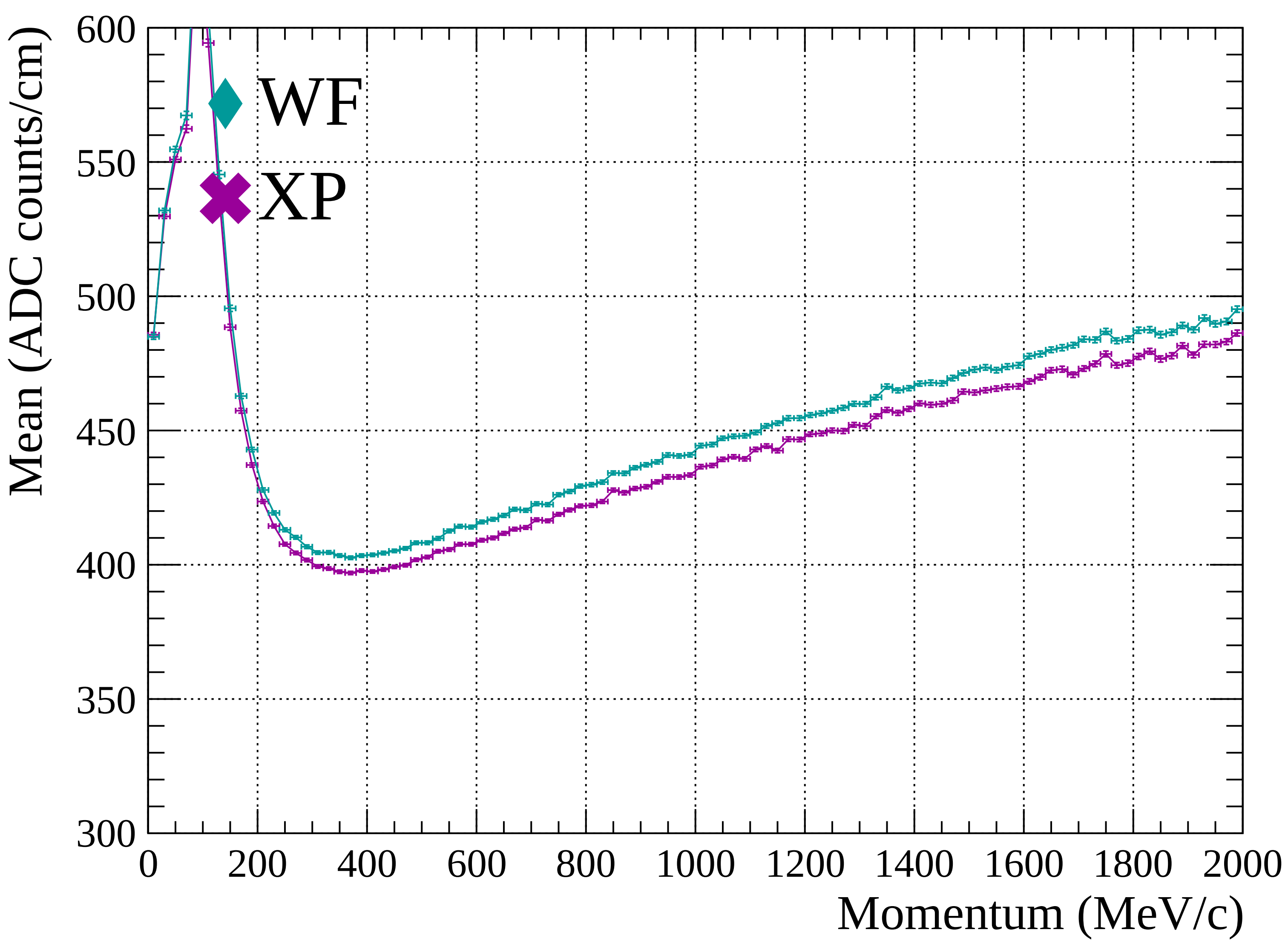
×10
Count
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

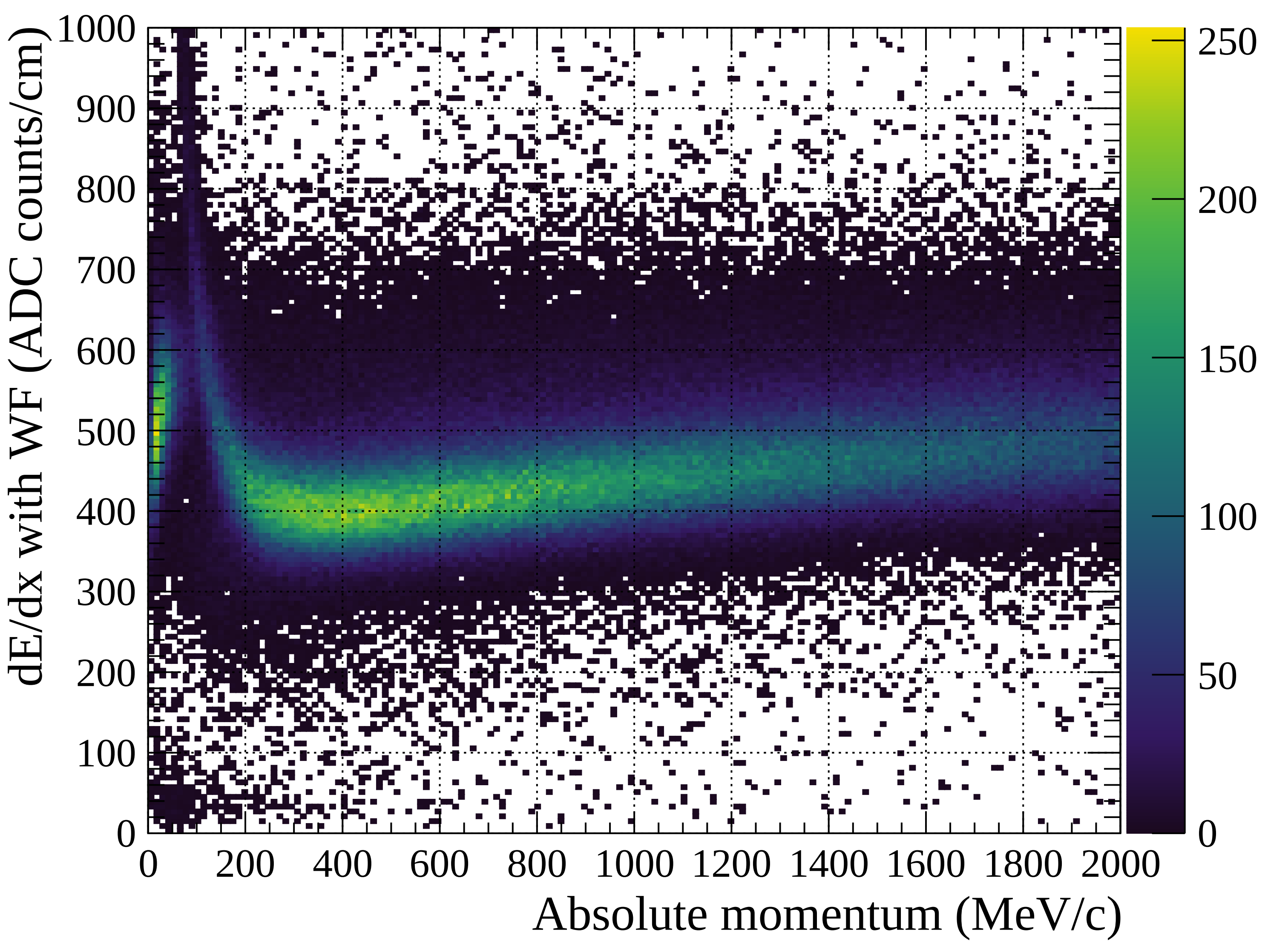


φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
×10
10
8
6
4
2
0

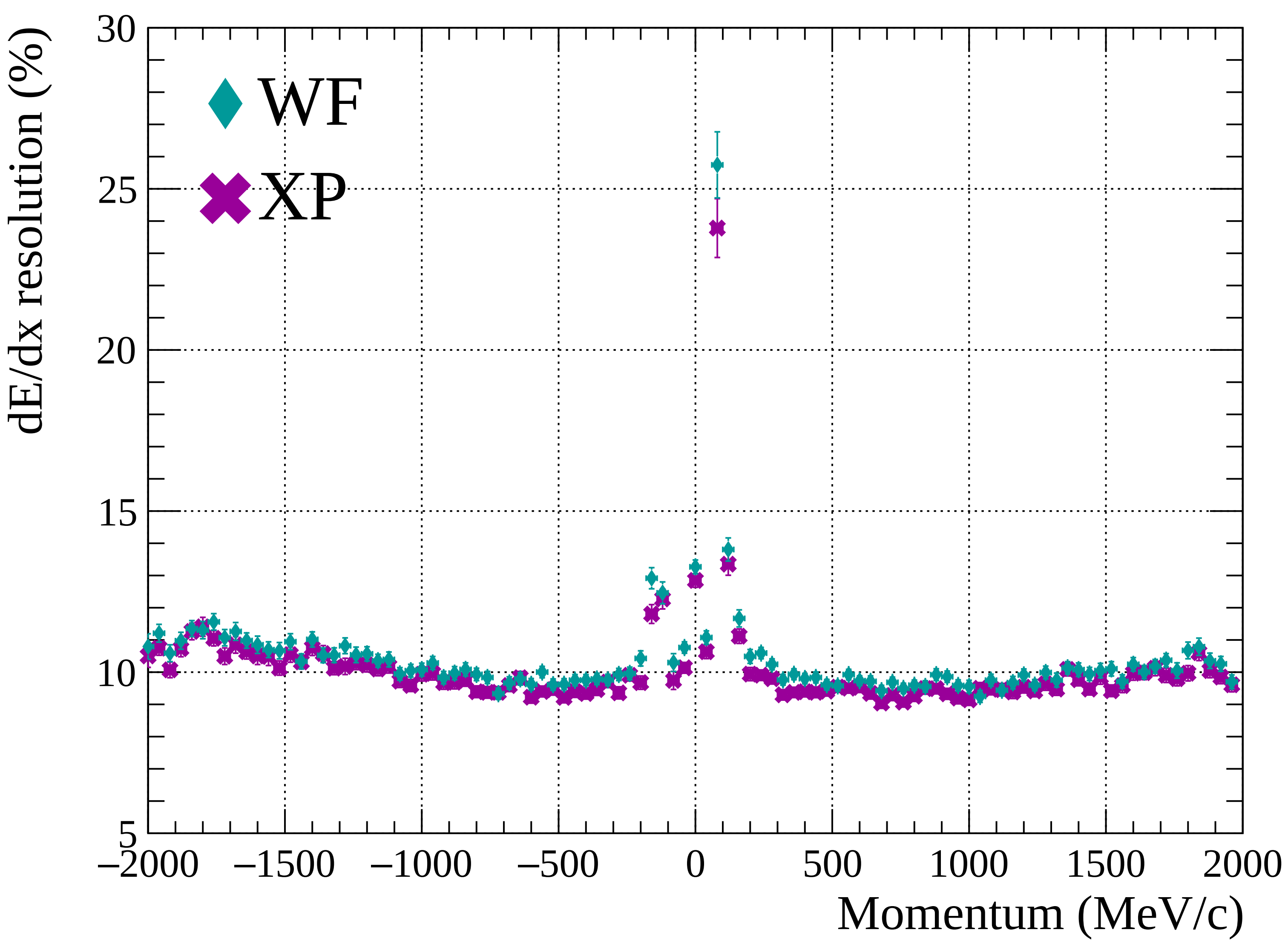


dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000

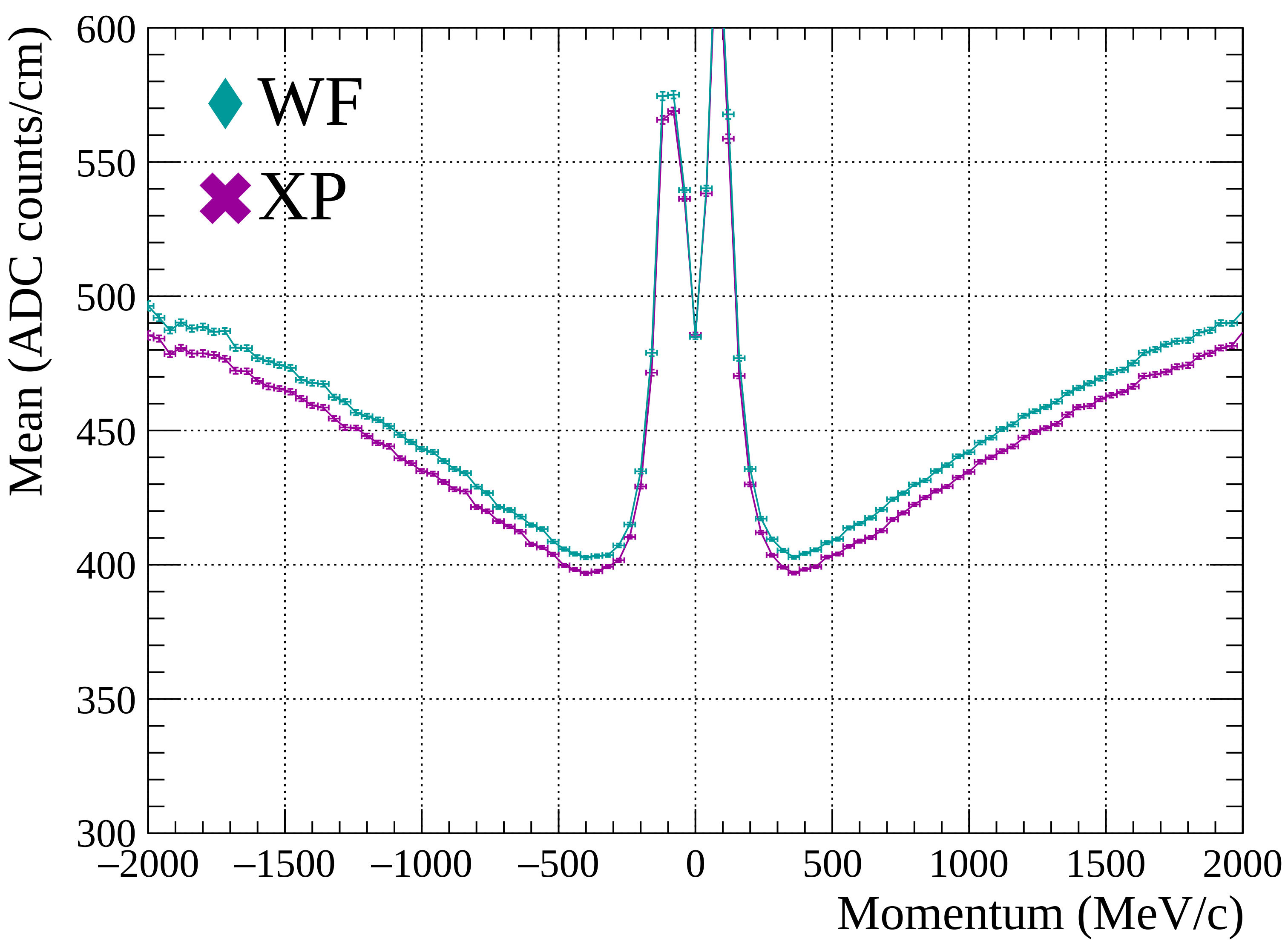

Mean (ADC counts/cm)
Momentum (MeV/c)
WF
XP
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000



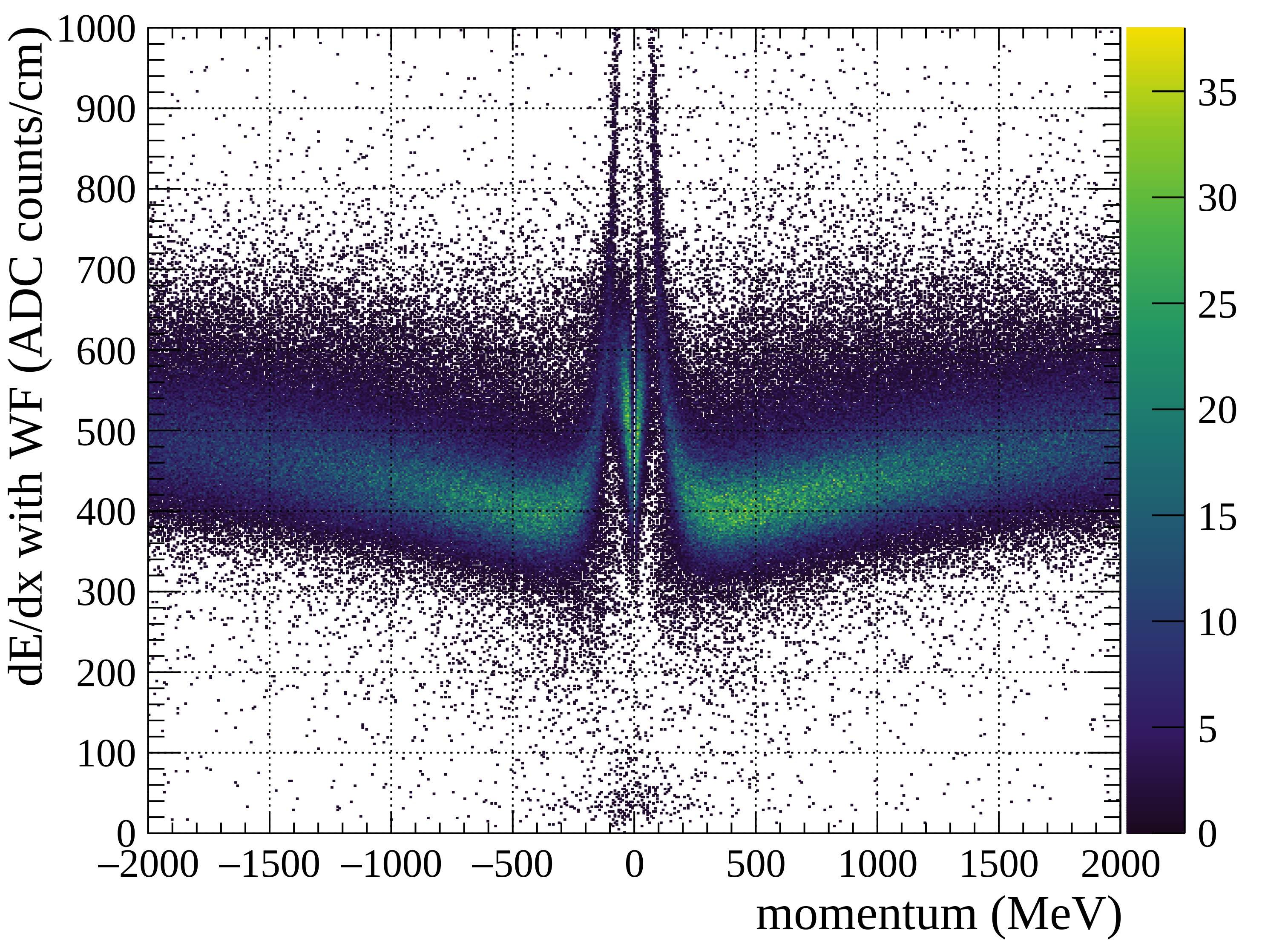
dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
50
100
150
200
250

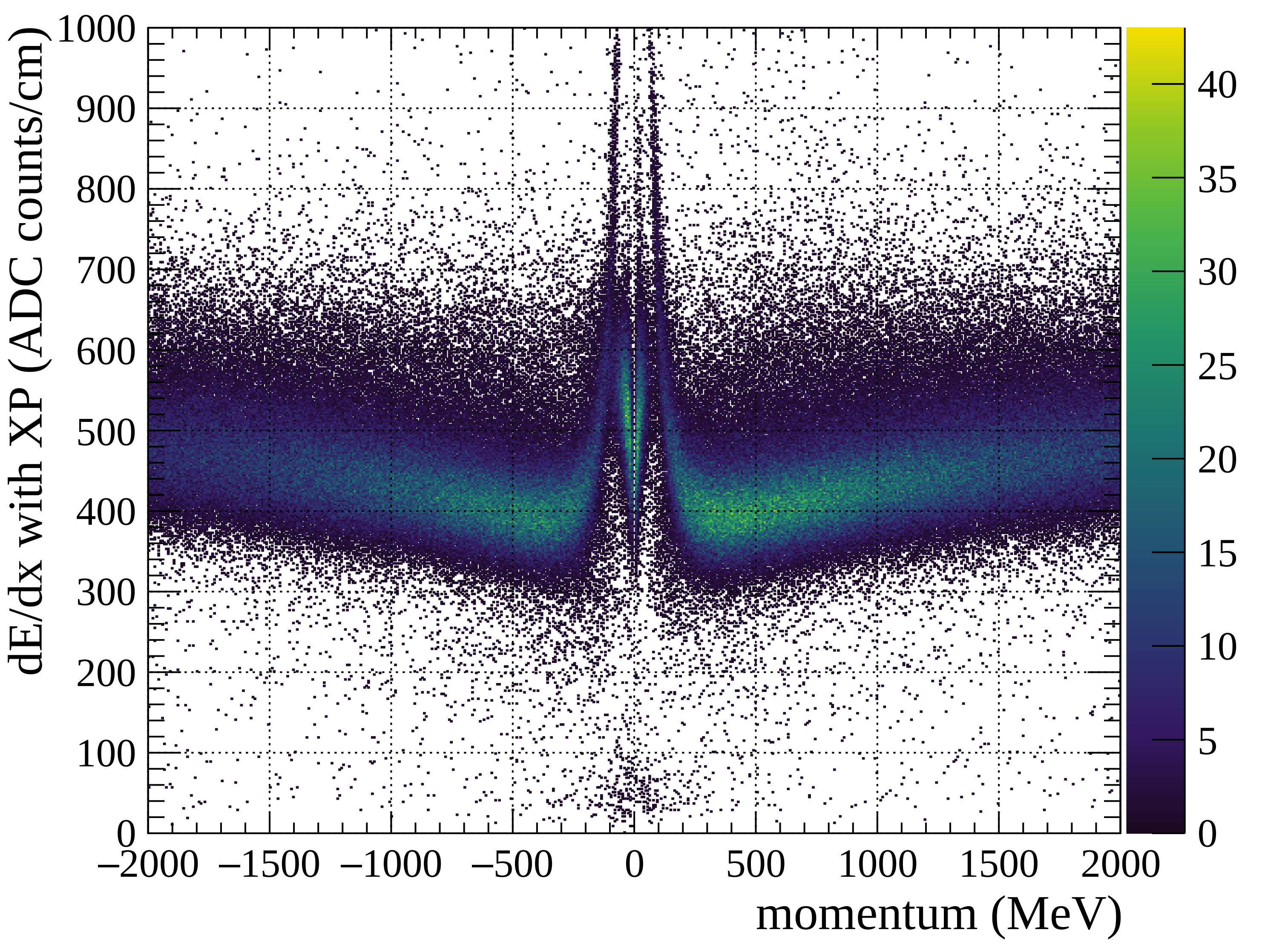


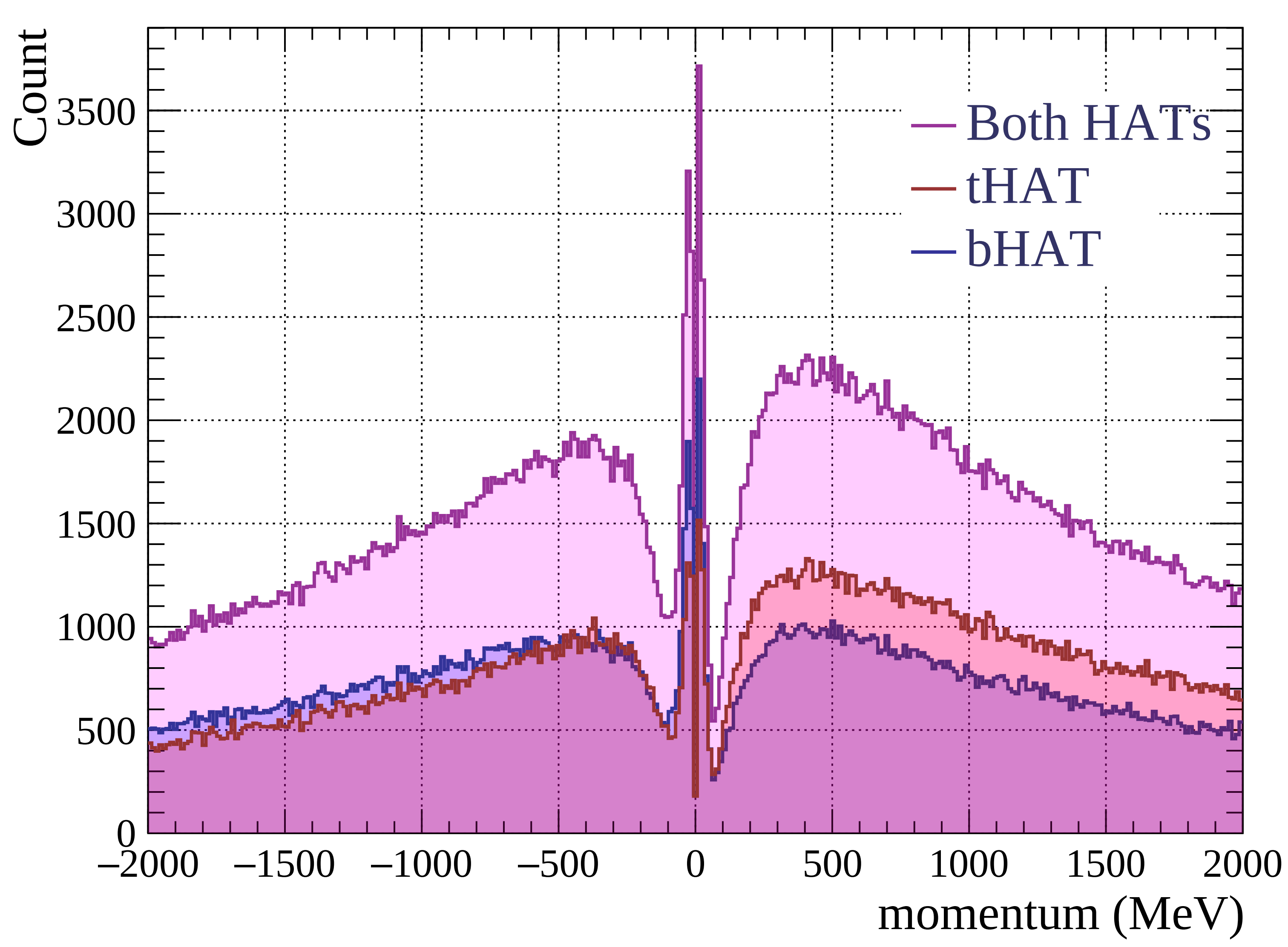
WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000



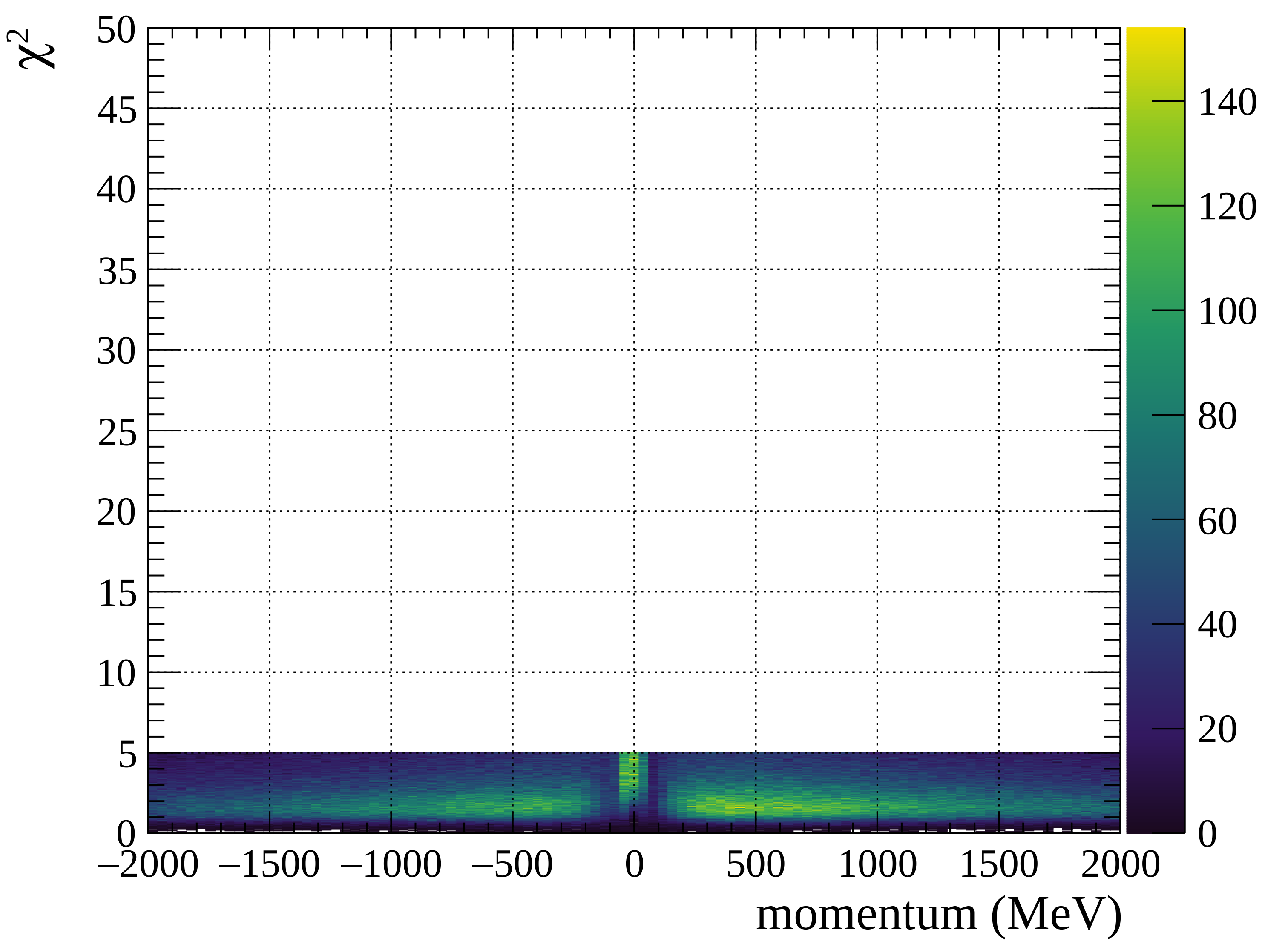
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000


dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
5
10
15
20
25
30
35

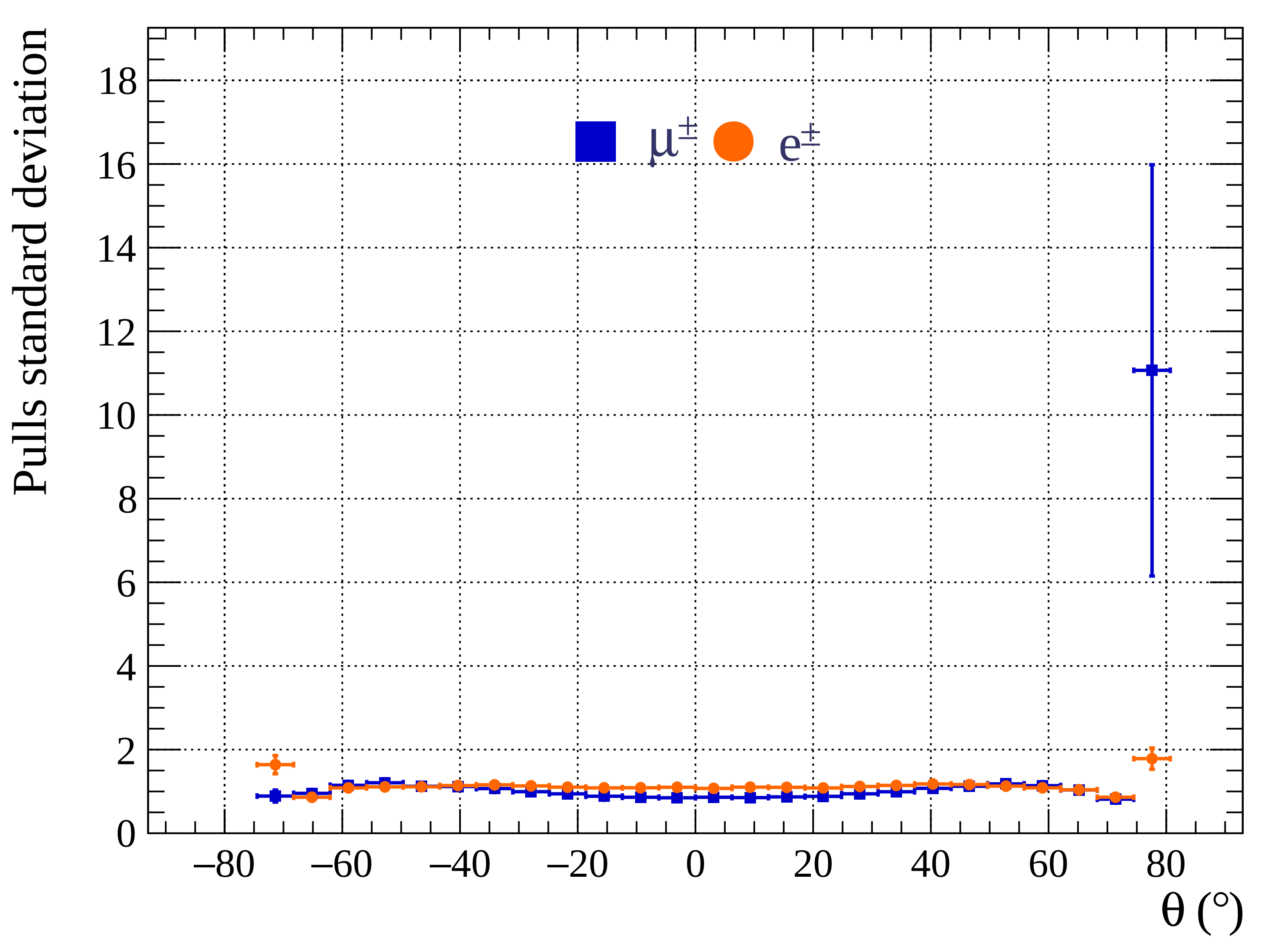

dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
5
10
15
20
25
30
35
40



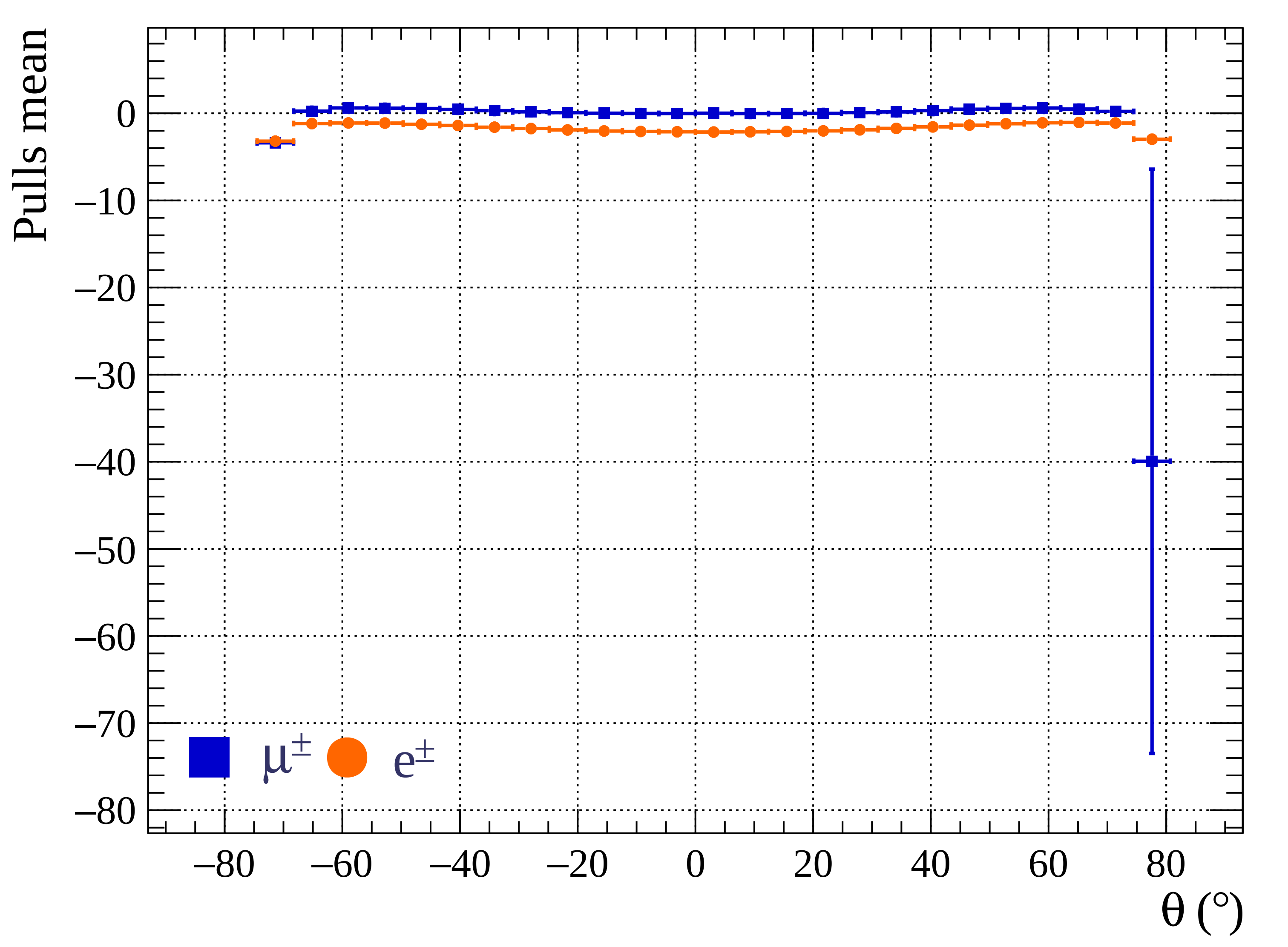
Count
3500
3000
2500
2000
1500
1000
500
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT

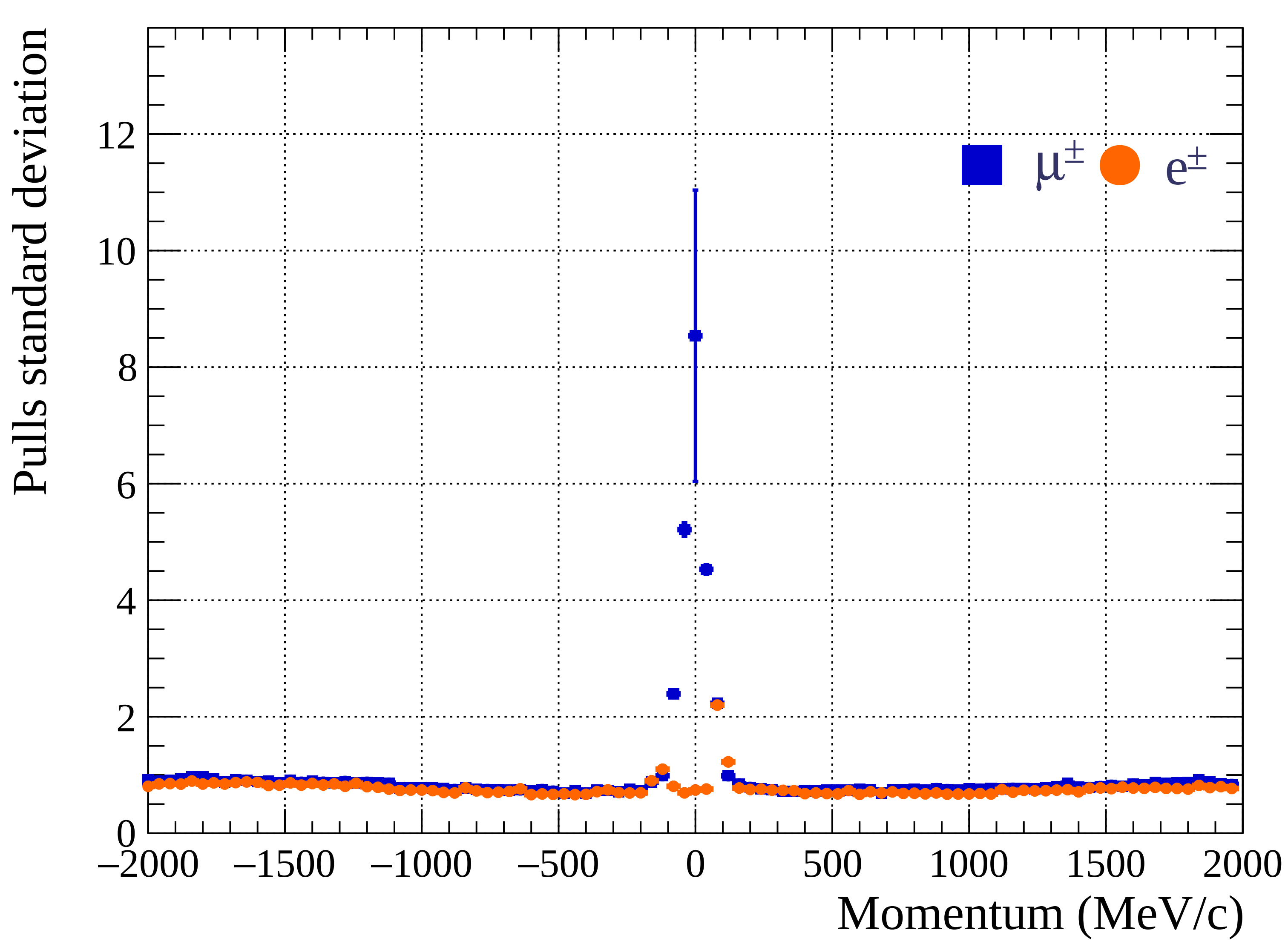


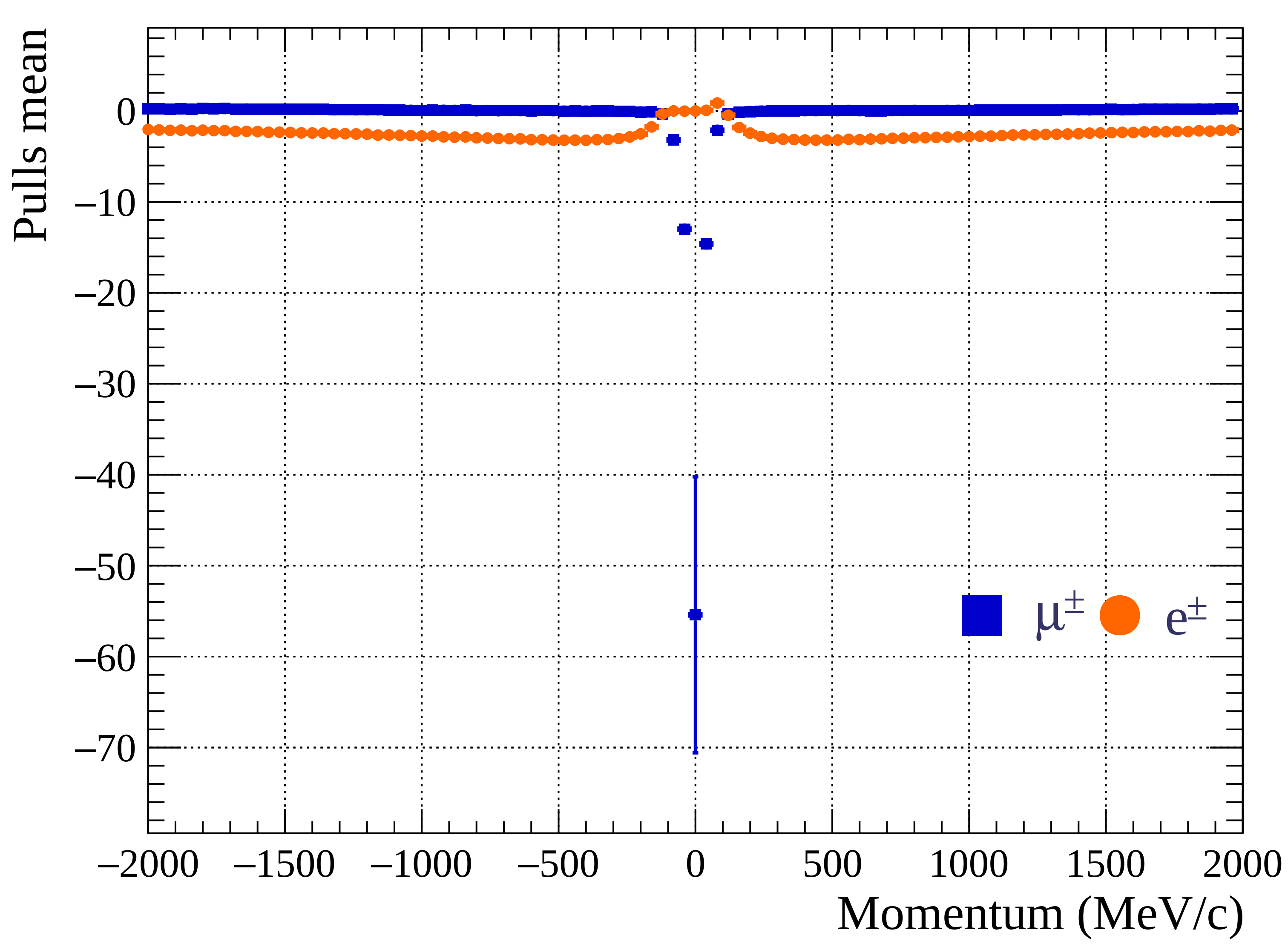
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
140
120
100
80
60
40
20
0



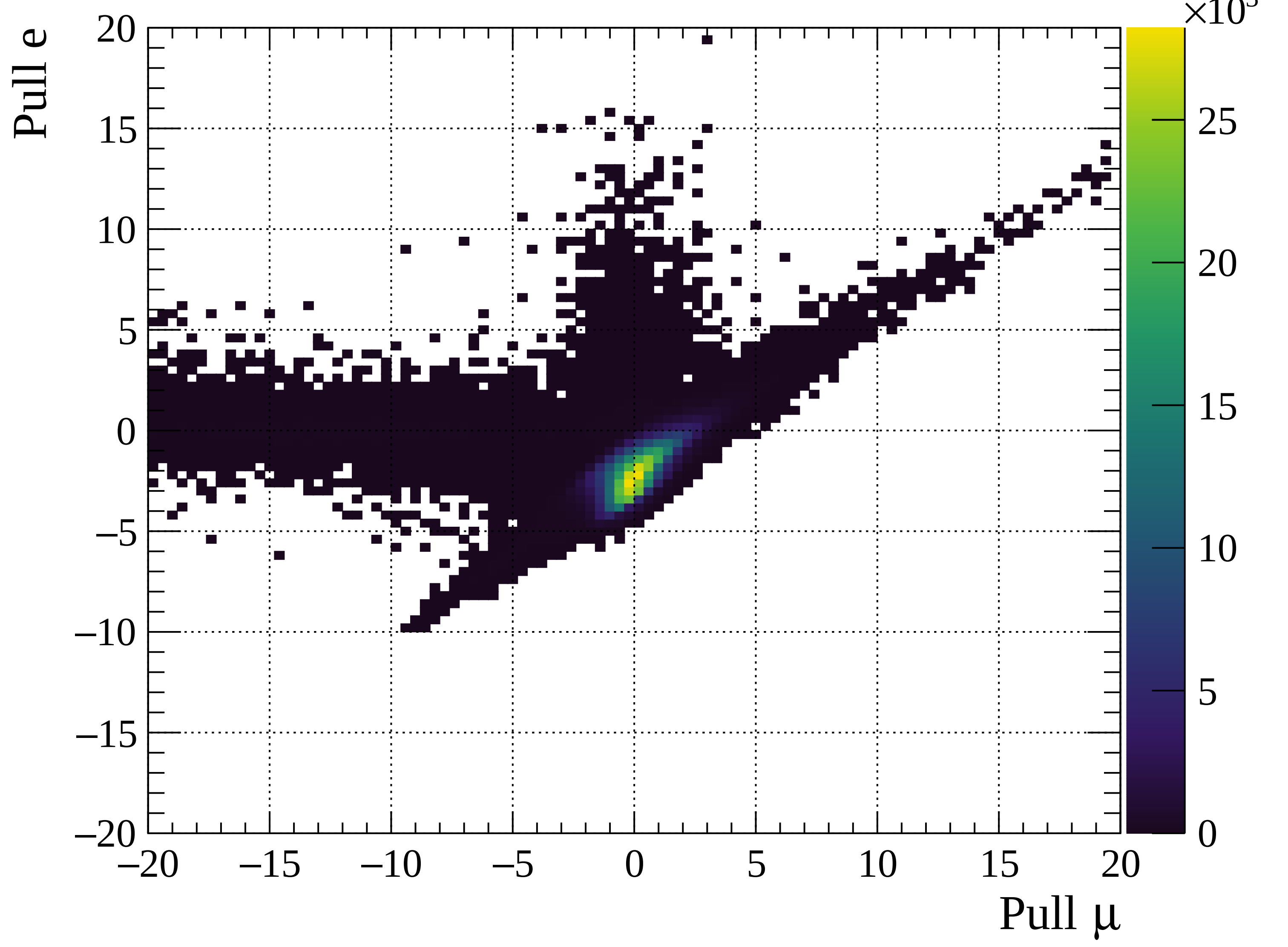
Pulls standard deviation
μ±
e±
θ (°)
0
2
4
6
8
10
12
14
16
18
−80
−60
−40
−20
0
20
40
60
80


Pulls mean
0
−10
−20
−30
−40
−50
−60
−70
−80
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

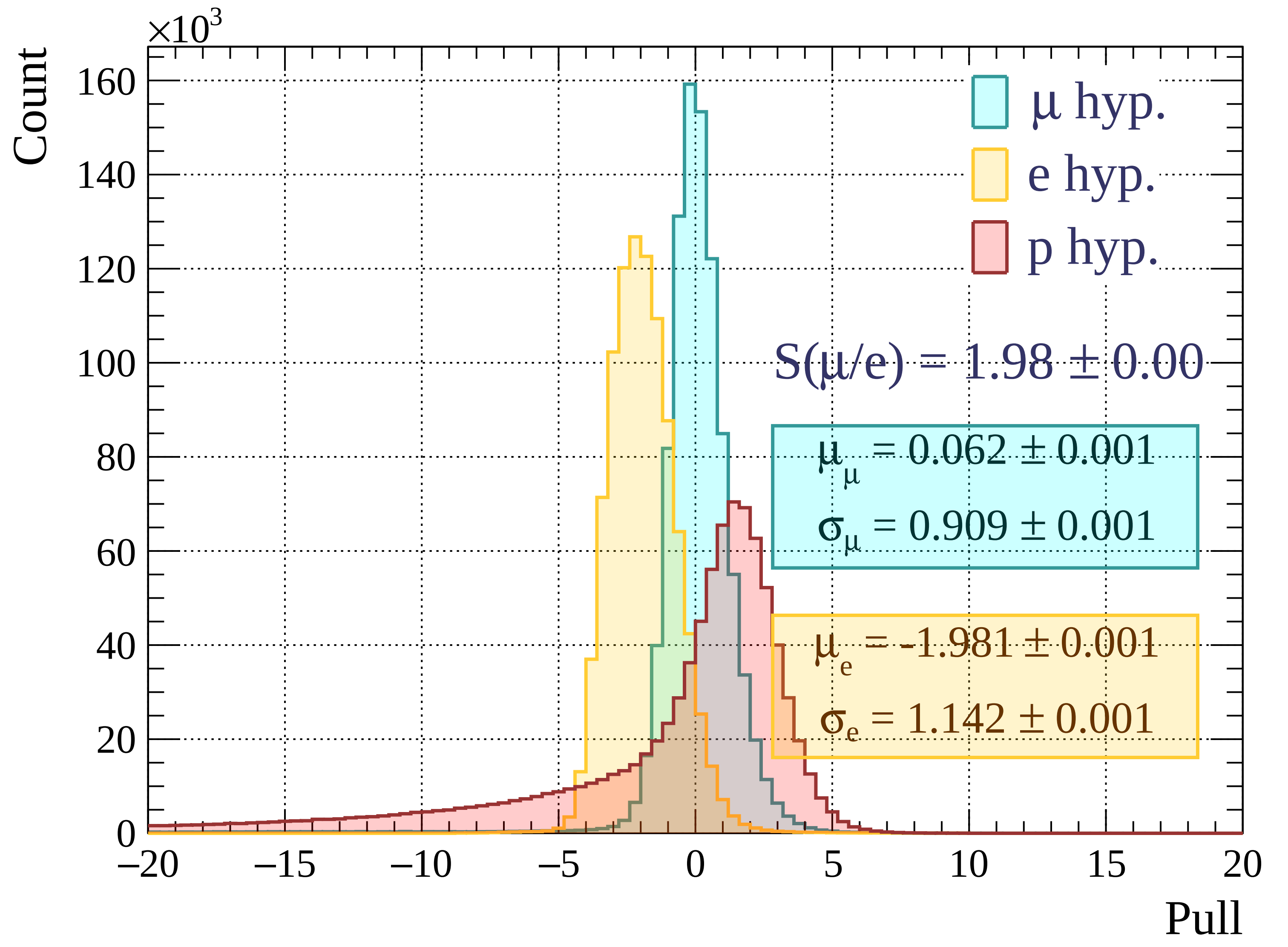

Pulls standard deviation
Momentum (MeV/c)
μ±
e±
0
2
4
6
8
10
12
−2000
−1500
−1000
−500
0
500
1000
1500
2000



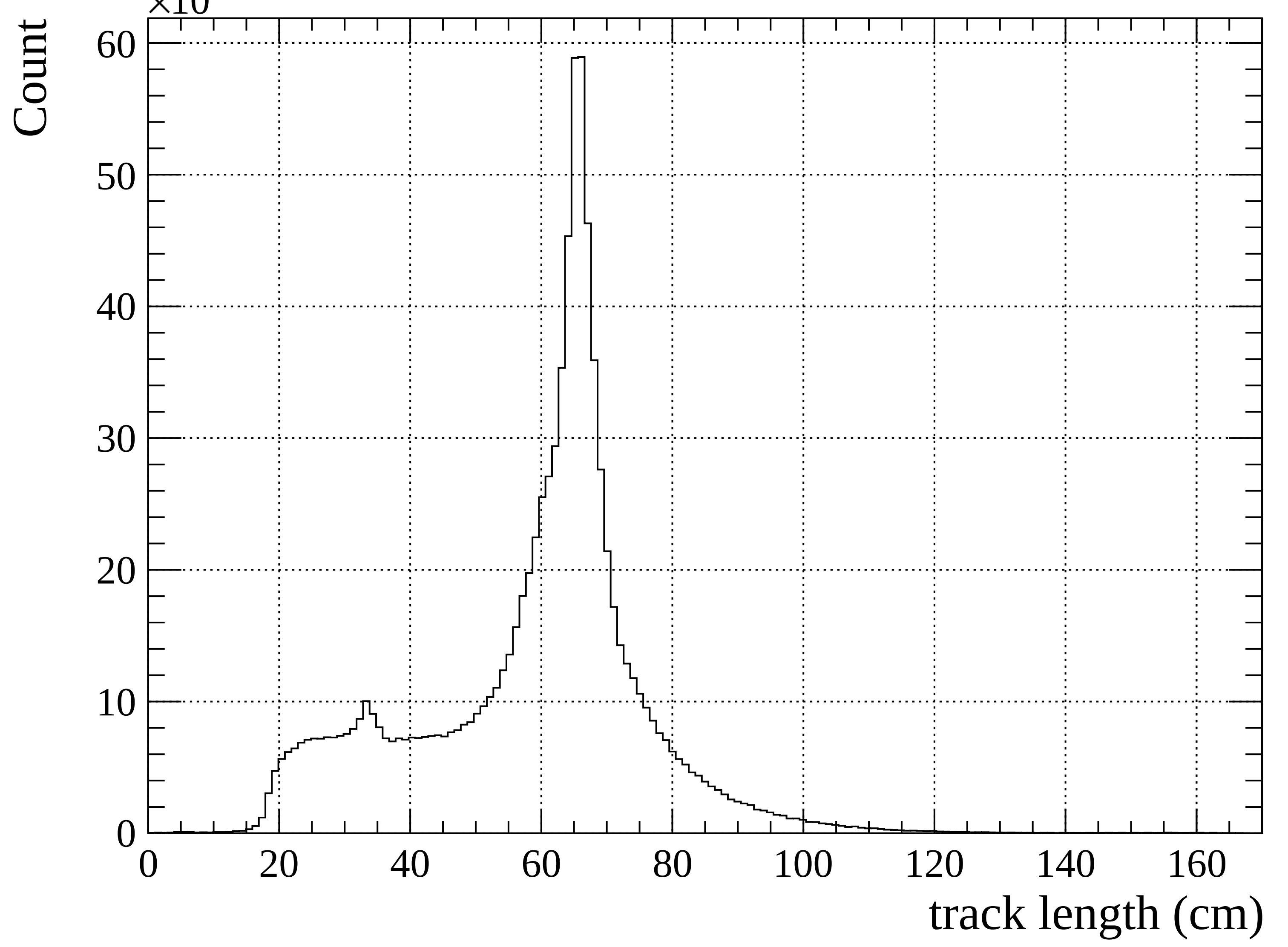
Pulls mean
0
−10
−20
−30
−40
−50
−60
−70
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±



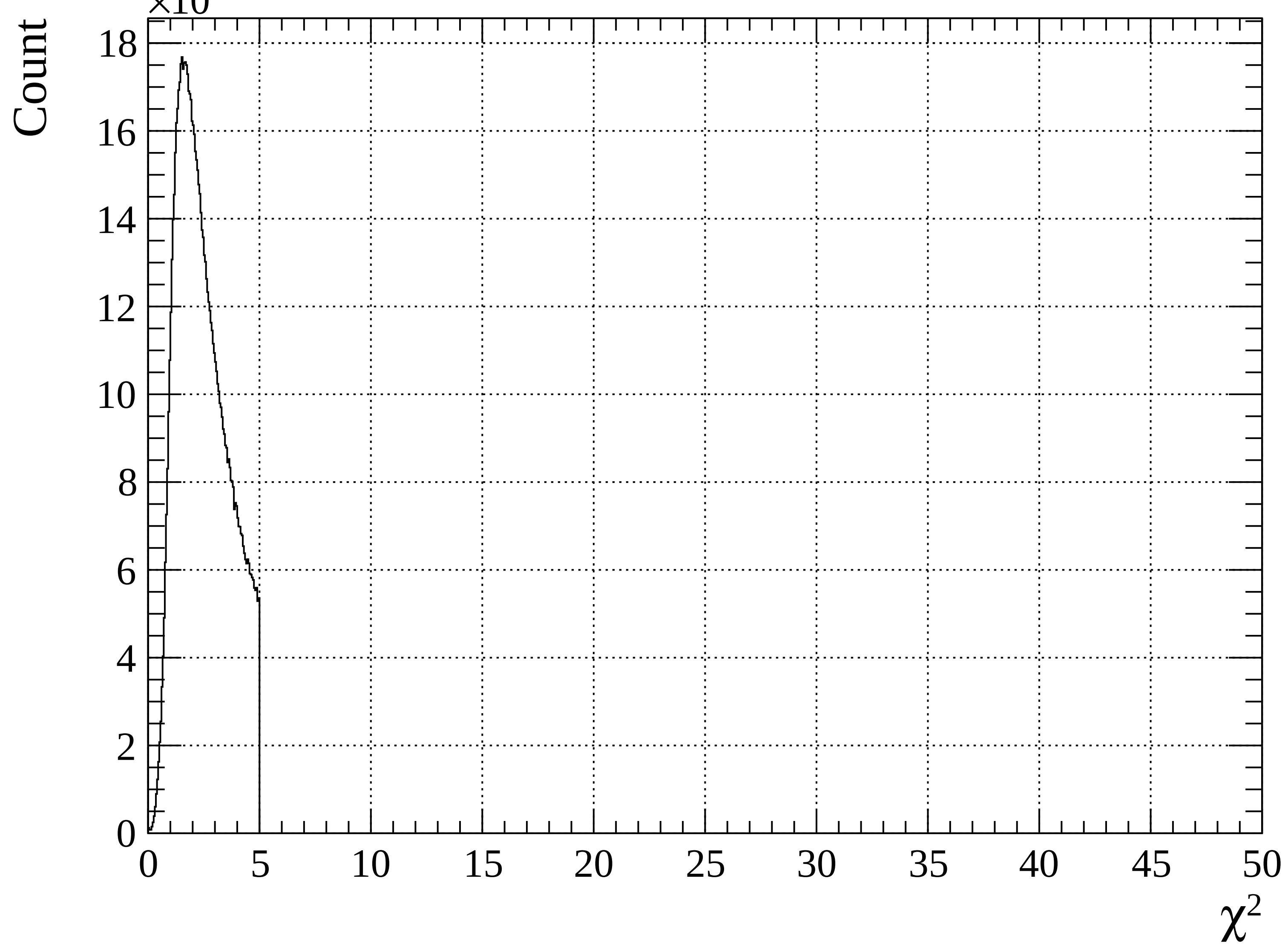
Pull e
Pull μ
×10³
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
25
20
15
10
5
0



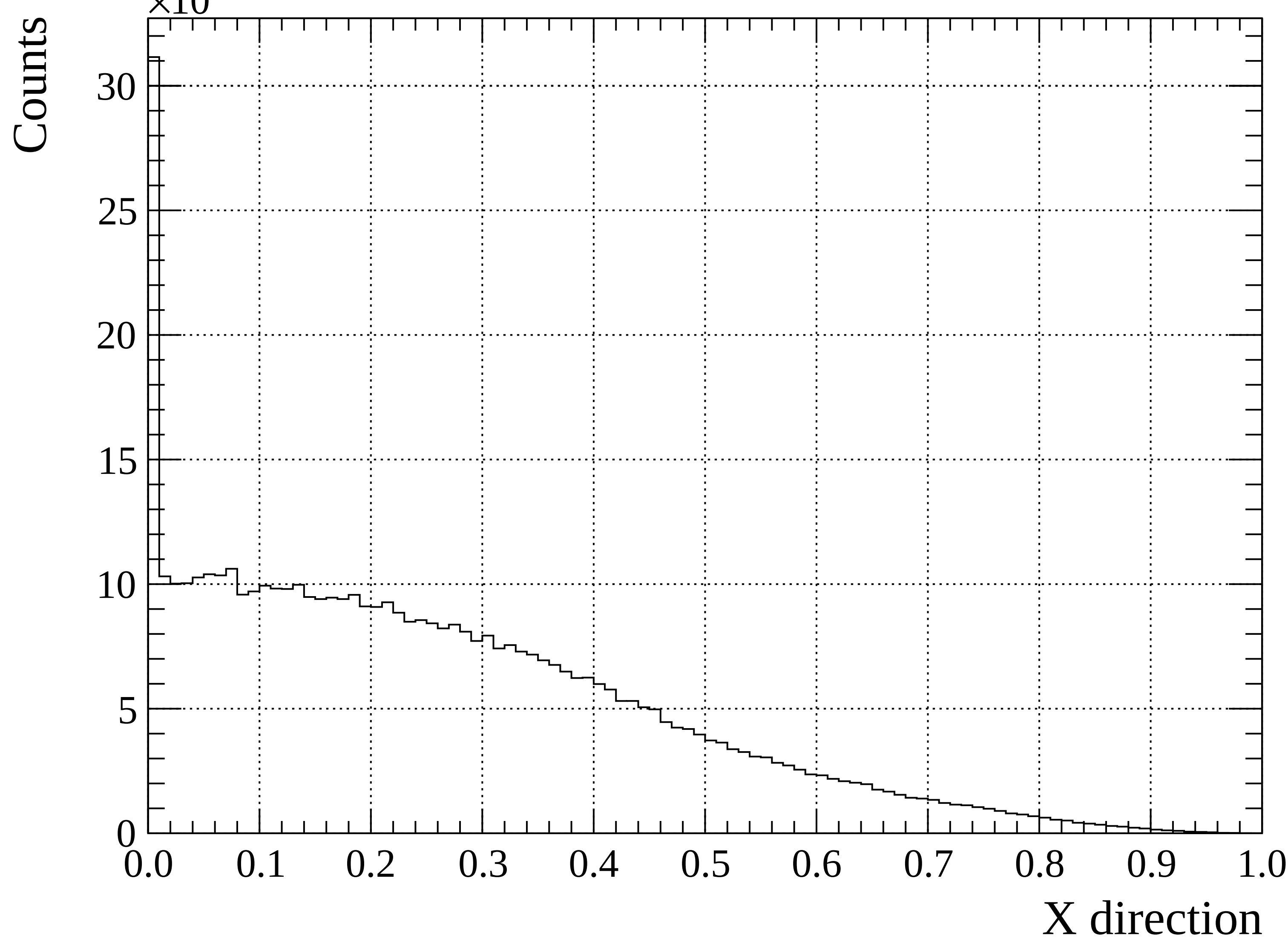
×10^3
Count
160
140
120
100
80
60
40
20
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 1.98 ± 0.00
μ_μ = 0.062 ± 0.001
σ_μ = 0.909 ± 0.001
μ_e = -1.981 ± 0.001
σ_e = 1.142 ± 0.001

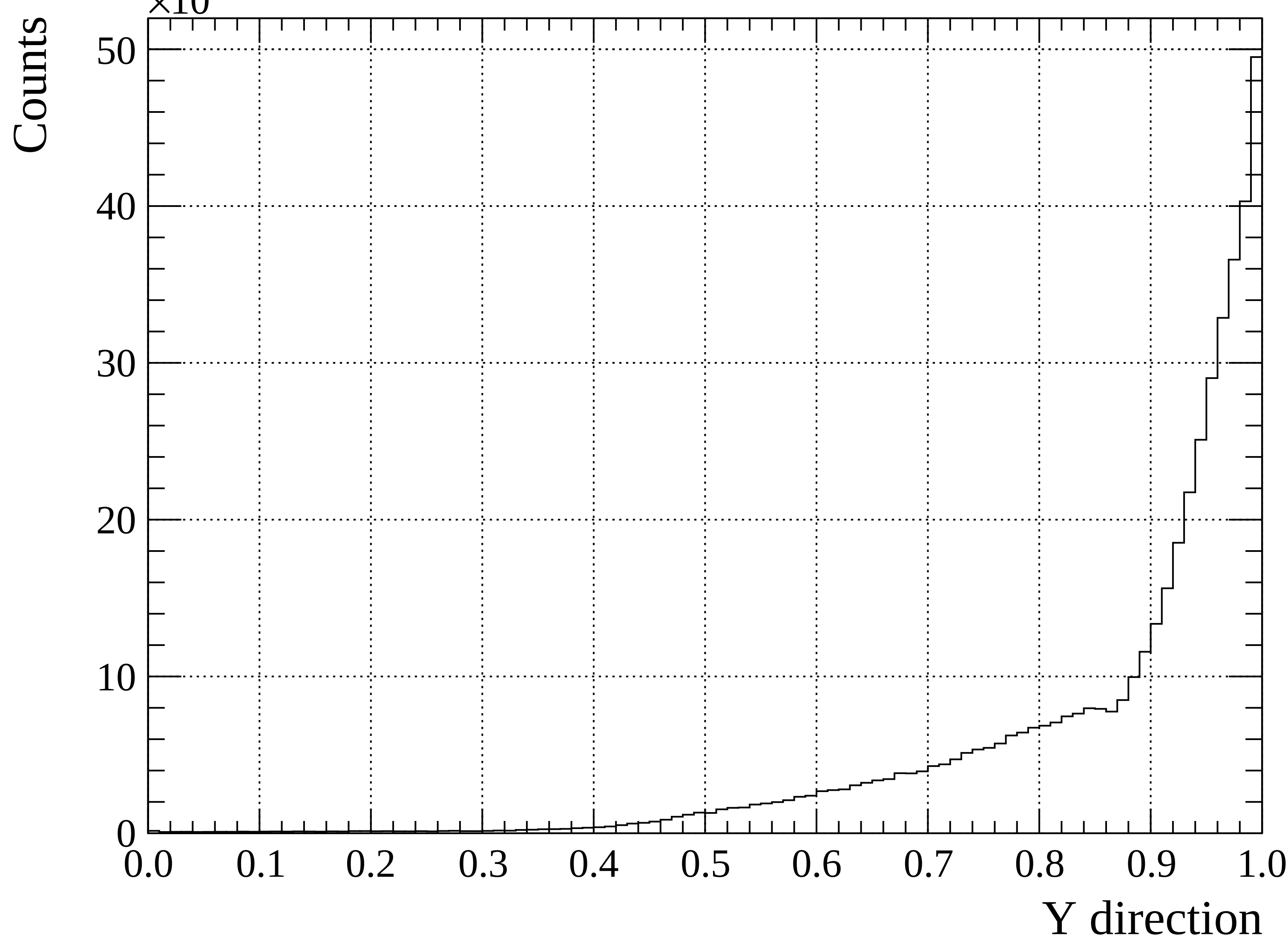


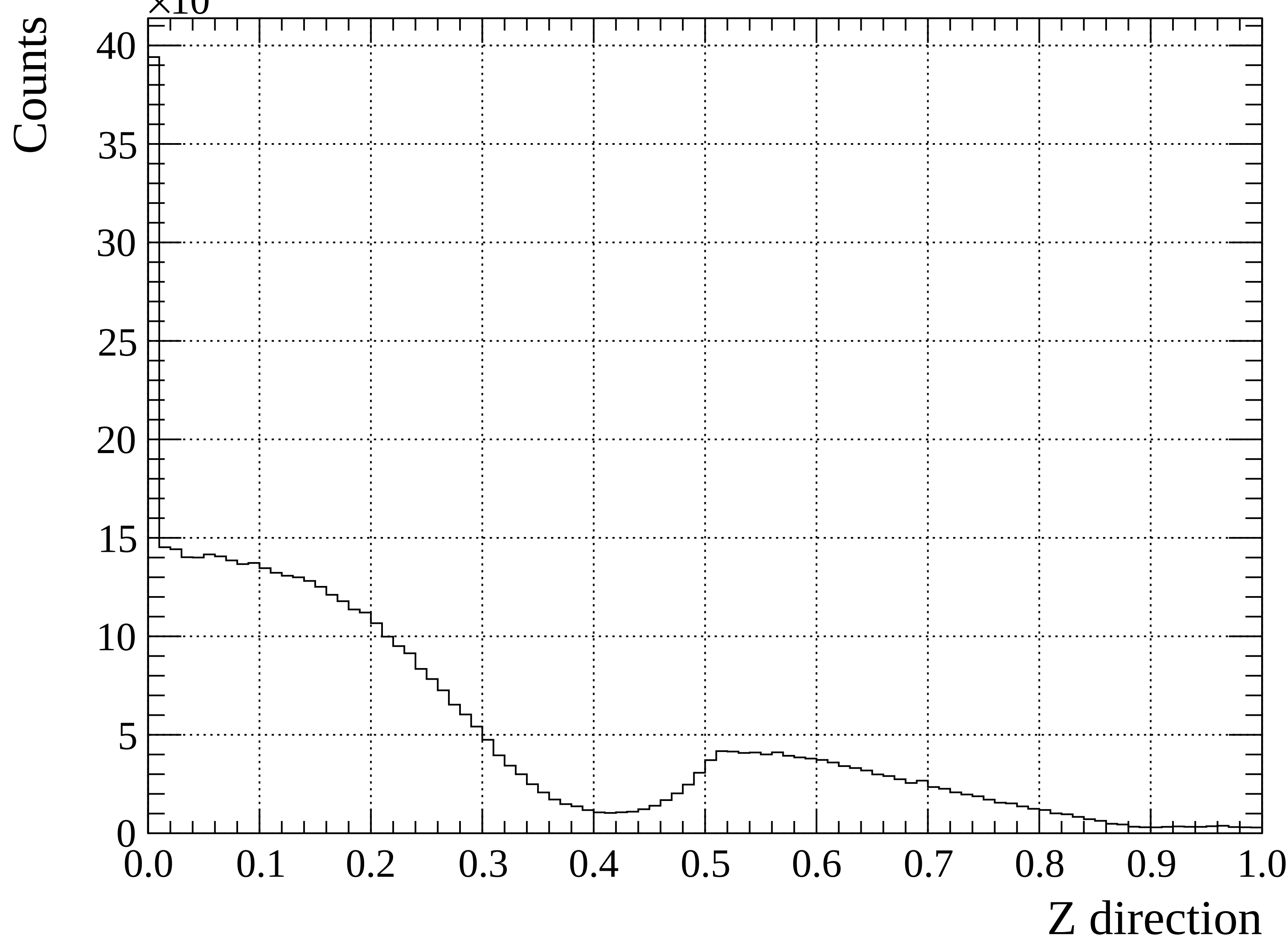
×10
Count
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
160
track length (cm)

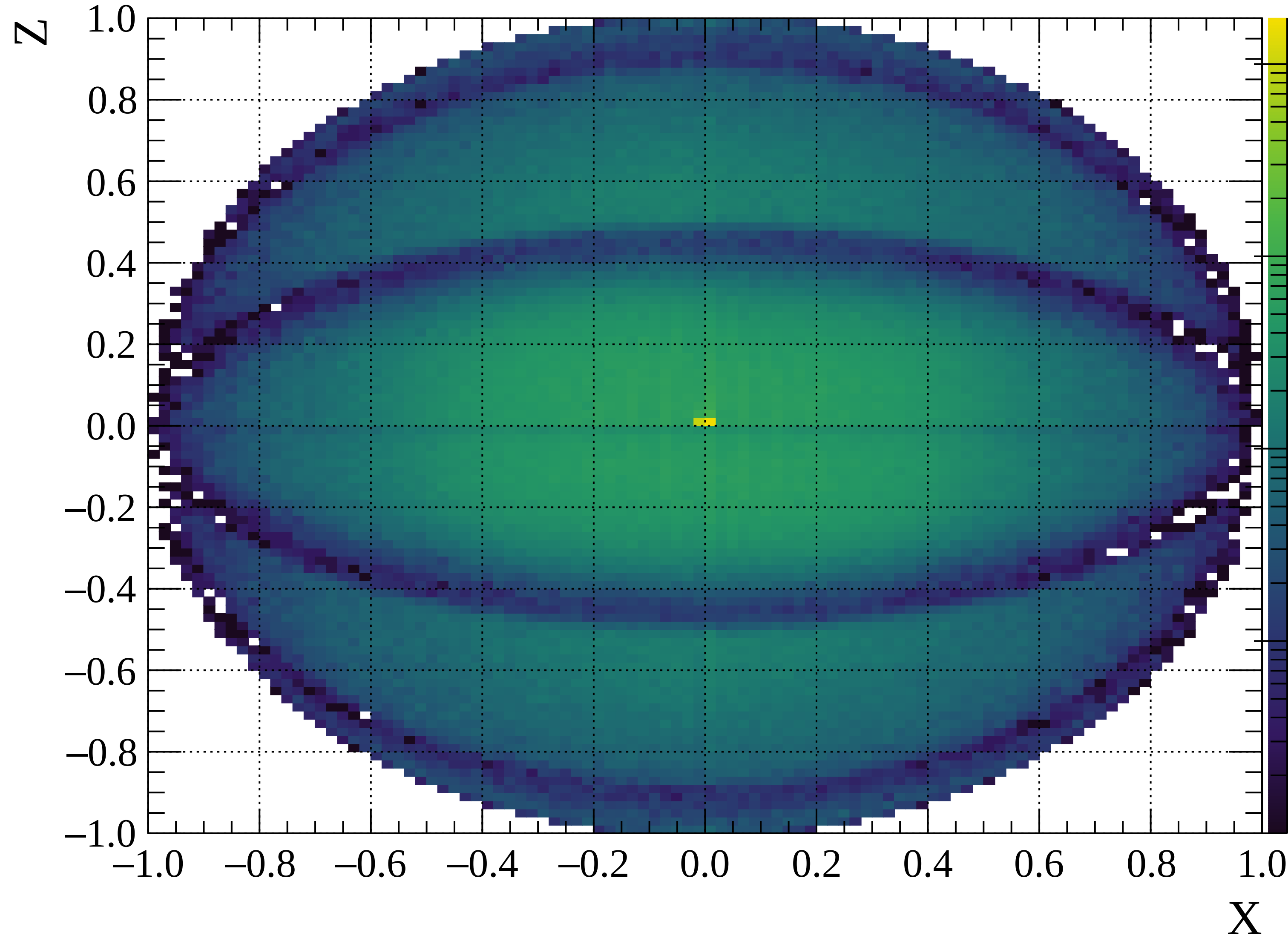


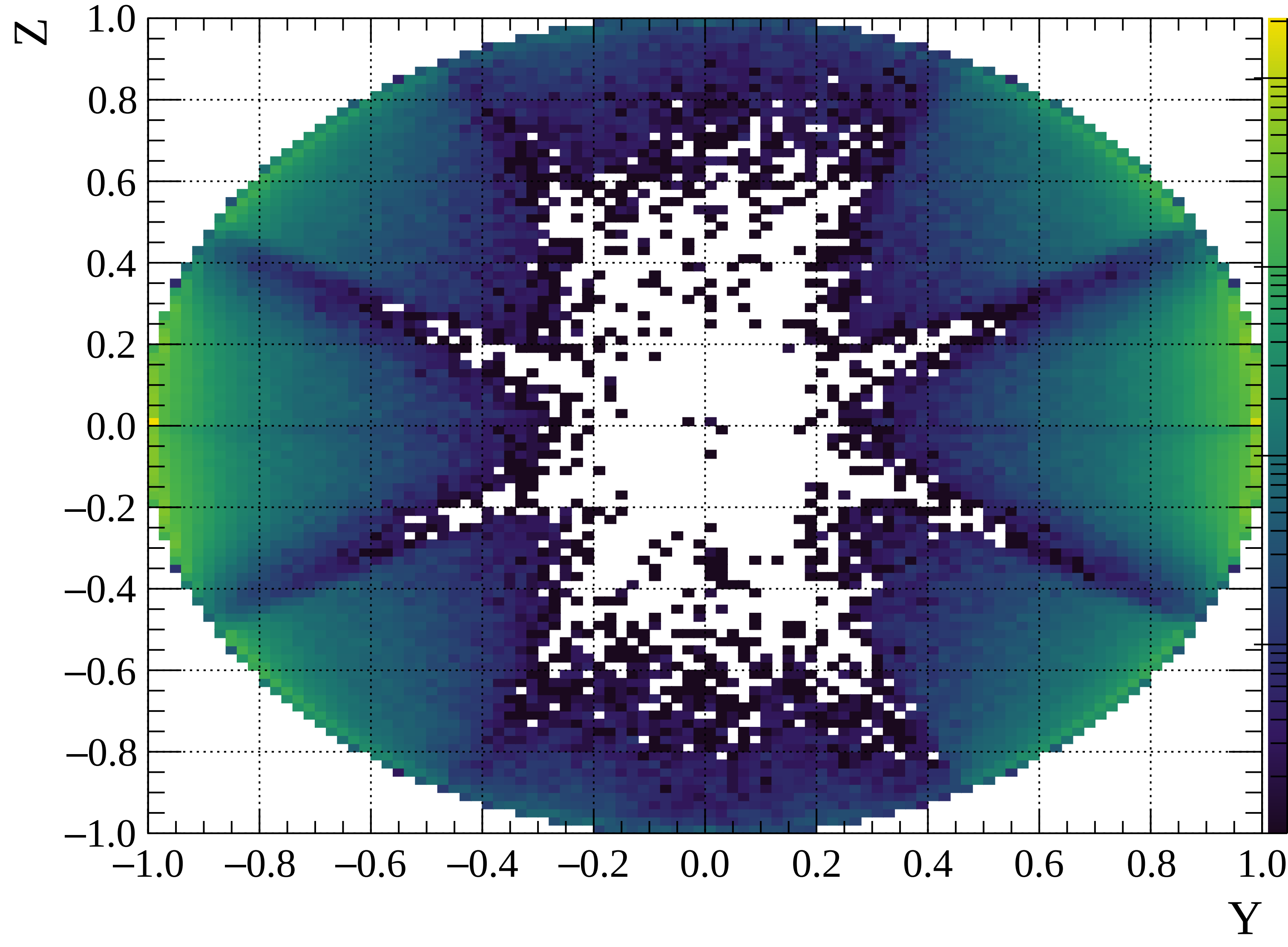
Count
×10
18
16
14
12
10
8
6
4
2
0
0
5
10
15
20
25
30
35
40
45
50
χ^2


×10
Counts
0
5
10
15
20
25
30
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction

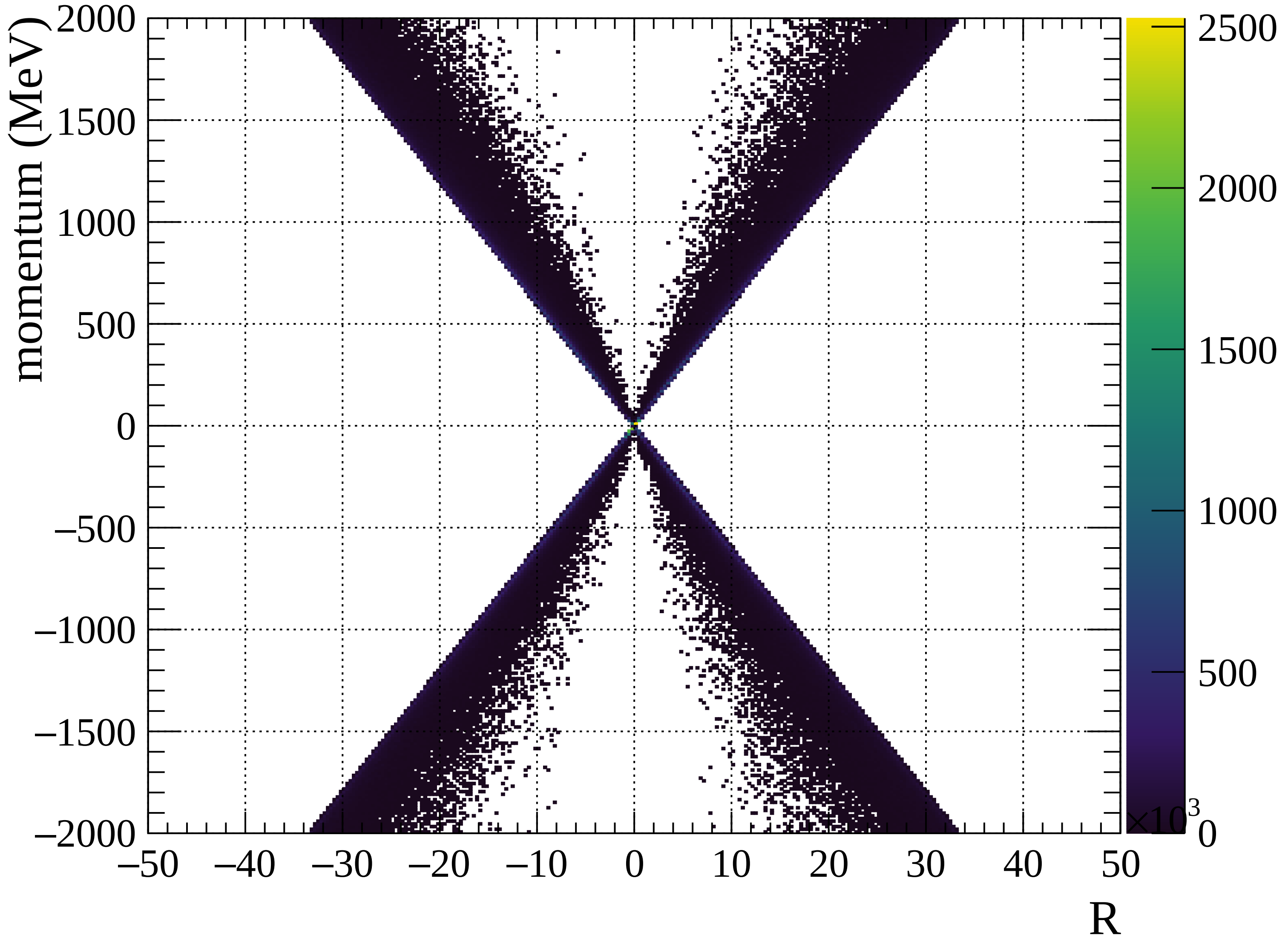

Counts
×10
50
40
30
20
10
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction


×10
Counts
0
5
10
15
20
25
30
35
40
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction

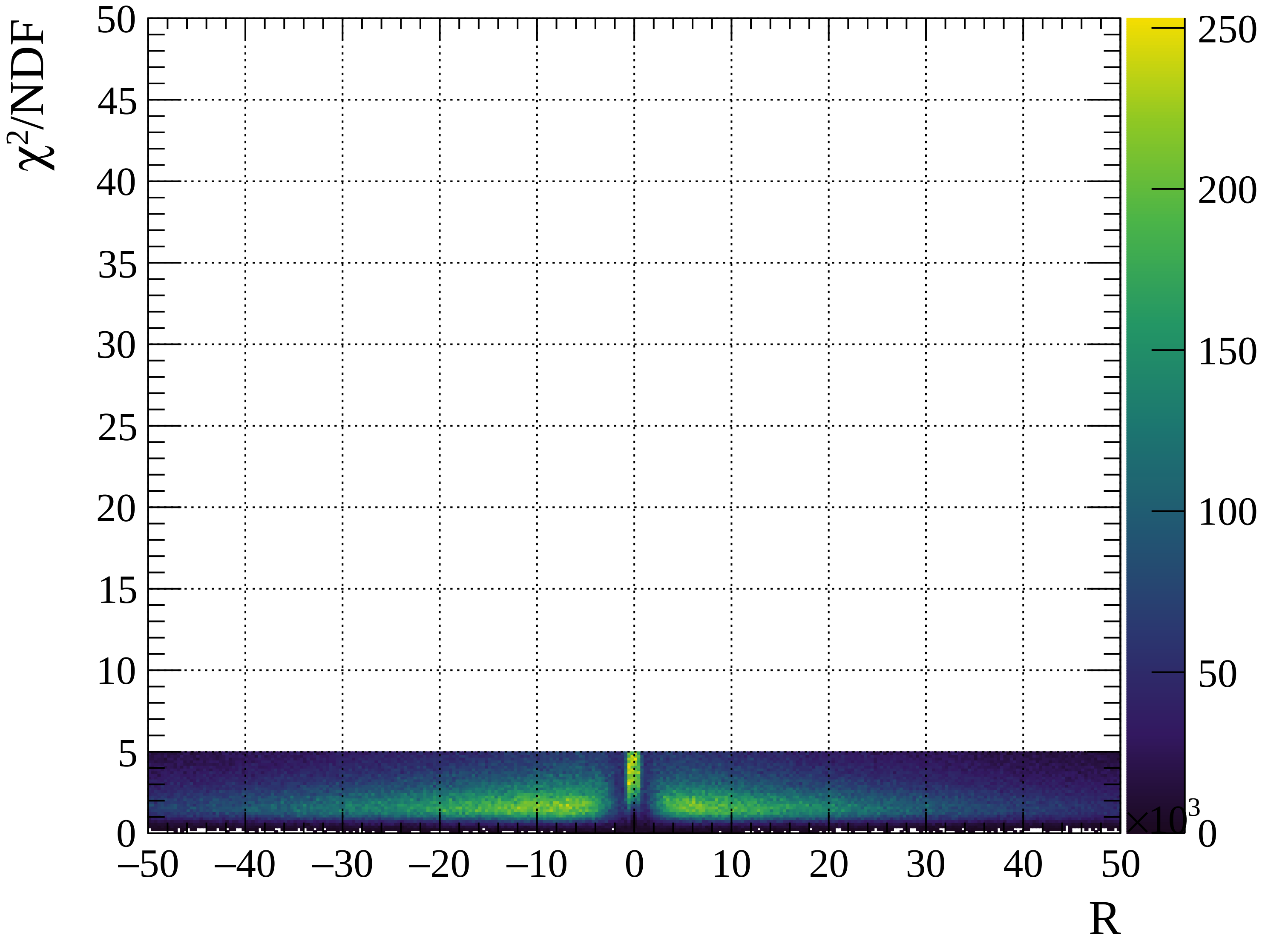

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X



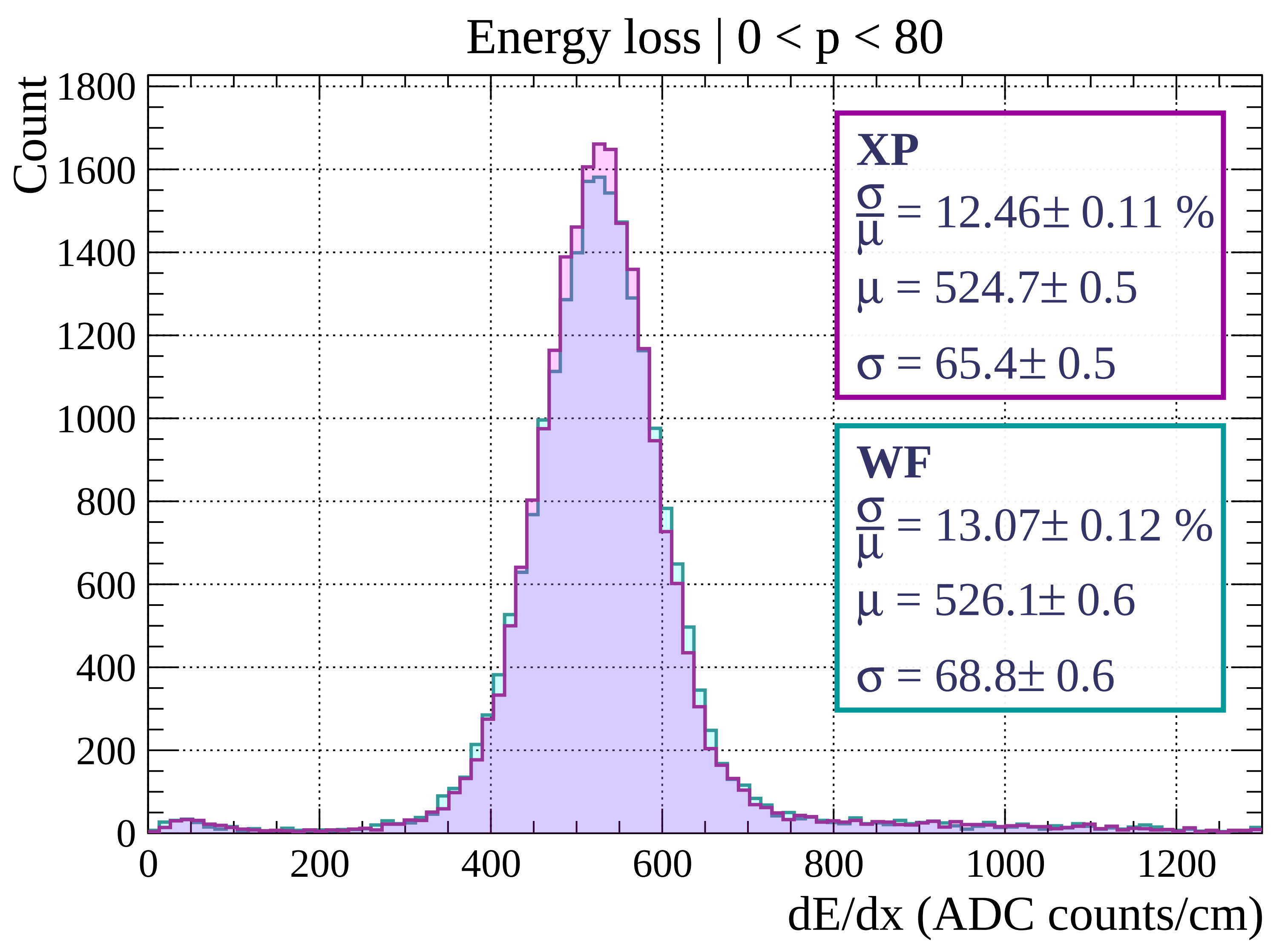
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y



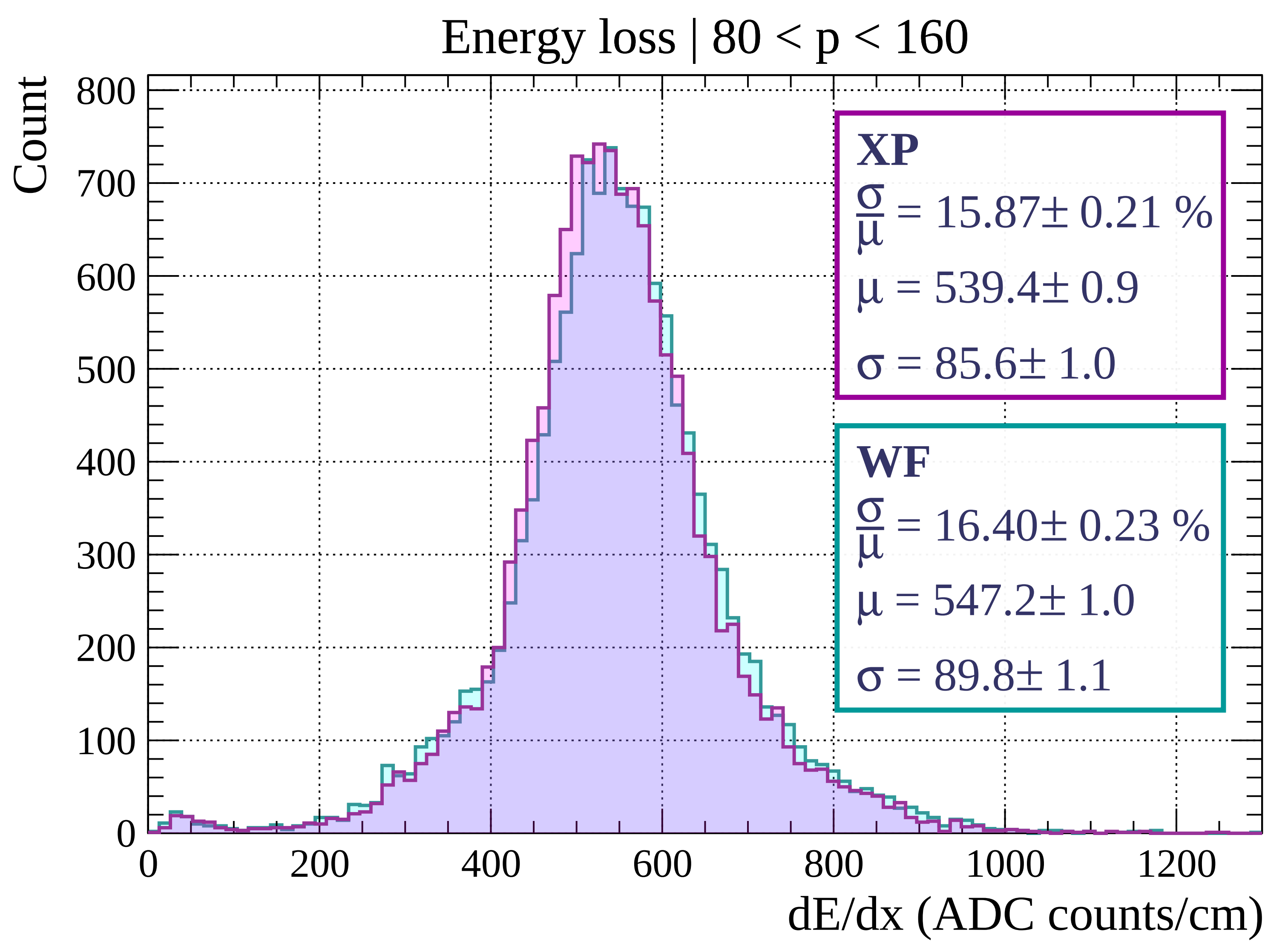
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
2500
2000
1500
1000
500
0
×10³

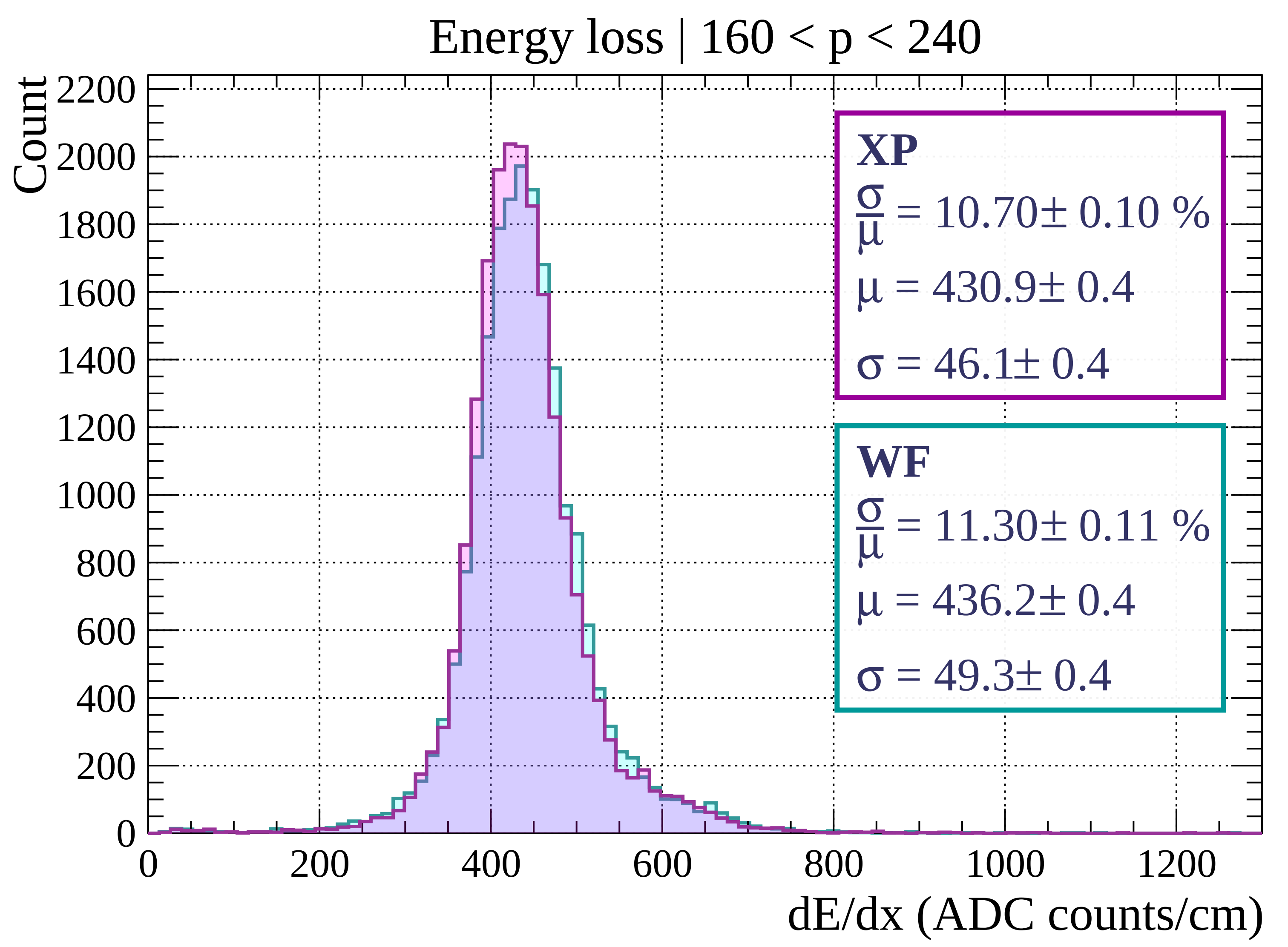


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
250
200
150
100
50
0
$\times 10^3$

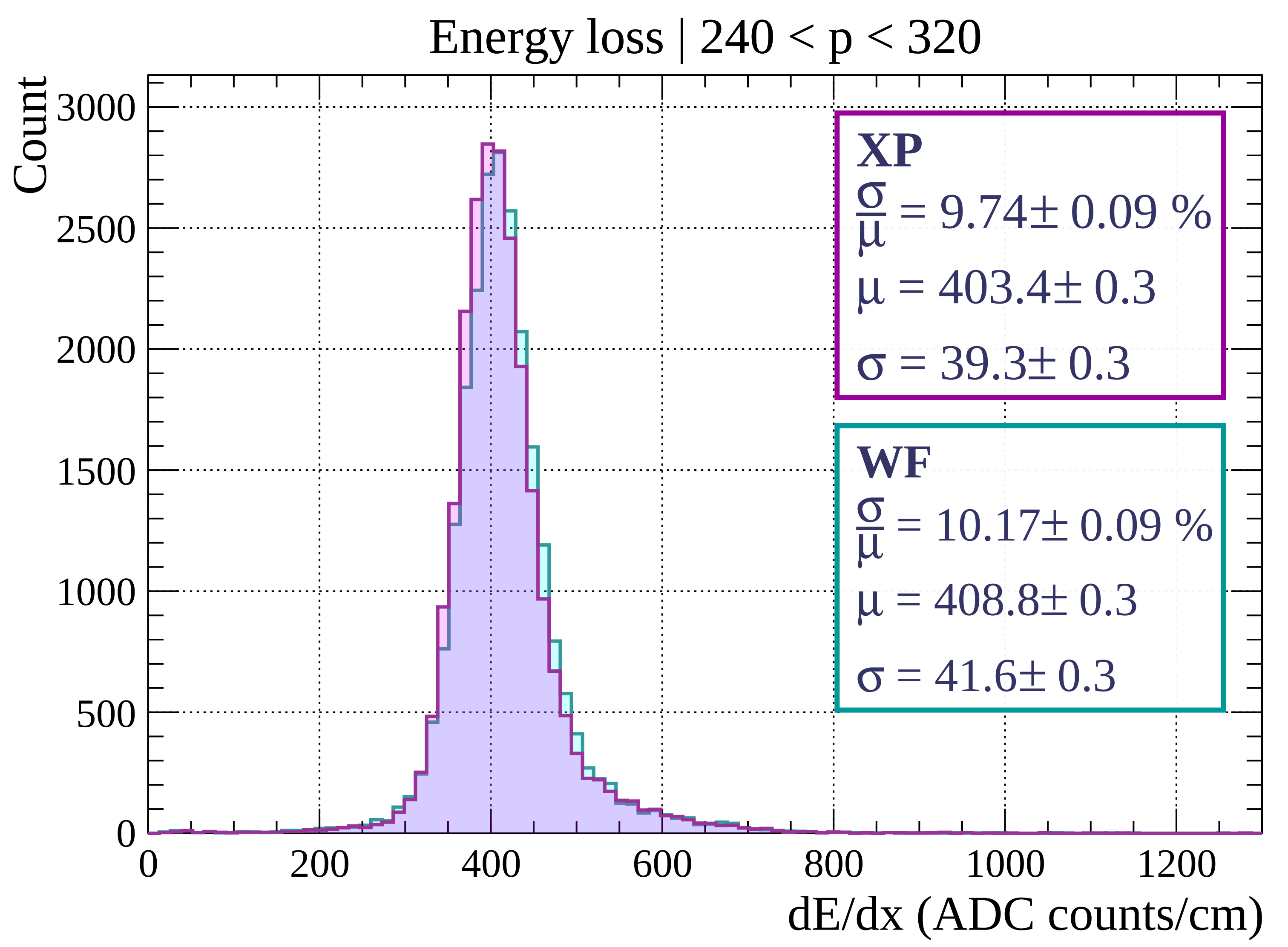


Energy loss | 0 < p < 80
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 12.46± 0.11 %
μ = 524.7± 0.5
σ = 65.4± 0.5
WF
$\frac{\sigma}{\mu}$ = 13.07± 0.12 %
μ = 526.1± 0.6
σ = 68.8± 0.6
0
200
400
600
800
1000
1200
1400
1600
1800

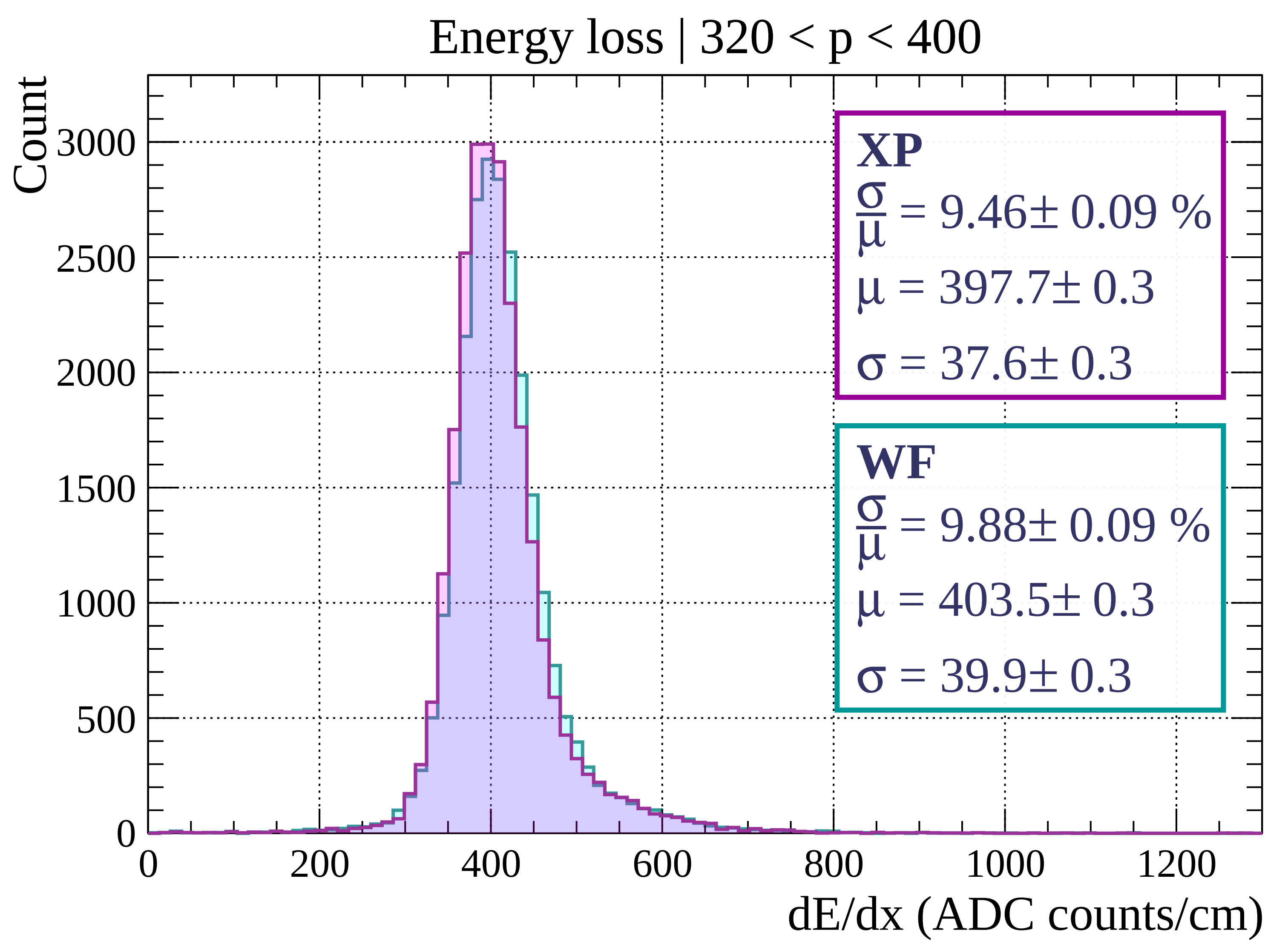

Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 15.87± 0.21 %
μ = 539.4± 0.9
σ = 85.6± 1.0
WF
σ/μ = 16.40± 0.23 %
μ = 547.2± 1.0
σ = 89.8± 1.1



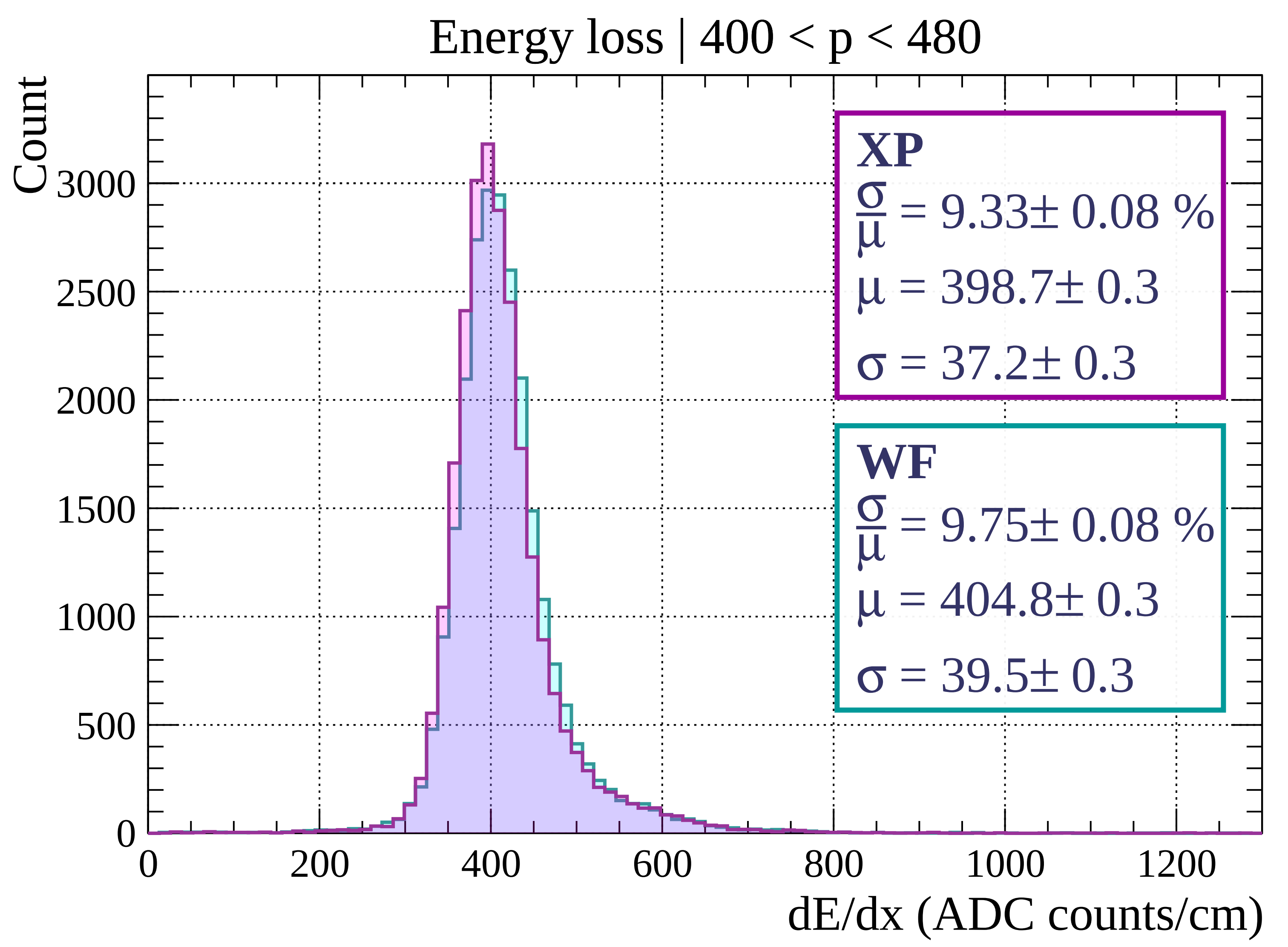
Energy loss | 160 < p < 240
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.70± 0.10 %
μ = 430.9± 0.4
σ = 46.1± 0.4
WF
$\frac{\sigma}{\mu}$ = 11.30± 0.11 %
μ = 436.2± 0.4
σ = 49.3± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200



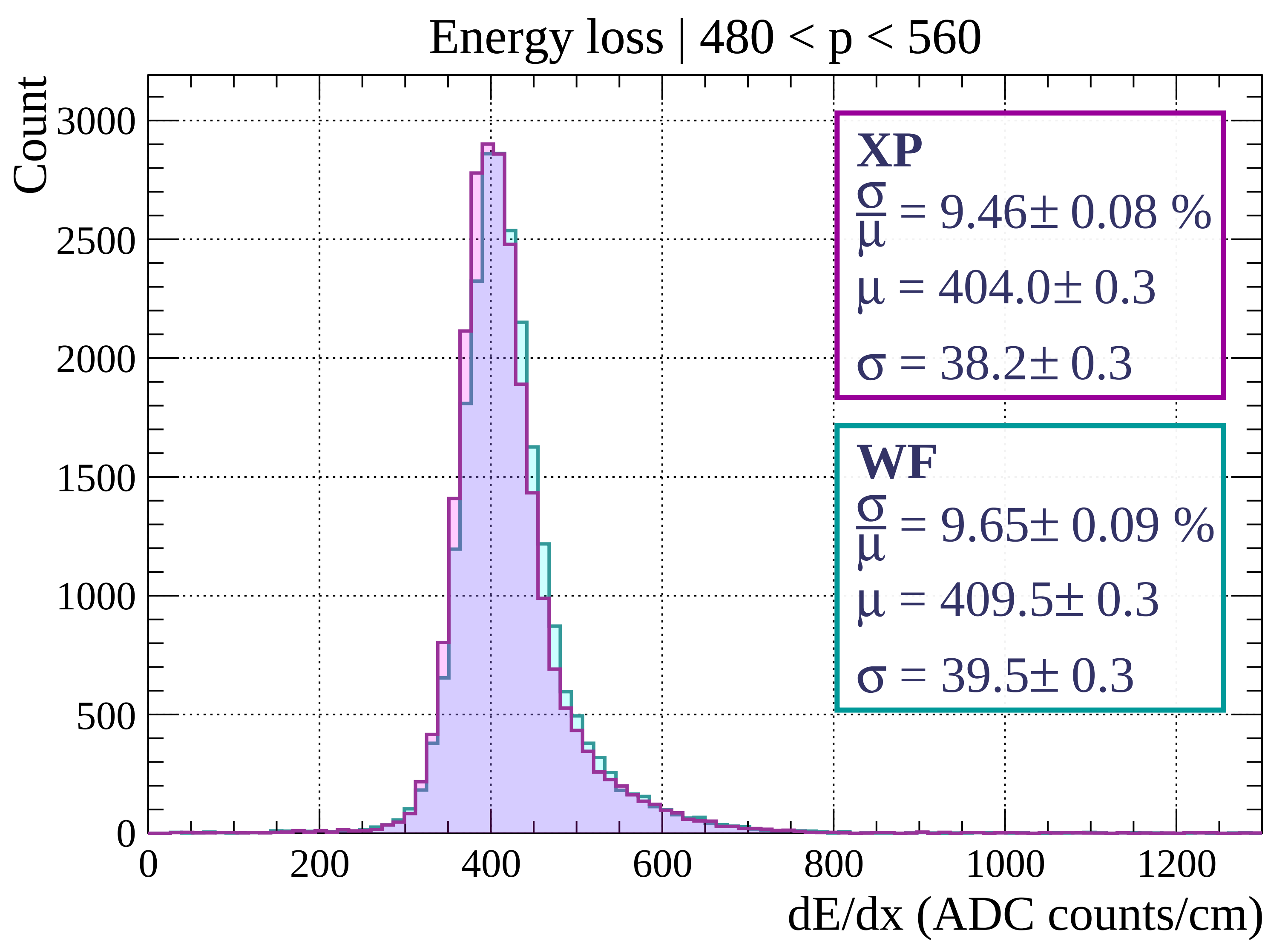
Energy loss | 240 < p < 320
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.74± 0.09 %
μ = 403.4± 0.3
σ = 39.3± 0.3
WF
σ/μ = 10.17± 0.09 %
μ = 408.8± 0.3
σ = 41.6± 0.3



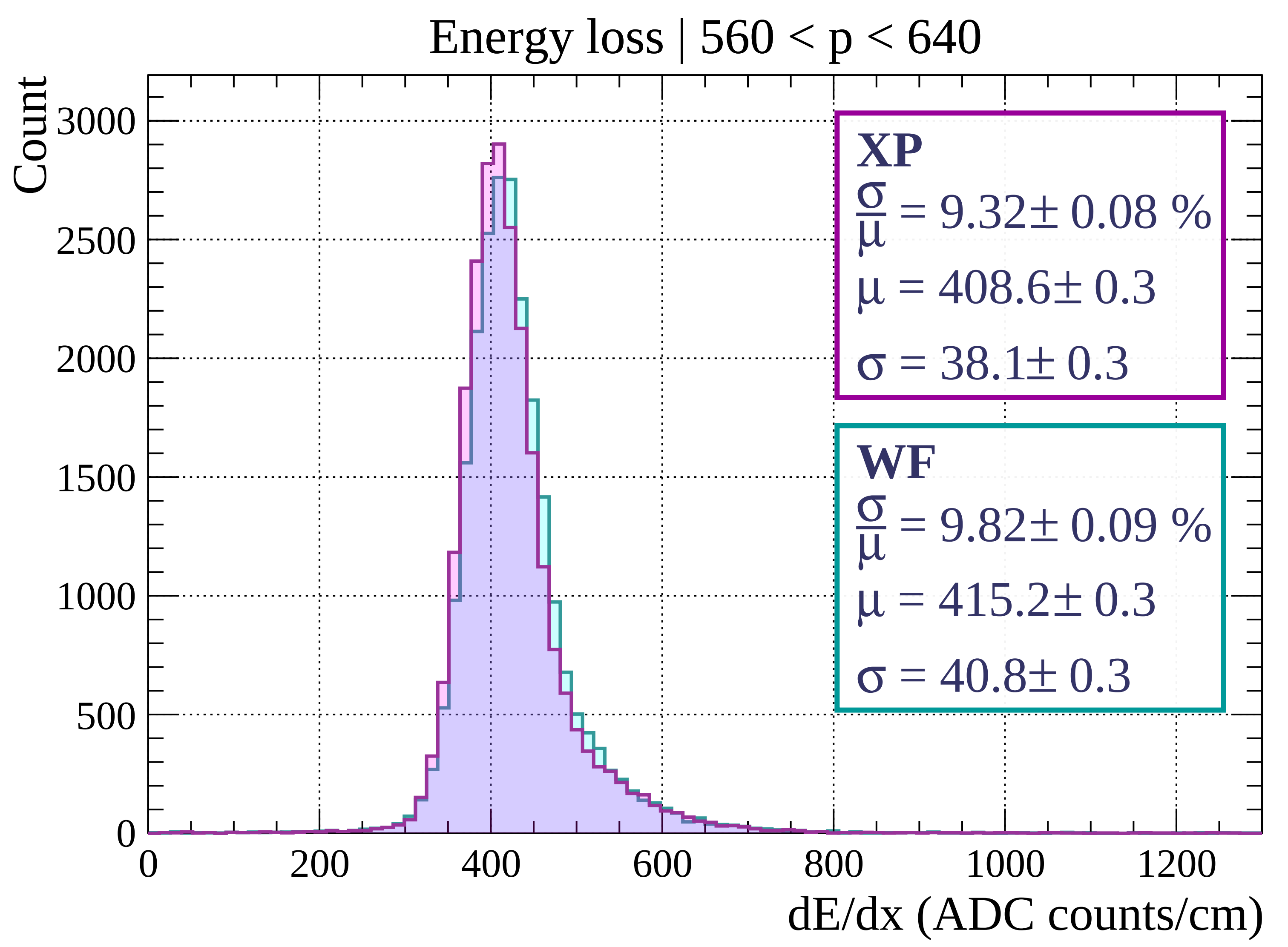
Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.46± 0.09 %
μ = 397.7± 0.3
σ = 37.6± 0.3
WF
$\frac{\sigma}{\mu}$ = 9.88± 0.09 %
μ = 403.5± 0.3
σ = 39.9± 0.3



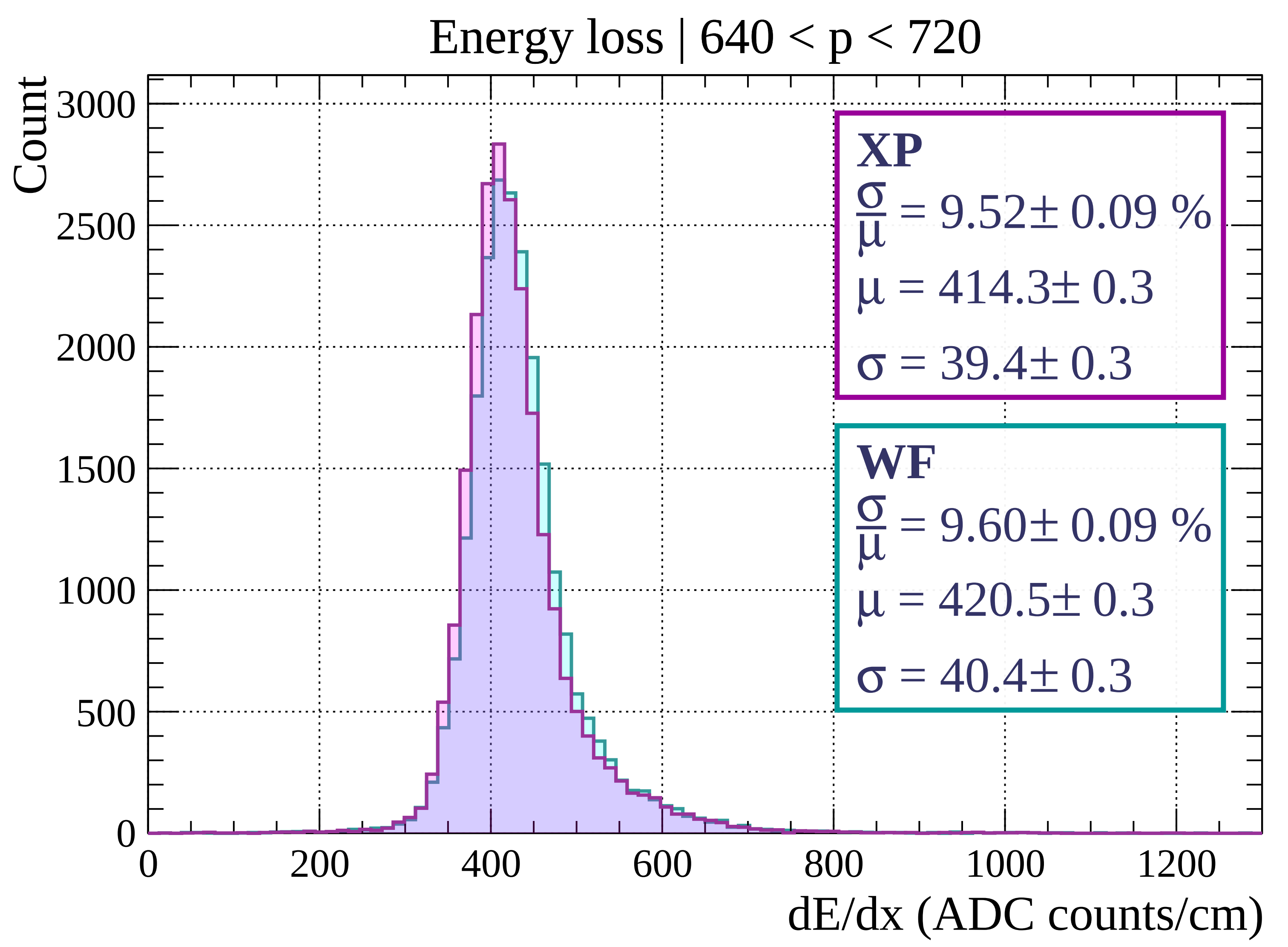
Energy loss | 400 < p < 480
Count
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.33 \pm 0.08$ %
$\mu = 398.7 \pm 0.3$
$\sigma = 37.2 \pm 0.3$
WF
$\frac{\sigma}{\mu} = 9.75 \pm 0.08$ %
$\mu = 404.8 \pm 0.3$
$\sigma = 39.5 \pm 0.3$

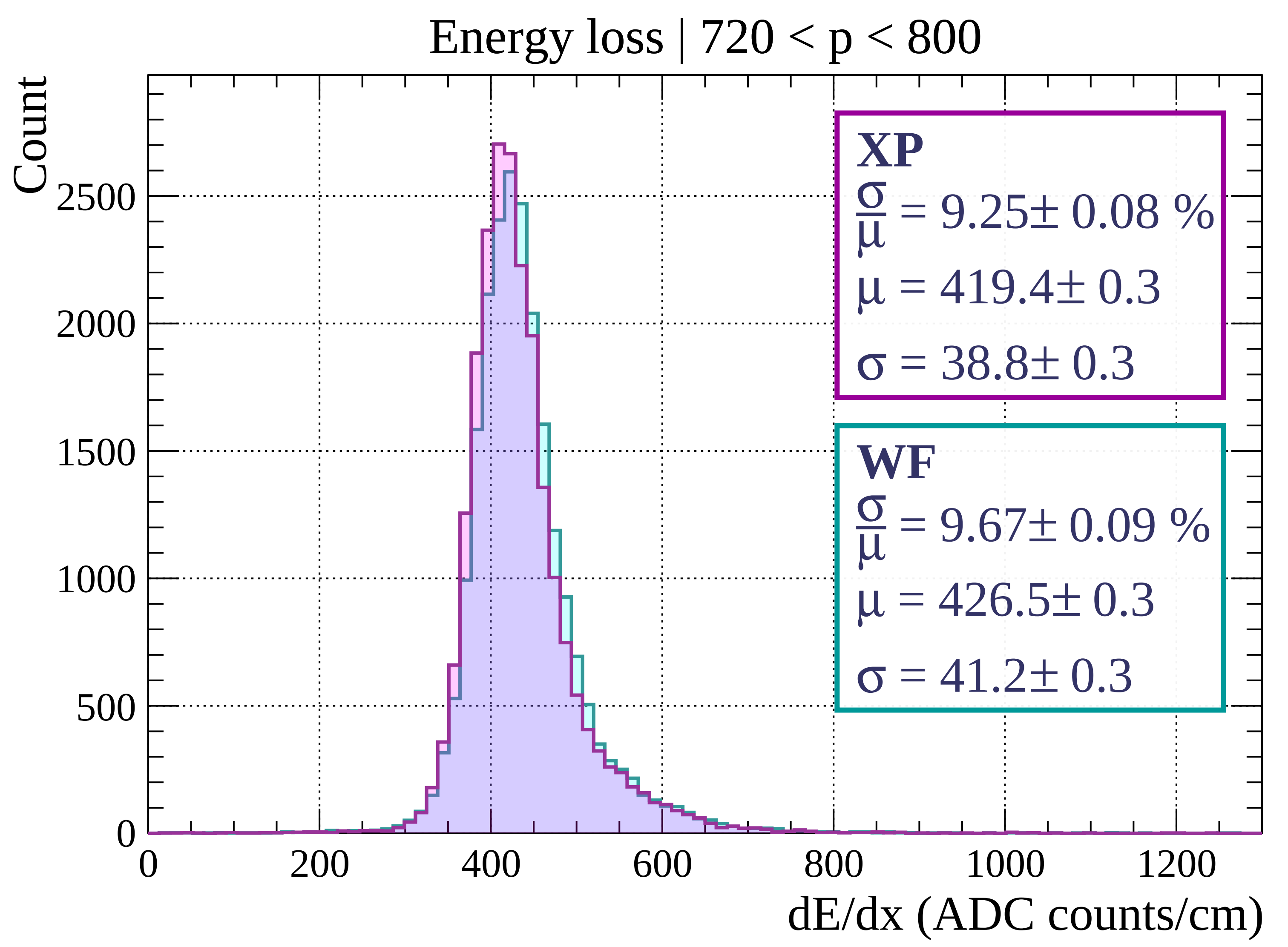


Energy loss | 480 < p < 560
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.46± 0.08 %
μ = 404.0± 0.3
σ = 38.2± 0.3
WF
$\frac{\sigma}{\mu}$ = 9.65± 0.09 %
μ = 409.5± 0.3
σ = 39.5± 0.3

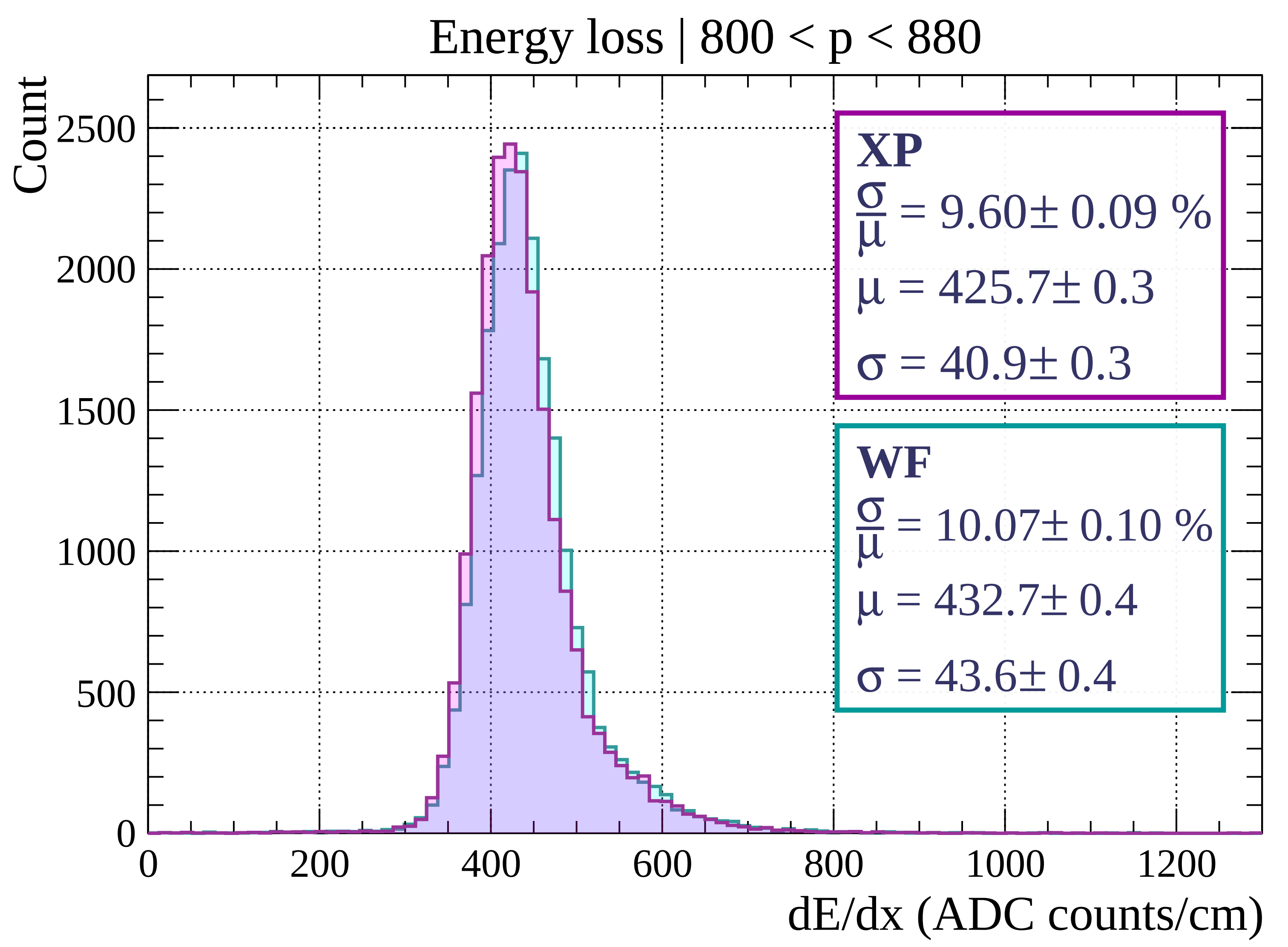


Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.32± 0.08 %
μ = 408.6± 0.3
σ = 38.1± 0.3
WF
$\frac{\sigma}{\mu}$ = 9.82± 0.09 %
μ = 415.2± 0.3
σ = 40.8± 0.3

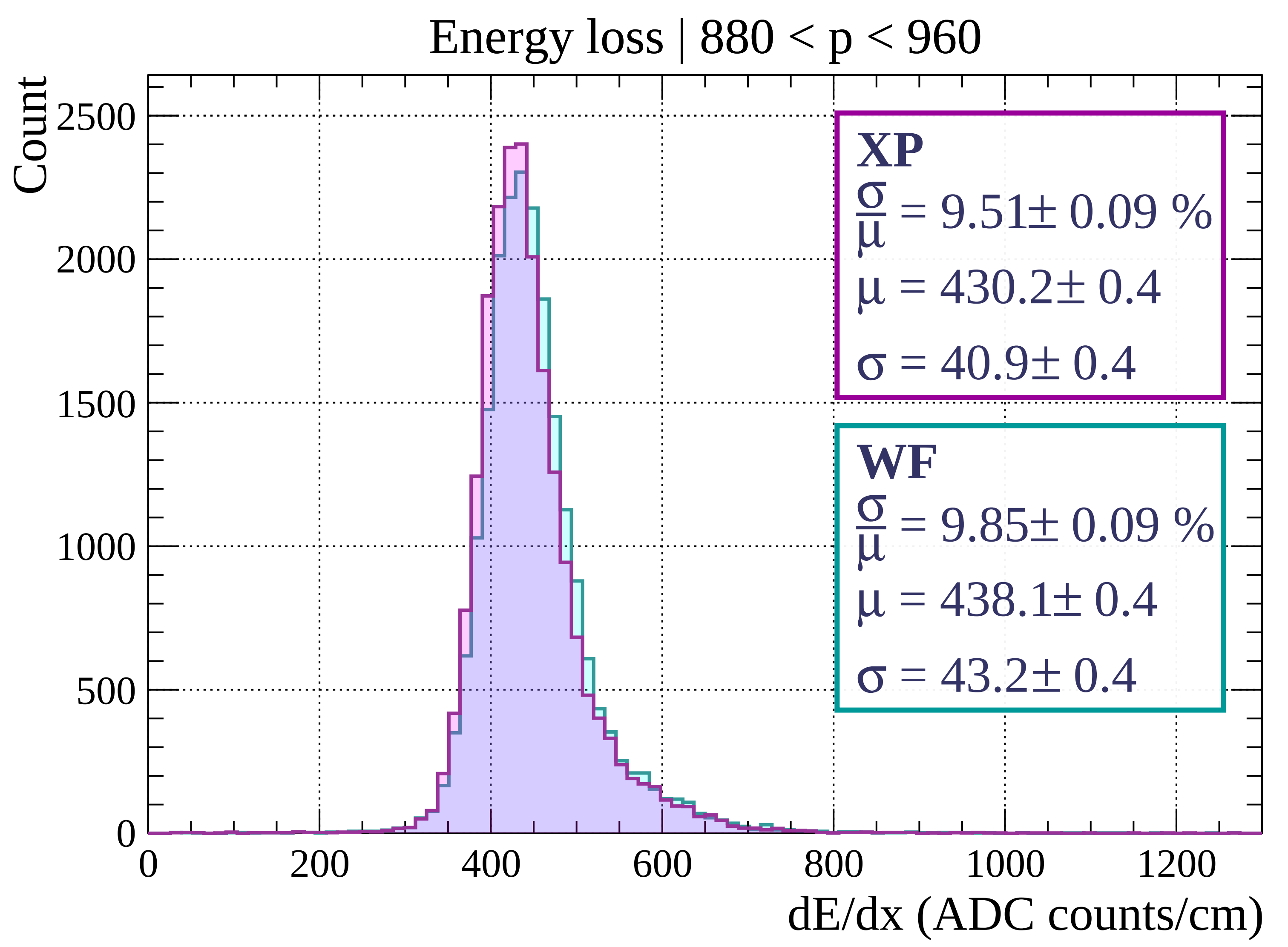

Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.52± 0.09 %
μ = 414.3± 0.3
σ = 39.4± 0.3
WF
$\frac{\sigma}{\mu}$ = 9.60± 0.09 %
μ = 420.5± 0.3
σ = 40.4± 0.3
0
500
1000
1500
2000
2500
3000
0
200
400
600
800
1000
1200



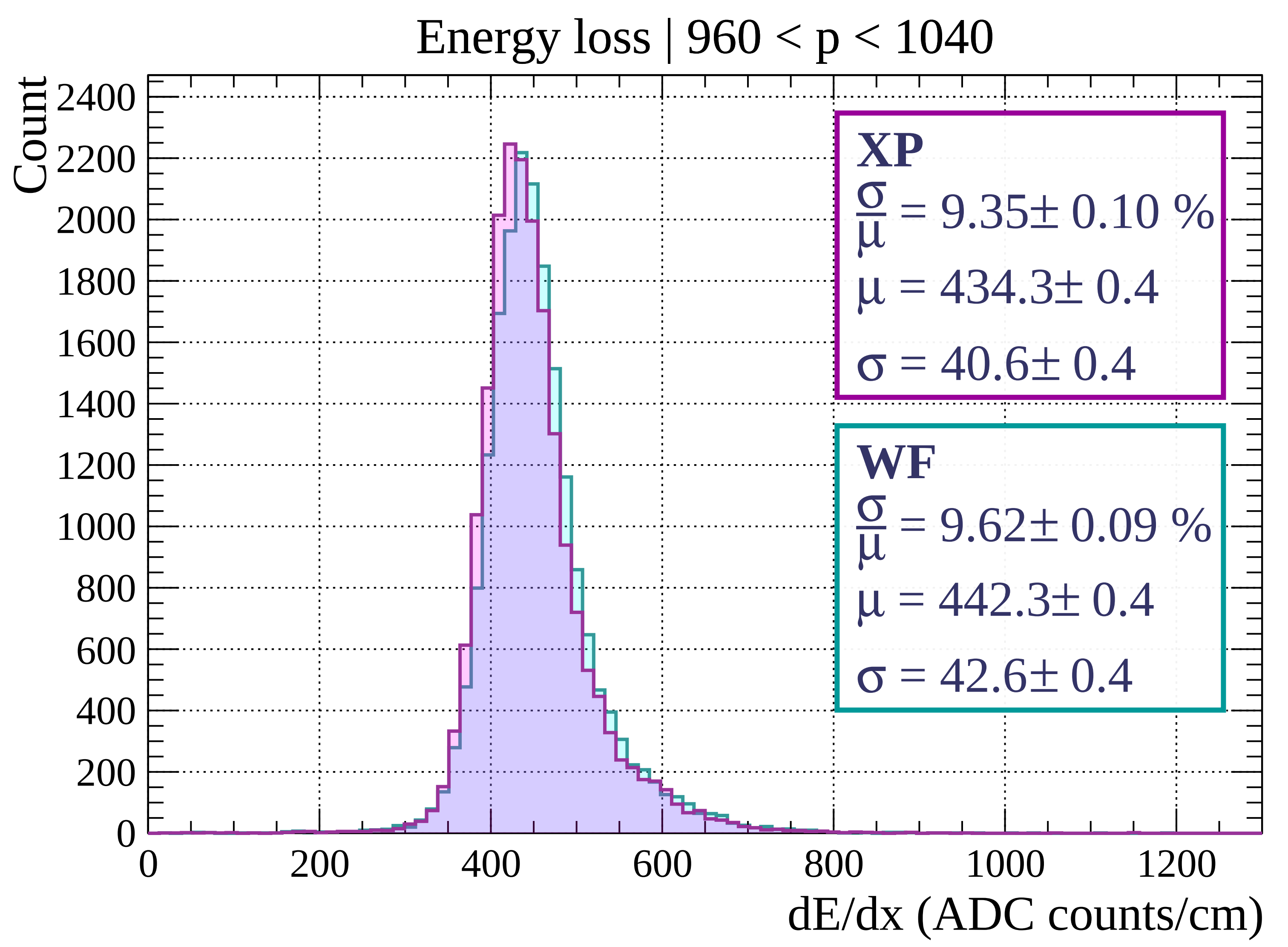
Energy loss | 720 < p < 800
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.25± 0.08 %
μ = 419.4± 0.3
σ = 38.8± 0.3
WF
σ/μ = 9.67± 0.09 %
μ = 426.5± 0.3
σ = 41.2± 0.3

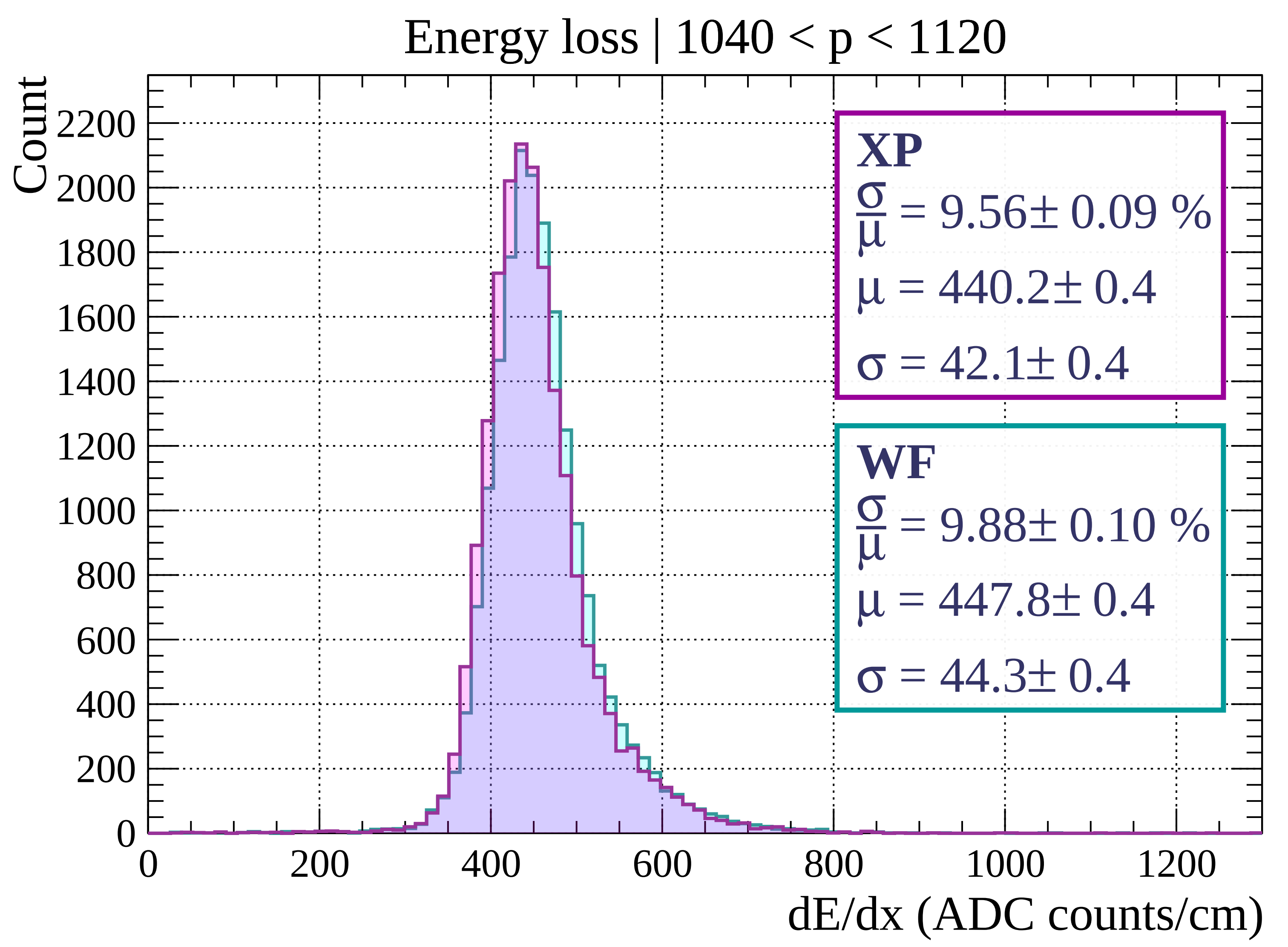


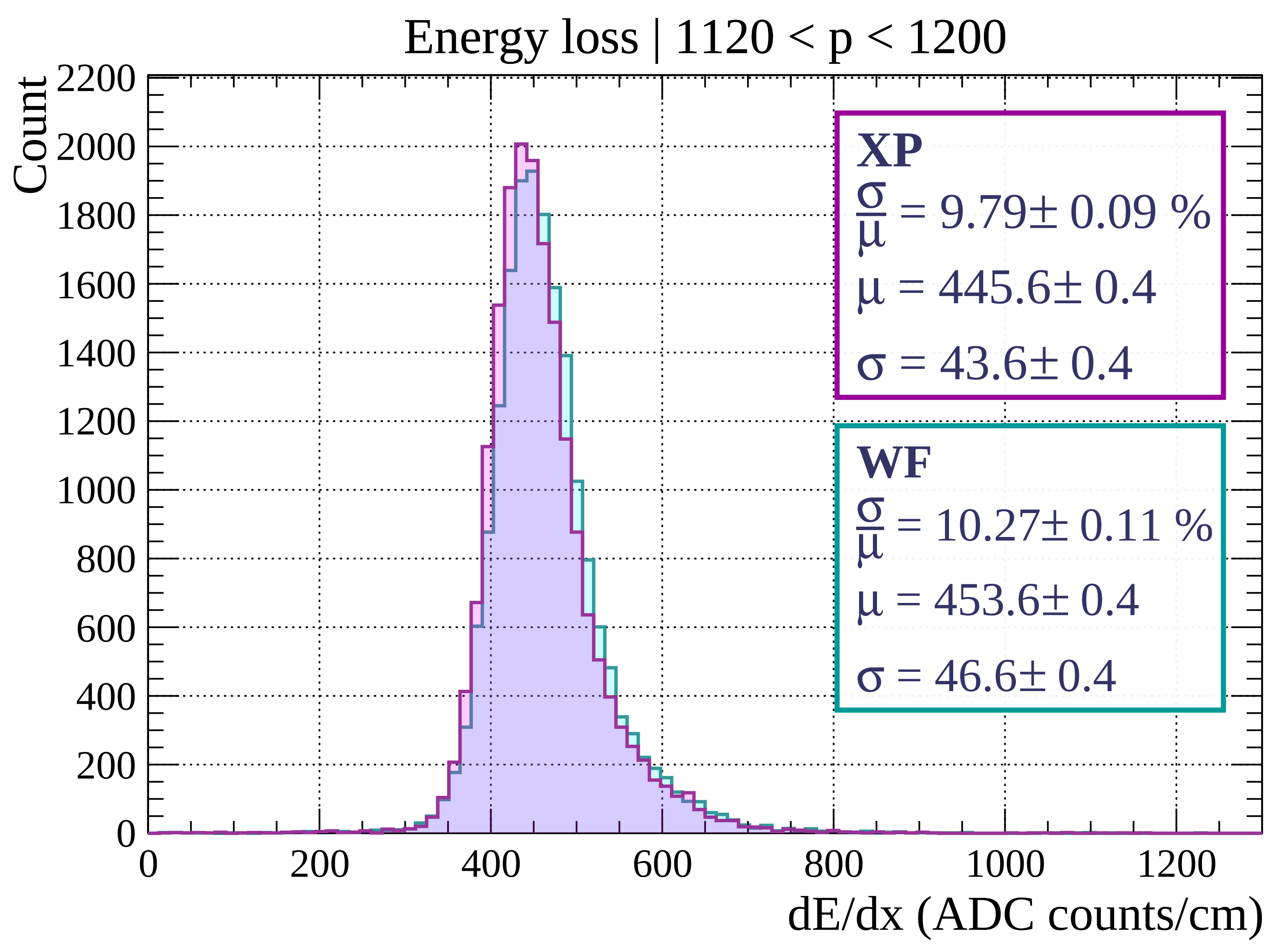
Energy loss | 800 < p < 880
Count
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.60± 0.09 %
μ = 425.7± 0.3
σ = 40.9± 0.3
WF
$\frac{\sigma}{\mu}$ = 10.07± 0.10 %
μ = 432.7± 0.4
σ = 43.6± 0.4



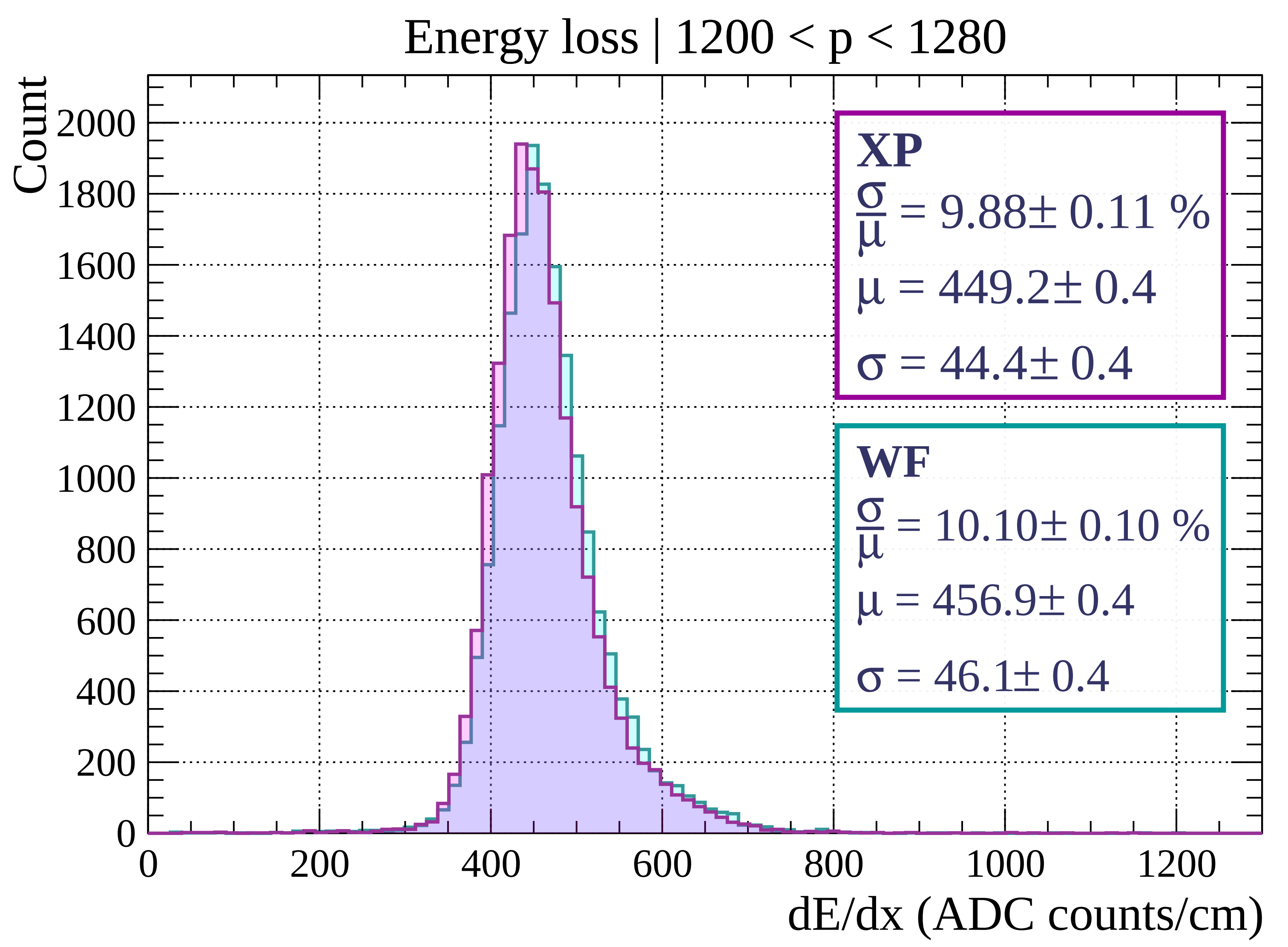
Energy loss | 880 < p < 960
Count
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.51± 0.09 %
μ = 430.2± 0.4
σ = 40.9± 0.4
WF
$\frac{\sigma}{\mu}$ = 9.85± 0.09 %
μ = 438.1± 0.4
σ = 43.2± 0.4

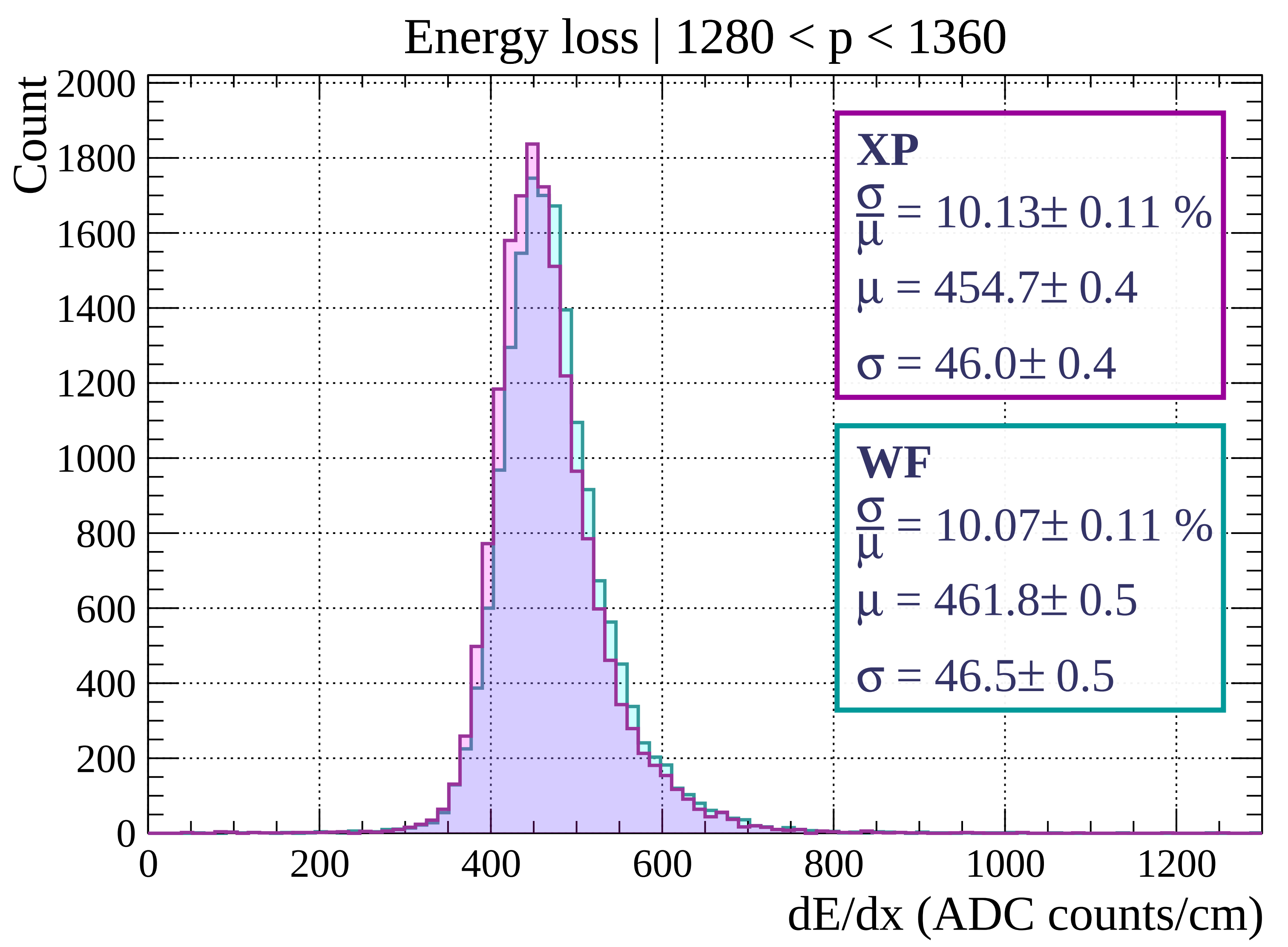

Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.35± 0.10 %
μ = 434.3± 0.4
σ = 40.6± 0.4
WF
σ/μ = 9.62± 0.09 %
μ = 442.3± 0.4
σ = 42.6± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400


Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
XP
σ/μ = 9.56 ± 0.09 %
μ = 440.2 ± 0.4
σ = 42.1 ± 0.4
WF
σ/μ = 9.88 ± 0.10 %
μ = 447.8 ± 0.4
σ = 44.3 ± 0.4

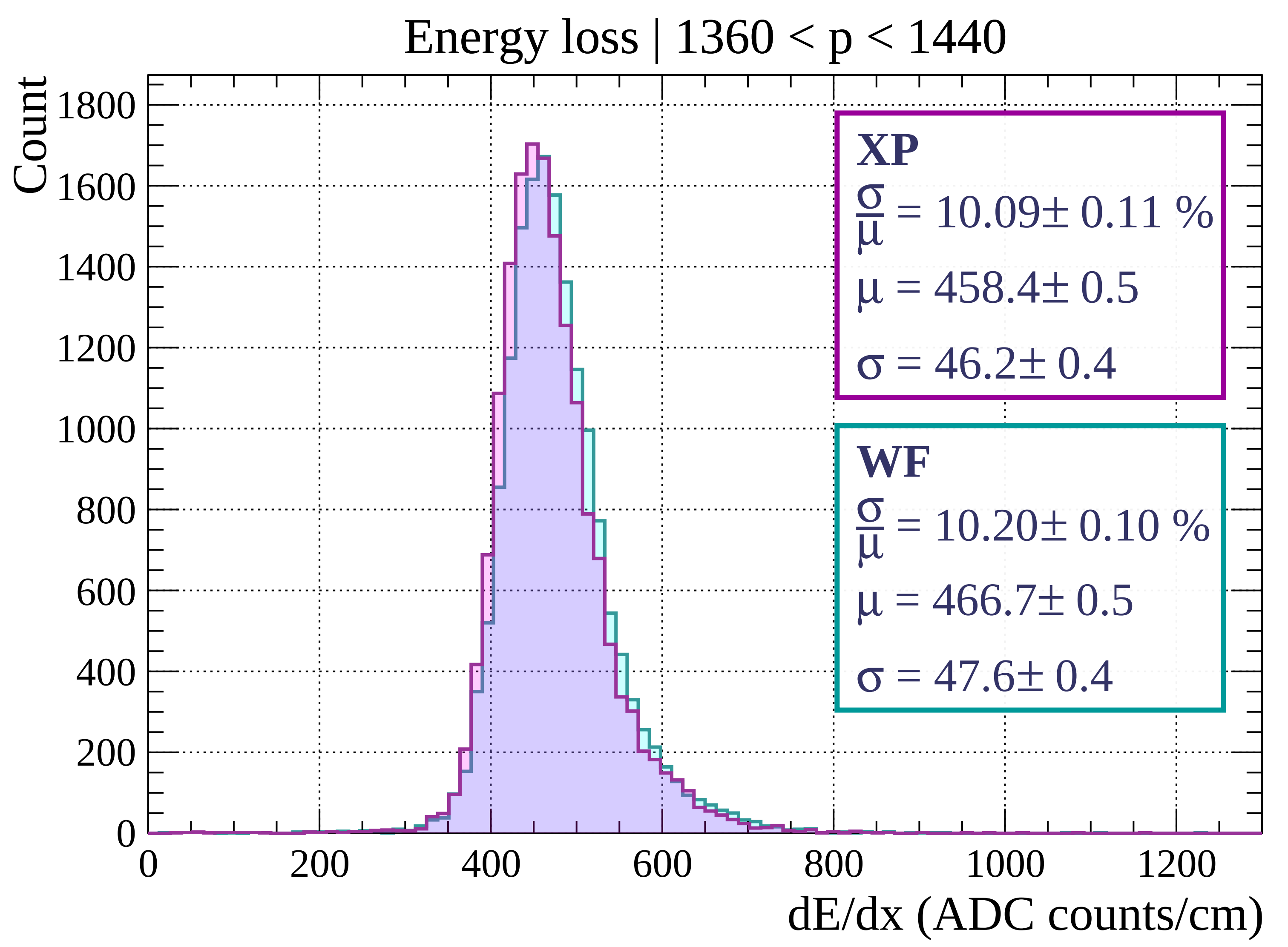


Energy loss | 1120 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.79± 0.09 %
μ = 445.6± 0.4
σ = 43.6± 0.4
WF
$\frac{\sigma}{\mu}$ = 10.27± 0.11 %
μ = 453.6± 0.4
σ = 46.6± 0.4

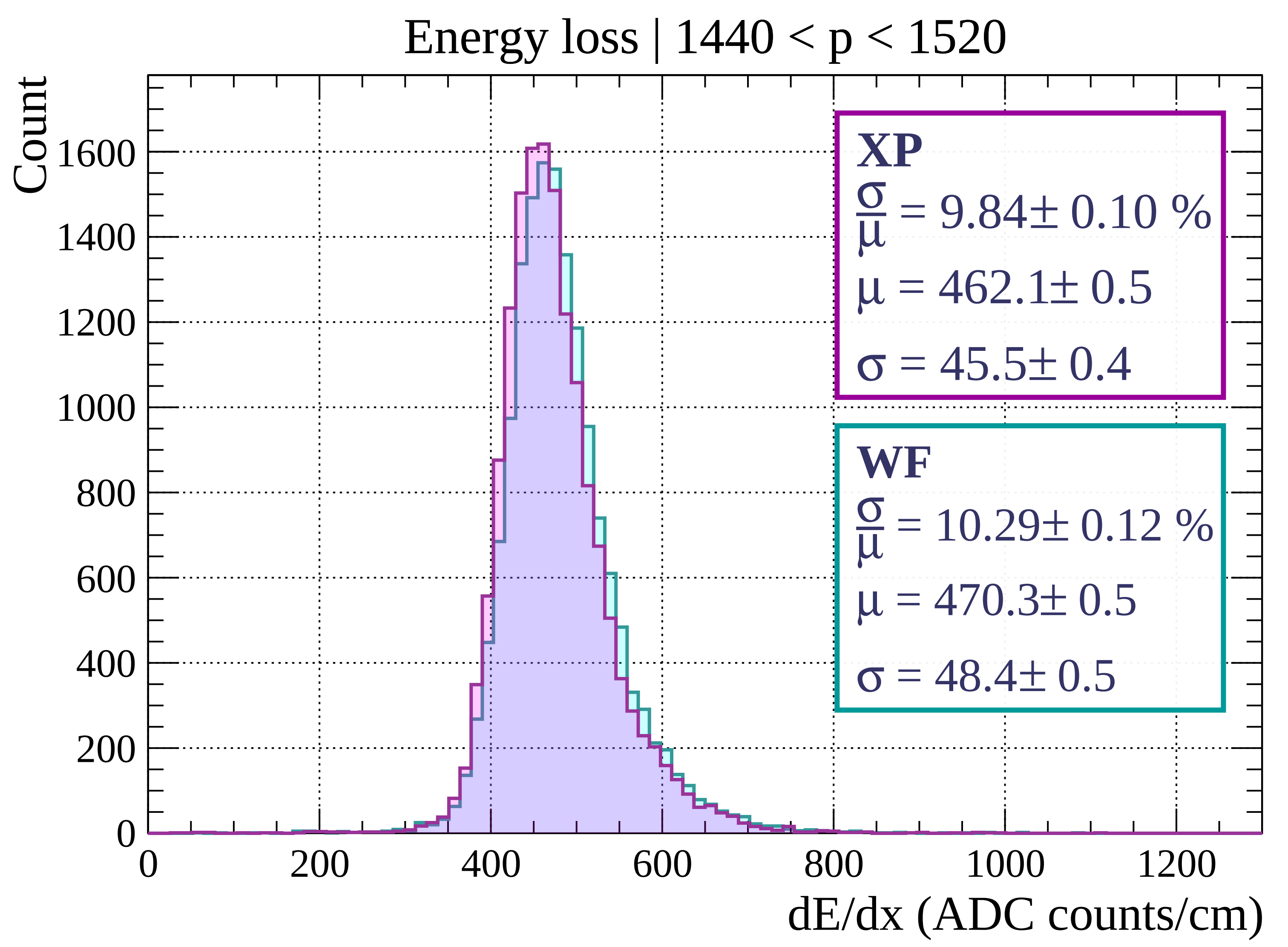

Energy loss | 1200 < p < 1280
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.88± 0.11 %
μ = 449.2± 0.4
σ = 44.4± 0.4
WF
$\frac{\sigma}{\mu}$ = 10.10± 0.10 %
μ = 456.9± 0.4
σ = 46.1± 0.4



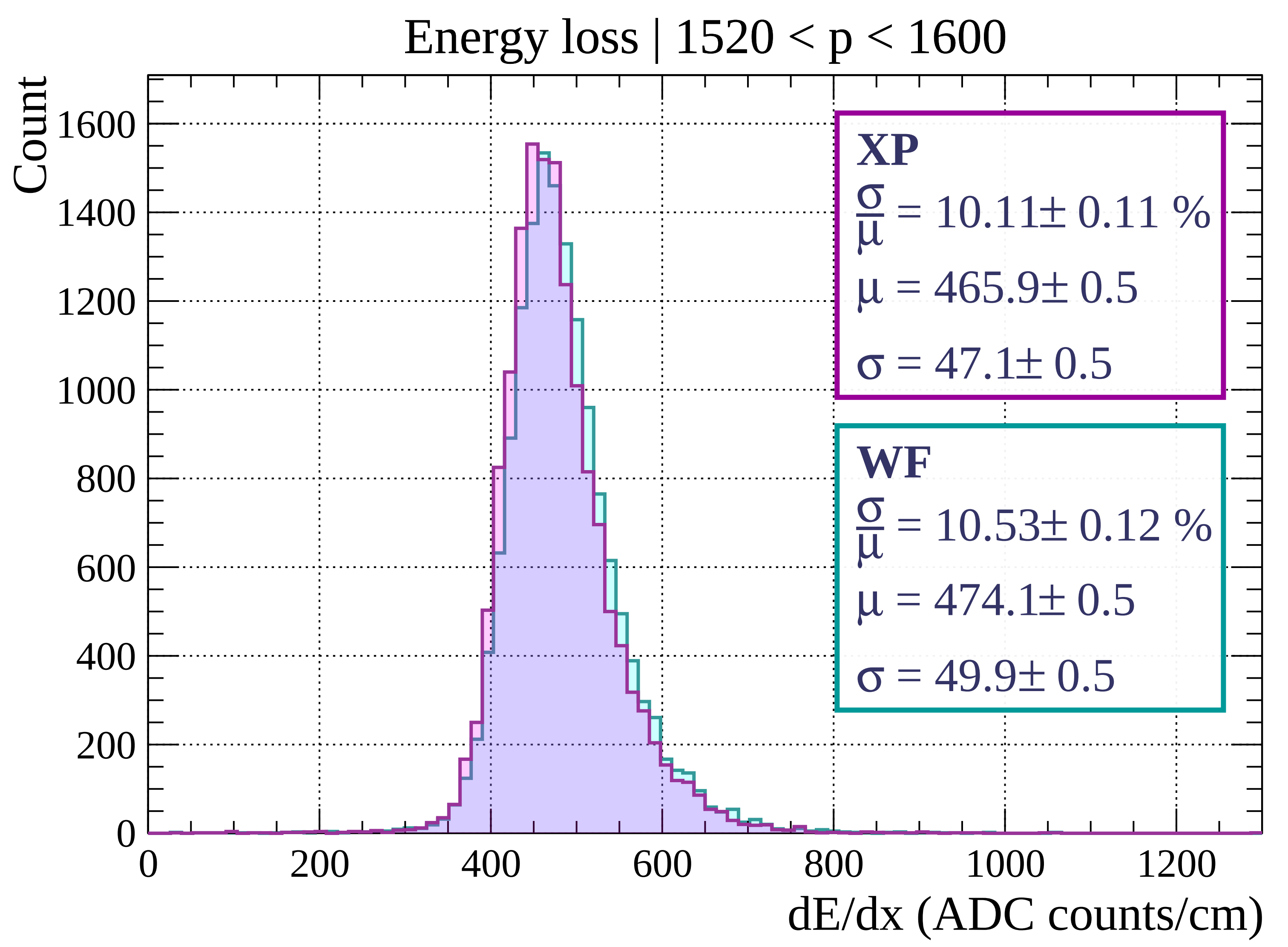
Energy loss | 1280 < p < 1360
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.13± 0.11 %
μ = 454.7± 0.4
σ = 46.0± 0.4
WF
σ/μ = 10.07± 0.11 %
μ = 461.8± 0.5
σ = 46.5± 0.5



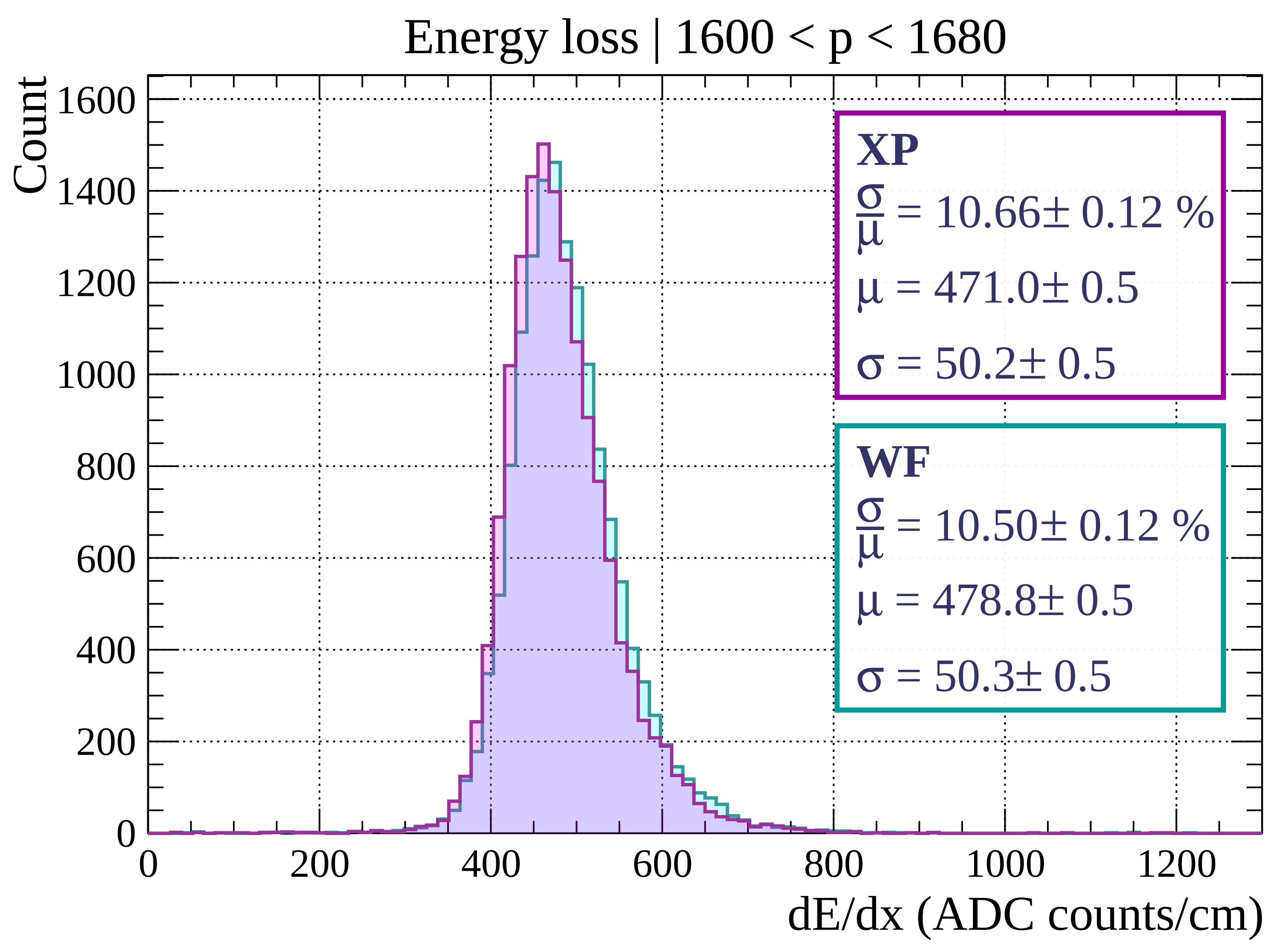
Energy loss | 1360 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
XP
σ/μ = 10.09± 0.11 %
μ = 458.4± 0.5
σ = 46.2± 0.4
WF
σ/μ = 10.20± 0.10 %
μ = 466.7± 0.5
σ = 47.6± 0.4



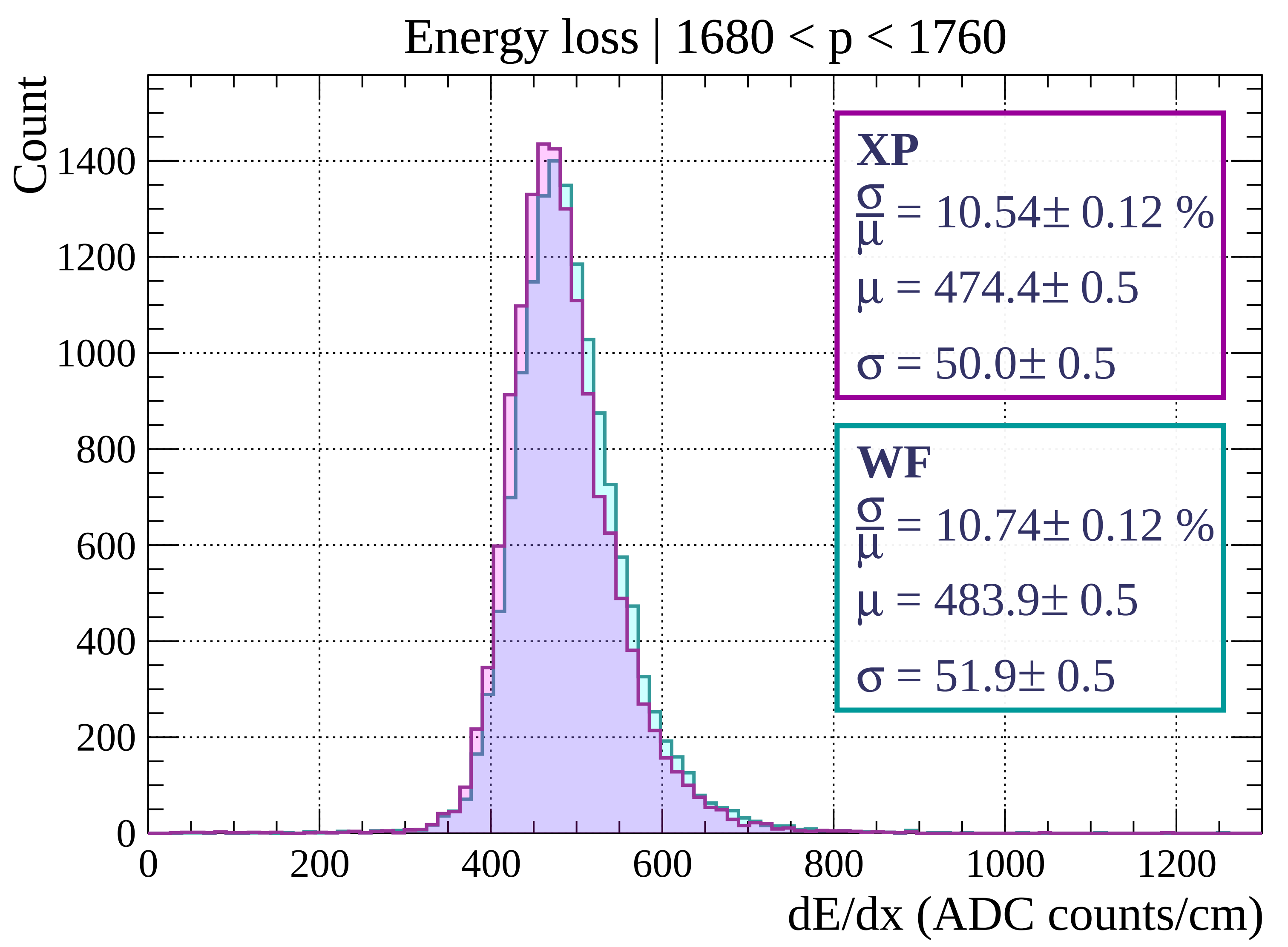
Energy loss | 1440 < p < 1520
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.84± 0.10 %
μ = 462.1± 0.5
σ = 45.5± 0.4
WF
$\frac{\sigma}{\mu}$ = 10.29± 0.12 %
μ = 470.3± 0.5
σ = 48.4± 0.5

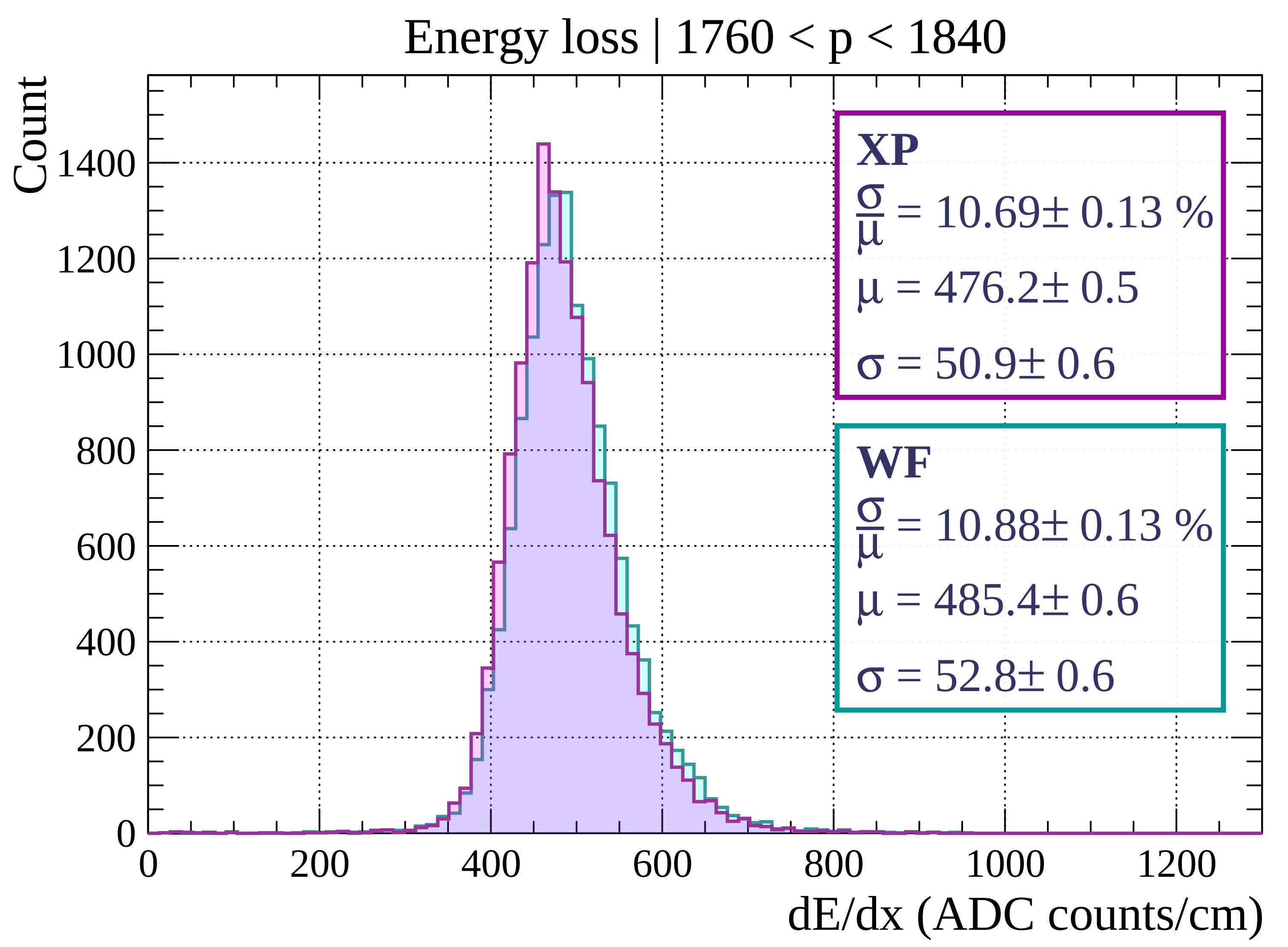


Energy loss | 1520 < p < 1600
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 10.11 \pm 0.11$ %
$\mu = 465.9 \pm 0.5$
$\sigma = 47.1 \pm 0.5$
WF
$\frac{\sigma}{\mu} = 10.53 \pm 0.12$ %
$\mu = 474.1 \pm 0.5$
$\sigma = 49.9 \pm 0.5$

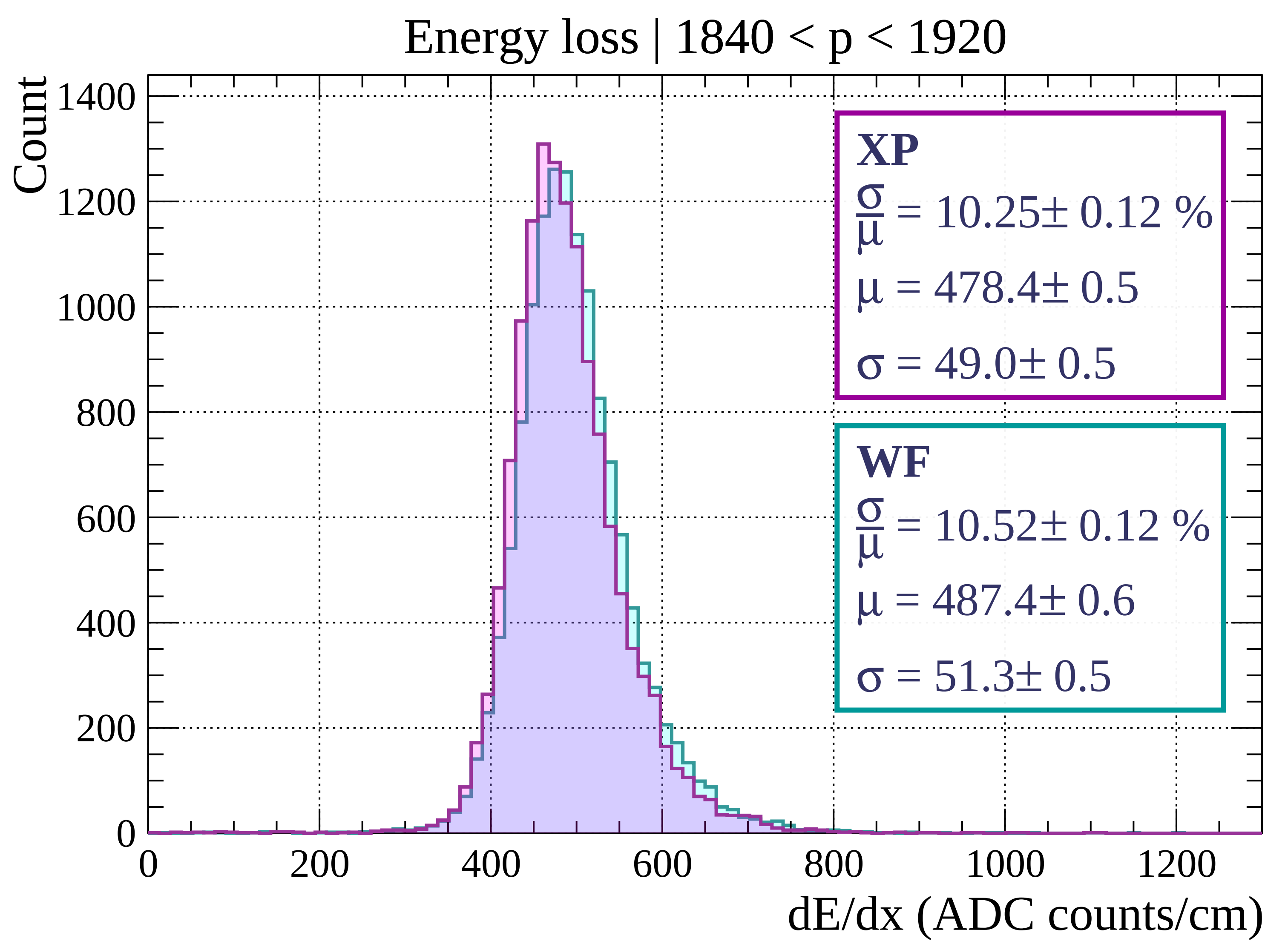


Energy loss | 1600 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
XP
$\frac{\sigma}{\mu}$ = 10.66± 0.12 %
μ = 471.0± 0.5
σ = 50.2± 0.5
WF
$\frac{\sigma}{\mu}$ = 10.50± 0.12 %
μ = 478.8± 0.5
σ = 50.3± 0.5

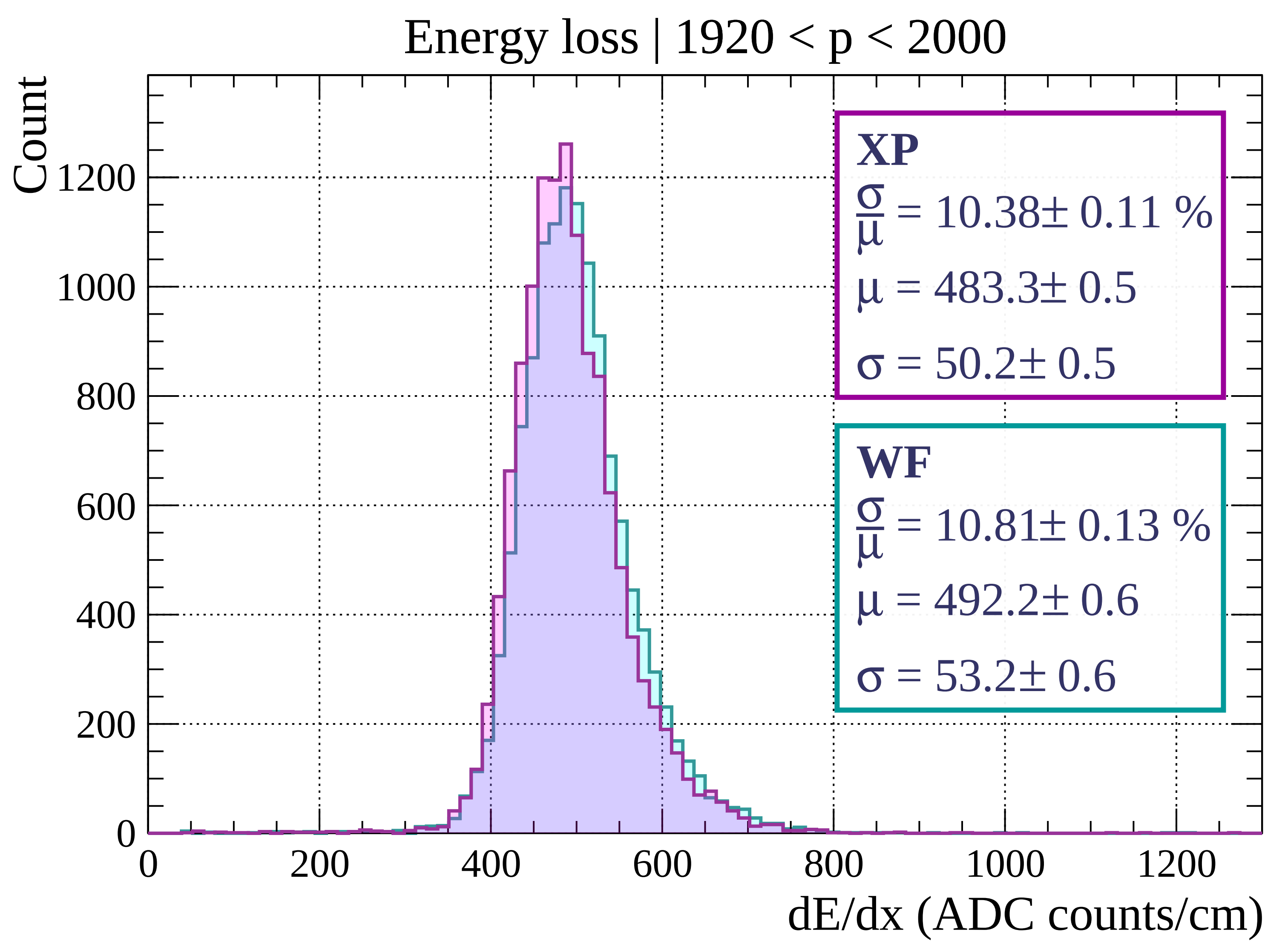

Energy loss | 1680 < p < 1760
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.54± 0.12 %
μ = 474.4± 0.5
σ = 50.0± 0.5
WF
σ/μ = 10.74± 0.12 %
μ = 483.9± 0.5
σ = 51.9± 0.5



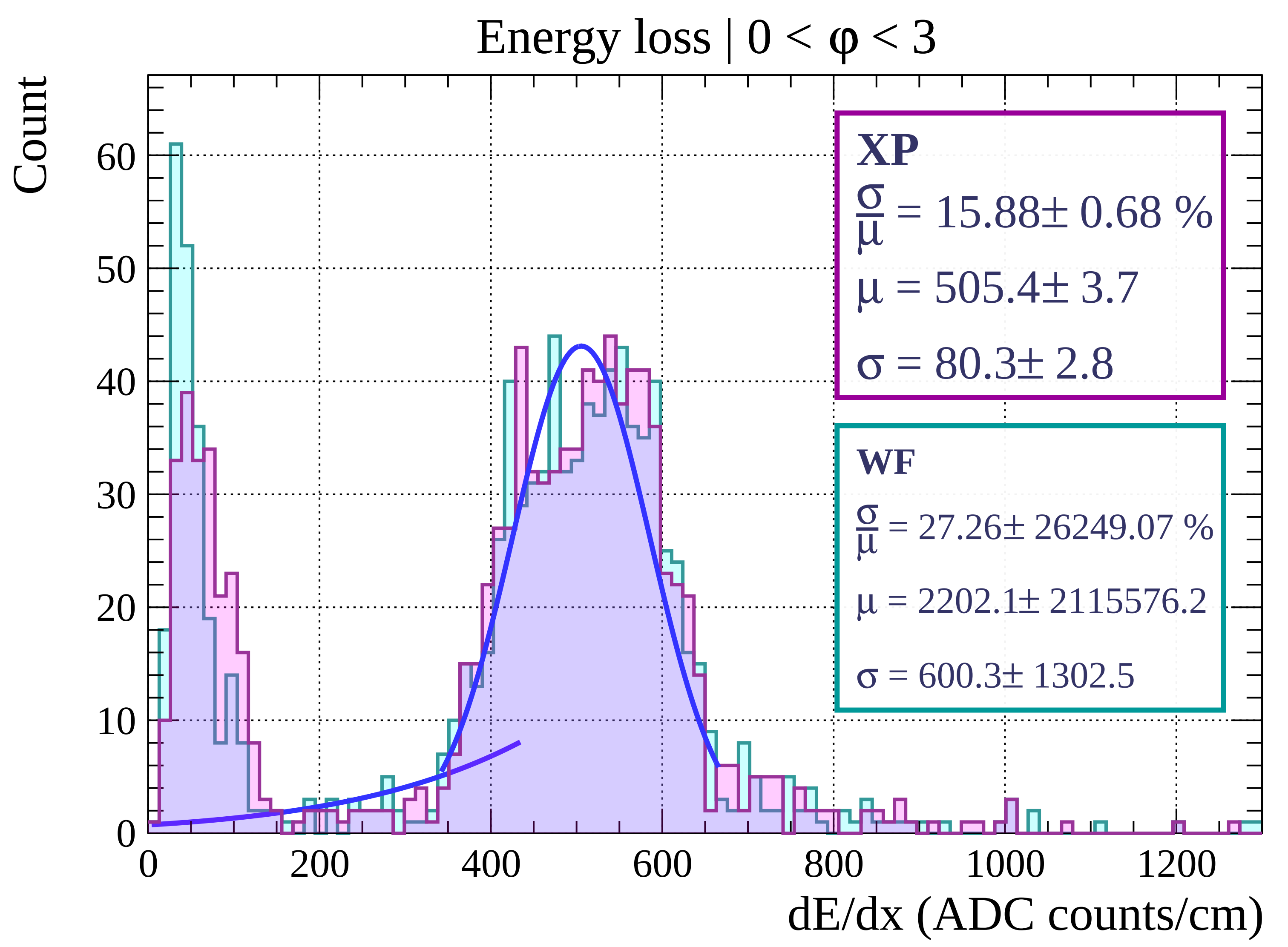
Energy loss | 1760 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.69± 0.13 %
μ = 476.2± 0.5
σ = 50.9± 0.6
WF
$\frac{\sigma}{\mu}$ = 10.88± 0.13 %
μ = 485.4± 0.6
σ = 52.8± 0.6



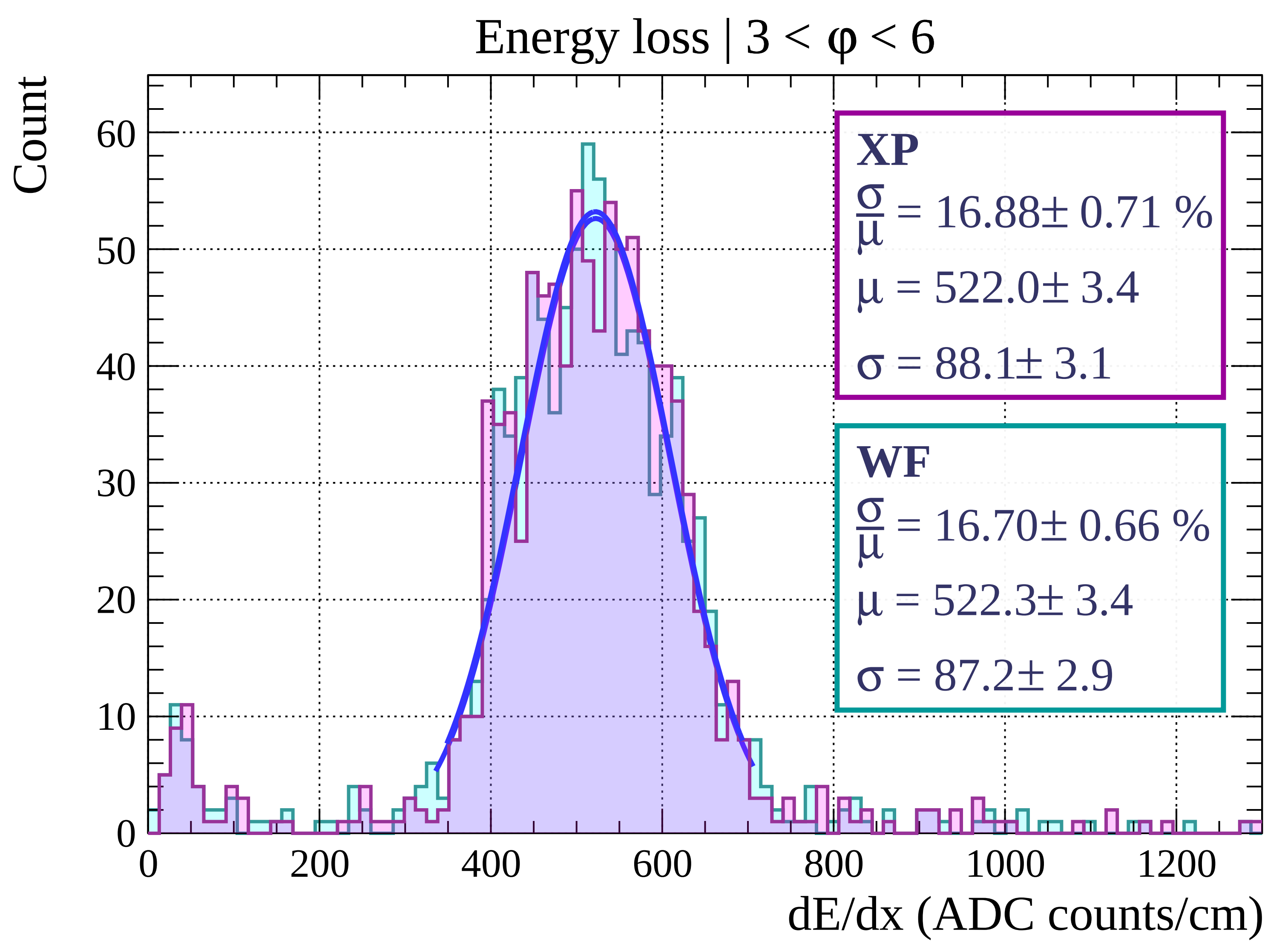
Energy loss | 1840 < p < 1920
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.25± 0.12 %
μ = 478.4± 0.5
σ = 49.0± 0.5
WF
σ/μ = 10.52± 0.12 %
μ = 487.4± 0.6
σ = 51.3± 0.5



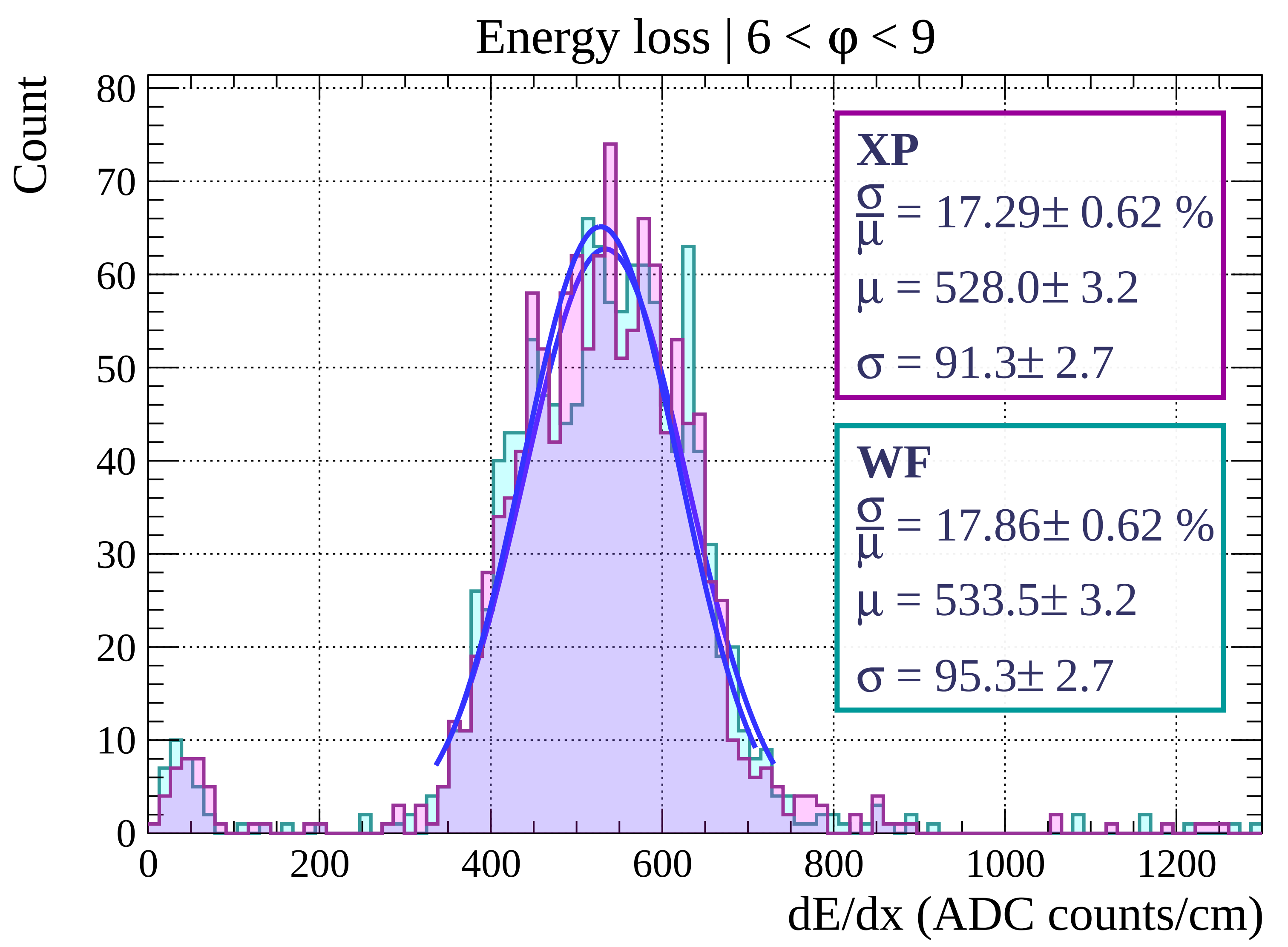
Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 10.38± 0.11 %
μ = 483.3± 0.5
σ = 50.2± 0.5
WF
σ/μ = 10.81± 0.13 %
μ = 492.2± 0.6
σ = 53.2± 0.6



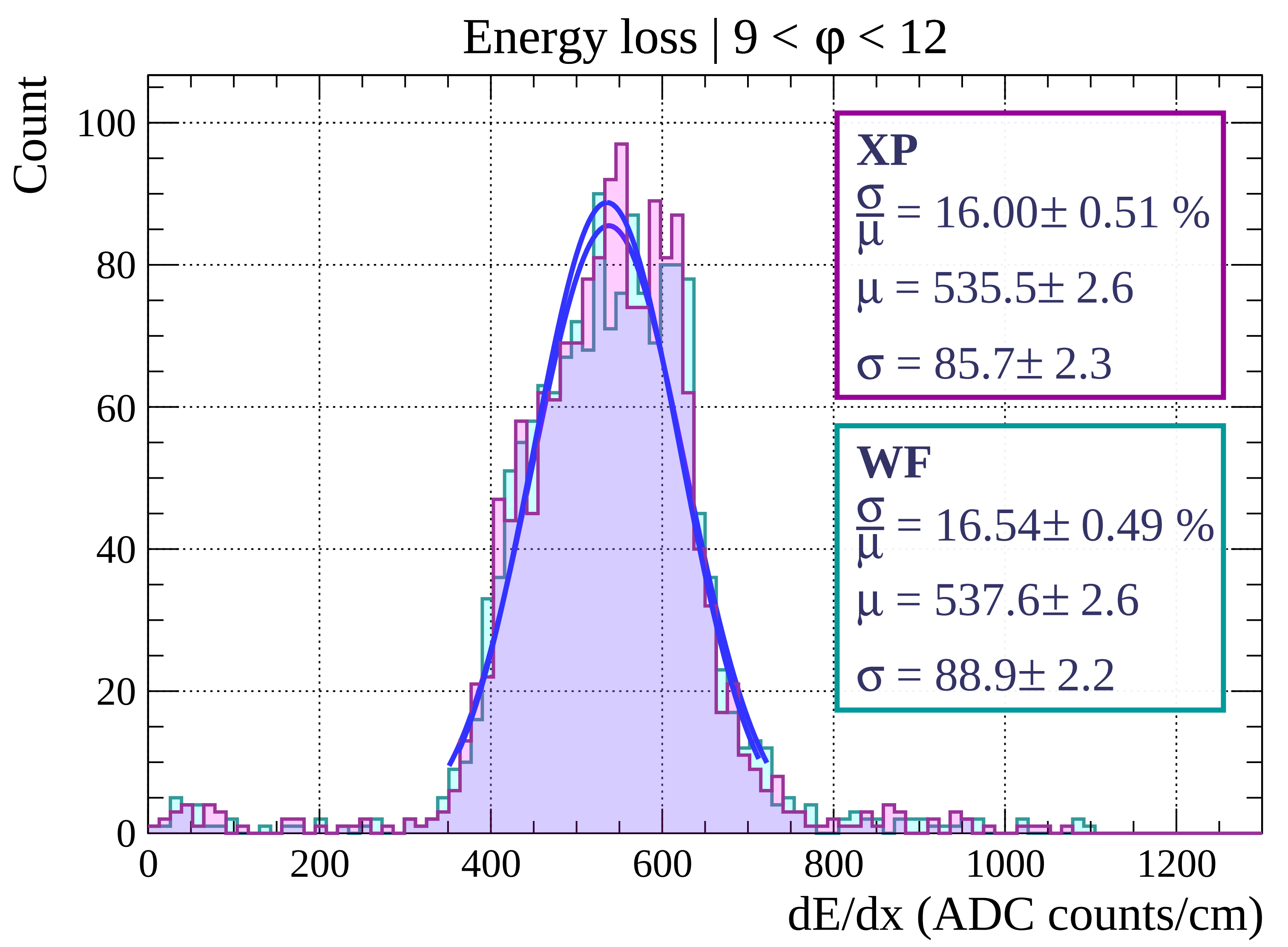
Energy loss | 0 < φ < 3
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
200
400
600
800
1000
1200
XP
σ/μ = 15.88± 0.68 %
μ = 505.4± 3.7
σ = 80.3± 2.8
WF
σ/μ = 27.26± 26249.07 %
μ = 2202.1± 2115576.2
σ = 600.3± 1302.5

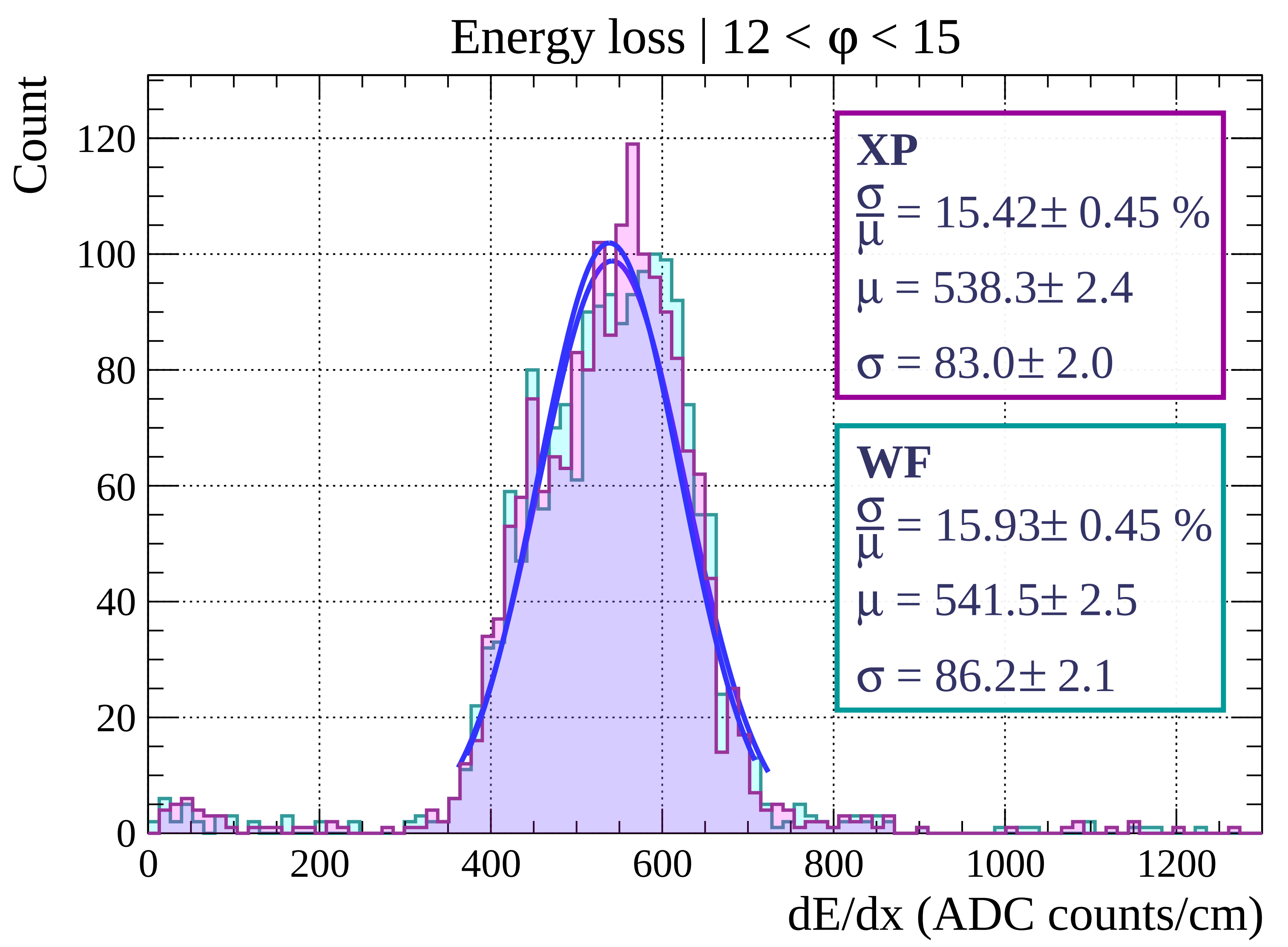


Energy loss | 3 < φ < 6
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 16.88± 0.71 %
μ = 522.0± 3.4
σ = 88.1± 3.1
WF
σ/μ = 16.70± 0.66 %
μ = 522.3± 3.4
σ = 87.2± 2.9

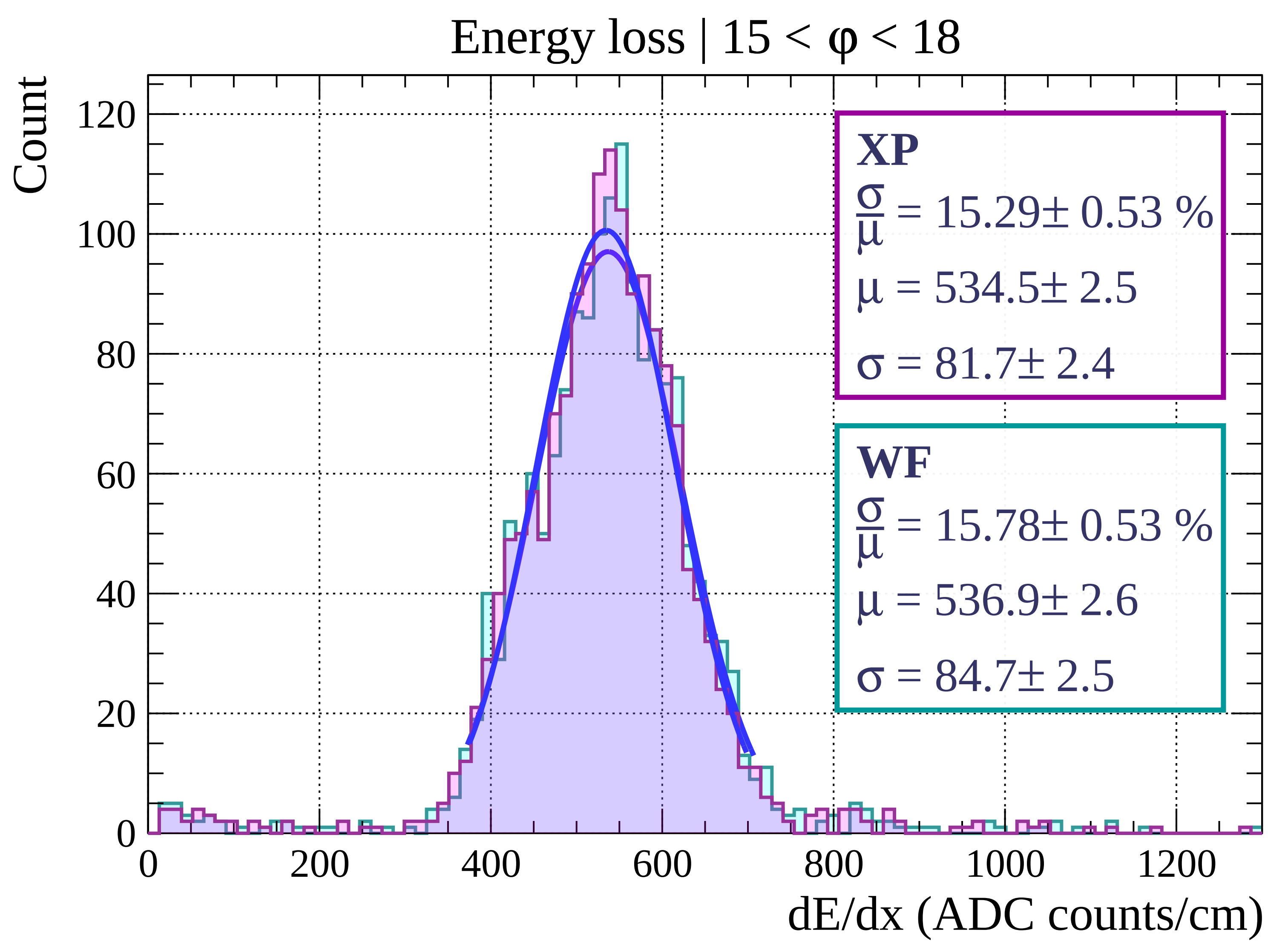


Energy loss | 6 < φ < 9
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 17.29± 0.62 %
μ = 528.0± 3.2
σ = 91.3± 2.7
WF
σ/μ = 17.86± 0.62 %
μ = 533.5± 3.2
σ = 95.3± 2.7

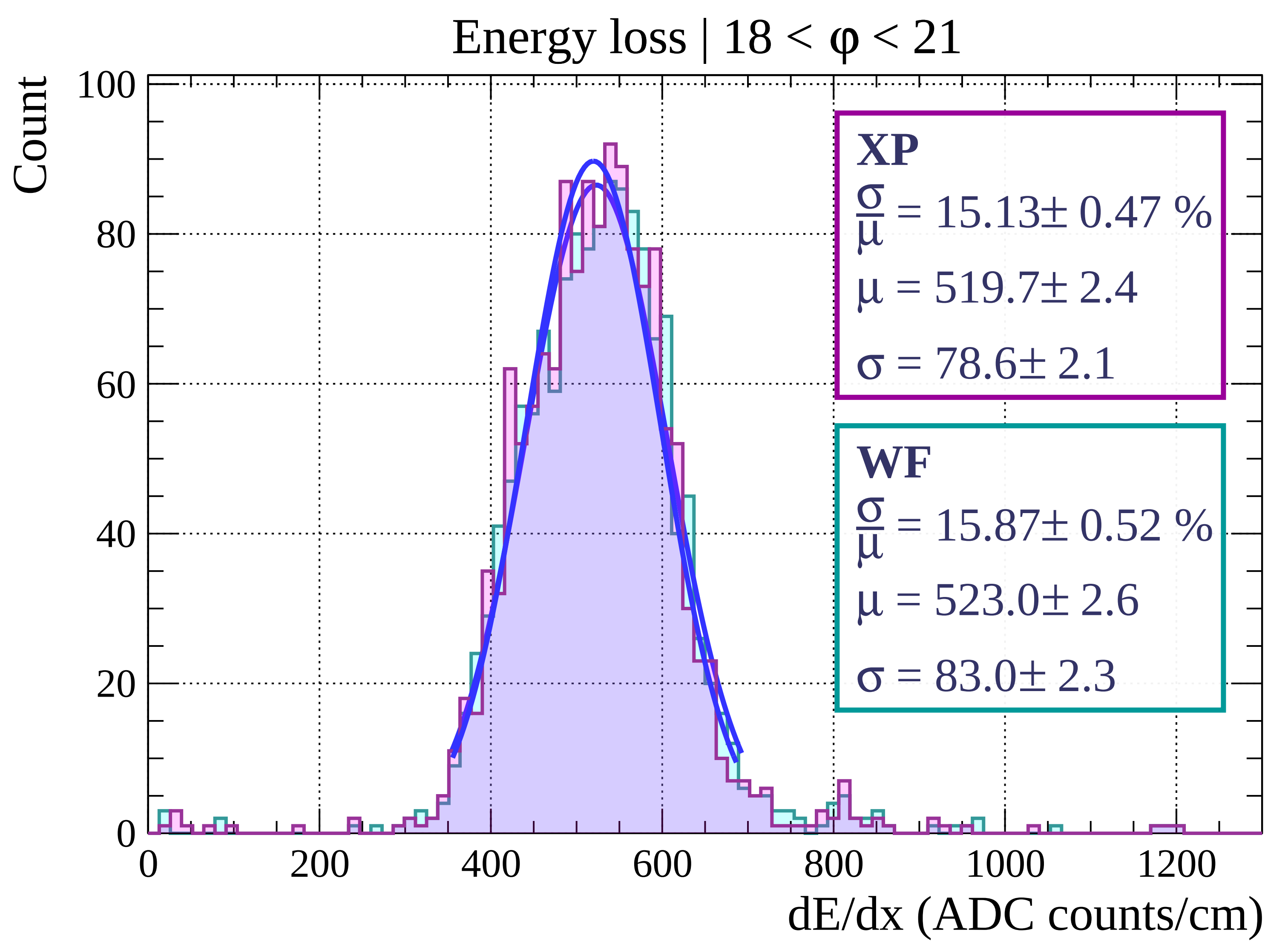

Energy loss | 9 < φ < 12
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 16.00± 0.51 %
μ = 535.5± 2.6
σ = 85.7± 2.3
WF
σ/μ = 16.54± 0.49 %
μ = 537.6± 2.6
σ = 88.9± 2.2



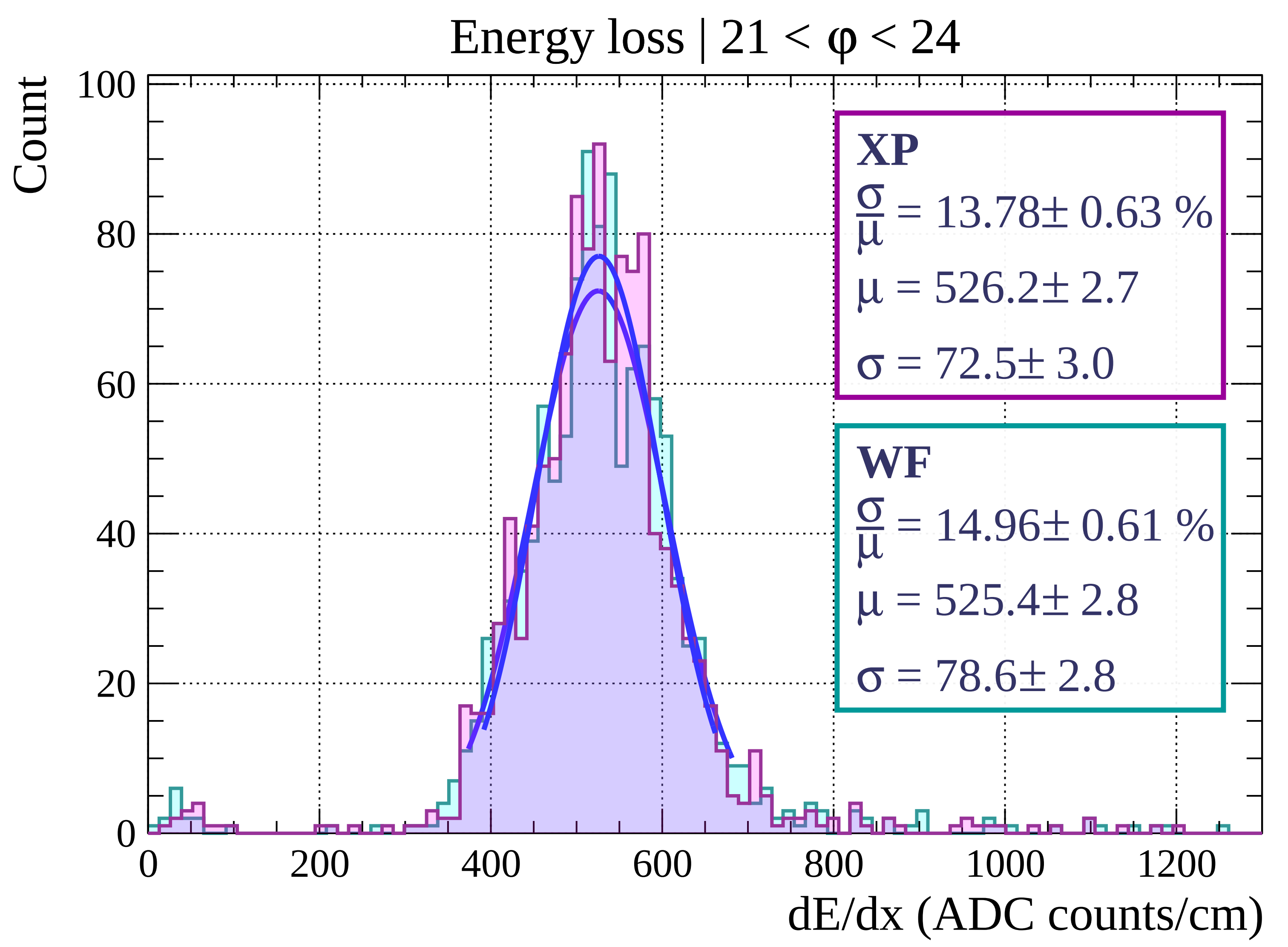
Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 15.42± 0.45 %
μ = 538.3± 2.4
σ = 83.0± 2.0
WF
σ/μ = 15.93± 0.45 %
μ = 541.5± 2.5
σ = 86.2± 2.1



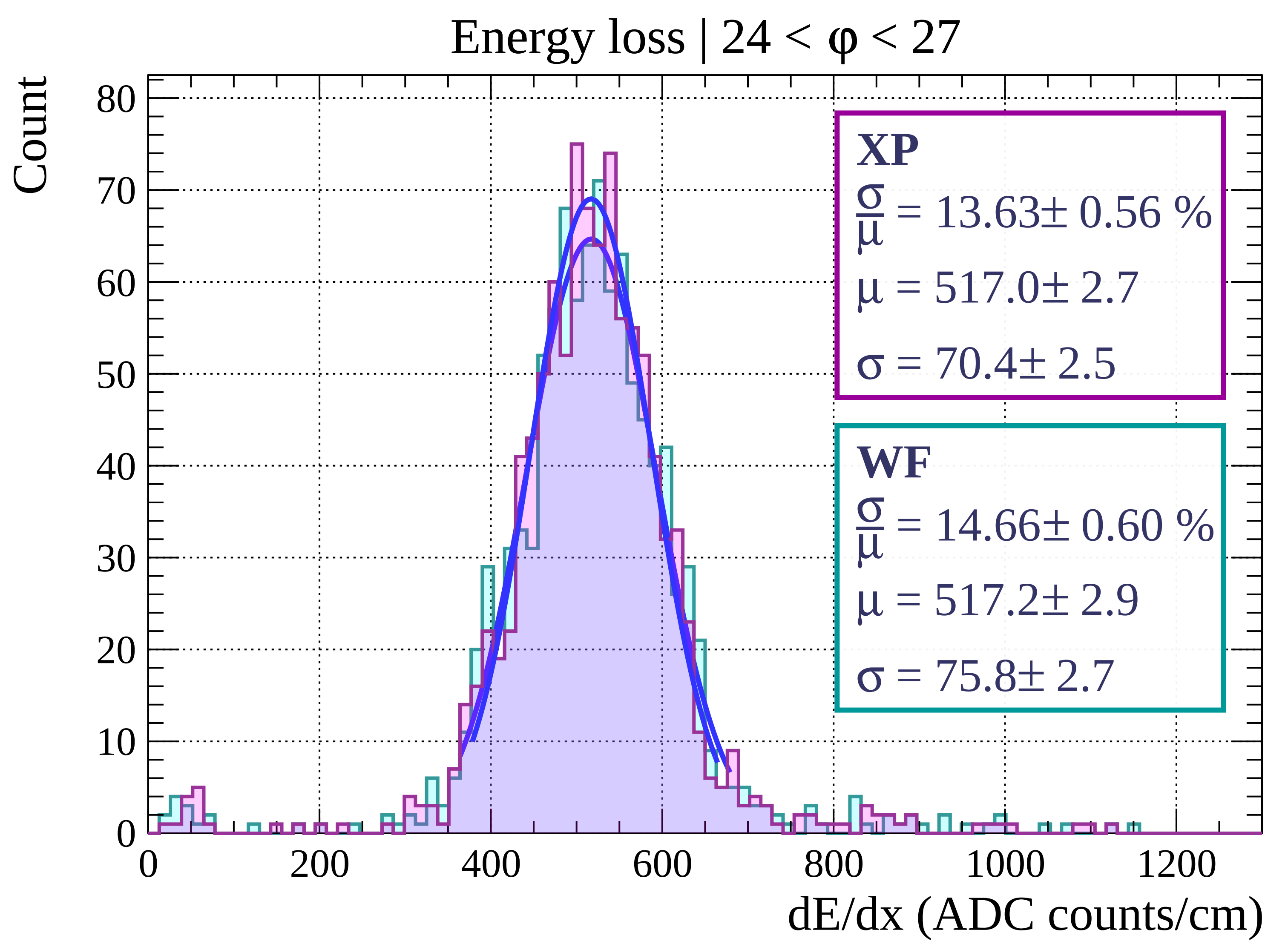
Energy loss | 15 < φ < 18
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 15.29± 0.53 %
μ = 534.5± 2.5
σ = 81.7± 2.4
WF
σ/μ = 15.78± 0.53 %
μ = 536.9± 2.6
σ = 84.7± 2.5

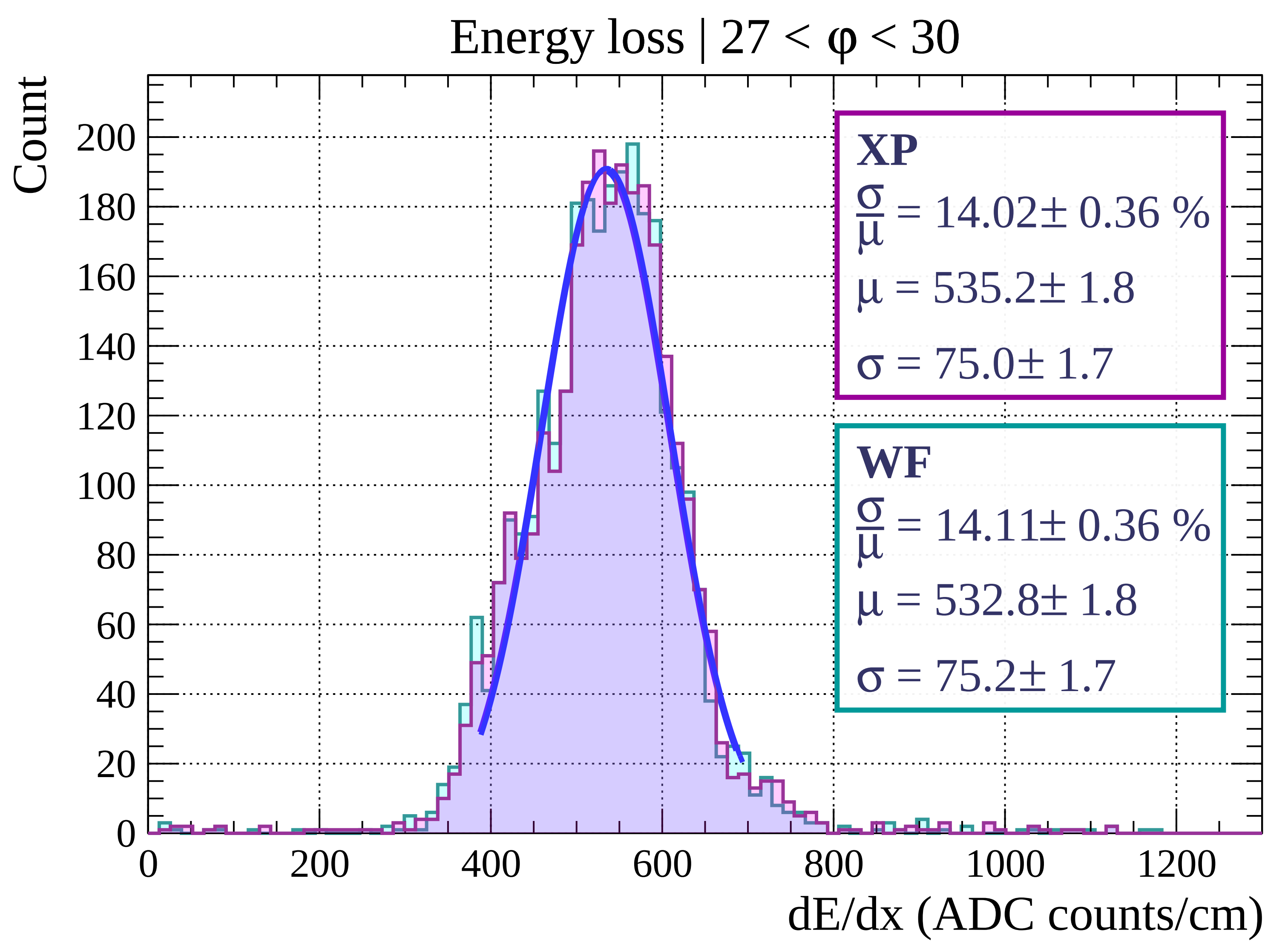


Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 15.13± 0.47 %
μ = 519.7± 2.4
σ = 78.6± 2.1
WF
σ/μ = 15.87± 0.52 %
μ = 523.0± 2.6
σ = 83.0± 2.3

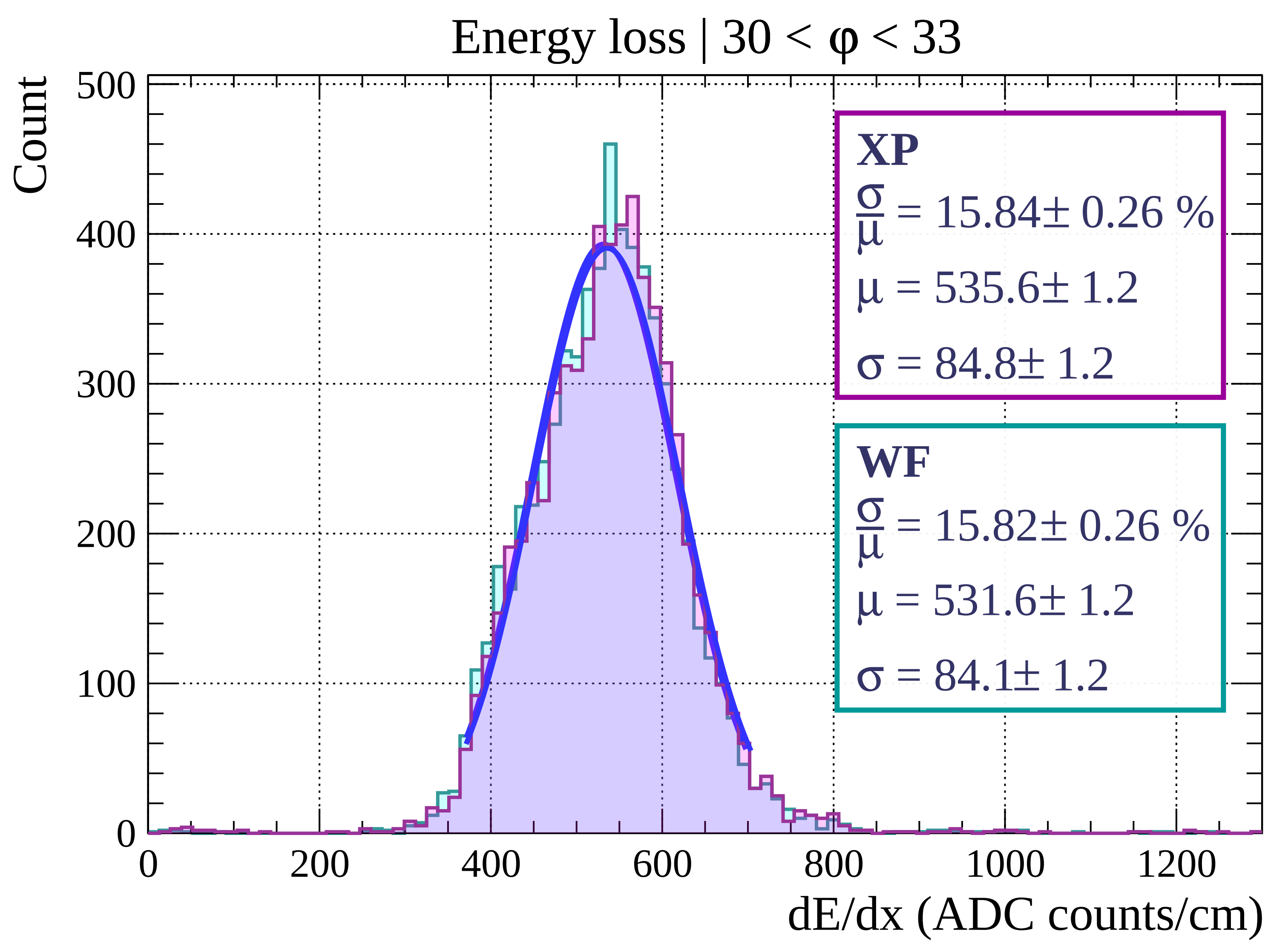


Energy loss | 21 < φ < 24
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.78± 0.63 %
μ = 526.2± 2.7
σ = 72.5± 3.0
WF
σ/μ = 14.96± 0.61 %
μ = 525.4± 2.8
σ = 78.6± 2.8

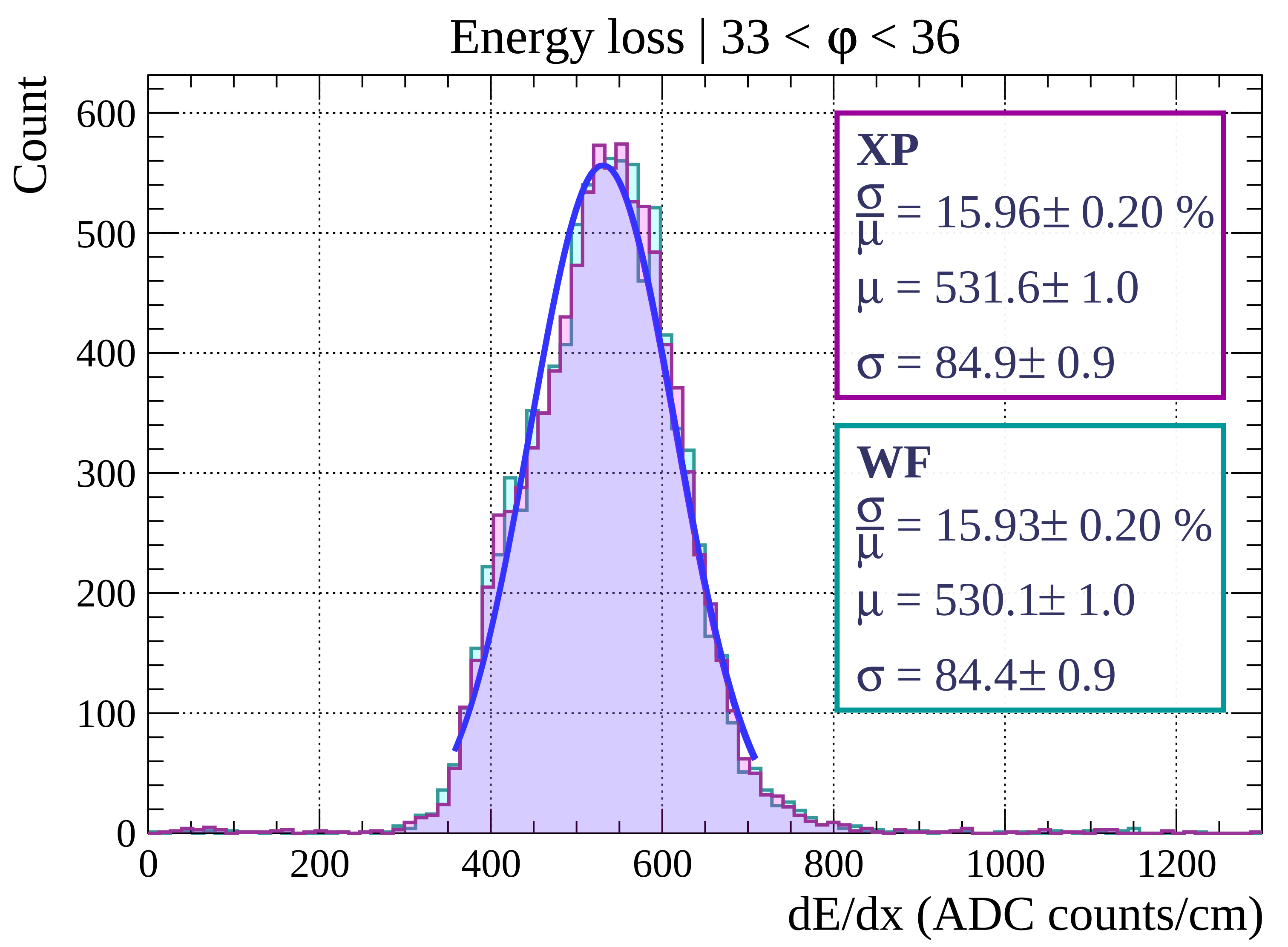

Energy loss | 24 < φ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP
σ/μ = 13.63± 0.56 %
μ = 517.0± 2.7
σ = 70.4± 2.5
WF
σ/μ = 14.66± 0.60 %
μ = 517.2± 2.9
σ = 75.8± 2.7



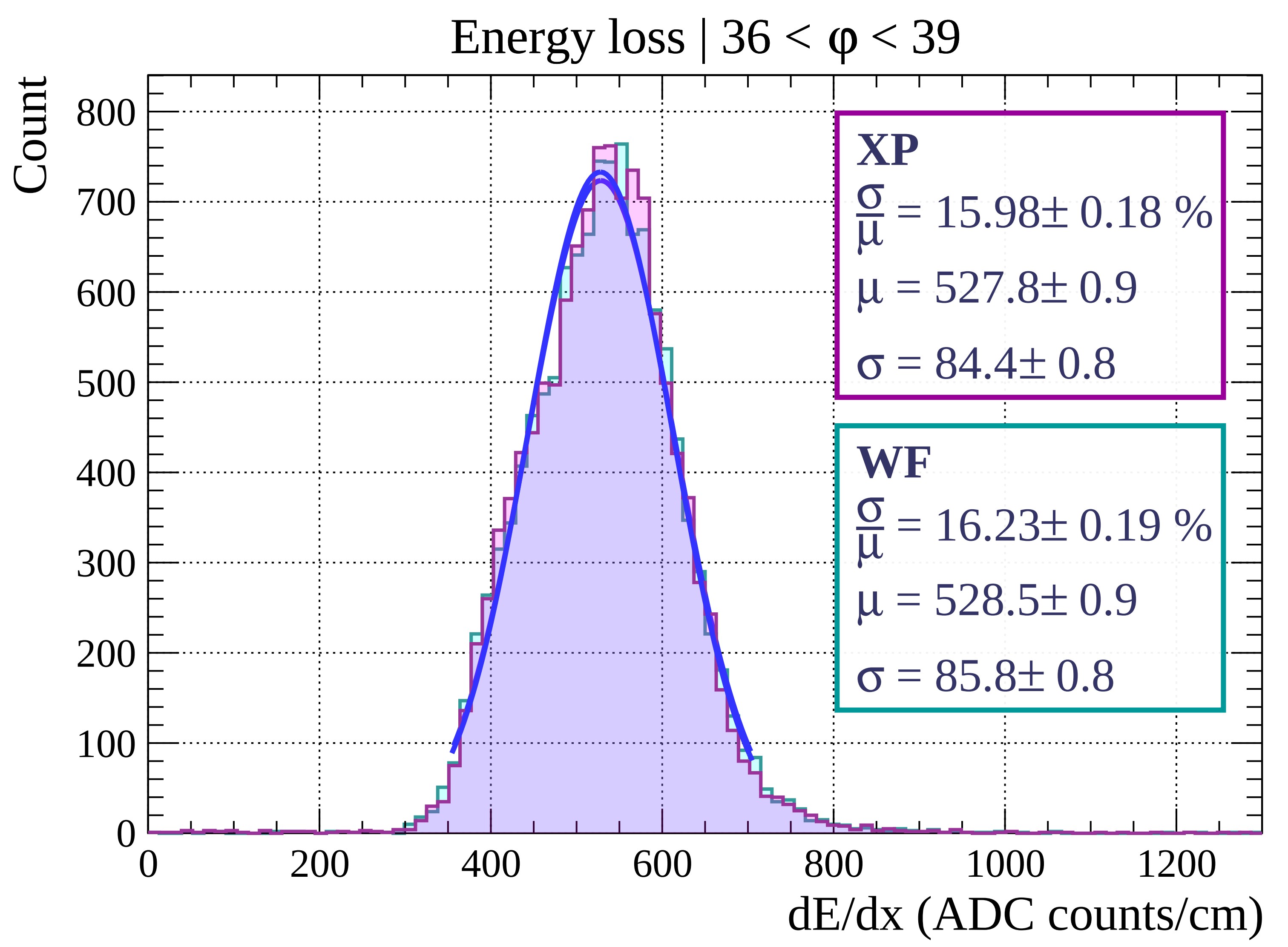
Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.02± 0.36 %
μ = 535.2± 1.8
σ = 75.0± 1.7
WF
σ/μ = 14.11± 0.36 %
μ = 532.8± 1.8
σ = 75.2± 1.7
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200



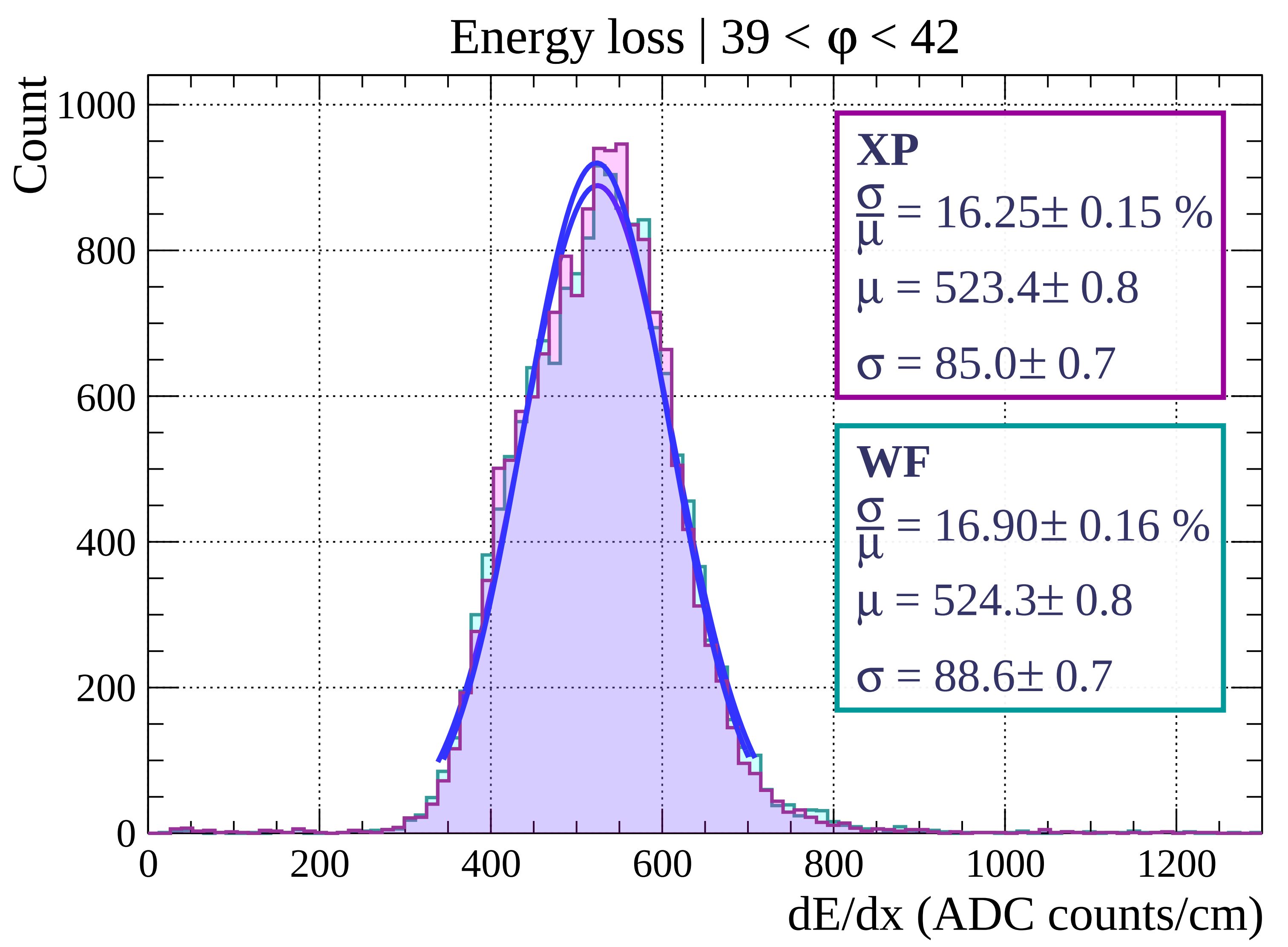
Energy loss | 30 < φ < 33
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.84± 0.26 %
μ = 535.6± 1.2
σ = 84.8± 1.2
WF
σ/μ = 15.82± 0.26 %
μ = 531.6± 1.2
σ = 84.1± 1.2

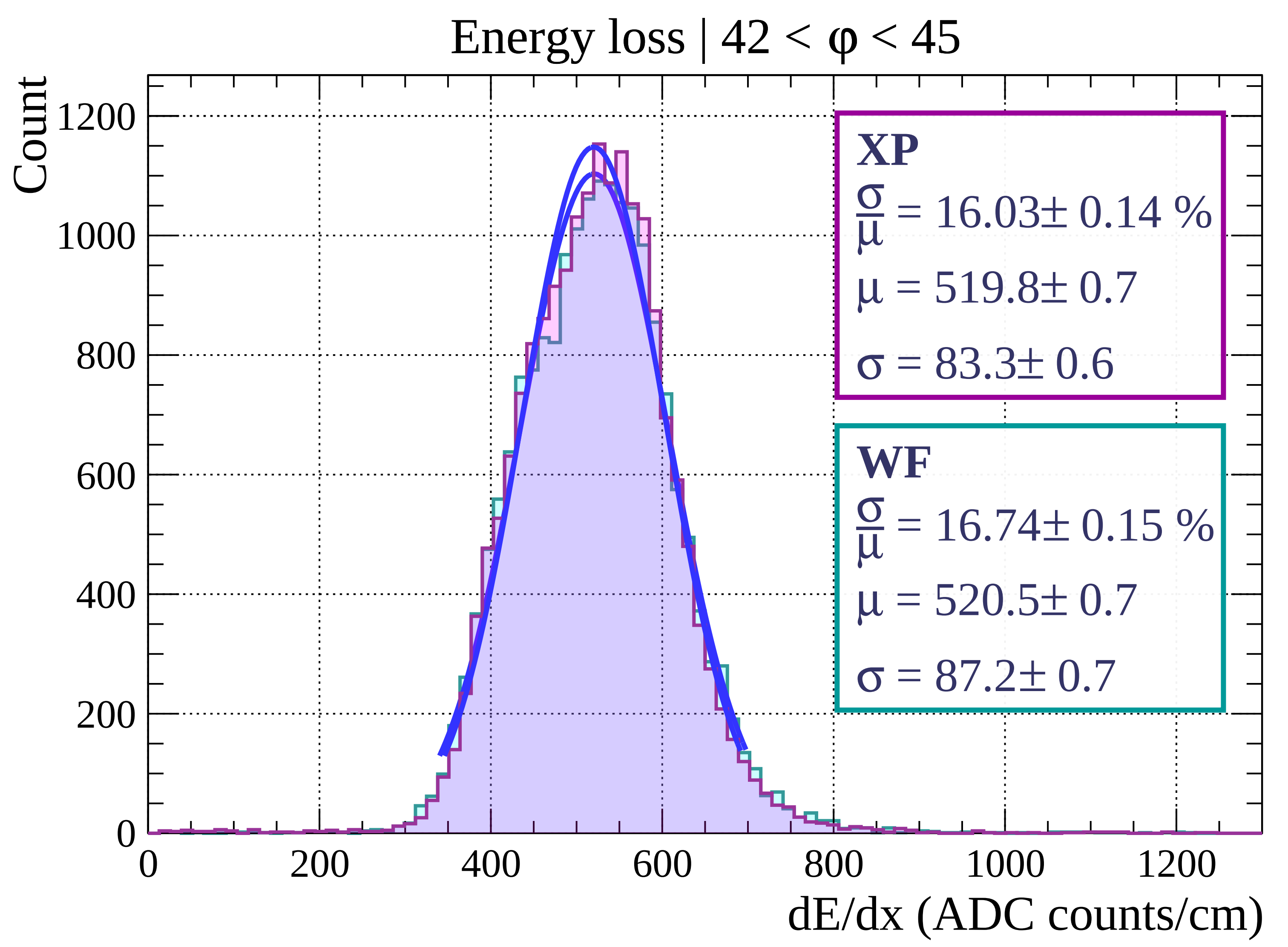


Energy loss | 33 < φ < 36
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 15.96± 0.20 %
μ = 531.6± 1.0
σ = 84.9± 0.9
WF
σ/μ = 15.93± 0.20 %
μ = 530.1± 1.0
σ = 84.4± 0.9

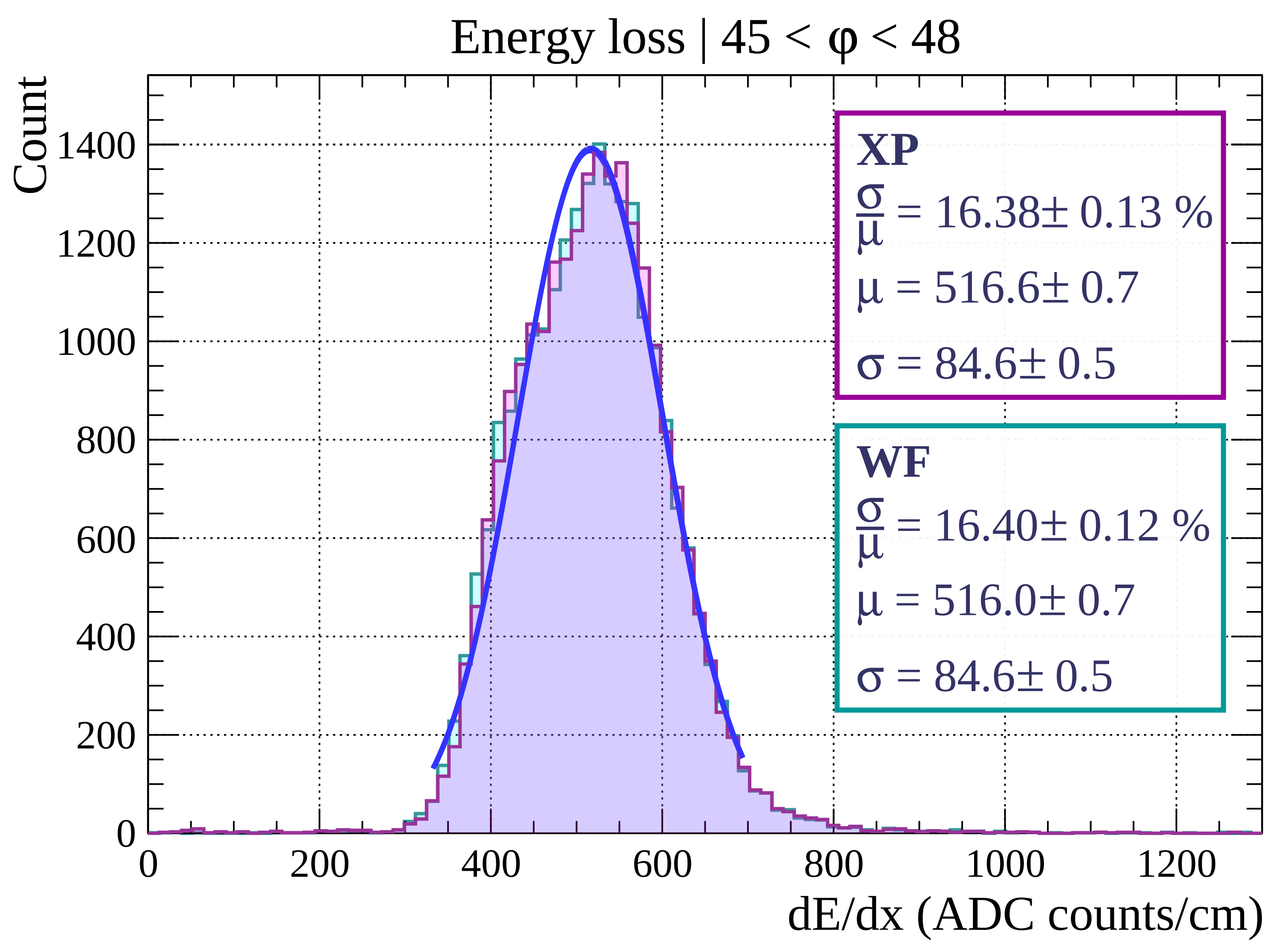


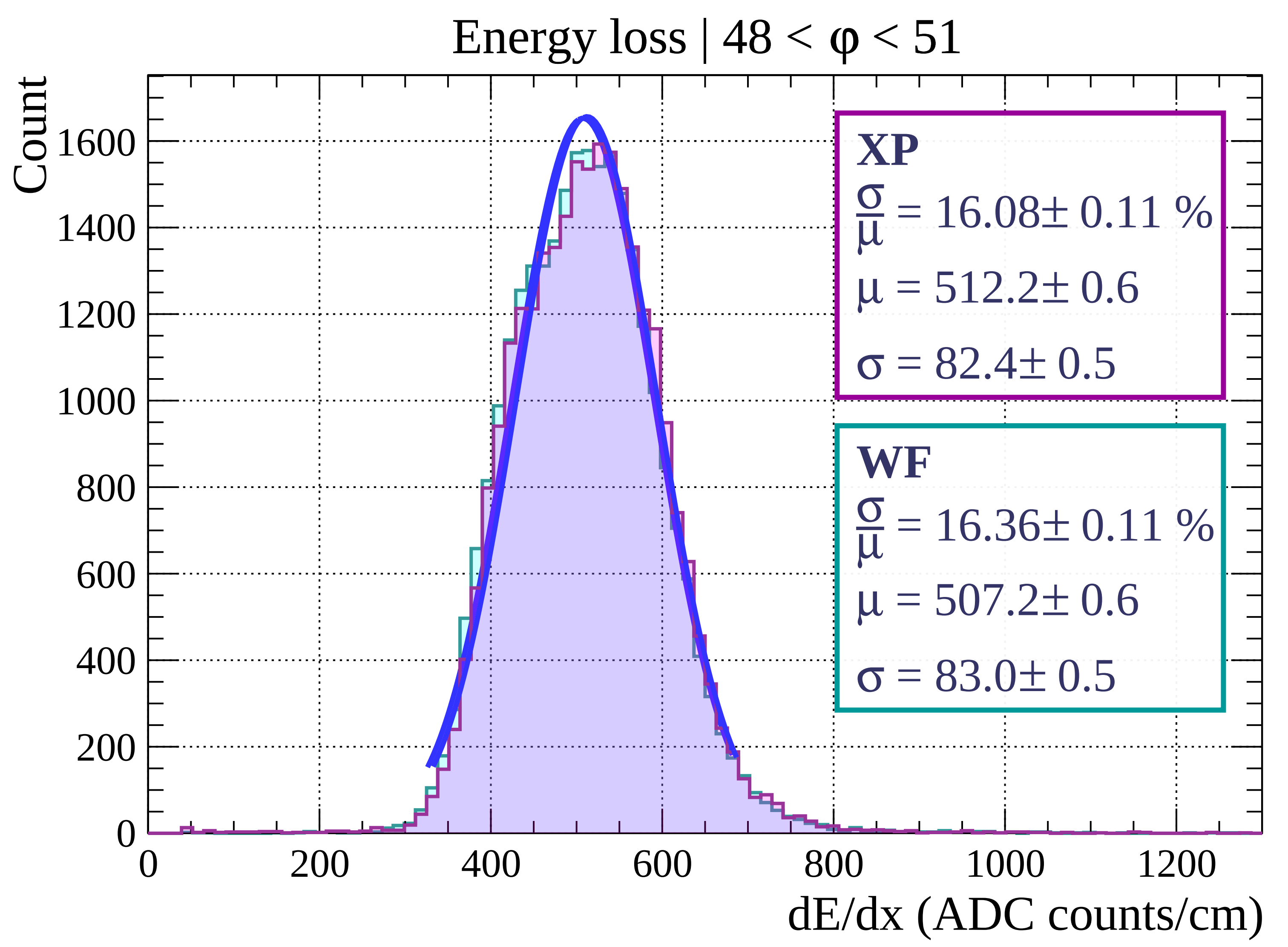
Energy loss | 36 < φ < 39
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 15.98± 0.18 %
μ = 527.8± 0.9
σ = 84.4± 0.8
WF
σ/μ = 16.23± 0.19 %
μ = 528.5± 0.9
σ = 85.8± 0.8


Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 16.25± 0.15 %
μ = 523.4± 0.8
σ = 85.0± 0.7
WF
σ/μ = 16.90± 0.16 %
μ = 524.3± 0.8
σ = 88.6± 0.7

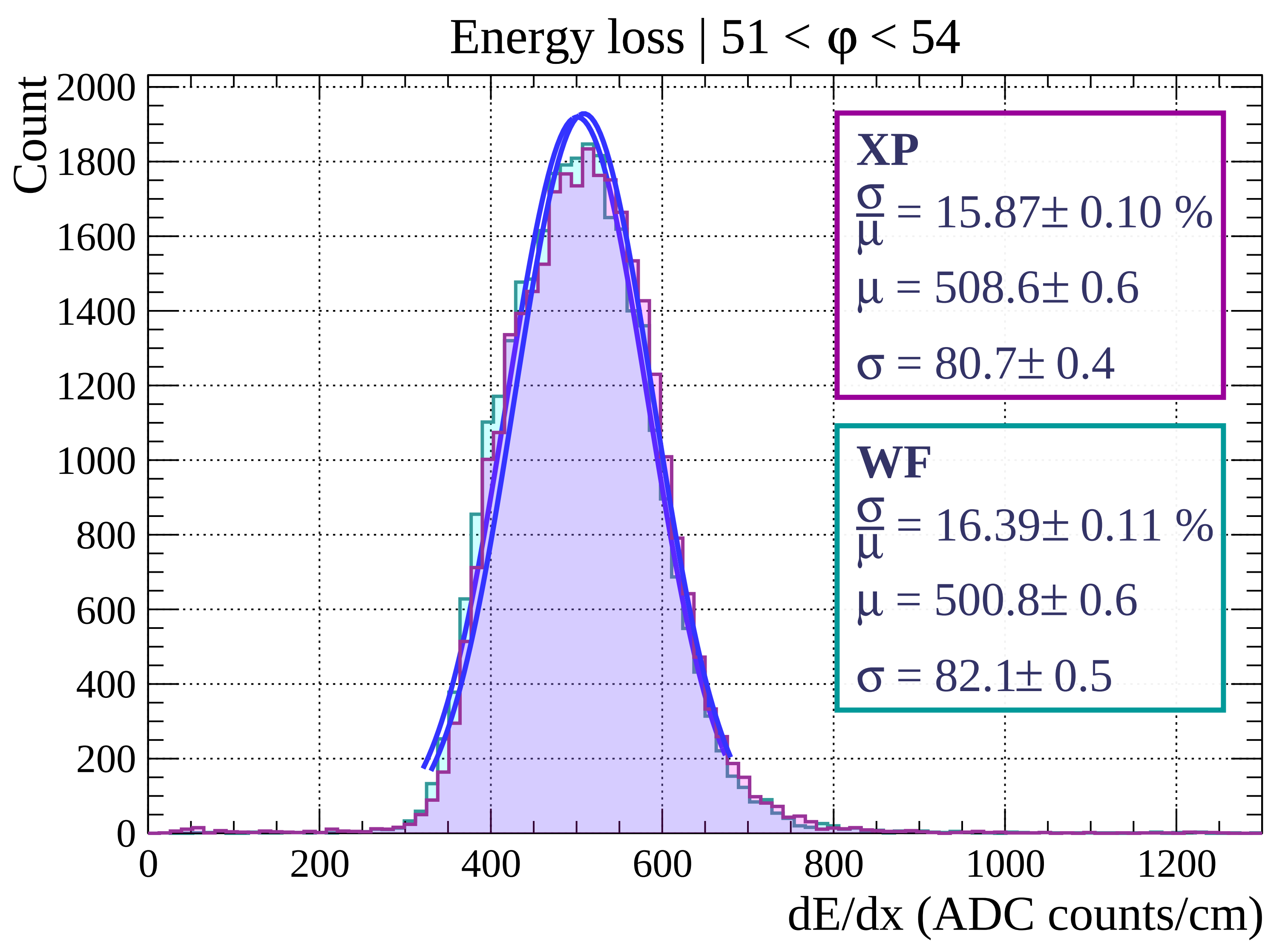


Energy loss | 42 < φ < 45
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 16.03± 0.14 %
μ = 519.8± 0.7
σ = 83.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 16.74± 0.15 %
μ = 520.5± 0.7
σ = 87.2± 0.7

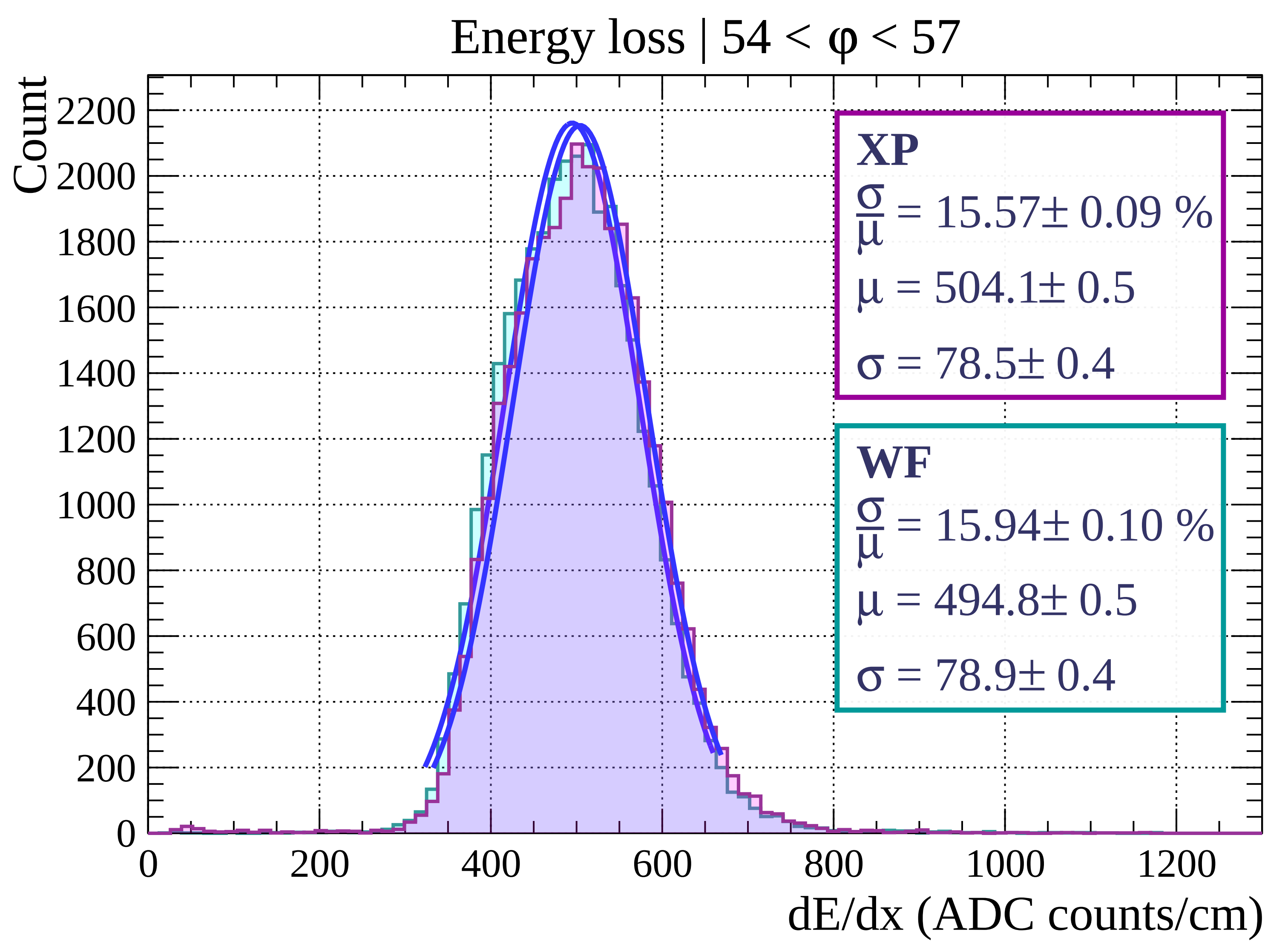

Energy loss | 45 < φ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP
$\frac{\sigma}{\mu}$ = 16.38± 0.13 %
μ = 516.6± 0.7
σ = 84.6± 0.5
WF
$\frac{\sigma}{\mu}$ = 16.40± 0.12 %
μ = 516.0± 0.7
σ = 84.6± 0.5



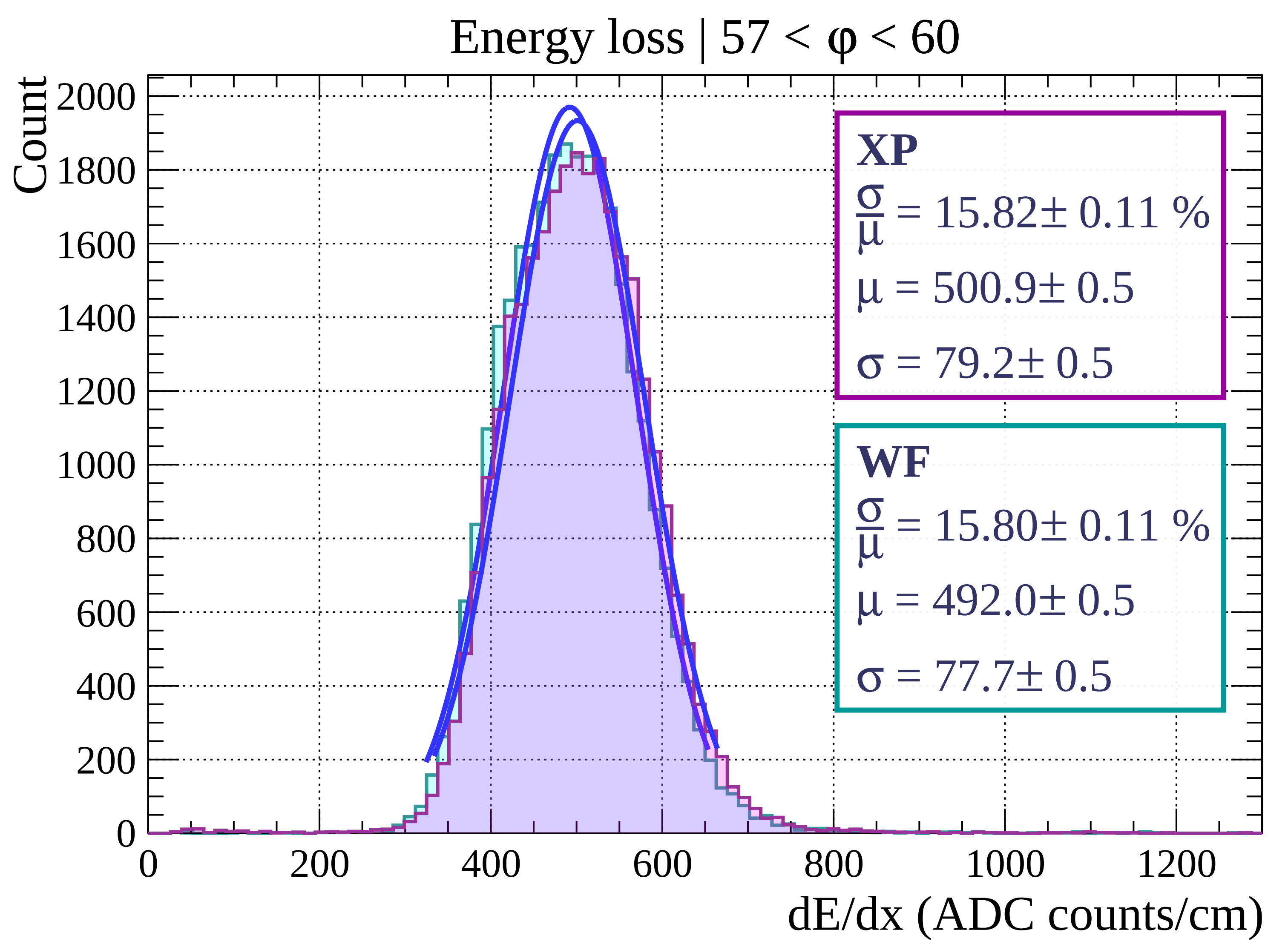
Energy loss | 48 < φ < 51
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.08± 0.11 %
μ = 512.2± 0.6
σ = 82.4± 0.5
WF
σ/μ = 16.36± 0.11 %
μ = 507.2± 0.6
σ = 83.0± 0.5



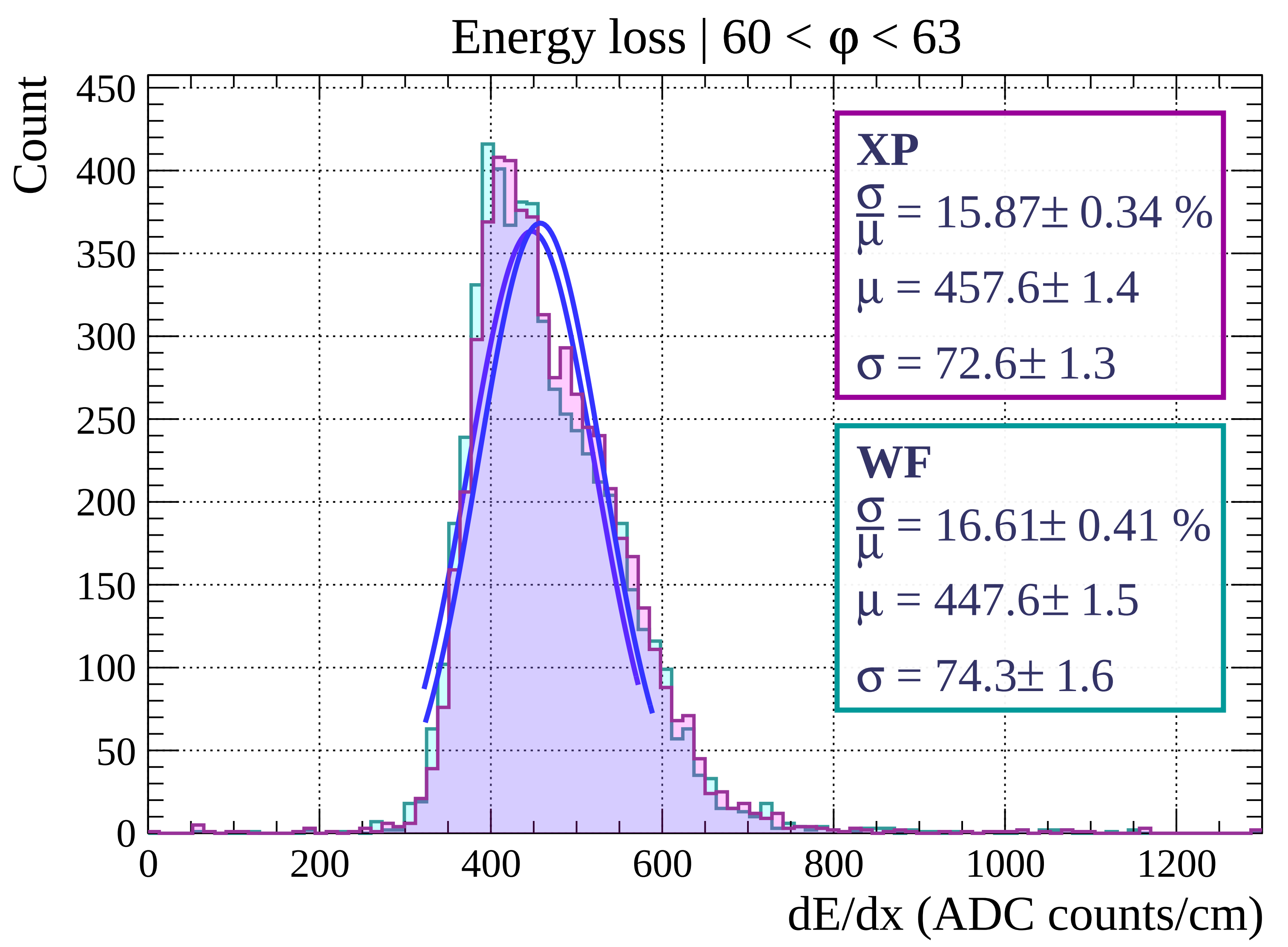
Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.87± 0.10 %
μ = 508.6± 0.6
σ = 80.7± 0.4
WF
σ/μ = 16.39± 0.11 %
μ = 500.8± 0.6
σ = 82.1± 0.5



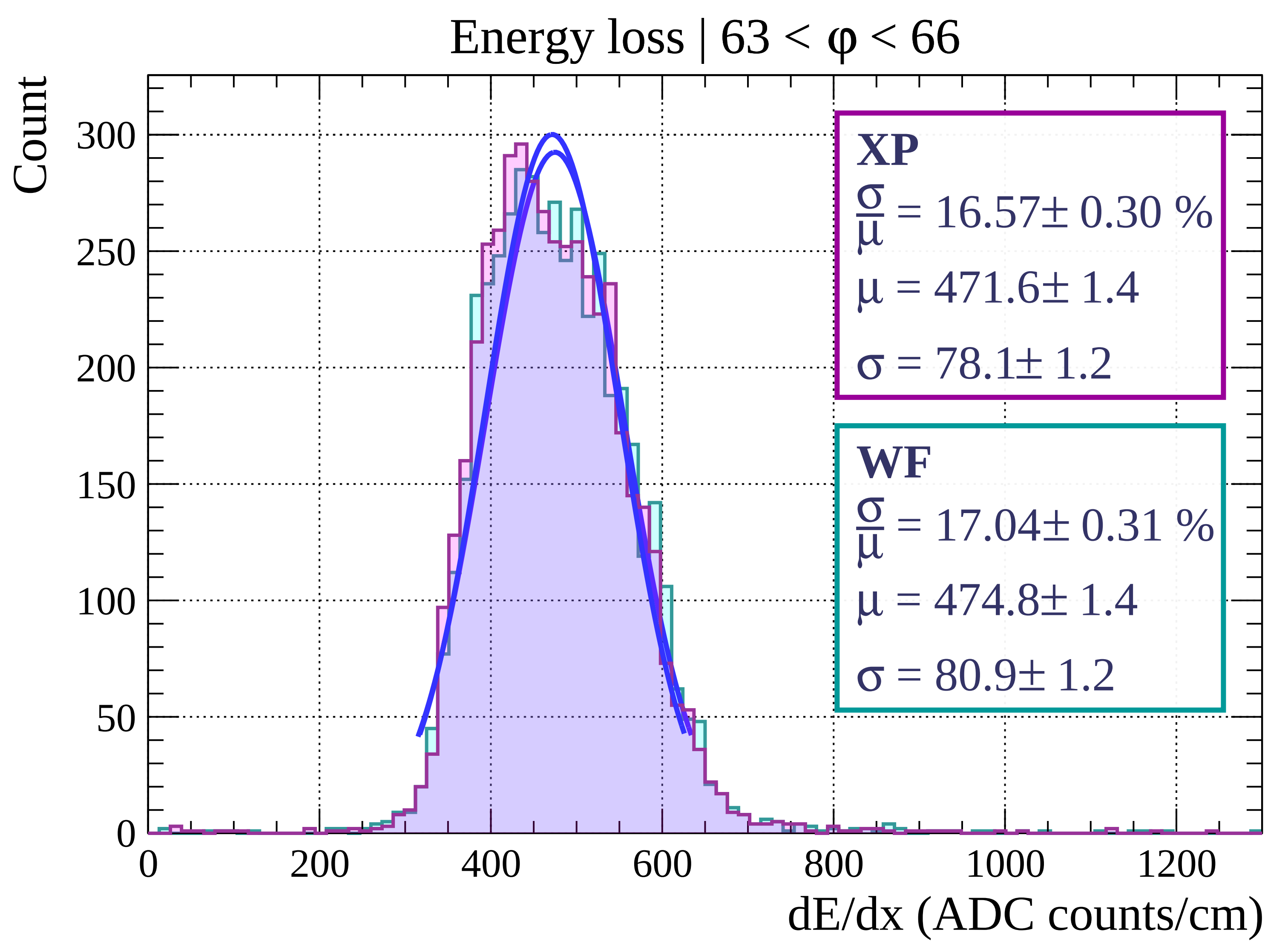
Energy loss | 54 < φ < 57
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.57± 0.09 %
μ = 504.1± 0.5
σ = 78.5± 0.4
WF
σ/μ = 15.94± 0.10 %
μ = 494.8± 0.5
σ = 78.9± 0.4



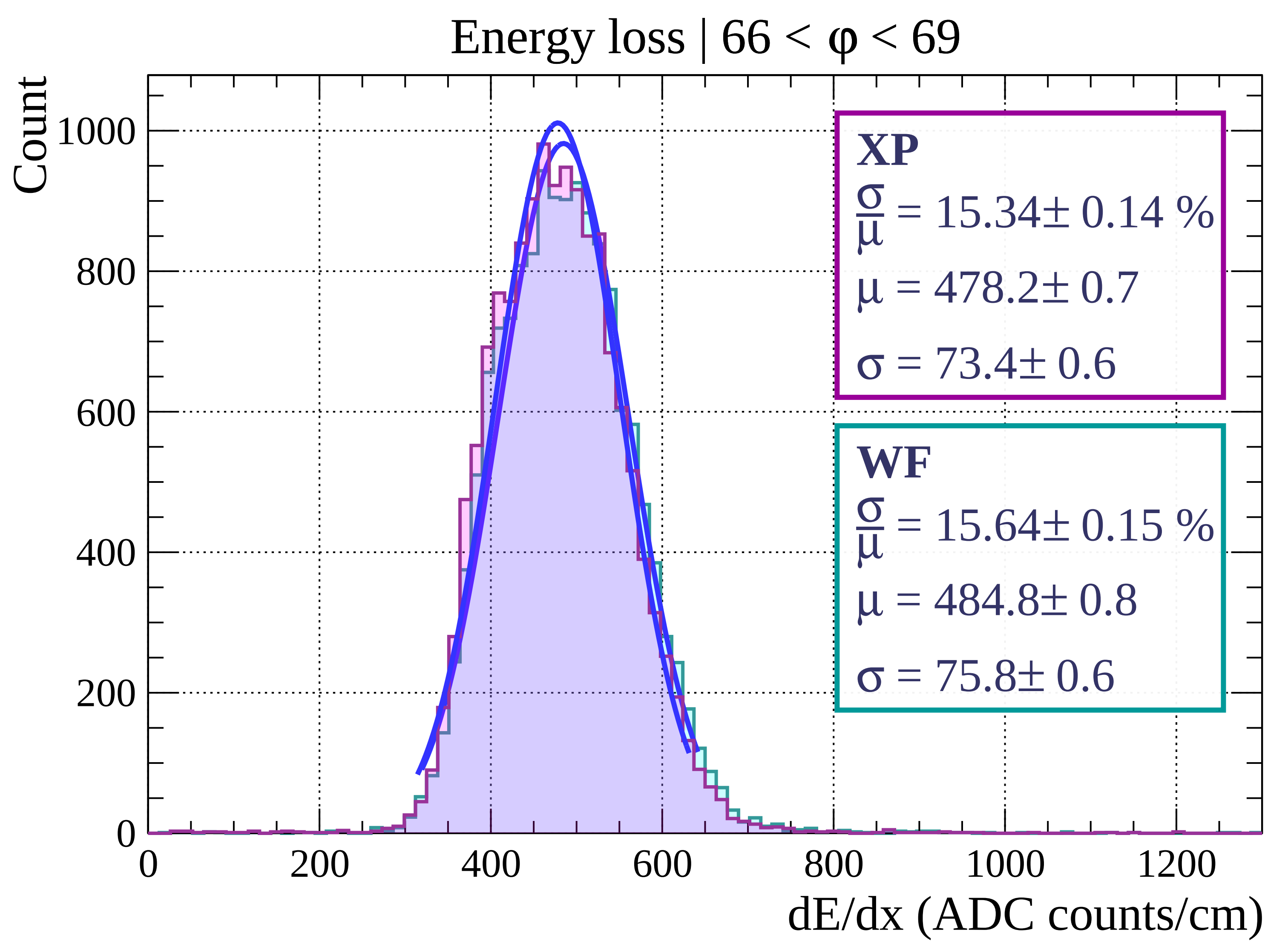
Energy loss | 57 < φ < 60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP
σ/μ = 15.82± 0.11 %
μ = 500.9± 0.5
σ = 79.2± 0.5
WF
σ/μ = 15.80± 0.11 %
μ = 492.0± 0.5
σ = 77.7± 0.5

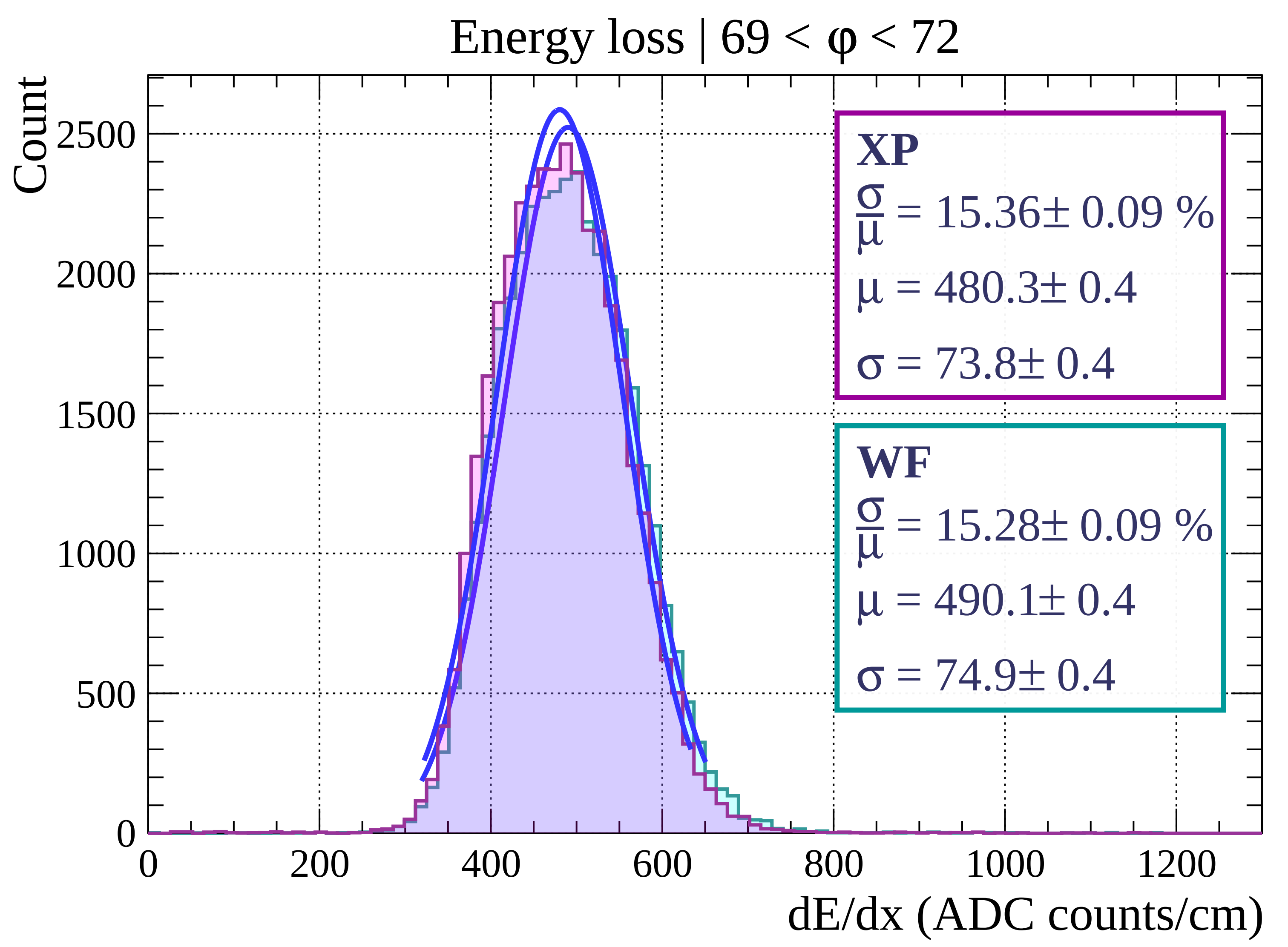


Energy loss | 60 < φ < 63
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.87± 0.34 %
μ = 457.6± 1.4
σ = 72.6± 1.3
WF
σ/μ = 16.61± 0.41 %
μ = 447.6± 1.5
σ = 74.3± 1.6

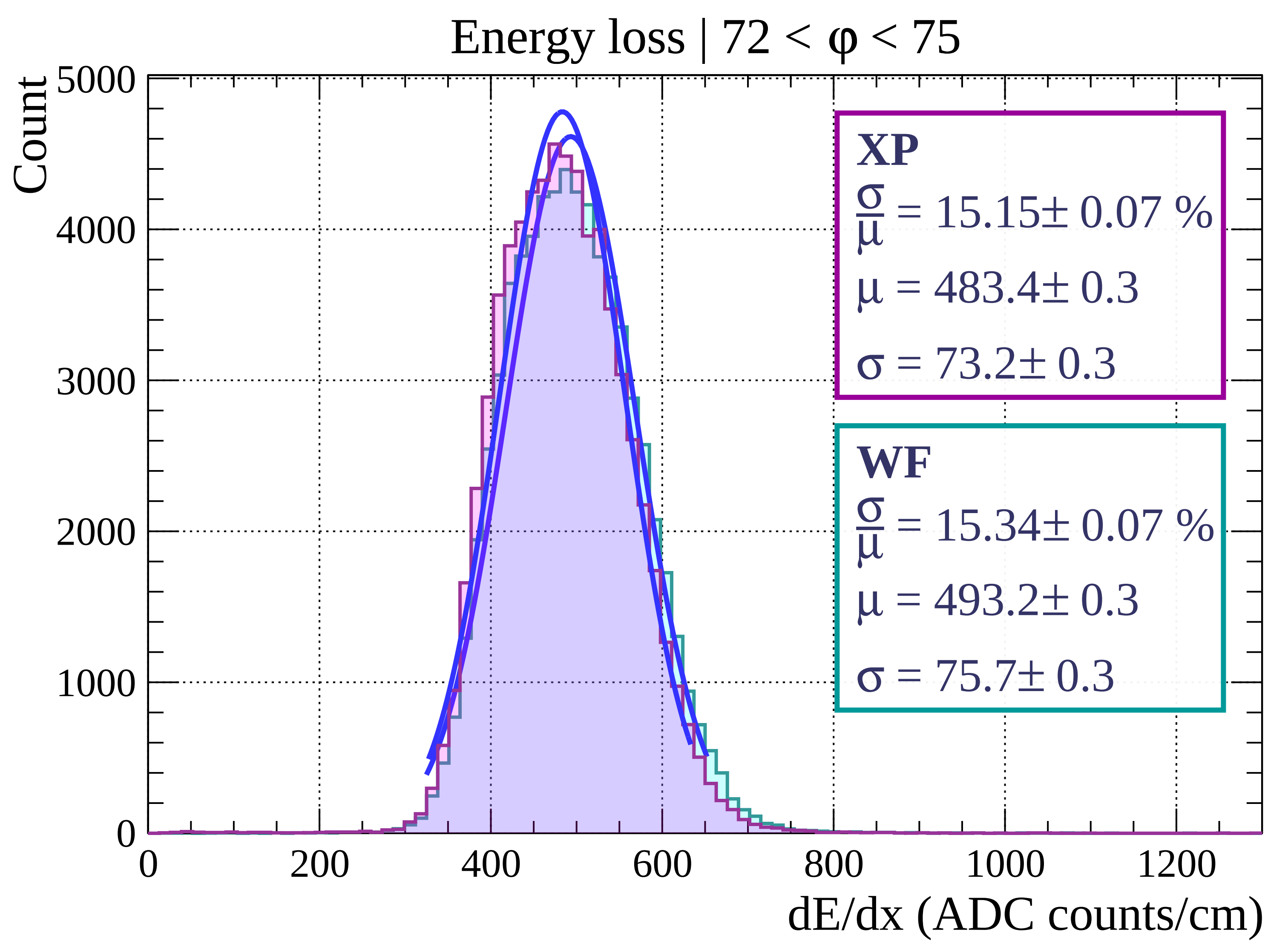


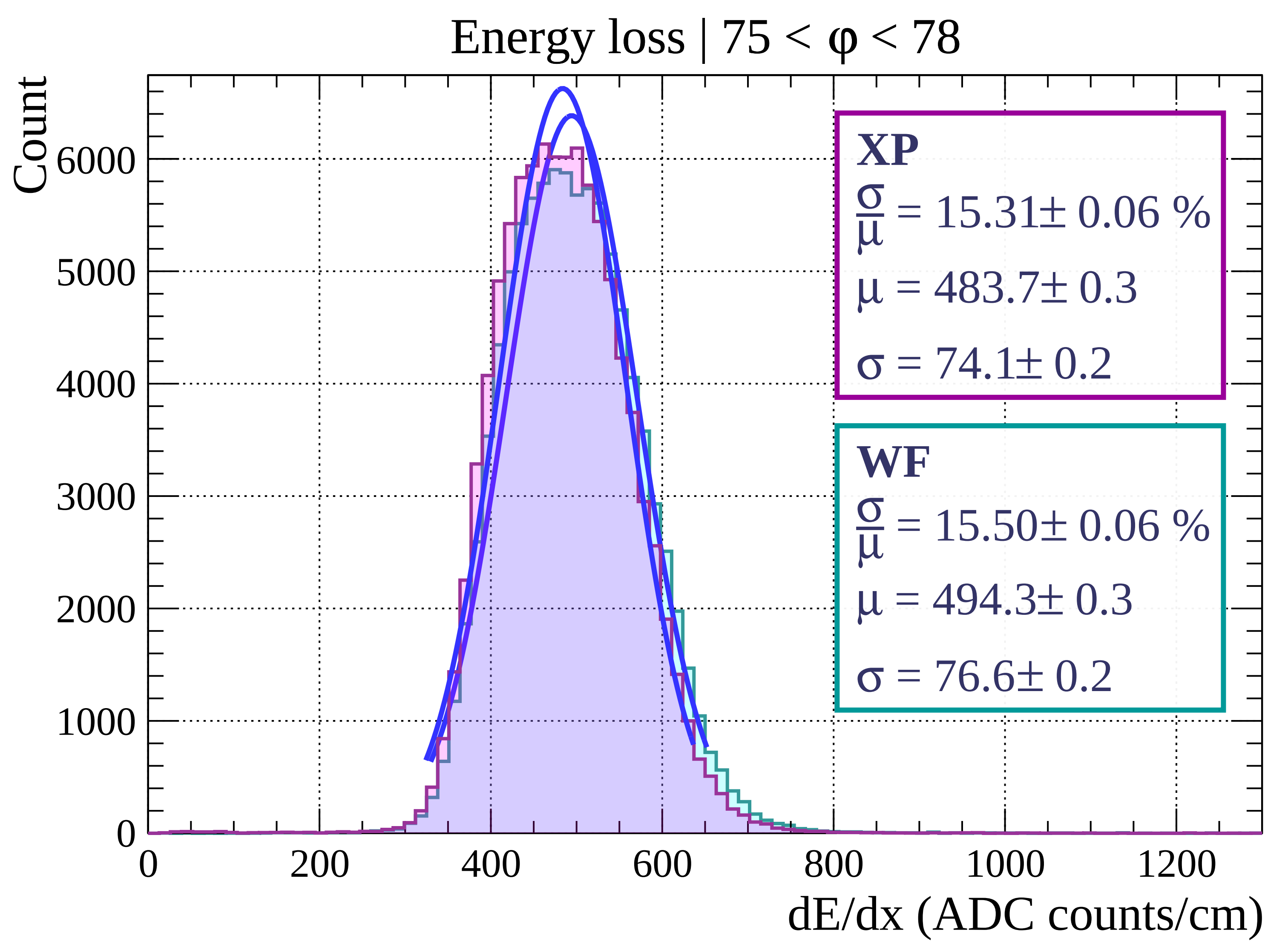
Energy loss | 63 < φ < 66
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.57± 0.30 %
μ = 471.6± 1.4
σ = 78.1± 1.2
WF
σ/μ = 17.04± 0.31 %
μ = 474.8± 1.4
σ = 80.9± 1.2


Energy loss | 66 < φ < 69
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 15.34± 0.14 %
μ = 478.2± 0.7
σ = 73.4± 0.6
WF
σ/μ = 15.64± 0.15 %
μ = 484.8± 0.8
σ = 75.8± 0.6

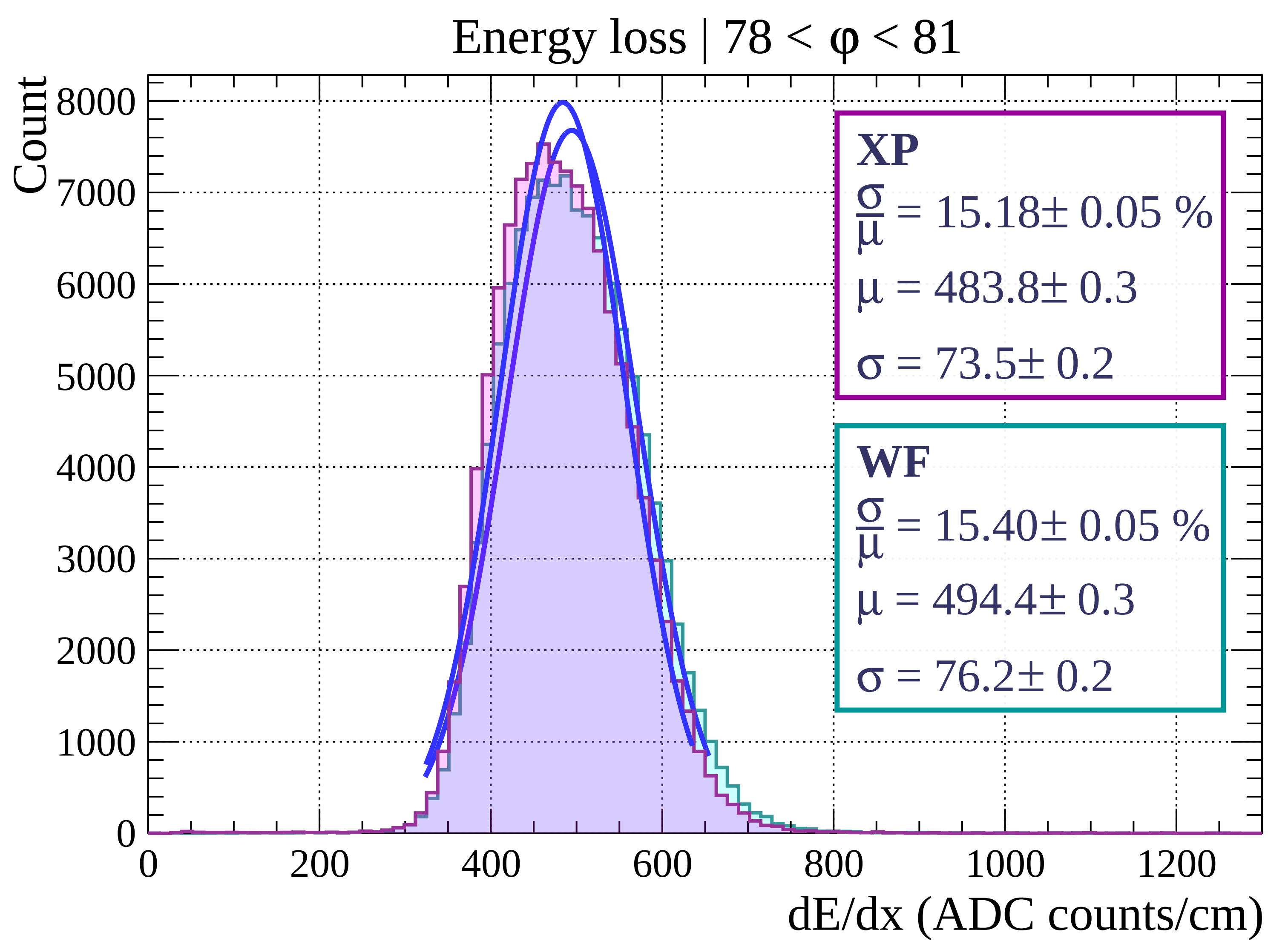


Energy loss | 69 < φ < 72
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
0
200
400
600
800
1000
1200
XP
σ/μ = 15.36± 0.09 %
μ = 480.3± 0.4
σ = 73.8± 0.4
WF
σ/μ = 15.28± 0.09 %
μ = 490.1± 0.4
σ = 74.9± 0.4

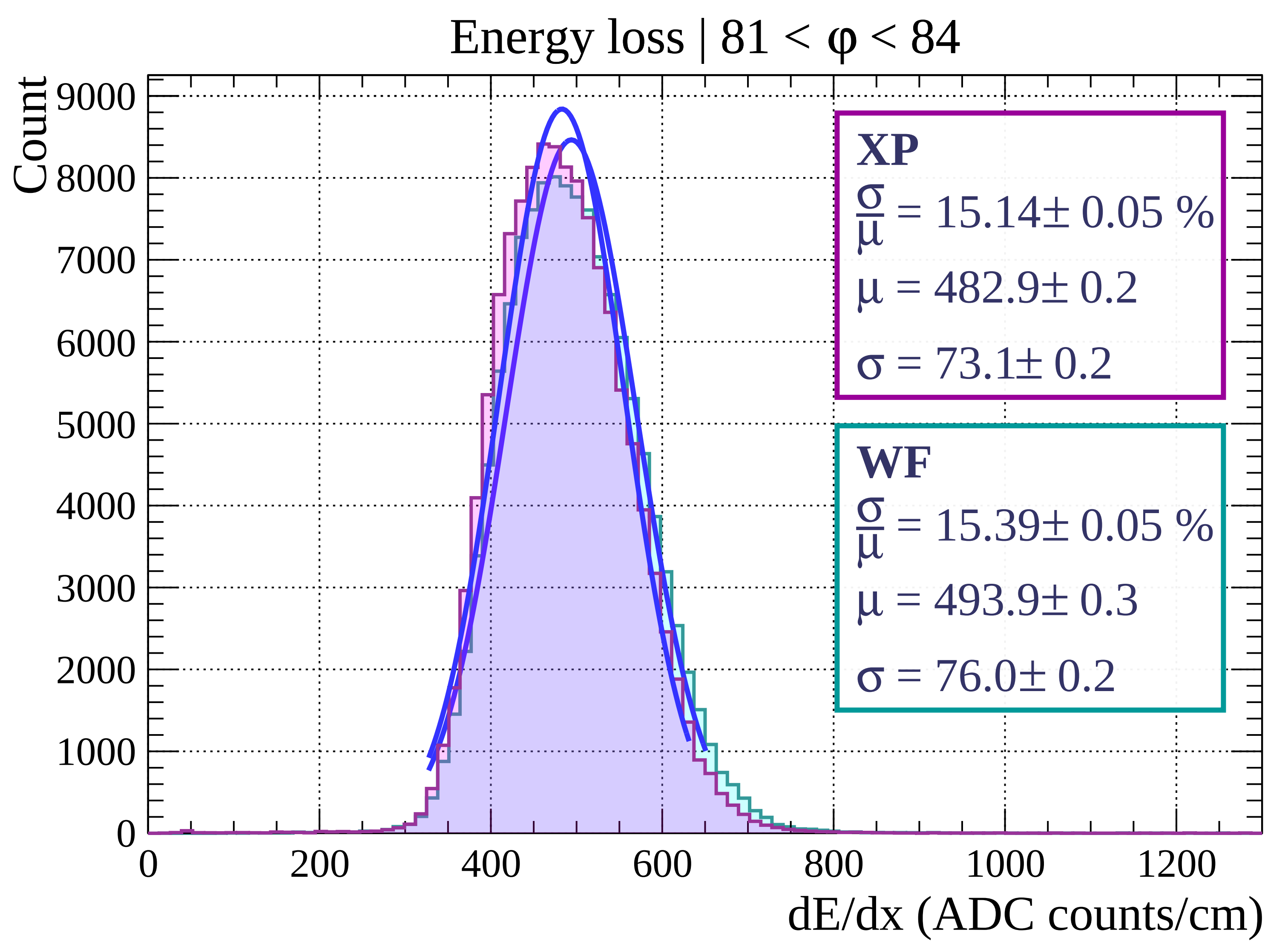

Energy loss | 72 < φ < 75
Count
dE/dx (ADC counts/cm)
0
1000
2000
3000
4000
5000
0
200
400
600
800
1000
1200
XP
σ/μ = 15.15± 0.07 %
μ = 483.4± 0.3
σ = 73.2± 0.3
WF
σ/μ = 15.34± 0.07 %
μ = 493.2± 0.3
σ = 75.7± 0.3



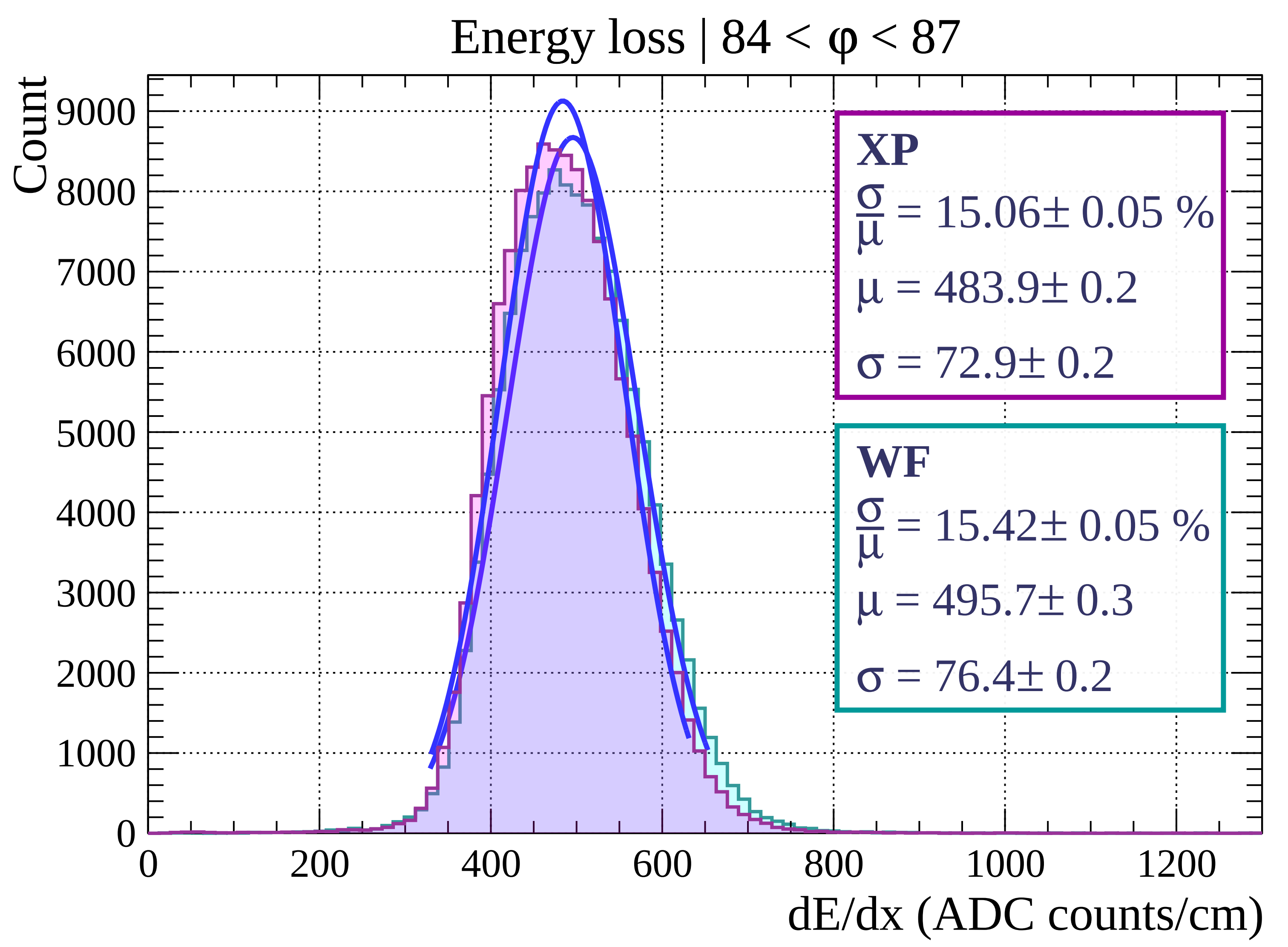
Energy loss | 75 < φ < 78
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.31± 0.06 %
μ = 483.7± 0.3
σ = 74.1± 0.2
WF
σ/μ = 15.50± 0.06 %
μ = 494.3± 0.3
σ = 76.6± 0.2

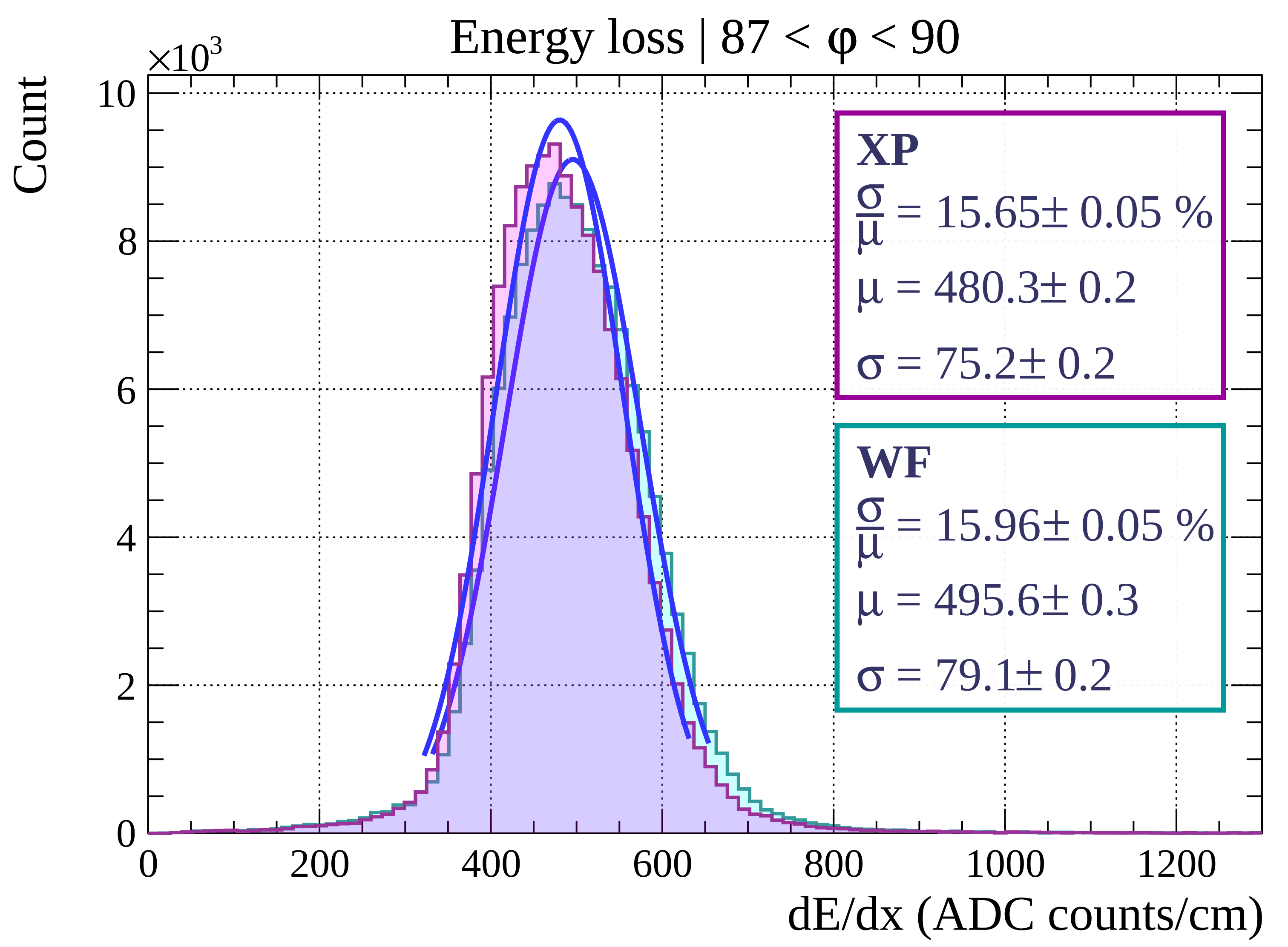


Energy loss | 78 < φ < 81
Count
8000
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.18± 0.05 %
μ = 483.8± 0.3
σ = 73.5± 0.2
WF
σ/μ = 15.40± 0.05 %
μ = 494.4± 0.3
σ = 76.2± 0.2

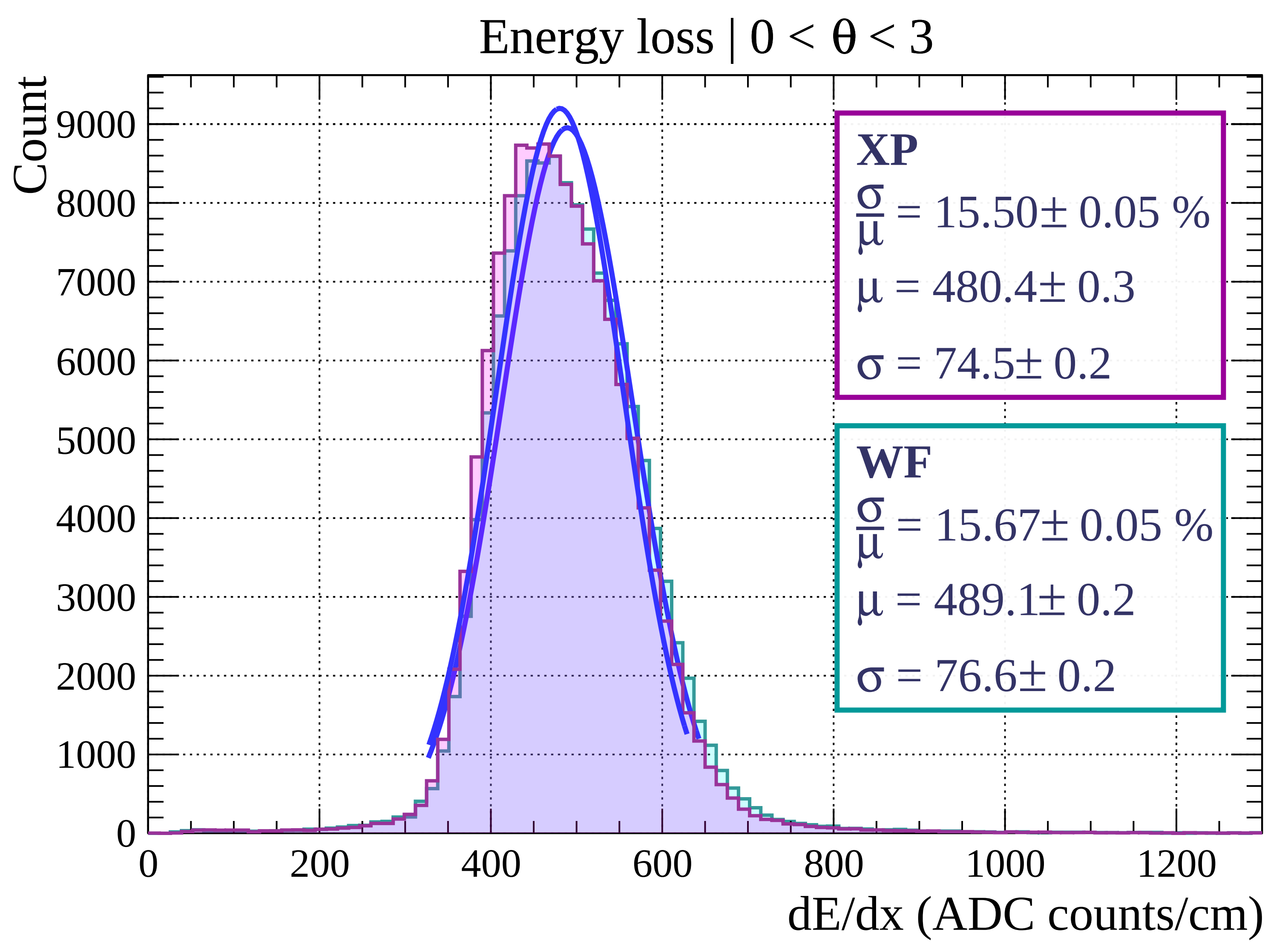


Energy loss | 81 < φ < 84
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
6000
7000
8000
9000
XP
σ/μ = 15.14± 0.05 %
μ = 482.9± 0.2
σ = 73.1± 0.2
WF
σ/μ = 15.39± 0.05 %
μ = 493.9± 0.3
σ = 76.0± 0.2

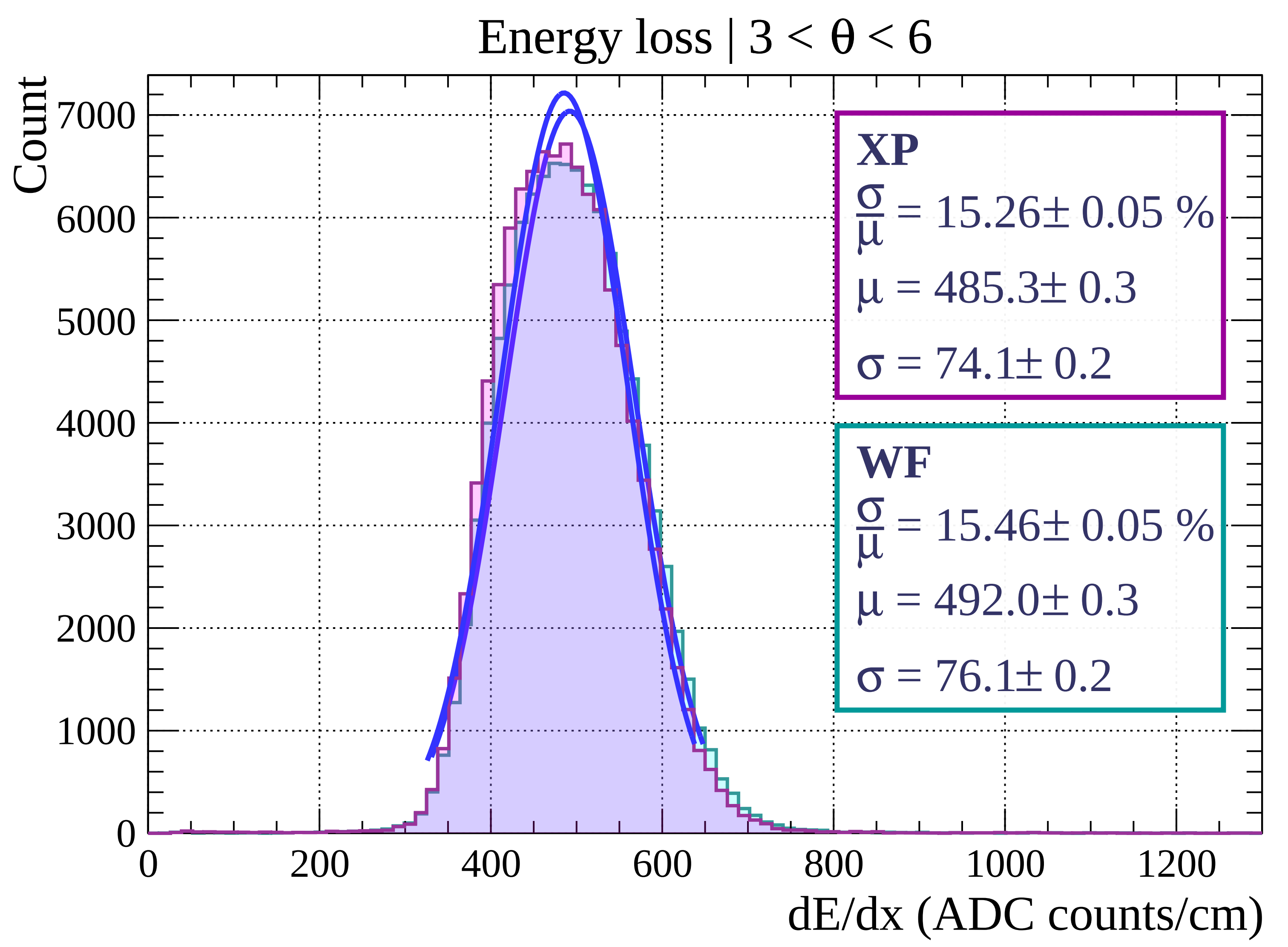

Energy loss | 84 < φ < 87
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
6000
7000
8000
9000
XP
$\frac{\sigma}{\mu}$ = 15.06± 0.05 %
μ = 483.9± 0.2
σ = 72.9± 0.2
WF
$\frac{\sigma}{\mu}$ = 15.42± 0.05 %
μ = 495.7± 0.3
σ = 76.4± 0.2

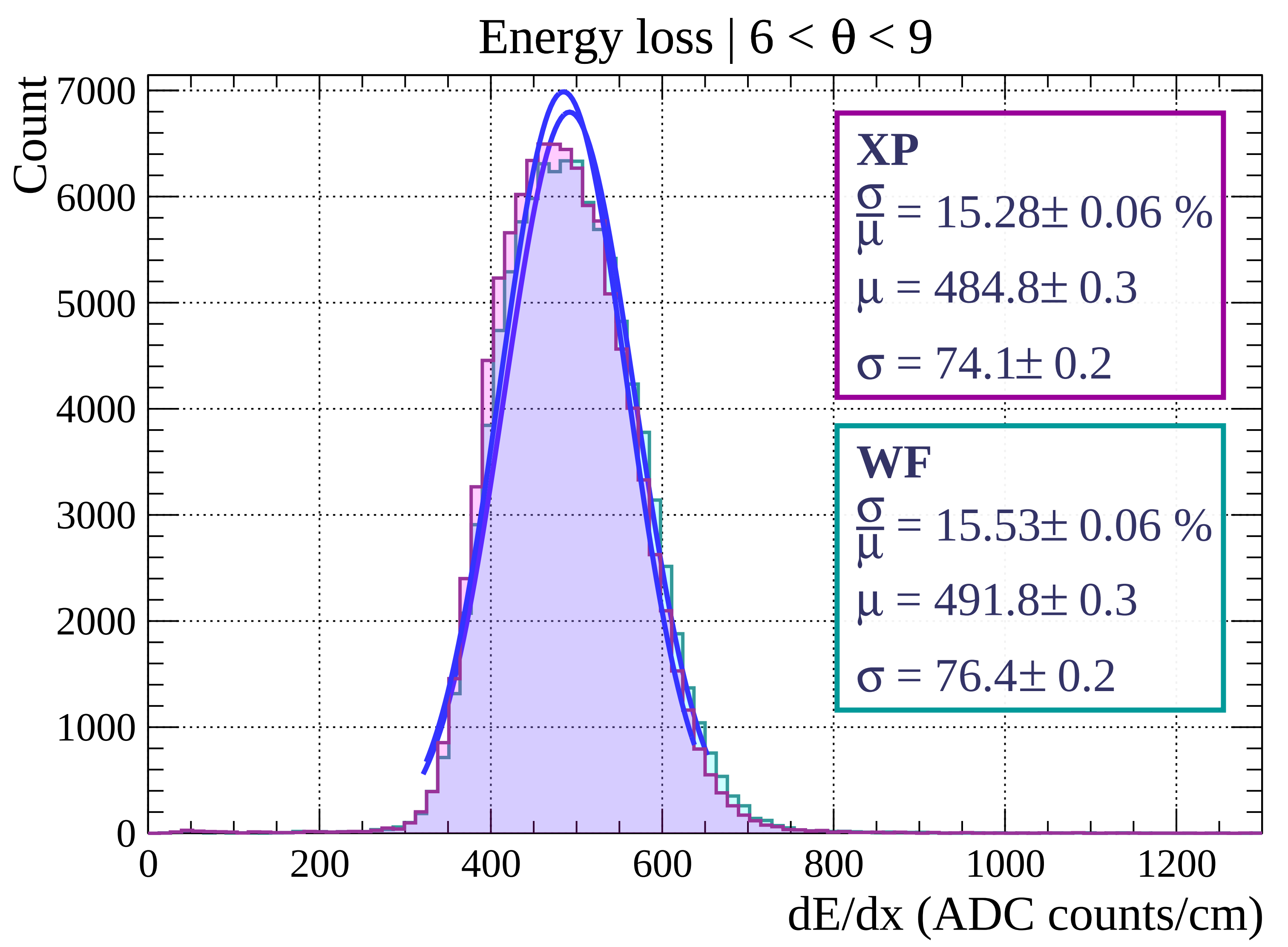


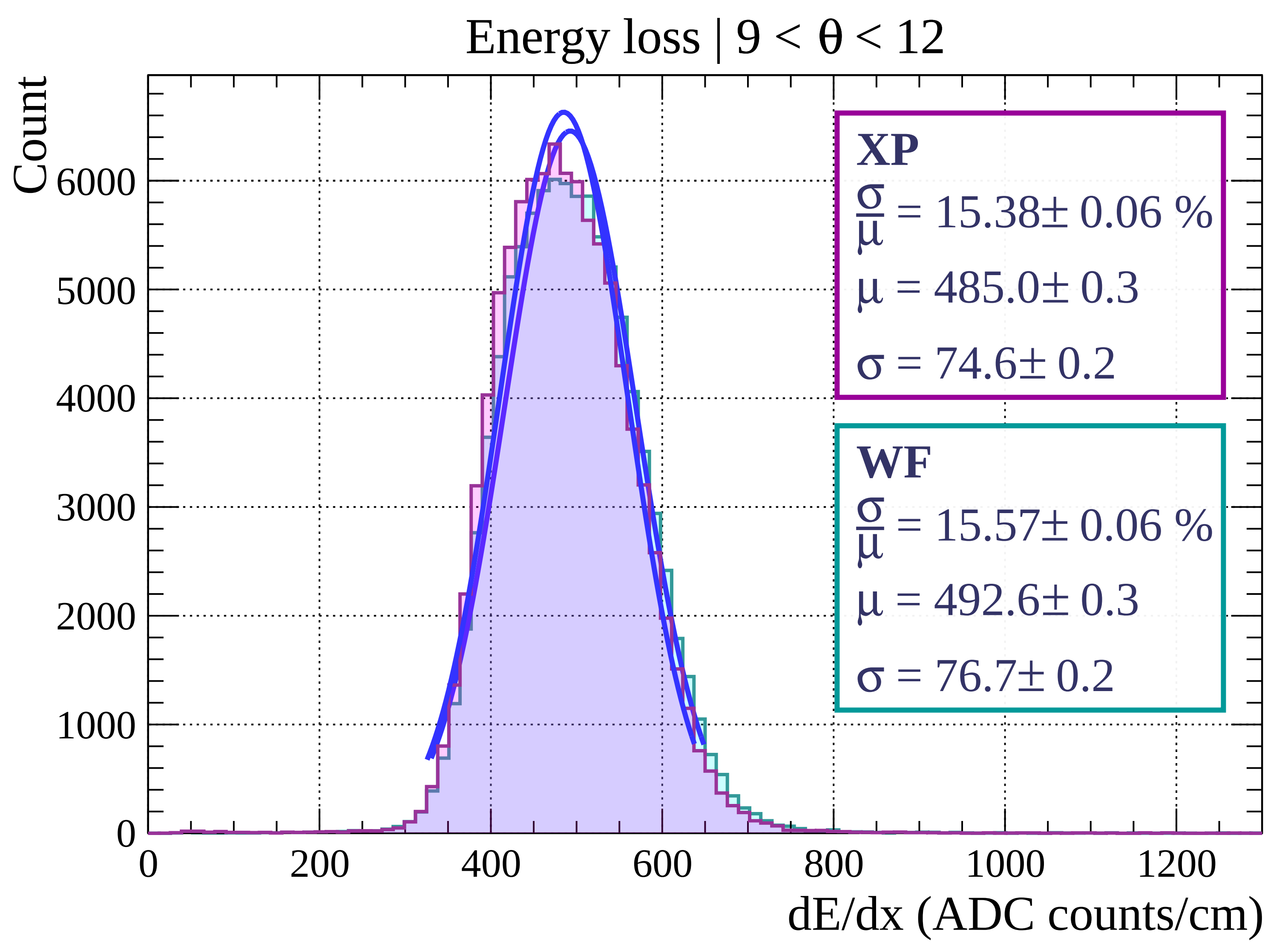
Energy loss | 87 < φ < 90
×10^3
Count
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.65± 0.05 %
μ = 480.3± 0.2
σ = 75.2± 0.2
WF
σ/μ = 15.96± 0.05 %
μ = 495.6± 0.3
σ = 79.1± 0.2



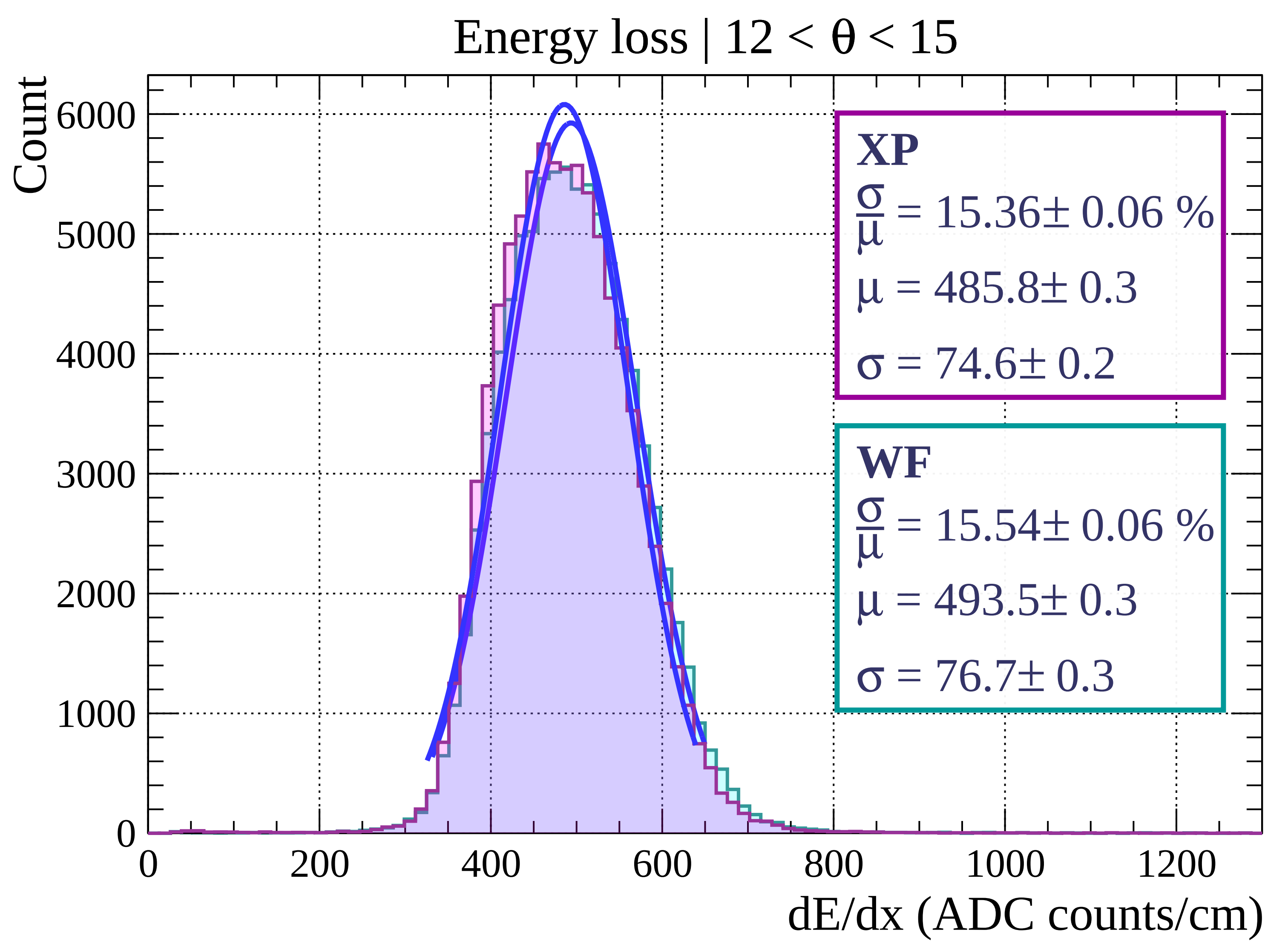
Energy loss | 0 < θ < 3
Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.50± 0.05 %
μ = 480.4± 0.3
σ = 74.5± 0.2
WF
σ/μ = 15.67± 0.05 %
μ = 489.1± 0.2
σ = 76.6± 0.2


Energy loss | 3 < θ < 6
Count
dE/dx (ADC counts/cm)
0
1000
2000
3000
4000
5000
6000
7000
200
400
600
800
1200
XP
σ/μ = 15.26± 0.05 %
μ = 485.3± 0.3
σ = 74.1± 0.2
WF
σ/μ = 15.46± 0.05 %
μ = 492.0± 0.3
σ = 76.1± 0.2

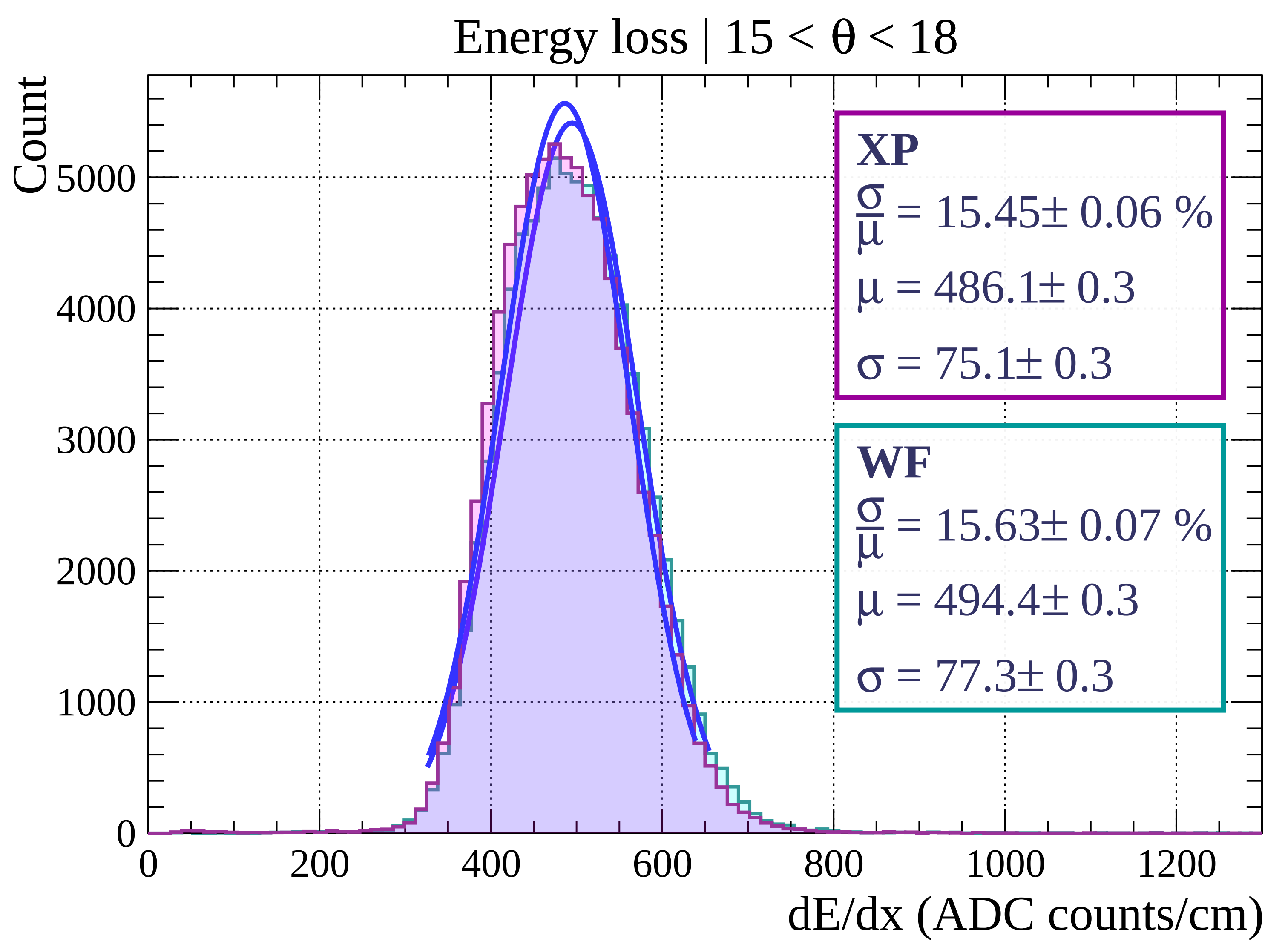

Energy loss | 6 < θ < 9
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1000
2000
3000
4000
5000
6000
7000
XP
σ/μ = 15.28± 0.06 %
μ = 484.8± 0.3
σ = 74.1± 0.2
WF
σ/μ = 15.53± 0.06 %
μ = 491.8± 0.3
σ = 76.4± 0.2



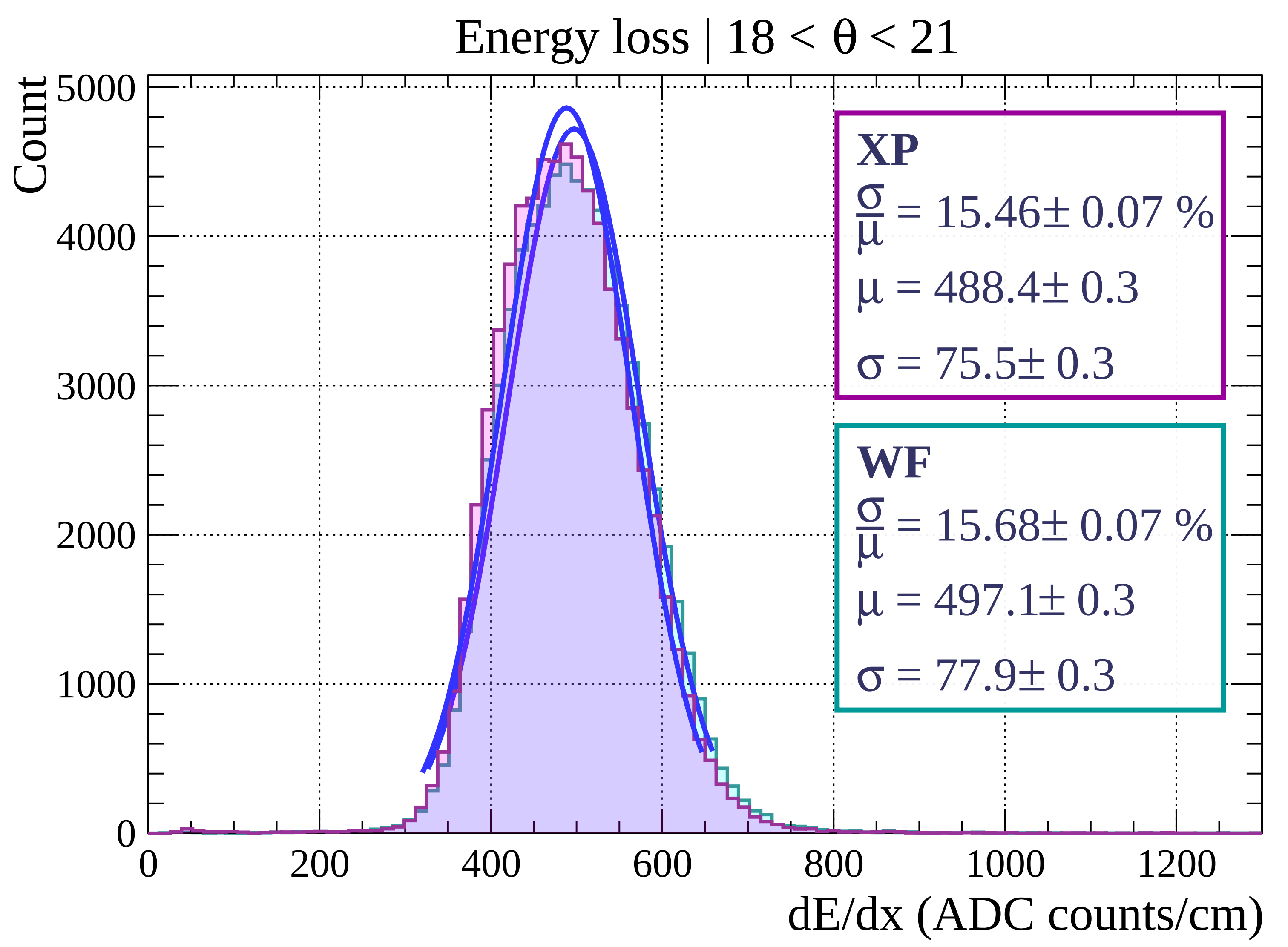
Energy loss | 9 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
6000
XP
$\frac{\sigma}{\mu}$ = 15.38± 0.06 %
μ = 485.0± 0.3
σ = 74.6± 0.2
WF
$\frac{\sigma}{\mu}$ = 15.57± 0.06 %
μ = 492.6± 0.3
σ = 76.7± 0.2



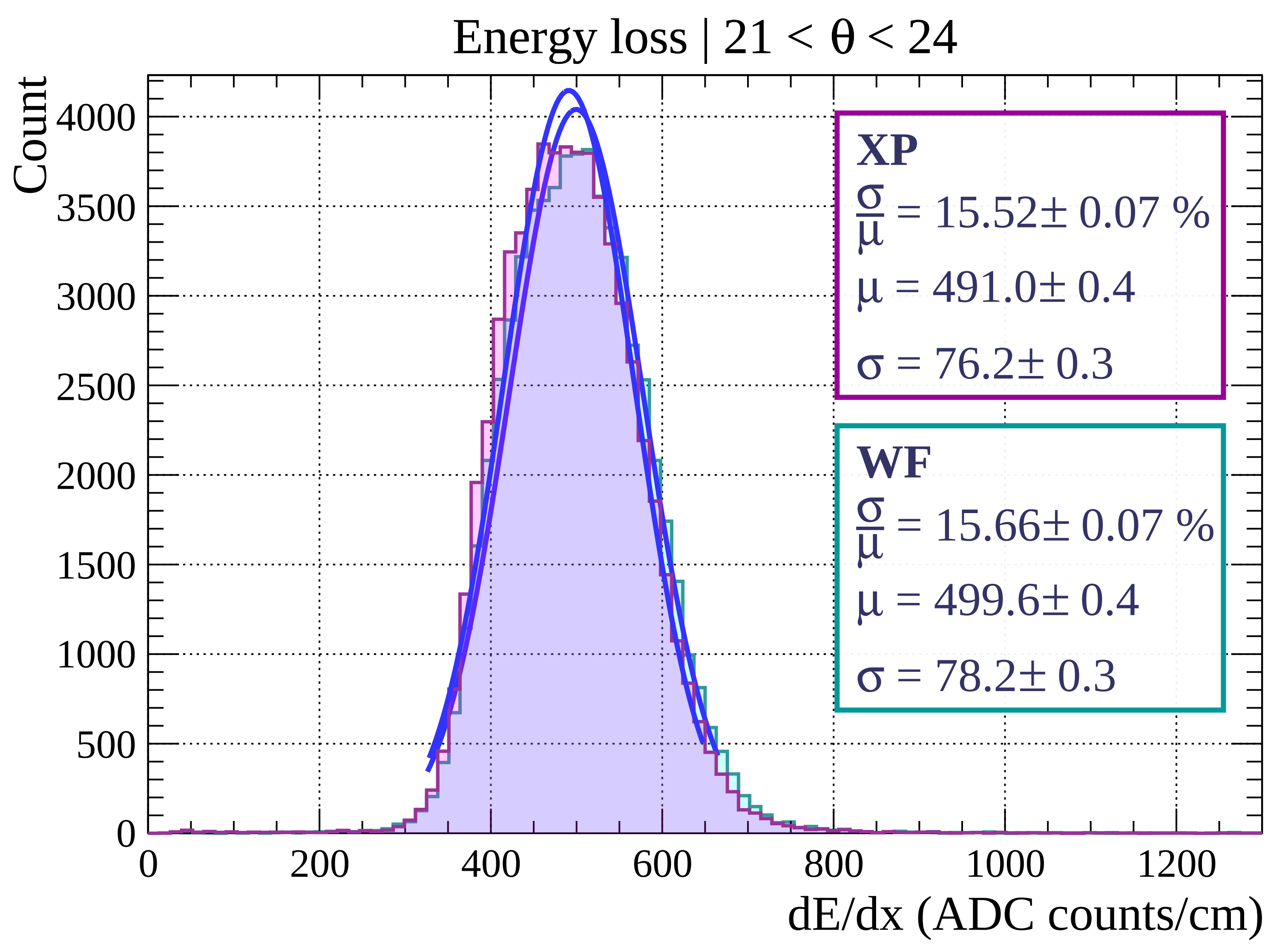
Energy loss | 12 < θ < 15
Count
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.36± 0.06 %
μ = 485.8± 0.3
σ = 74.6± 0.2
WF
σ/μ = 15.54± 0.06 %
μ = 493.5± 0.3
σ = 76.7± 0.3



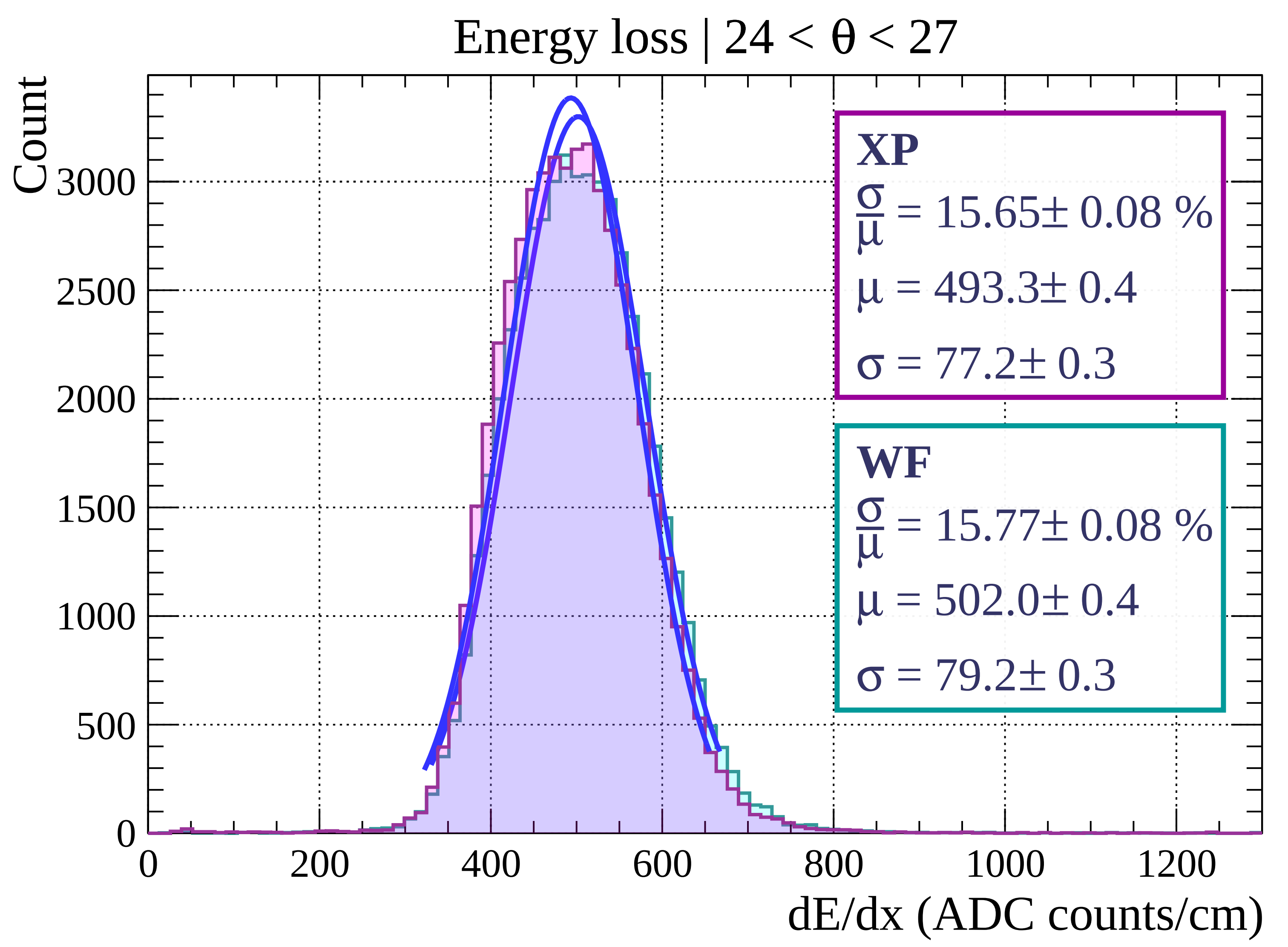
Energy loss | 15 < θ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1000
2000
3000
4000
5000
XP
$\frac{\sigma}{\mu}$ = 15.45± 0.06 %
μ = 486.1± 0.3
σ = 75.1± 0.3
WF
$\frac{\sigma}{\mu}$ = 15.63± 0.07 %
μ = 494.4± 0.3
σ = 77.3± 0.3



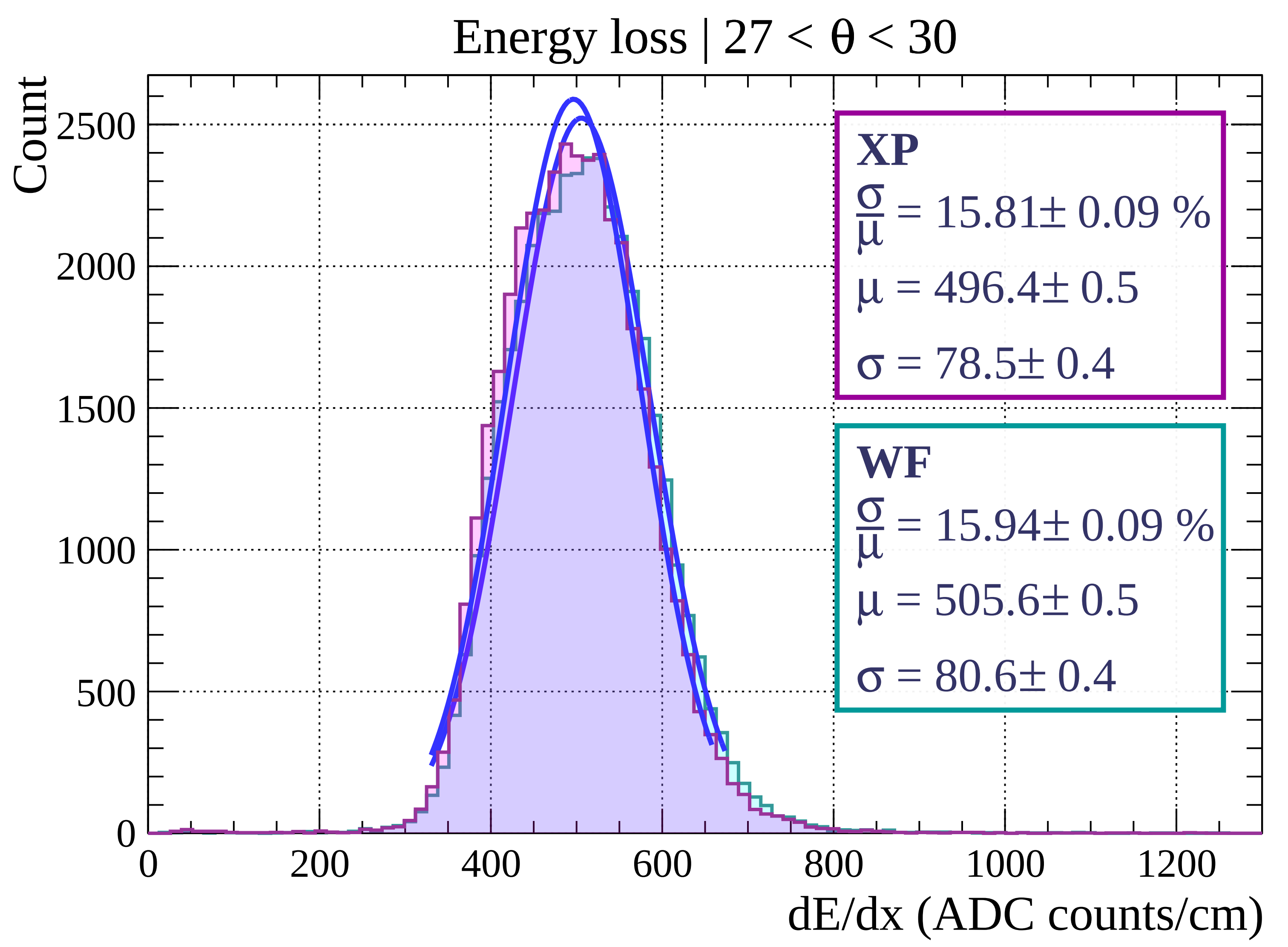
Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
0
1000
2000
3000
4000
5000
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.46± 0.07 %
μ = 488.4± 0.3
σ = 75.5± 0.3
WF
$\frac{\sigma}{\mu}$ = 15.68± 0.07 %
μ = 497.1± 0.3
σ = 77.9± 0.3



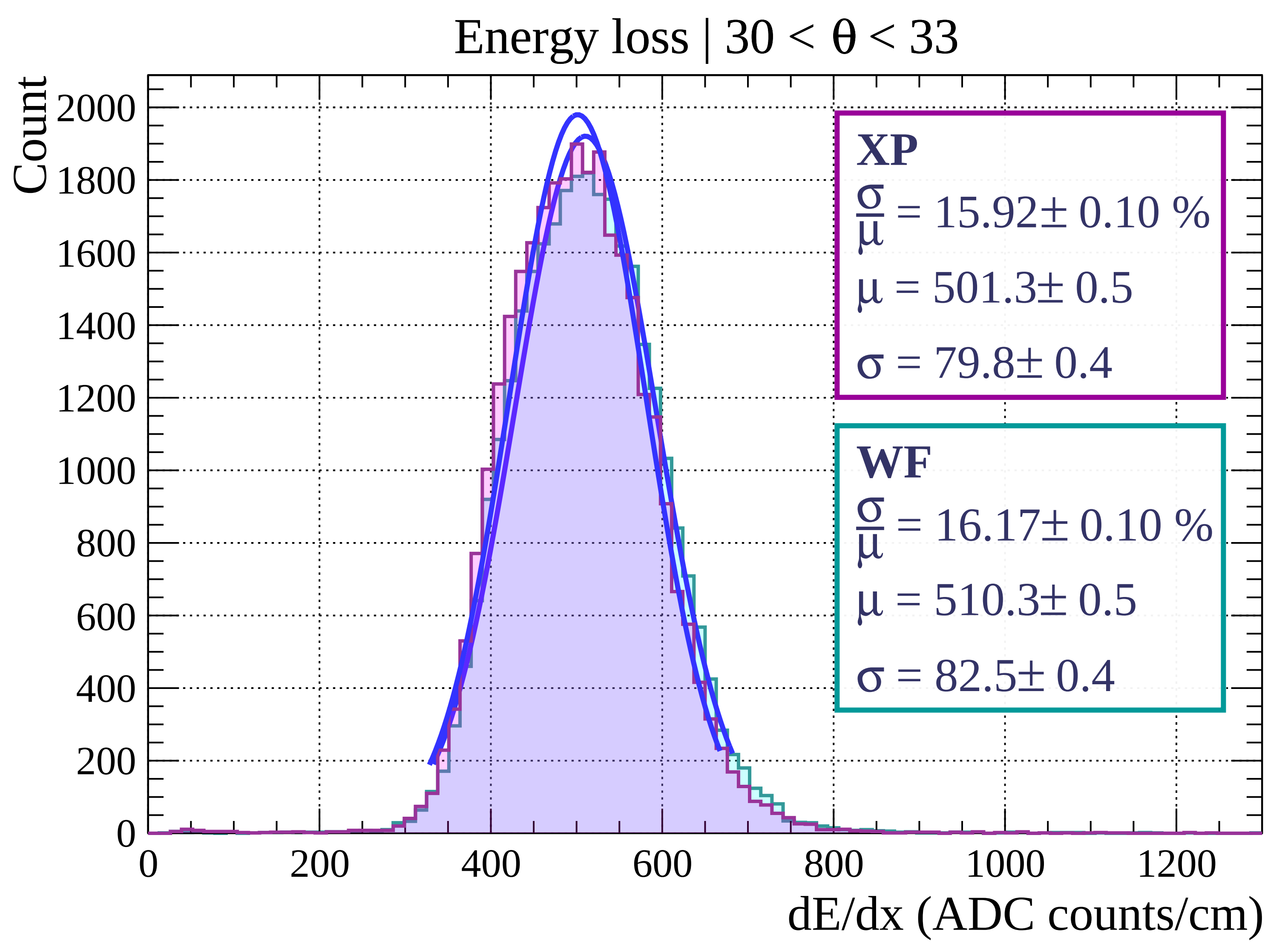
Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.52± 0.07 %
μ = 491.0± 0.4
σ = 76.2± 0.3
WF
σ/μ = 15.66± 0.07 %
μ = 499.6± 0.4
σ = 78.2± 0.3
0
500
1000
1500
2000
2500
3000
3500
4000
0
200
400
600
800
1000
1200

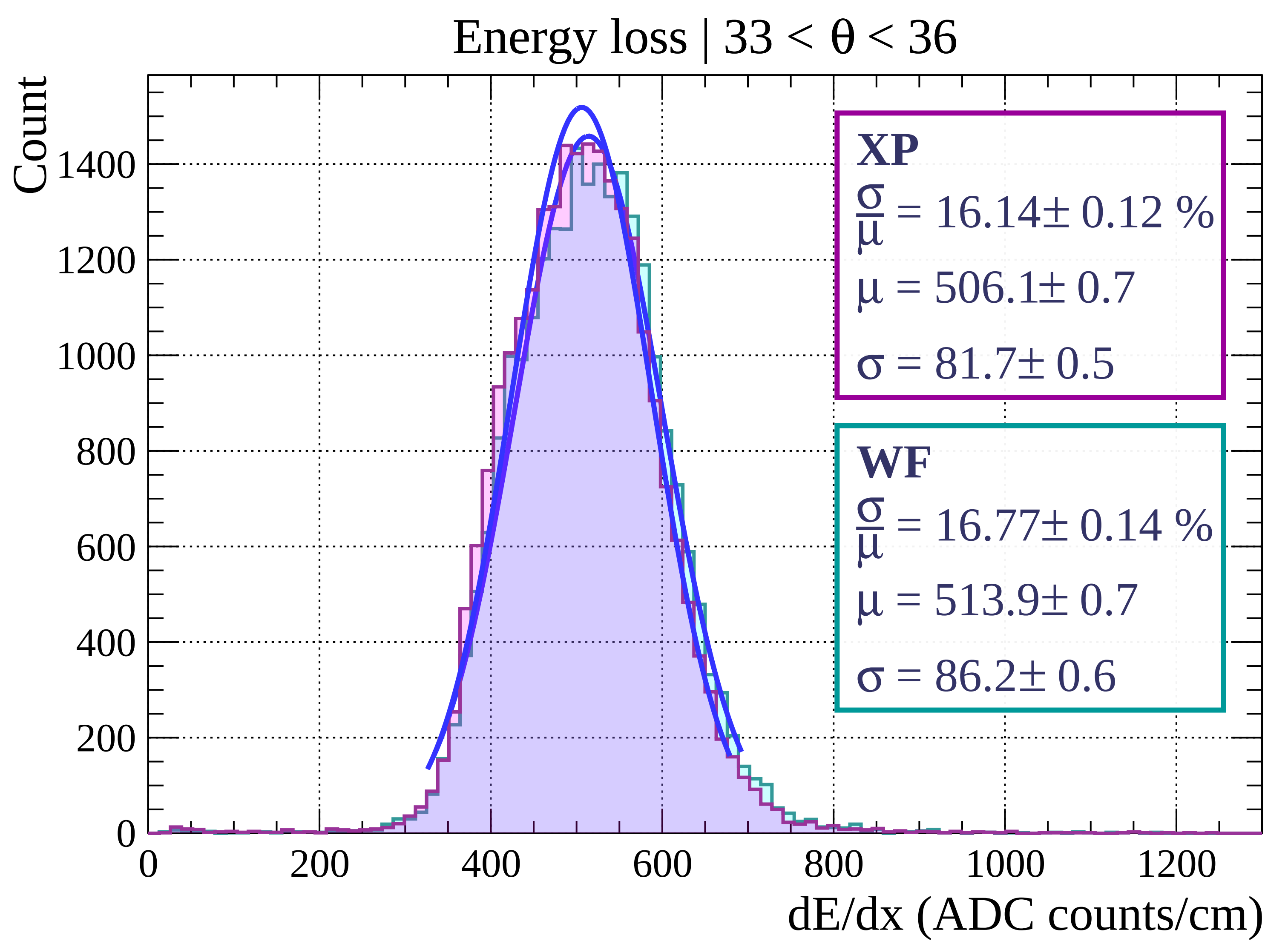


Energy loss | 24 < θ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
XP
σ/μ = 15.65± 0.08 %
μ = 493.3± 0.4
σ = 77.2± 0.3
WF
σ/μ = 15.77± 0.08 %
μ = 502.0± 0.4
σ = 79.2± 0.3

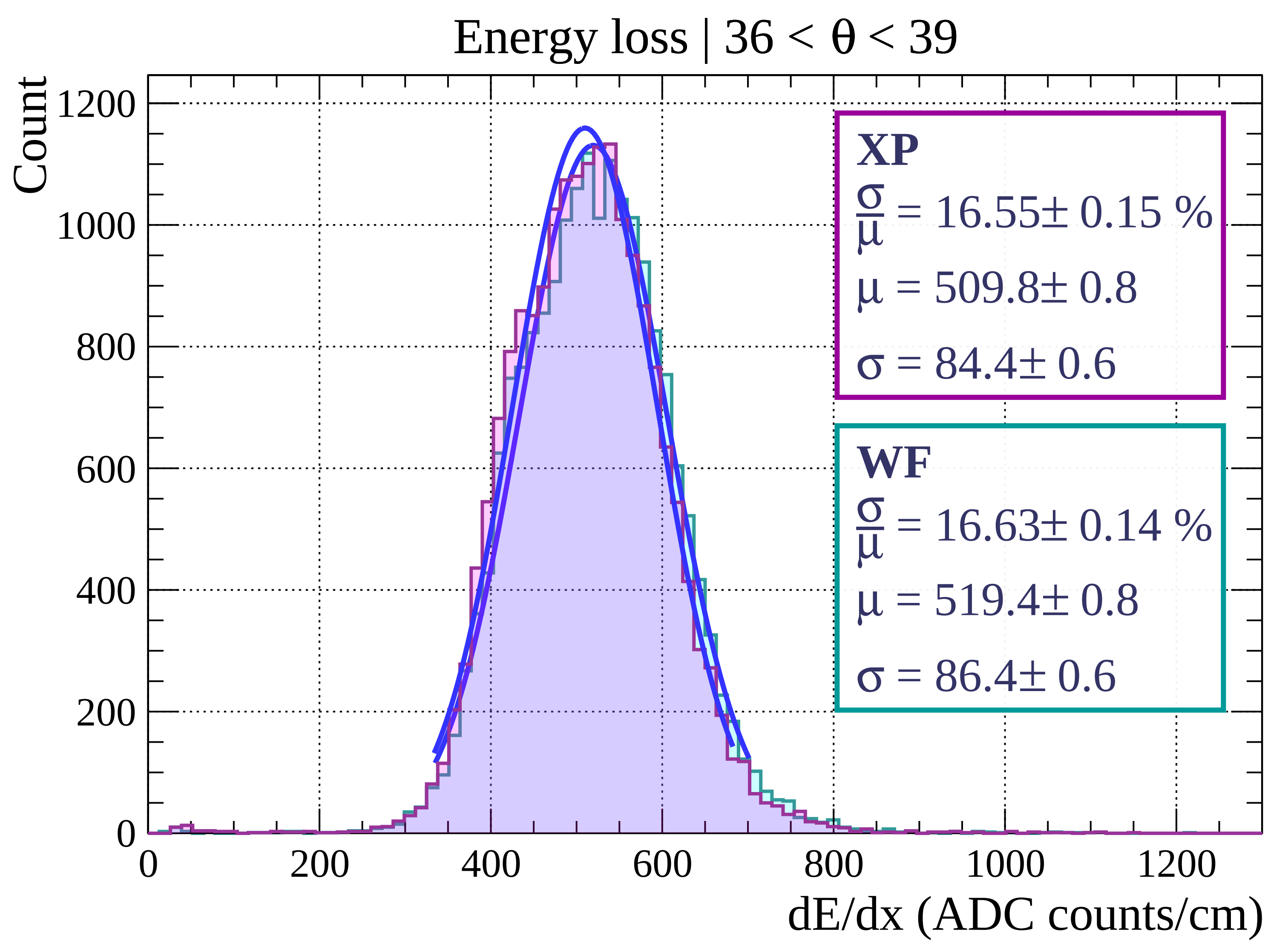


Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
0
200
400
600
800
1000
1200
XP
σ/μ = 15.81± 0.09 %
μ = 496.4± 0.5
σ = 78.5± 0.4
WF
σ/μ = 15.94± 0.09 %
μ = 505.6± 0.5
σ = 80.6± 0.4

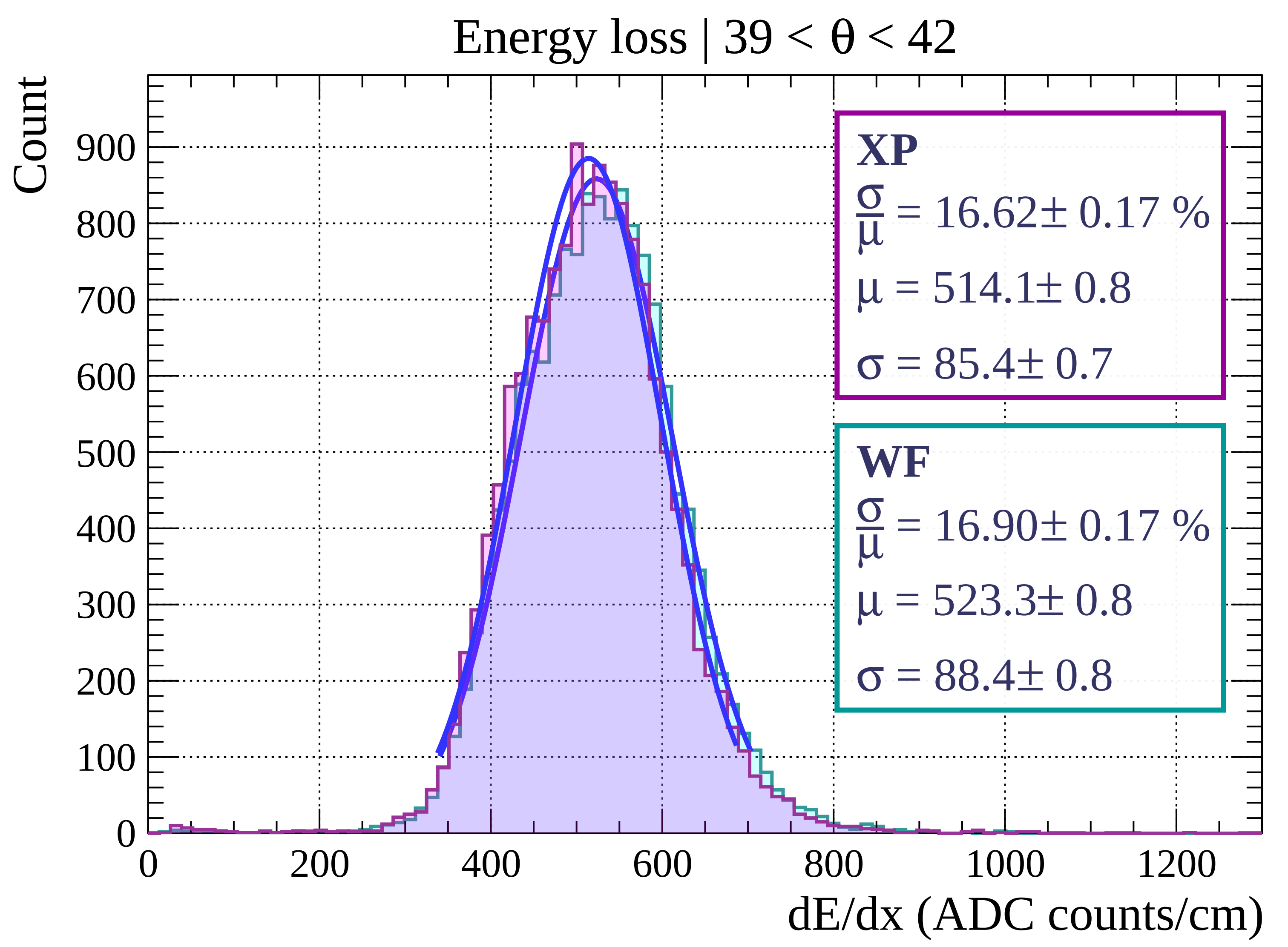

Energy loss | 30 < θ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
XP
σ/μ = 15.92± 0.10 %
μ = 501.3± 0.5
σ = 79.8± 0.4
WF
σ/μ = 16.17± 0.10 %
μ = 510.3± 0.5
σ = 82.5± 0.4

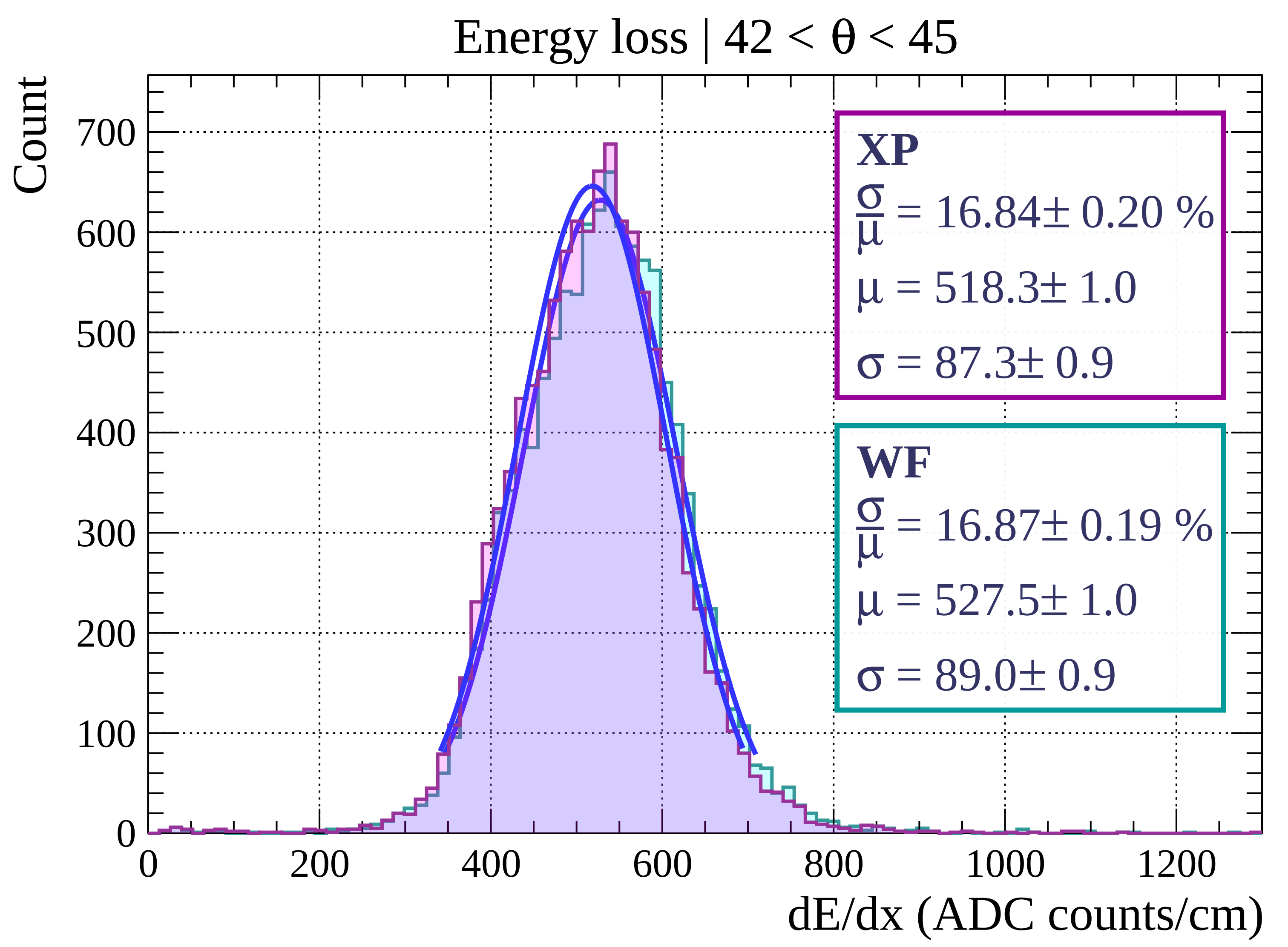


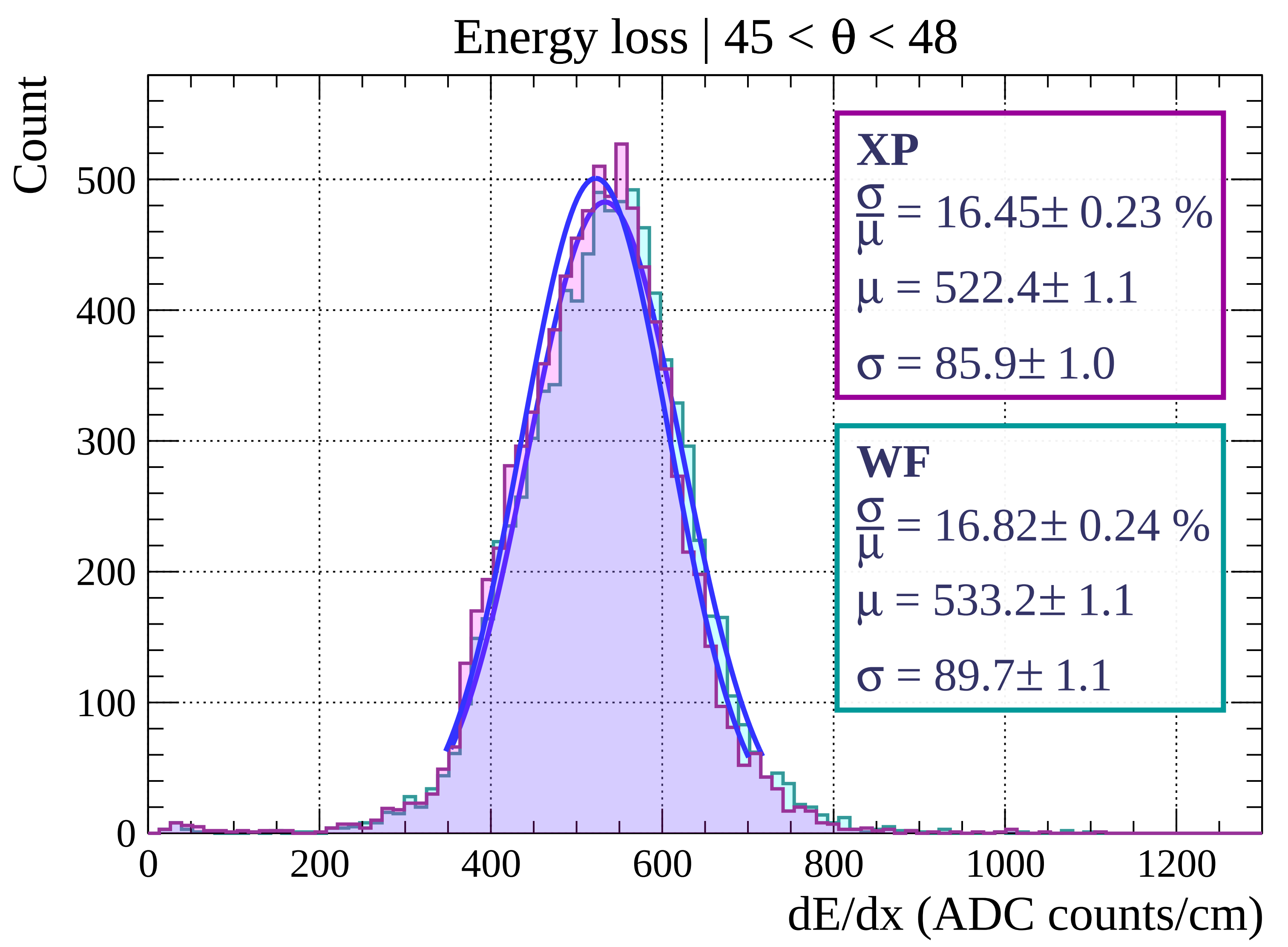
Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP
σ/μ = 16.14± 0.12 %
μ = 506.1± 0.7
σ = 81.7± 0.5
WF
σ/μ = 16.77± 0.14 %
μ = 513.9± 0.7
σ = 86.2± 0.6



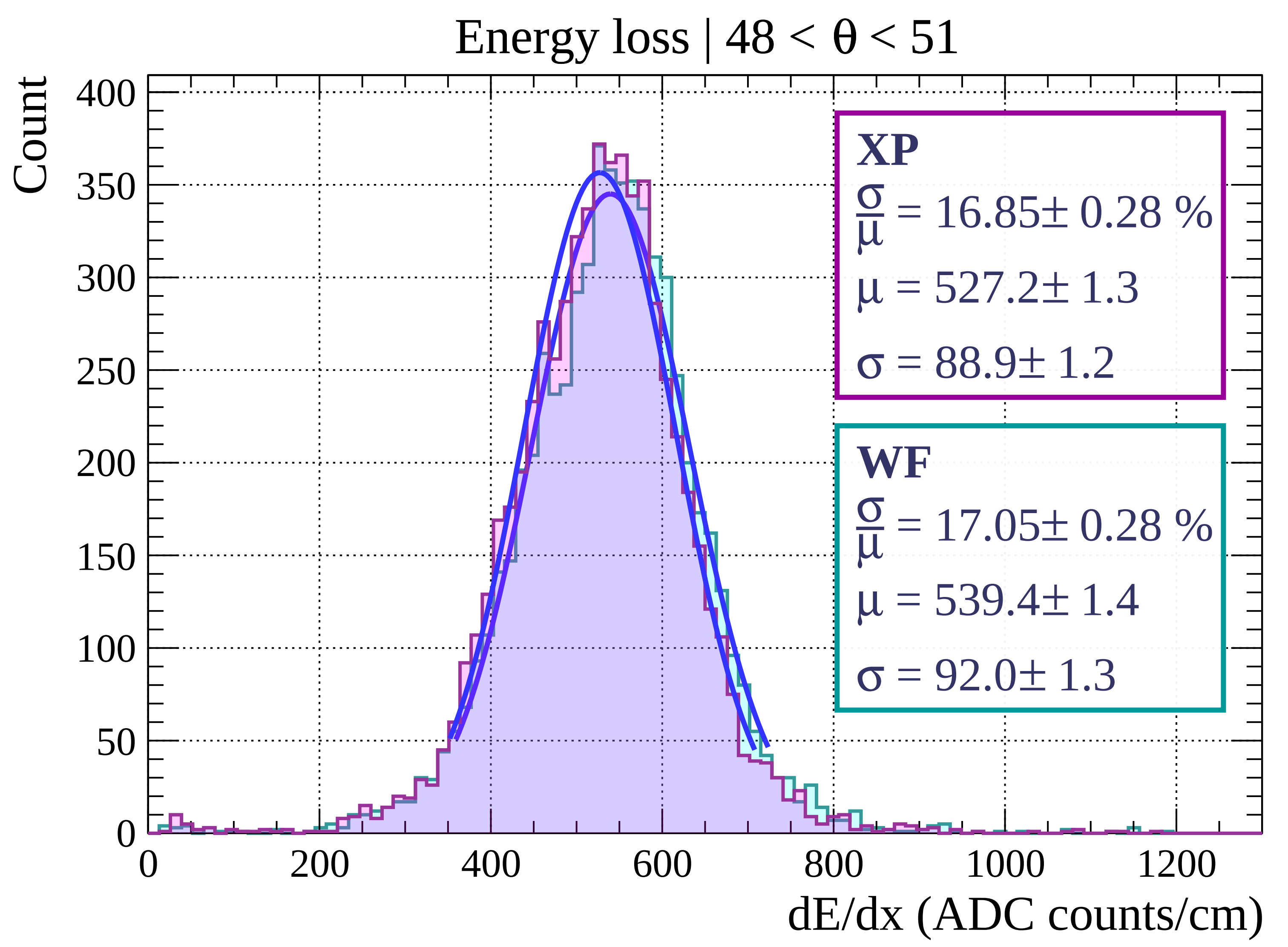
Energy loss | 36 < θ < 39
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.55± 0.15 %
μ = 509.8± 0.8
σ = 84.4± 0.6
WF
σ/μ = 16.63± 0.14 %
μ = 519.4± 0.8
σ = 86.4± 0.6
0
200
400
600
800
1000
1200


Energy loss | 39 < θ < 42
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 16.62± 0.17 %
μ = 514.1± 0.8
σ = 85.4± 0.7
WF
σ/μ = 16.90± 0.17 %
μ = 523.3± 0.8
σ = 88.4± 0.8

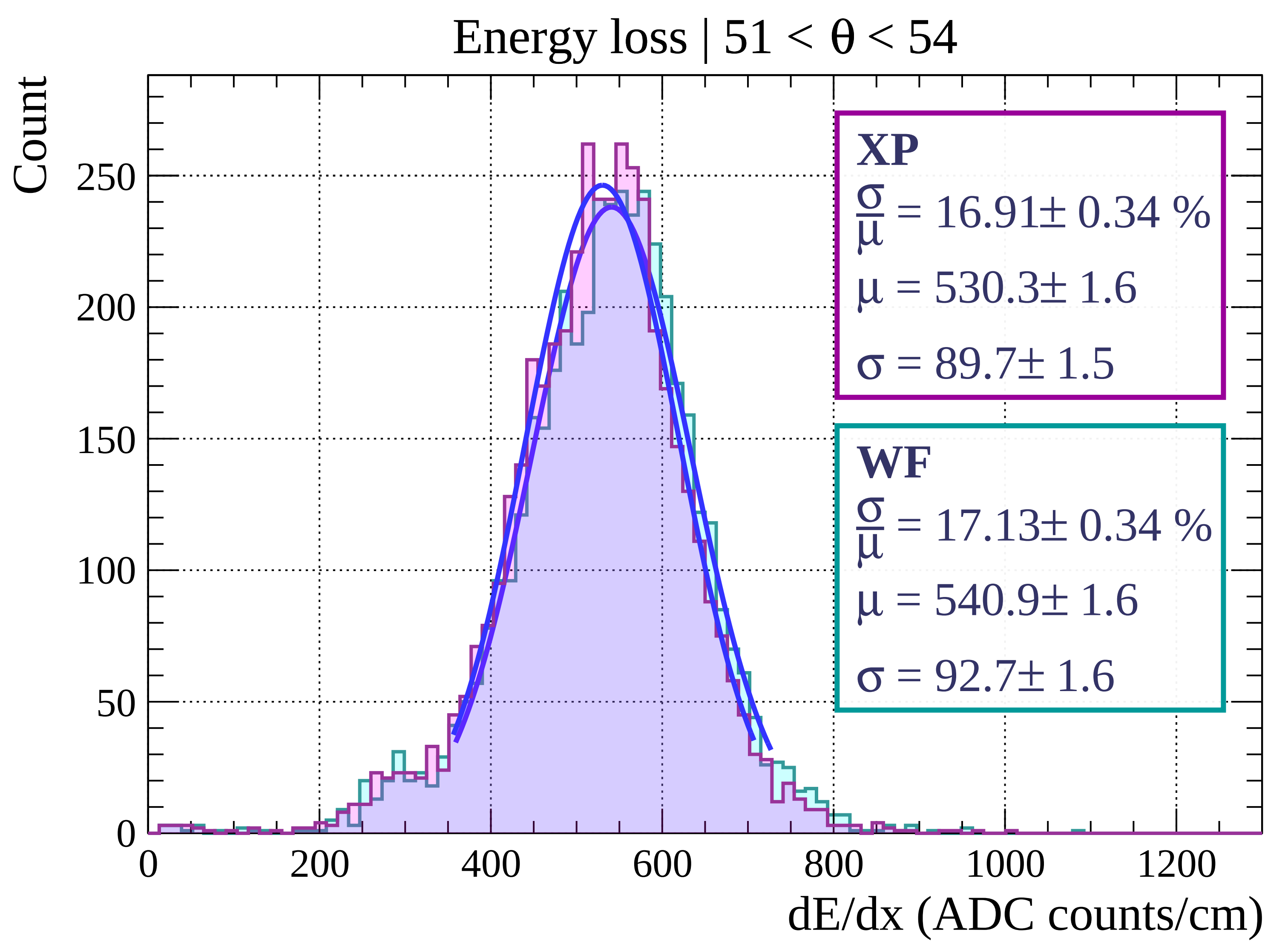

Energy loss | 42 < θ < 45
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.84± 0.20 %
μ = 518.3± 1.0
σ = 87.3± 0.9
WF
σ/μ = 16.87± 0.19 %
μ = 527.5± 1.0
σ = 89.0± 0.9
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200



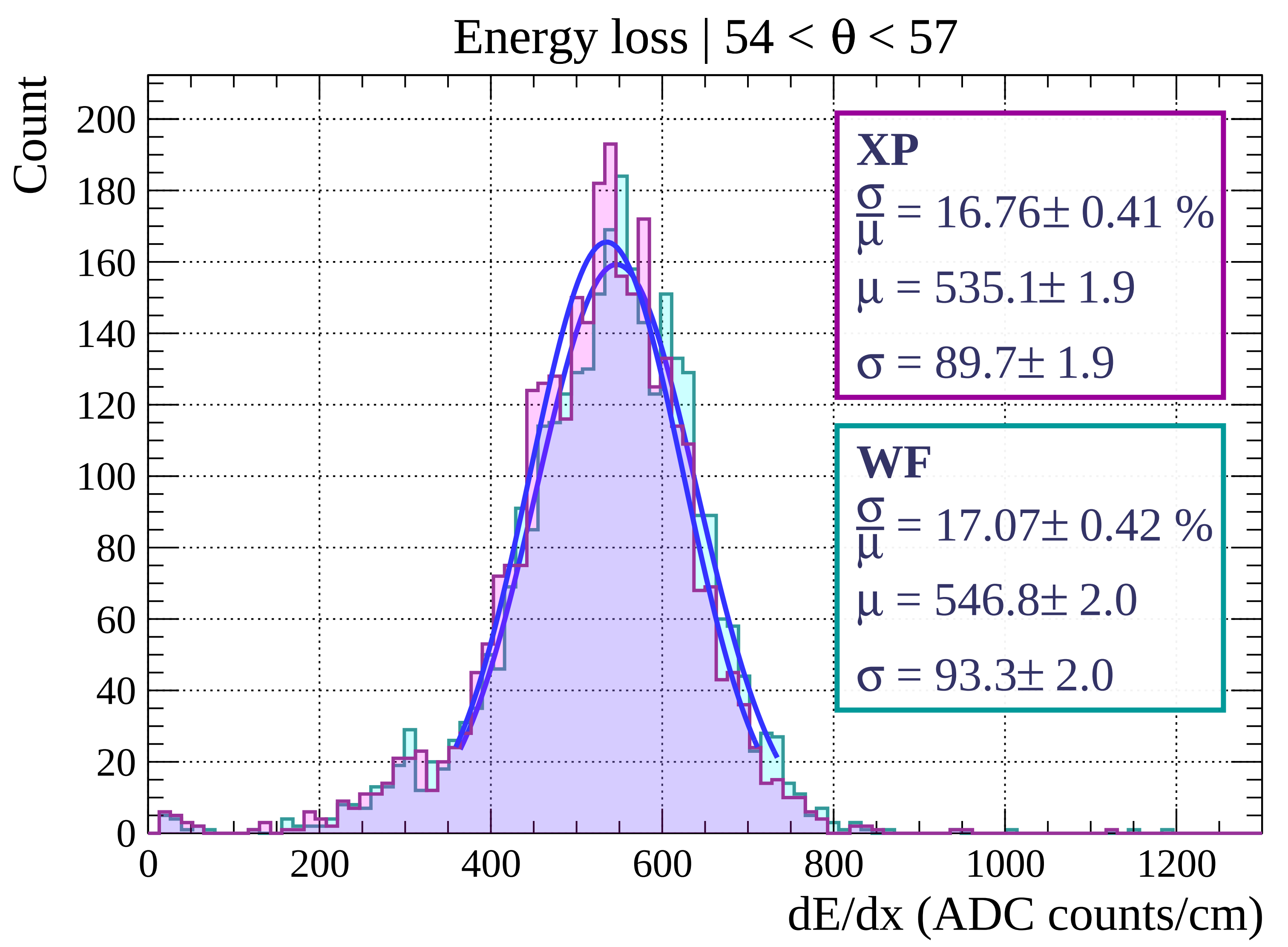
Energy loss | 45 < θ < 48
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 16.45± 0.23 %
μ = 522.4± 1.1
σ = 85.9± 1.0
WF
σ/μ = 16.82± 0.24 %
μ = 533.2± 1.1
σ = 89.7± 1.1



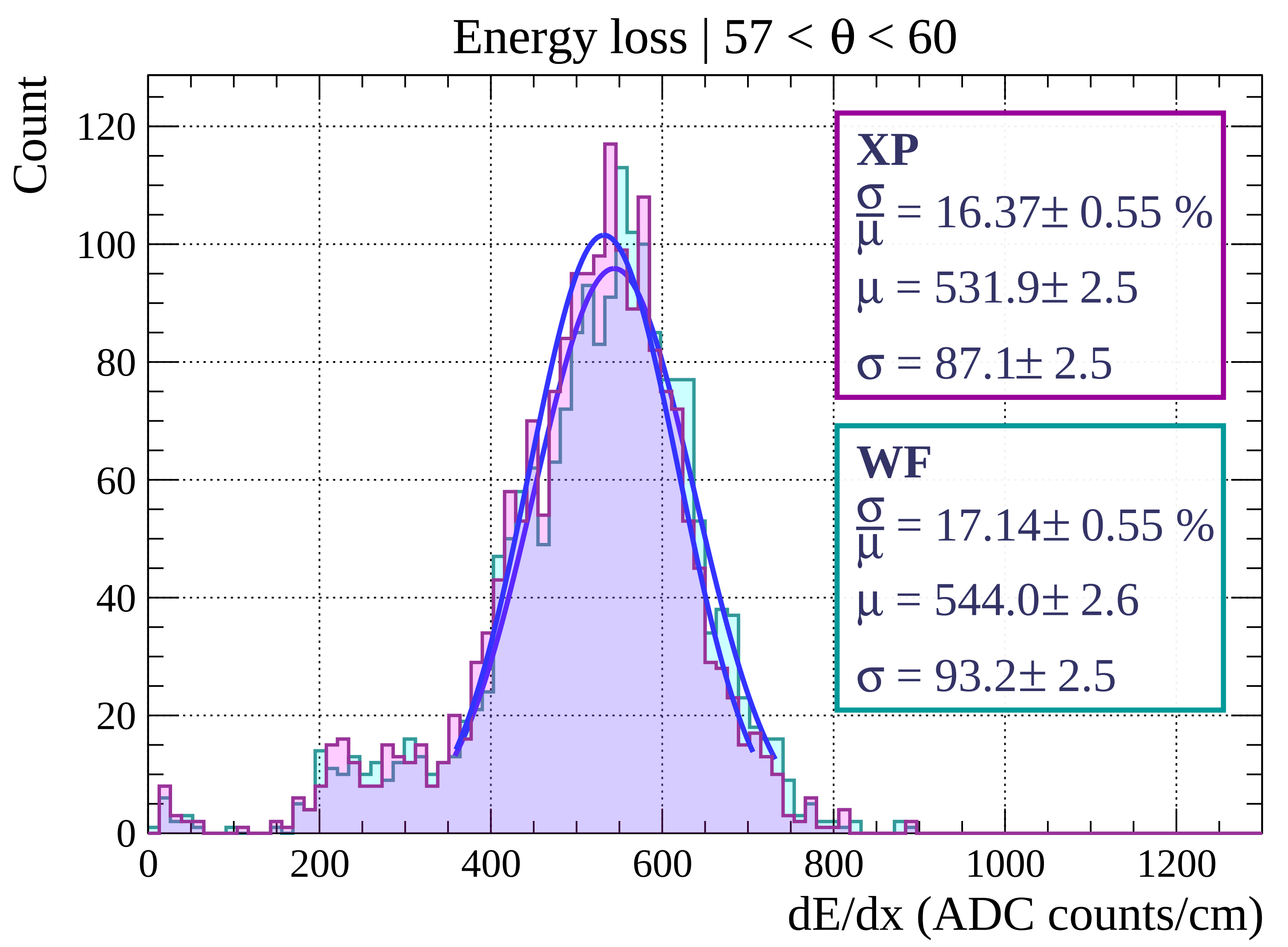
Energy loss | 48 < θ < 51
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.85± 0.28 %
μ = 527.2± 1.3
σ = 88.9± 1.2
WF
σ/μ = 17.05± 0.28 %
μ = 539.4± 1.4
σ = 92.0± 1.3



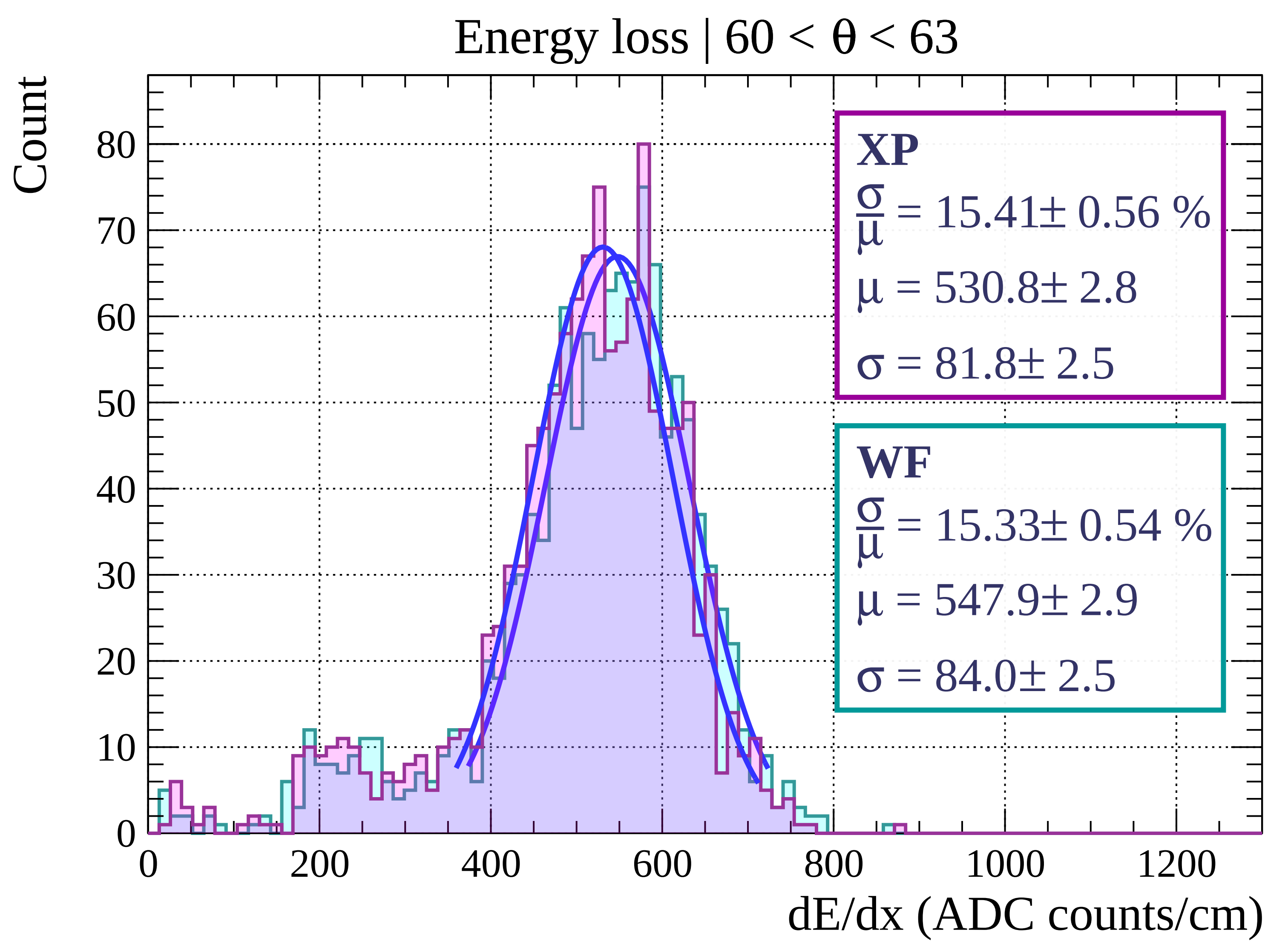
Energy loss | 51 < θ < 54
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.91± 0.34 %
μ = 530.3± 1.6
σ = 89.7± 1.5
WF
σ/μ = 17.13± 0.34 %
μ = 540.9± 1.6
σ = 92.7± 1.6



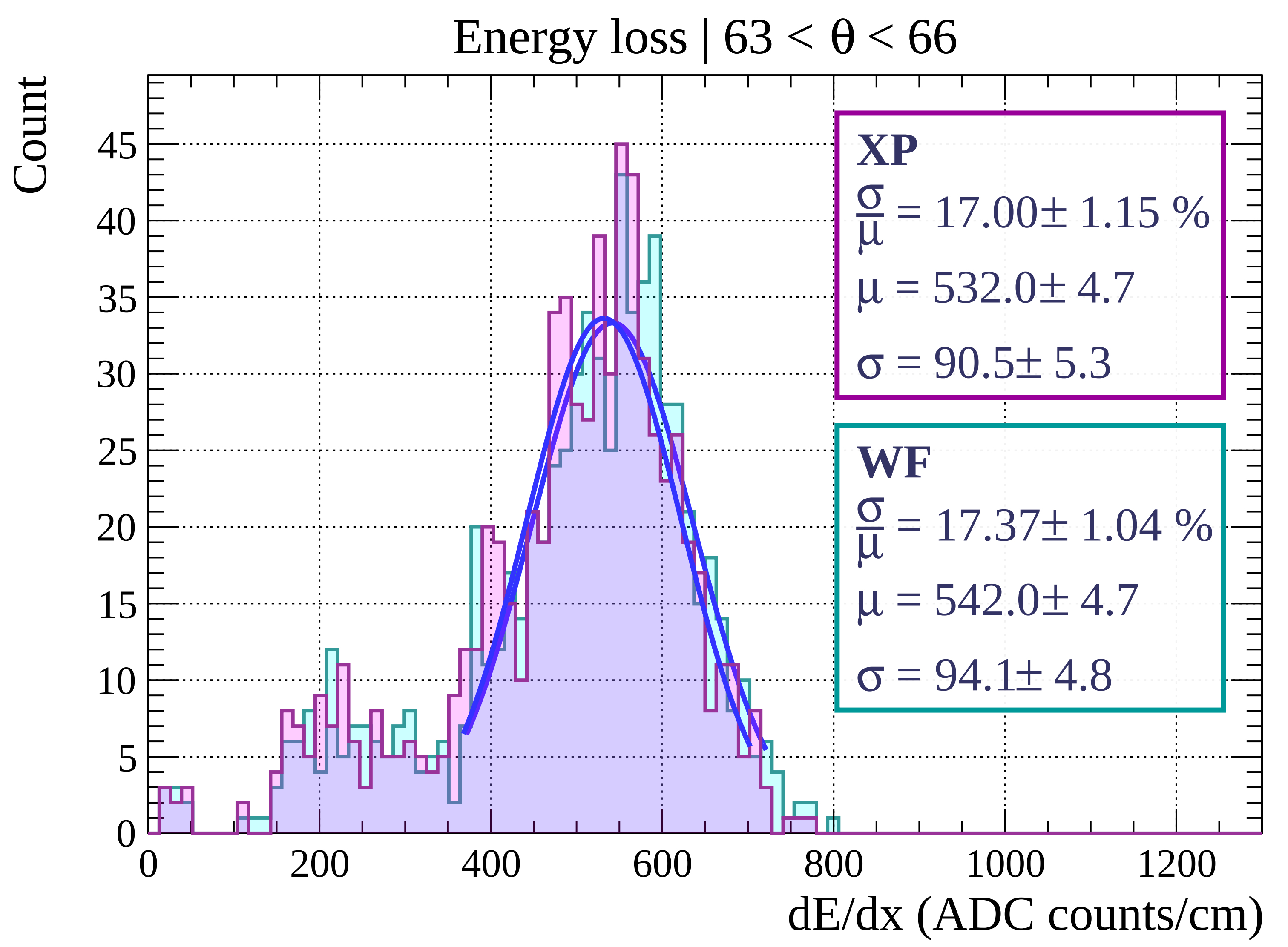
Energy loss | 54 < θ < 57
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.76± 0.41 %
μ = 535.1± 1.9
σ = 89.7± 1.9
WF
σ/μ = 17.07± 0.42 %
μ = 546.8± 2.0
σ = 93.3± 2.0

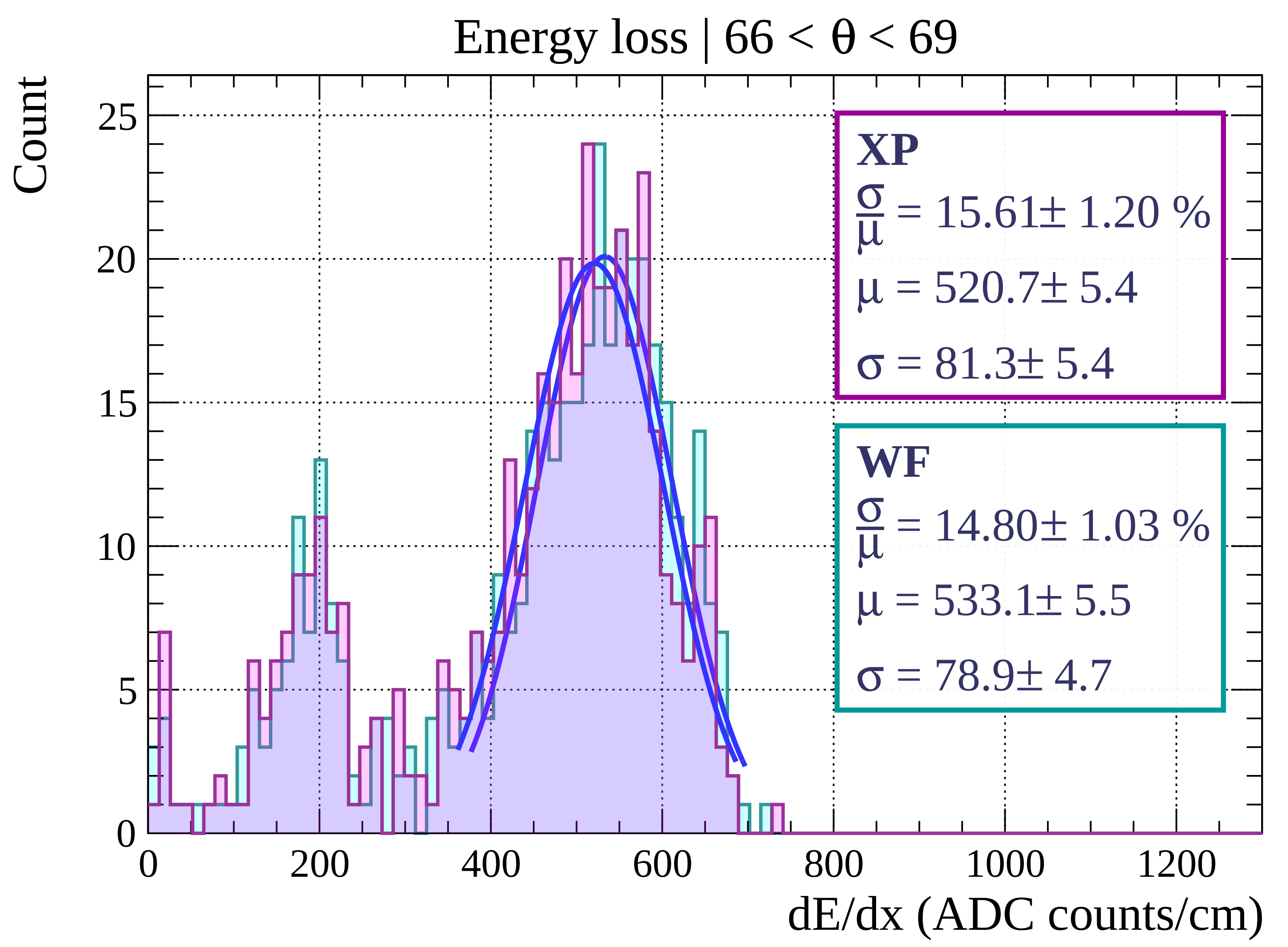


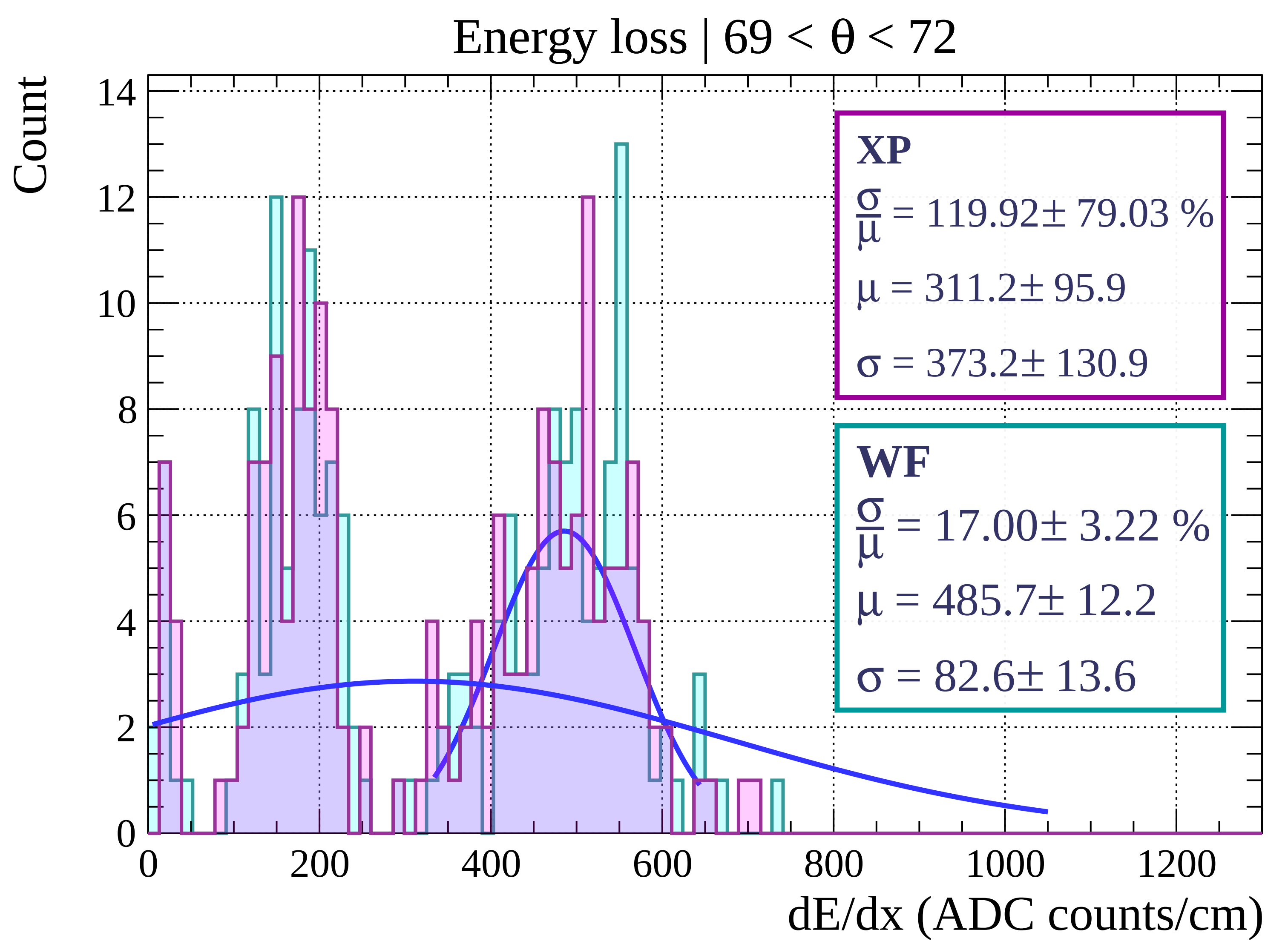
Energy loss | 57 < θ < 60
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 16.37± 0.55 %
μ = 531.9± 2.5
σ = 87.1± 2.5
WF
σ/μ = 17.14± 0.55 %
μ = 544.0± 2.6
σ = 93.2± 2.5



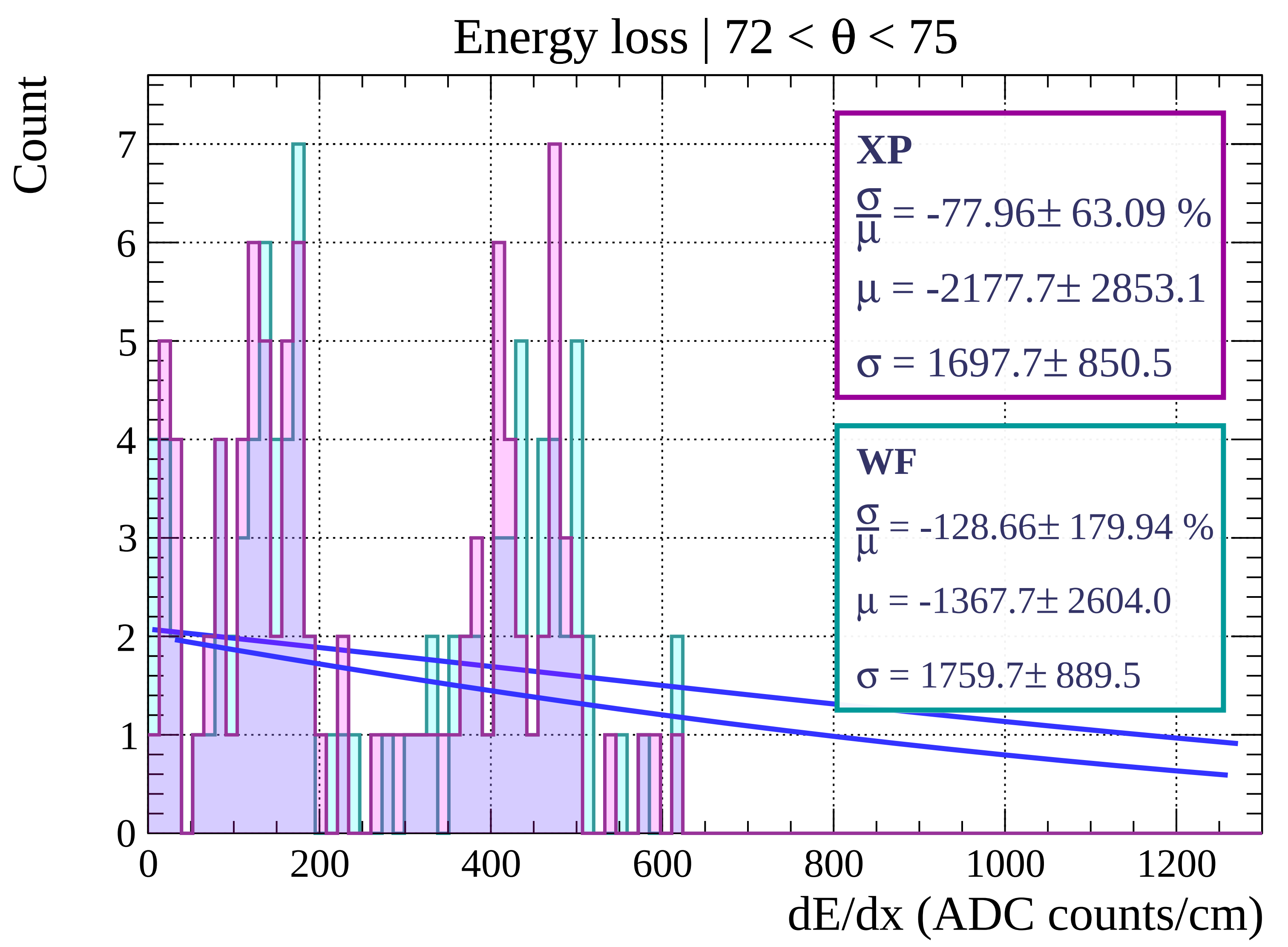
Energy loss | 60 < θ < 63
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
σ/μ = 15.41± 0.56 %
μ = 530.8± 2.8
σ = 81.8± 2.5
WF
σ/μ = 15.33± 0.54 %
μ = 547.9± 2.9
σ = 84.0± 2.5


Energy loss | 63 < θ < 66
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 17.00± 1.15 %
μ = 532.0± 4.7
σ = 90.5± 5.3
WF
σ/μ = 17.37± 1.04 %
μ = 542.0± 4.7
σ = 94.1± 4.8

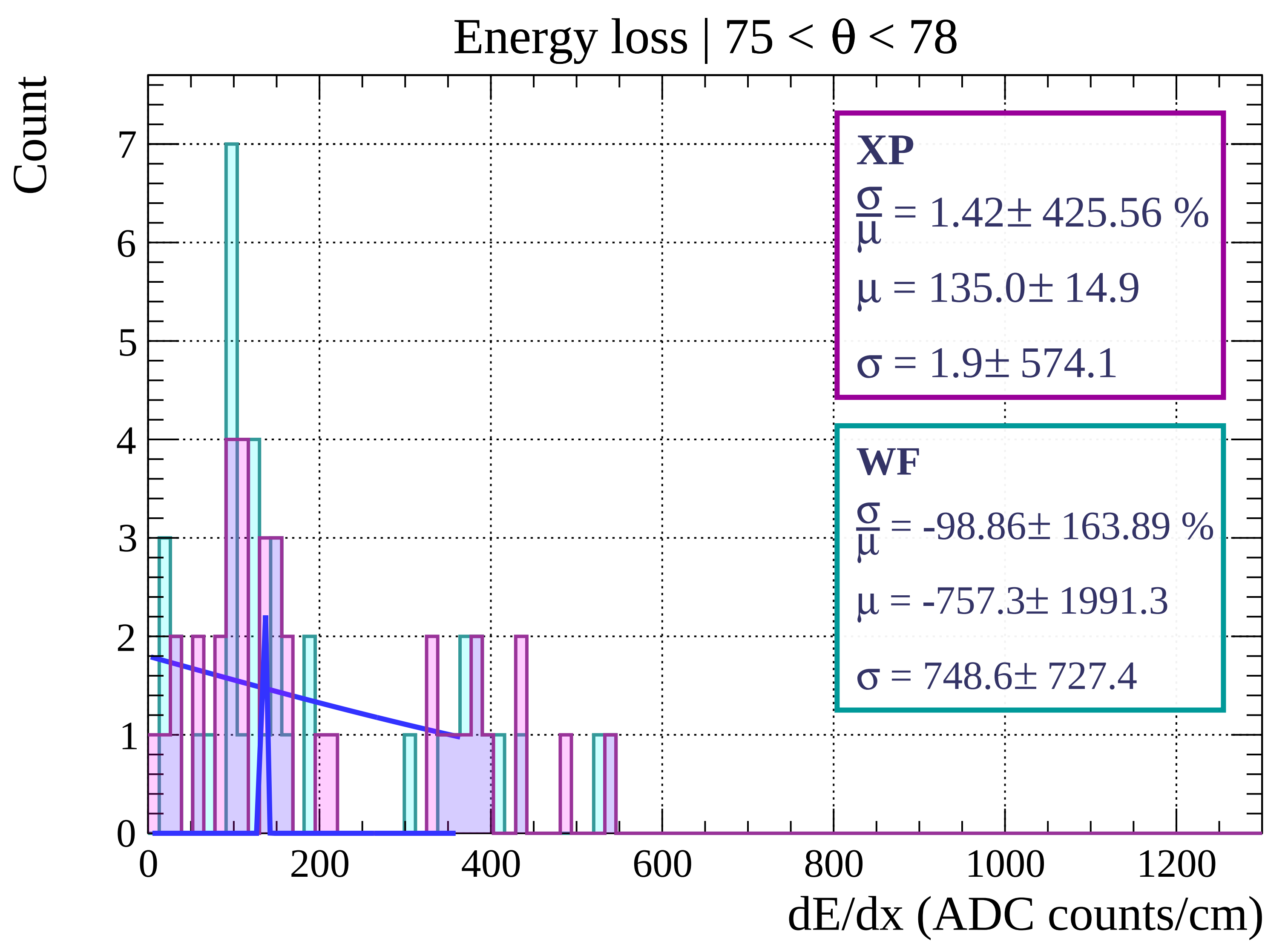

Energy loss | 66 < θ < 69
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
σ/μ = 15.61± 1.20 %
μ = 520.7± 5.4
σ = 81.3± 5.4
WF
σ/μ = 14.80± 1.03 %
μ = 533.1± 5.5
σ = 78.9± 4.7



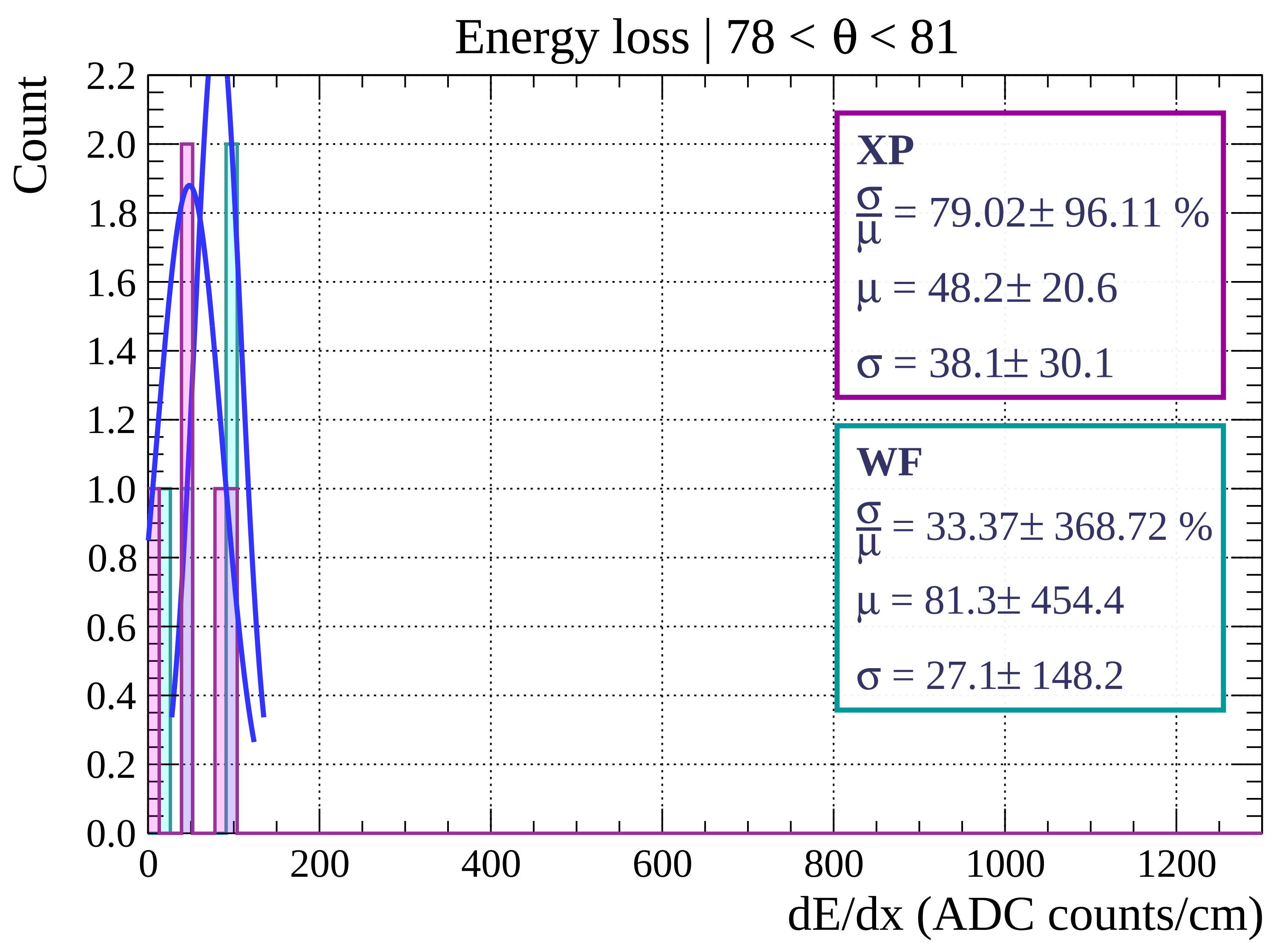
Energy loss | 69 < θ < 72
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
XP
σ/μ = 119.92± 79.03 %
μ = 311.2± 95.9
σ = 373.2± 130.9
WF
σ/μ = 17.00± 3.22 %
μ = 485.7± 12.2
σ = 82.6± 13.6



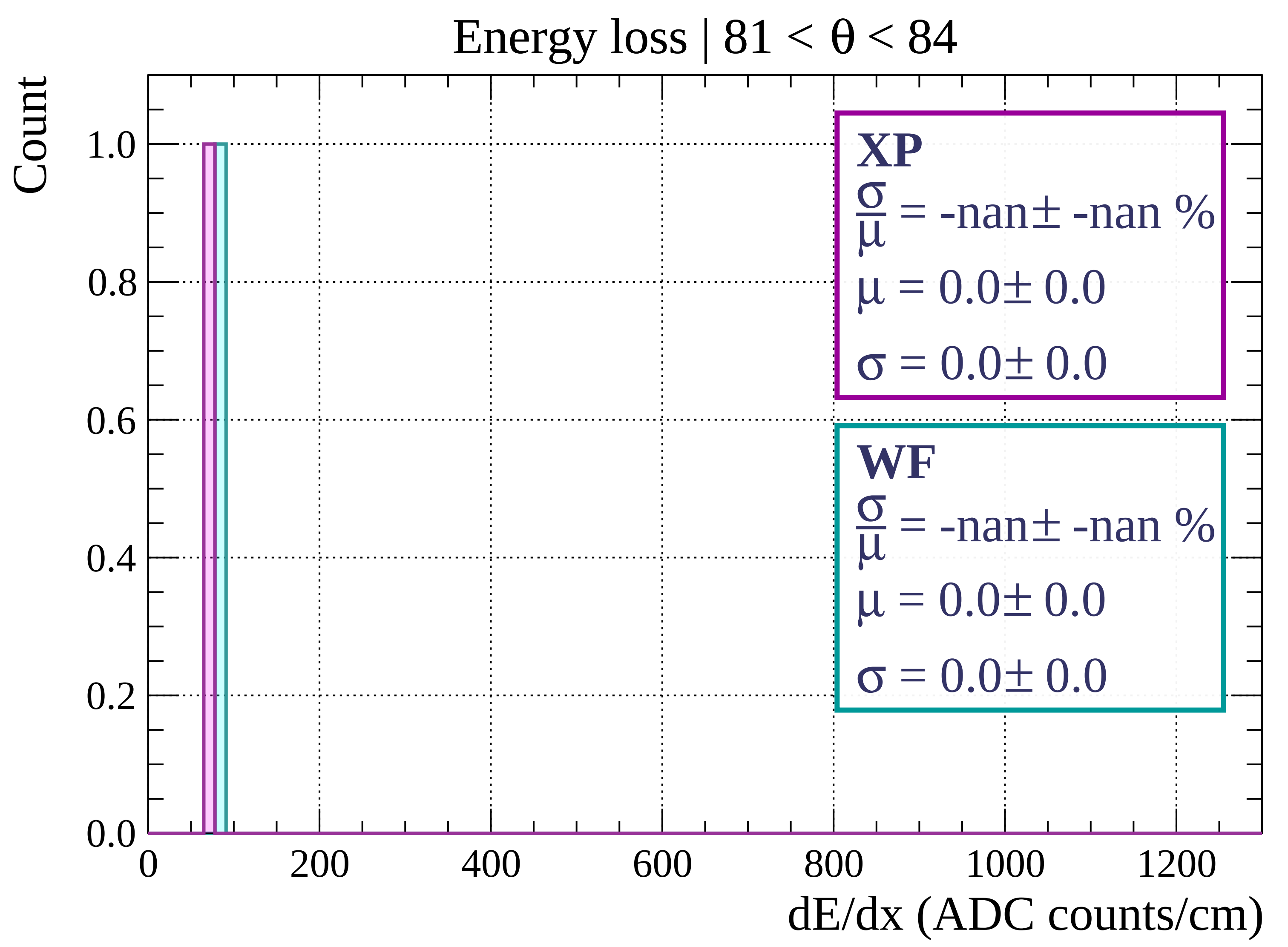
Energy loss | 72 < θ < 75
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -77.96± 63.09 %
μ = -2177.7± 2853.1
σ = 1697.7± 850.5
WF
$\frac{\sigma}{\mu}$ = -128.66± 179.94 %
μ = -1367.7± 2604.0
σ = 1759.7± 889.5

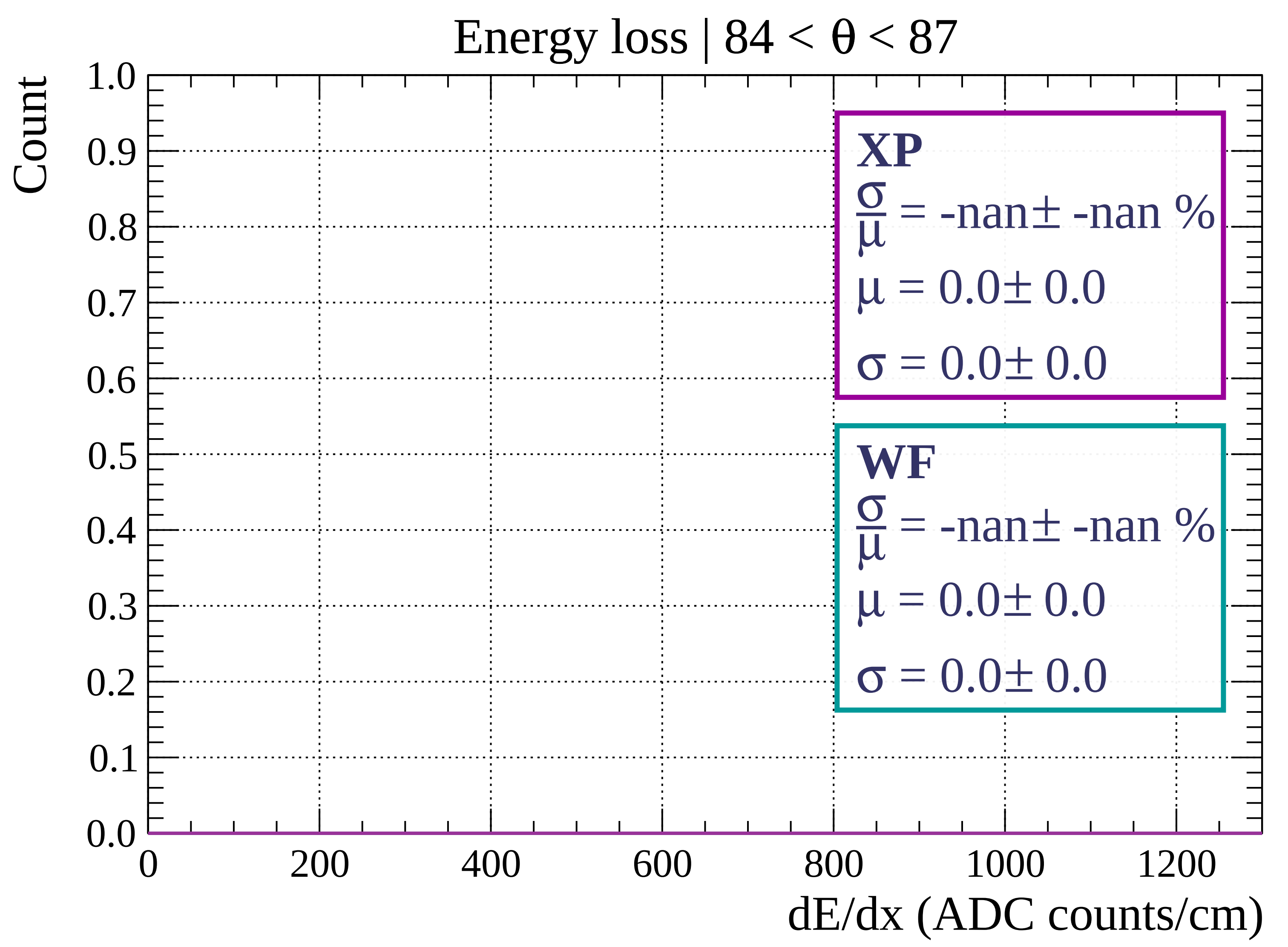


Energy loss | 75 < θ < 78
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
σ/μ = 1.42± 425.56 %
μ = 135.0± 14.9
σ = 1.9± 574.1
WF
σ/μ = -98.86± 163.89 %
μ = -757.3± 1991.3
σ = 748.6± 727.4

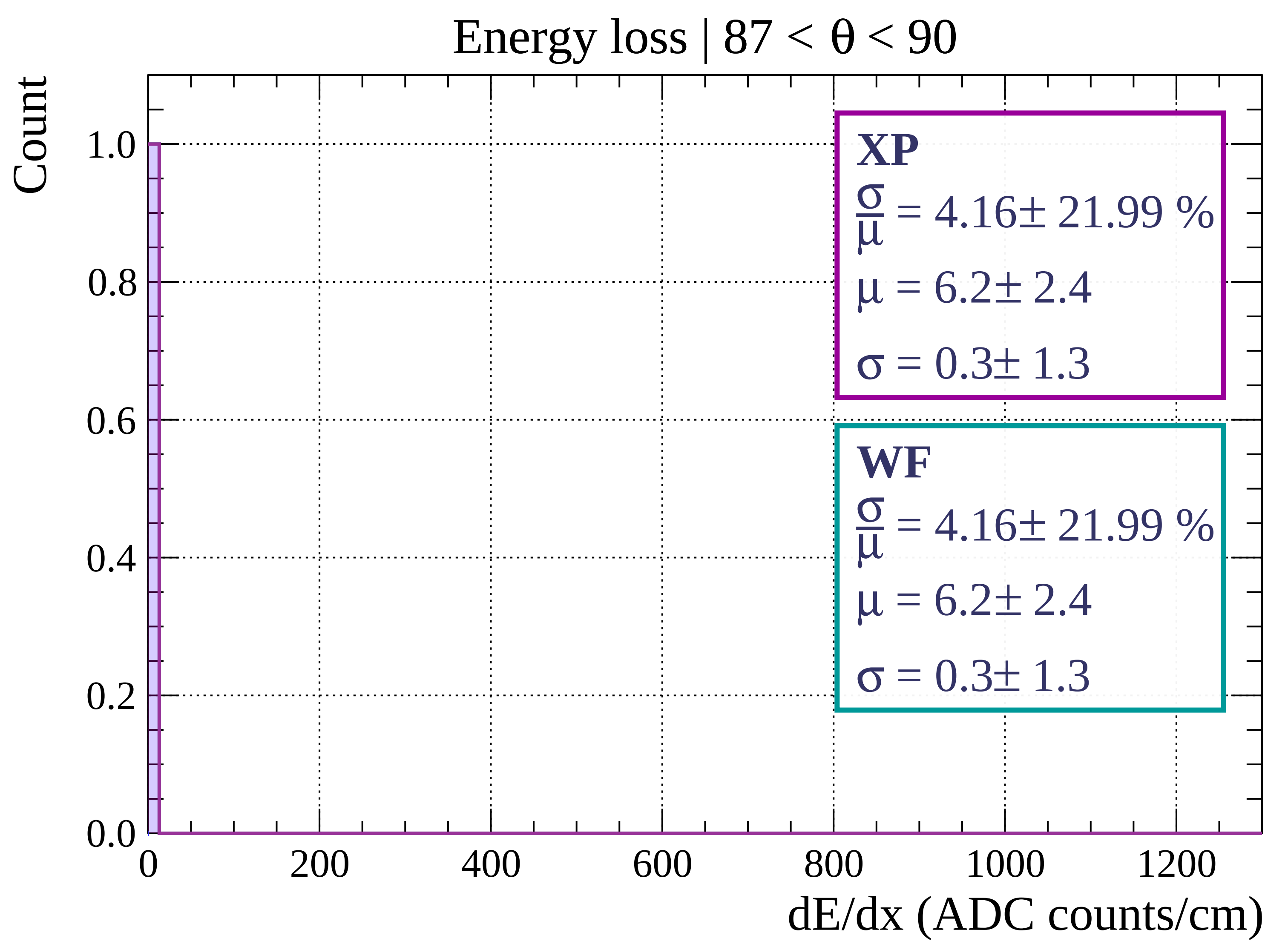


Energy loss | 78 < θ < 81
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 79.02± 96.11 %
μ = 48.2± 20.6
σ = 38.1± 30.1
WF
σ/μ = 33.37± 368.72 %
μ = 81.3± 454.4
σ = 27.1± 148.2

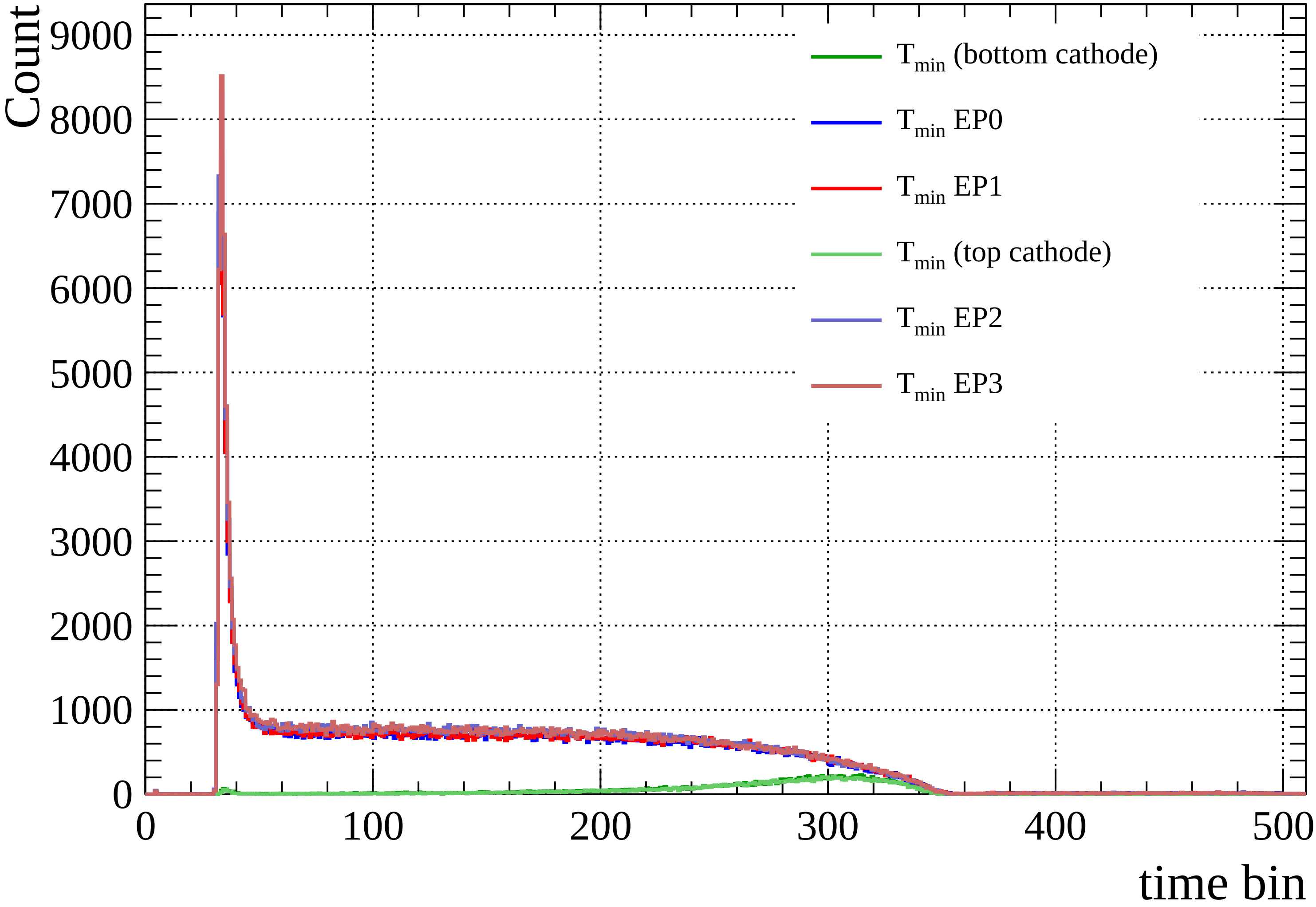

Energy loss | 81 < θ < 84
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



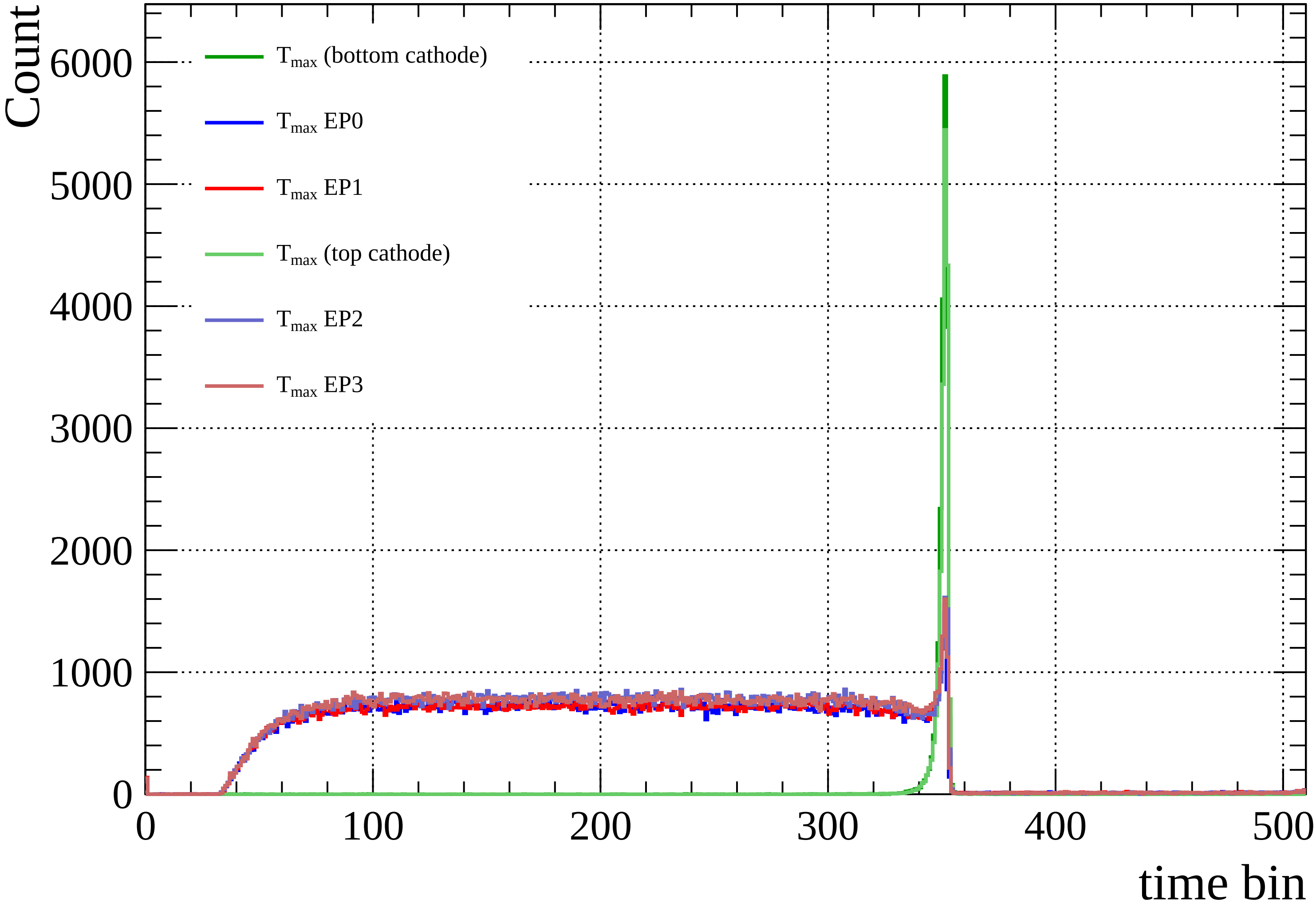
Energy loss | 84 < θ < 87
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 87 < θ < 90
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 4.16± 21.99 %
μ = 6.2± 2.4
σ = 0.3± 1.3
WF
$\frac{\sigma}{\mu}$ = 4.16± 21.99 %
μ = 6.2± 2.4
σ = 0.3± 1.3



Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
1000
2000
3000
4000
5000
6000
0
100
200
300
400
500